

PERSPECTIVES ON INVARIANT THEORY: SCHUR DUALITY, MULTIPLICITY-FREE ACTIONS AND BEYOND

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Thanks again, Ilya

Abstract. Multiplicity-free actions have recently become appreciated in invariant theory as a class of actions which permit simple and detailed analysis. The role of multiplicity-free actions in classical invariant theory is studied, and is found to be pervasive. Skew-symmetric multiplicity-free actions are also considered. A variety of examples is given.

Prolog

The goal of this essay is to present the notion of *multiplicity-free action* as an organizing principle in invariant theory. We will try to show, first, that the main theorems of Classical Invariant Theory, as formulated for example in Weyl's book [We], can be understood in terms of multiplicity-free actions. Whereas Classical Invariant Theory was concerned essentially exclusively with group actions on polynomial algebras, or, dually, symmetric algebras, it is now common to consider actions on general varieties, and we will consider multiplicity-free actions in this context. Likewise, though it is not yet standard to do so, there are now many reasons for considering the case of skew-symmetric variables, i.e., actions on exterior algebras, and we will also investigate which of these are multiplicity-free. We also wish to show that the notion of multiplicity-free action is relevant to more than just the Fundamental Theorems of invariant theory. We will cite and explain briefly a variety of examples from the more recent literature, and we treat a few in some detail. Among the examples considered are the "branching laws" describing

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the restrictions of representations from GL_n to GL_{n-1} , or O_n to O_{n-1} [Pr1], [Zb]; Schwarz's Fundamental Theorems for G_2 and $Spin_7$ [Schw 1,2]; an analog due to R. Huang [Hn] of the First Fundamental Theorem for the variety of lines in 3-space; the invariants of 4 arbitrary subspaces of a vector space; the de Rham cohomology and invariant differential operators on classical symmetric spaces [He3]; various "stability theorems" from the recent literature; and an analysis of several actions of particular interest. We also give applications to the problem of decomposing tensor products.

This paper is intended to complement the author's earlier work [Hw 1]. That paper took the Fundamental Theorems of Classical Invariant Theory for granted, and was concerned with reinterpreting them and extending their scope. This essay, while concerned with further extending the range of application of the Fundamental Theorems of Classical Invariant Theory, is also concerned with foundational questions, and in particular provides new proofs of the Fundamental Theorems, along with a discussion of the relation between the classical (meaning Weyl's) approach to the theory and the one presented here. Our point of view is that the highest weight theory of representations of reductive groups has been established as a standard piece of mathematics, and it is desirable to base invariant theory on it. The theory of multiplicity-free actions seems to be an optimal tool for accomplishing this. We note in particular that the Capelli Identity, Weyl's heavy use of which bothered some readers, is completely unneeded here. On the contrary, we are able to give a simple interpretation of the role it played in Weyl's treatment (see §2.7). (We have given elsewhere [Hw1][Hu] a conceptual interpretation of the identity itself.)

We should note that other authors have recently given treatments of the foundations of invariant theory, or parts of it [Kr], [Vu], [Po], [GKR]. The account of Zhelobenko [Zb], though not specifically concerned with invariant theory, in particular foreshadows this one. We hope that the reader will find here more than just a repetition of the discussion of those papers.

We should also note that a vigorous study and classification of multiplicity-free actions has developed over the last 15 years [Krm], [Br1], [Br2], [BLV], [VK], [GS], [Kn], etc. We do not attempt to review that literature here. Rather, we emphasize examples, and wish to show how the theory of multiplicity-free actions can elucidate situations which are not multiplicity-free or do not appear to be so on the surface.

My hope at the start of this project was to give an essentially elementary, self-contained treatment of this topic, complete at least in its basic aspects. I now feel I have failed on all counts: the extent of background knowledge needed was too great, and new avenues and new examples to be examined kept presenting themselves. I have tried to fill in some background material in appendices, but the paper is still not self-contained. Perhaps some readers will pursue the examples further.

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1: Basic Theory.

1.1: Highest weight theory.

We recall for the convenience of the reader the basic features of the highest weight theory of representations of reductive groups, in its geometric form. We begin with some basic notions of representation theory.

1.1.1: We hope the reader is familiar with the notion of representation of a group G . By representation we mean a homomorphism

$$\rho: G \rightarrow GL(V)$$

from G to the group of invertible matrices on some complex vector space V . We denote a representation by (ρ, V) , or simply by ρ , or by V , if the other element of the pair (ρ, V) is clear from the context.

1.1.2: We assume the following terminology is familiar: subrepresentation, quotient representation, irreducible representation, constituent or subquotient, composition series, direct sum of representations, intertwining operator or G -mapping or G -morphism, equivalence of representations, isotypic subspace, multiplicity of an irreducible representation in a reducible one, contragredient representation, tensor product of representations.

1.1.3: We also assume that the reader is familiar with the basic facts about the symmetric algebra of a vector space (cf. [Jac 2], [La]). We review them briefly. The symmetric algebra of a vector space V is the free commutative algebra generated by V . We denote it by $S(V)$. We denote the polynomial algebra on V (i.e., the algebra of polynomial functions on V) by $\mathcal{P}(V)$. The natural inclusion $V^* \subseteq \mathcal{P}(V)$ of the dual of V as linear functions on V defines an isomorphism

$$(1.1.3.1) \quad S(V^*) \simeq \mathcal{P}(V).$$

Each of $S(V)$ and $\mathcal{P}(V)$ has a standard grading:

$$(1.1.3.2) \quad S(V) \simeq \sum_{n \geq 0} S^n(V) \quad \mathcal{P}(V) \simeq \sum_{n \geq 0} \mathcal{P}^n(V).$$

The isomorphism (1.1.3.1) preserves homogeneous components. There is a natural pairing between $S(V)$ and $S(V^*)$ extending the pairing between V and V^* , and inducing isomorphisms

$$(1.1.3.3) \quad S^n(V^*) \simeq (S^n(V))^*.$$

If $S(V)$ is identified with the algebra of constant-coefficient differential operators on V (as explained in §A2.1.1), then this pairing is given by

$$(1.1.3.4) \quad (D, \rho) = D(\rho)(0), \quad D \in S(V), \rho \in \mathcal{P}(V).$$

The action of $GL(V)$ on V extends uniquely to an action of $GL(V)$ on $S(V)$ and on $\mathcal{P}(V)$ by graded algebra automorphisms. The pairing (1.1.3.4) is preserved by this action

In what follows, we have attempted to distinguish between $S(V)$ and $\mathcal{P}(V)$. However, the reader is warned to be on the lookout for places where they are confused.

1.1.4: The matrix coefficients of a representation are the functions

$$\varphi_{\lambda, v}(g) = \lambda(\rho(g)v), \quad v \in V, \lambda \in V^*, g \in G.$$

1.1.5: Matrix coefficients connect a general representation of G with specific representations on spaces of functions on G . For a function $f: G \rightarrow \mathbb{C}$, we define $R_h(f)$ and $L_h(f)$, the right and left translations of f by h , by the formulas

$$R_h(f)(g) = f(g h) \quad L_h(f)(g) = f(h^{-1}g).$$

The mappings

$$R: h \rightarrow R_h \quad \text{and} \quad L: h \rightarrow L_h$$

define representations of G on the space $\mathcal{C}(G)$ of all functions on G , and on various subspaces which are invariant for the R_h or the L_h . The representations R and L are called the right regular representation and the left regular representation respectively.

1.1.6: It is easy to verify the formulas

$$L_h(\varphi_{\lambda, v}) = \varphi_{\rho^*(h)(\lambda), v} \quad R_h(\varphi_{\lambda, \rho(h)(v)}).$$

These mean that the mappings

$$\Phi_\lambda: v \rightarrow \varphi_{\lambda, v} \quad \text{and} \quad \Phi_v^*: \lambda \rightarrow \varphi_{\lambda, v}$$

respectively define intertwining operators from ρ to R and from ρ^* to L . Here ρ^* denotes the representation of G on V^* contragredient to ρ .

1.1.7: We note a simple case of Frobenius Reciprocity. Let $H \subseteq G$ be a subgroup. Given a representation (ρ, V) of G , let V^H denote the space of vectors which are invariant under all operators $\rho(h)$ for h in H , i.e., the space of H -invariants. Given $\lambda \in (V^*)^H$, we can see easily from the formulas of §1.1.6 that the image of the map Φ_λ consists of functions which are invariant under left translation by H , i.e., consists of functions on the coset space $H \backslash G$. Conversely, if $T: V \rightarrow \mathcal{C}(H \backslash G)$ is an intertwining operator, and if we set

$$\lambda_T(v) = T(v)(1_G),$$

where 1_G is the identity element of G , then λ_T is an H -invariant linear functional on V . It is easy to check that the maps $\lambda \rightarrow \Phi_\lambda$ and $T \rightarrow \lambda_T$ are mutually inverse isomorphisms. In particular, if ρ is irreducible, then we may say that the multiplicity of ρ in $\mathcal{C}(H \backslash G)$ is equal to $\dim(V^*)^H$.

1.1.8: We will work in the context of algebraic groups, but we advise the reader not to be overly concerned with the finer points of the theory of algebraic groups. We will be mostly occupied with the classical examples: $GL_n(\mathbb{C})$ itself; $O_n(\mathbb{C})$, the

orthogonal group, the isometry group of a non-degenerate symmetric bilinear form (an inner product) on $\mathbb{C}^n$, or $Sp_{2n}(\mathbb{C})$, the symplectic group, the isometry group of a non-degenerate skew symmetric bilinear form (a symplectic form) on $\mathbb{C}^{2n}$. Any general fact quoted below can be explicitly verified for these examples. We will discuss the example of $GL_n(\mathbb{C})$ in more detail in the next section. Since all our groups will be defined over $\mathbb{C}$ unless otherwise specified, we will often drop the $\mathbb{C}$ from our notation.

Let G be a reductive algebraic subgroup of $GL_n(\mathbb{C})$, the group of invertible matrices with complex entries. Reductive can be taken to mean that the action of G on $\mathbb{C}^n$ is completely reducible, i.e., $\mathbb{C}^n$ is a direct sum of subspaces on which G acts irreducibly. We will assume for convenience that G is connected. (Of course, the orthogonal group O_n is not connected. We will comment when the facts recited here need modification to apply to O_n .) By the regular functions on G , we mean the algebra of functions on G generated by the matrix entries of the elements of G , and by the inverse of determinant. We denote the algebra of regular functions on G by $\mathcal{R}(G)$.

1.1.9: When discussing representations of algebraic groups, we usually consider only rational representations. A representation of G is called rational if its matrix coefficients are regular functions on the group. This essentially means that they are determined by holomorphic extension from a compact subgroup. This is well illustrated by $\mathbb{C}^\times$, the multiplicative group of complex numbers. Since $\mathbb{C}^\times$ is abelian, its irreducible representations will be one dimensional, and will be specified by characters, that is, homomorphisms into $\mathbb{C}^\times$. Rationality requires that the homomorphism should be complex analytic as a function of $z \in \mathbb{C}^\times$. Thus, a rational homomorphism will be determined by its restriction to $\mathbb{T}$, the unit circle in $\mathbb{C}^\times$. The theory of Fourier series tells us that the characters of $\mathbb{T}$ are parametrized by the integers and are just the restrictions to $\mathbb{T}$ of the power mappings $z \rightarrow z^n$. Thus, the rational characters of $\mathbb{C}^\times$ are exactly the integer power mappings. On the other hand, there are continuous characters of $\mathbb{C}^\times$ which are trivial on $\mathbb{T}$ but non-trivial on $\mathbb{R}^\times$; these can never be rational.

1.1.10: A basic fact is that a rational representation of a reductive group is always completely reducible: it is a direct sum of irreducible representations. This follows from the fact that such a representation is a holomorphic extension of its restriction to a compact subgroup. This point of view was exploited by Weyl, who called it the "unitarian trick". In any case, we will assume this fact as known.

The theory of the highest weight describes the collection of irreducible rational representations of G . Its basic features are summarized in §§1.1.11 through 1.1.16. Convenient references are [Hu], [Ja3].

1.1.11: Any two maximal connected solvable subgroups of G are conjugate. Any one of them is called a Borel subgroup. For Borel subgroup of GL_n , we may take the group of upper triangular matrices. The fact that this is the unique maximal

connected solvable subgroup of $GL_n(\mathbb{C})$ is essentially the content of Lie's theorem (or, in this context, the Lie-Kolchin Theorem) [Ja3], [So], [Sr].

1.1.12: Let (ρ, V) be a (rational) representation of G . The Lie-Kolchin Theorem says that $\rho(B)$ must have an eigenvector, that is, there exists a vector v_B in V such that

$$\rho(b)v_B = \psi(b)v_B$$

for all b in B . The function $\psi: B \rightarrow \mathbb{C}^\times$ is easily seen to be a character of B , and is called the weight of v_B .

1.1.13: If the representation ρ is irreducible, then the vector v_B is unique up to multiples. It is called the highest weight vector of ρ , and the B -eigencharacter $\psi = \psi_\rho$ is called the highest weight of ρ . The uniqueness of v_B is essentially a consequence of the fact we can find another Borel subgroup B' which is complementary to B in the sense that the double coset $B'B$ is open dense in G . In fact, the generic B' will be complementary to B in this sense. For $G = GL_n$, we may take for B' the lower triangular matrices.

1.1.14: Furthermore, the highest weight $\psi = \psi_\rho$ of v_B determines ρ . Thus, the highest weight theory gives us an embedding of $\hat{G}$, the set of (equivalence classes of) irreducible (rational) representations of G , into $\hat{B}$, the group of characters of B . To complete this description of $\hat{G}$, we need to specify with more precision the image of the highest weight mapping.

1.1.15: The Borel subgroup B is a semidirect product

$$B = A \cdot U$$

where U is the commutator subgroup of B , and A is a maximal torus in B (and in G - it is a Cartan subgroup of G). Thus, A is an abelian, reductive group. Its irreducible representations are all one dimensional and given by characters, homomorphisms from A to $\mathbb{C}^\times$. Under the operation of pointwise multiplication, the set $\hat{A}$ of characters is a group. In fact, it is isomorphic to $\mathbb{Z}^m$ for $m = \dim A$. Since U is the commutator subgroup of B , every character of B is trivial on U , hence is determined by its restriction to A . Thus, the highest weight map may be thought of as mapping $\hat{G}$ to $\hat{A}$. By considering tensor products of representations of G , one can see that the image of the highest weight mapping is a semigroup in $\hat{A}$. It is possible to characterize this subsemigroup of $\hat{A}$ by means of some positivity conditions, usually called dominance, and defined purely in terms of the structure of B . Let $\hat{A}^+$ denote the semigroup of dominant characters of A . Then we may summarize the highest weight theory by saying:

The highest weight mapping $\rho \rightarrow \psi_\rho$ defines a bijection between $\hat{G}$ and the semigroup $\hat{A}^+$ of dominant weights of A .

1.1.16: If G is simply connected (which implies that it is semisimple, i.e., its center is finite), we may add the complement that the semigroup $\hat{A}^+$ is free and that

its canonical generators form a basis over $\mathbb{Z}$ for $\hat{A}$. In this case, the representations whose highest weights are the fundamental generators for $\hat{A}^+$ are called the fundamental representations. For the classical groups, the fundamental representations have an elegant theory connected with the exterior algebra. See Appendix 4 for some details.

1.1.17: Consider a subgroup H in G . By $\mathcal{R}(H \setminus G)$, we mean the algebra of regular functions on G which are invariant under right translations by elements of H , that is, they factor to the quotient $H \setminus G$. (It may happen that $\mathcal{R}(H \setminus G)$ reduces to the constant functions even though $H \setminus G$ has positive dimension.) We call $\mathcal{R}(H \setminus G)$ the ring of regular functions on $H \setminus G$ and also the coordinate ring of $H \setminus G$. The right regular representation of G on $\mathcal{C}(H \setminus G)$ preserves $\mathcal{R}(H \setminus G)$, which is therefore a G -module. We have the following description of $\mathcal{R}(H \setminus G)$ as a G -module.

PROPOSITION (FROBENIUS RECIPROCITY). *As G -module, we have*

$$\mathcal{R}(H \setminus G) = \sum_{\sigma \in \hat{G}} (\dim(\sigma^*)^H) V_\sigma$$

where σ^H denotes the space of H -invariant vectors in $\sigma \in \hat{G}$, and σ^* is the representation contragredient to σ .

This follows from §§1.1.7 and 1.1.10 if we observe that subrepresentations of $\mathcal{R}(H \setminus G)$ must necessarily be rational.

1.1.18: Consider the case of $H = U$, the commutator subgroup of the Borel subgroup B . Since in the factorization of §1.1.15 the torus A is reductive, the span of the B -eigenspaces in any representation of G is the same as the space of U -invariant vectors. Hence the uniqueness of the highest weight vector in any irreducible representation V of G is equivalent to the fact that $\dim V^U = 1$. Hence the Frobenius Reciprocity statement of §1.1.17 says that the coordinate ring of $U \setminus G$ contains each irreducible representation of G exactly once:

$$\mathcal{R}(U \setminus G) = \sum_{\sigma \in \hat{G}} V_\sigma.$$

1.1.19: We can be slightly more precise about the decomposition described in §1.1.18. The torus A normalizes U , so it acts on $U \setminus G$ via left translations:

$$a(Ug) = Uag \quad a \in A, g \in G.$$

Clearly, this action of A commutes with the action of G by right translations. Hence the irreducible subspaces for G must be eigenspaces for A . In fact, it is not hard to identify the character by which A acts on the subspace $V_\sigma \in \mathcal{R}(U \setminus G)$. If we refer to §1.1.7 for the explicit form of the embedding of V_σ in $\mathcal{R}(U \setminus G)$, we see that it is given by Φ_{λ_0} , where λ_0 is the highest weight vector in σ^* . Thus, the formulas of §1.1.5 tell us that the space V_σ consists of eigenfunctions for the character ψ_{σ^*} of A , where ψ_{σ^*} denotes the highest weight of σ^* . We state this formally.

PROPOSITION. *The eigenspace decomposition of $\mathcal{R}(U \setminus G)$ resulting from the action of A on $U \setminus G$ by left translations provides a grading of $\mathcal{R}(U \setminus G)$ by $\hat{A}^+$. The A -eigenspace for the character ψ is the irreducible G -module dual to the module with highest weight ψ :*

$$\mathcal{R}(U \setminus G) = \sum_{\psi \in \hat{A}^+} V_{\psi}^*.$$

Here V_{ψ} is the irreducible module with highest weight ψ . The statement that this decomposition is a grading amounts to saying that multiplication in $\mathcal{R}(U \setminus G)$ satisfies

$$V_{\psi} V_{\phi} \subseteq V_{\psi + \phi}.$$

This is obvious when you take into account that the spaces involved are A -eigenspaces.

1.2: GL_n and diagram notation.

We describe the above general theory in the case of $G = GL_n$ and introduce some useful notation for the representations of that group.

1.2.1: As noted in §1.1, the upper triangular matrices may be taken as a typical Borel subgroup B . We have

$$B = AU$$

where A is the group of diagonal matrices (the diagonal torus), and U is the group of upper triangular matrices with diagonal entries all equal to one (the unipotent upper triangular matrices). Note that for an upper triangular matrix, the diagonal entries are the eigenvalues, so elements of U have all eigenvalues equal to one; such transformations are called *unipotent*. If we need to specify the n such that B or A or U are in GL_n , we will write B_n, A_n or U_n .

1.2.2: Following the example of §1.1.9, we see that the group $\hat{A}$ of characters of A is isomorphic to $\mathbb{Z}^n$. Precisely, given $\ell = (\ell_1, \ell_2, \ell_3, \dots, \ell_n)$ in $\mathbb{Z}^n$, we can define the character ψ_{ℓ} of A by the formula

$$\psi_{\ell}(a) = \prod_{j=1}^n a_j^{\ell_j},$$

where

$$a = \begin{vmatrix} a_1 & 0 & 0 & \cdots & 0 \\ 0 & a_2 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \vdots & \vdots & \vdots & a_n \end{vmatrix}$$

is a typical element of A . The correspondence $\ell \mapsto \psi_{\ell}$ establishes the desired isomorphism between $\mathbb{Z}^n$ and $\hat{A}$.

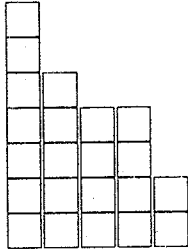
1.2.3: The semigroup $\hat{A}^+$ of highest weights corresponds by means of the isomorphism of §1.2.2 to the cone

$$C = \{\ell \in \mathbb{Z}^n : \ell_j \geq \ell_{j+1}\}.$$

Note that the group of permutation matrices, which is isomorphic to the symmetric group S_n on n letters, normalizes A and permutes the diagonal entries of the typical element a of A . The dual action on $\hat{A}$, transferred to $\mathbb{Z}^n$ by the isomorphism of §1.2.2, permutes the entries of $\ell \in \mathbb{Z}^n$. It is easy to see that any element of $\mathbb{Z}^n$ can be transformed by means of this action to a unique element of C .

1.2.4: Let 1_n be the element of $\mathbb{Z}^n$ each of whose entries is 1. The character ψ_{1_n} is the restriction to A of the determinant character $\det_n = \det$ of GL_n . If ρ is a representation of GL_n , with highest weight ψ_{ℓ} , then the representation $\rho \otimes \det$ has highest weight $\psi_{(1_n + \ell)}$. Thus, up to twisting (i.e., tensoring) by some power of $\det$, we may assume when we wish that all the components of ℓ for the highest weight vector ψ_{ℓ} are non-negative. Let C^+ denote the subcone of consisting of n -tuples with non-negative entries. We can see that C is invariant under translation by 1_n , and any element of C is contained in some translate of C^+ by some multiple of 1_n . Irreducible representations whose highest weights correspond to elements of C^+ are called polynomial representations, because their matrix coefficients will be polynomials on the $n \times n$ matrices, i.e., will not involve $(\det)^{-1}$. We can summarize this discussion by saying that any irreducible representation may be obtained by twisting a polynomial representation by a suitable power of $\det$.

1.2.5: It is often convenient to work with polynomial representations and to use for them a notation that ignores the n in GL_n . Thus, we will redefine C^+ to be the set of all decreasing sequences of non-negative integers with only finitely many non-zero terms. If ℓ_k is the last non-zero term of the element ℓ of C^+ , we will say ℓ has depth k . We can regard ℓ as defining a character of the diagonal torus A_n by means of the formula of §1.2.2 whenever $n \geq \text{depth}(\ell)$. An element of C^+ is often visualized as a Young or Ferrers Diagram [JK], [Ma], [Sc], which is an array of square boxes, arranged in left-justified rows, with the top row of length ℓ_1 , the second row of length ℓ_2 , and so forth. For example, the diagram corresponding to the sequence $(7, 5, 4, 4, 2, 0, 0, \dots)$ is



In recognition of this geometric mnemonic, we call this formalism which ignores n diagram notation.

1.2.6: We will often denote a diagram by a capital letter, such as D . When we do so, we will let ψ_D denote the character of A_n associated to (the sequence associated to) D . In using this notation, we always understand that n should be at least as large as the depth of D . The irreducible representation of GL_n corresponding to D (of depth at most n), i.e., the representation with highest weight ψ_D , will be denoted by ρ_n^D . We will make the convention that, if $\text{depth}(D) > n$, then $\rho_n^D = \{0\}$.

1.2.7: In keeping with the spirit of §1.2.5, let us now understand by 1_k the sequence which has exactly k 1s as its non-zero entries. These sequences form a basis for C^+ , in the sense that any element of C^+ is a unique linear combination of the 1_k , with non-negative integers for coefficients. We observe that, as long as $k \leq n$, the representation of GL_n with ψ_{1k} as its highest weight is the k -th exterior power $\Lambda^k(C^n)$. In the notation of §1.2.6, we can write

$$\rho_n^{1_k} = \Lambda^k(C^n).$$

Thus, the $\Lambda^k(C^n)$ for $1 \leq k \leq n-1$ are the fundamental representations of GL_n (more correctly, for SL_n). See Appendix 4.

We will not try to describe at this point the representations of GL_n in any detail, nor will we try to be any more explicit about the structure of $\mathcal{R}(U \setminus G)$ for the case of GL_n . These topics are discussed somewhat in §5.4.

1.2.8: Double diagrams. Sometimes it is useful to attach diagrams to non-polynomial representations of GL_n . If $\rho \in (GL_n)^\wedge$ has highest weight $\psi = \psi_t$, and the entries of $\ell = (l_1, l_2, l_3, \dots, l_n)$ are not all positive, we can attach to ρ a pair of diagrams as follows. For ℓ in $\mathbb{Z}^n$, let ℓ^* denote the sequence

$$(1.2.8.1) \quad \ell^* = (-l_n, -l_{n-1}, \dots, -l_2, -l_1)$$

resulting from changing the signs of all the entries of ℓ and then putting them in reverse order. Note that if ℓ belongs to the cone C , then so does ℓ^* . We remark that if ψ_t is the highest weight of $\rho \in (GL_n)^\wedge$, then ψ_{t^*} is the highest weight of the contragredient representation. Denote by ℓ^+ the n -tuple whose entries are equal to the entries of ℓ when these are positive, and equal to zero otherwise. Thus,

$$(1.2.8.2) \quad \ell^+ = (\max\{l_1, 0\}, \max\{l_2, 0\}, \dots, \max\{l_n, 0\}).$$

We may define ℓ^- analogously:

$$(1.2.8.3) \quad \ell^- = -((-\ell)^+) = (\min\{l_1, 0\}, \min\{l_2, 0\}, \dots, \min\{l_n, 0\}).$$

The reader may check that $\ell = \ell^+ + \ell^-$. It is also easy to verify that if ℓ is in the cone C , then both ℓ^+ and $(\ell^-)^*$ have decreasing, non-negative entries, and so may be considered to belong to the cone C^+ . We label the representation $\rho = \rho_{\psi_t}$ by the pair of diagrams (ℓ^+, ℓ^-) . Note that the sum of the depths of ℓ^+ and ℓ^- is at most n . We can clearly, reverse this process, and from a pair of diagrams (D_1, D_2) such that $\text{depth}(D_1) + \text{depth}(D_2)$ is not more than n , we can find a representation $\rho = \rho_{n, D_1, D_2}^D$ such that (D_1, D_2) are the pair of diagrams associated to ρ_{n, D_1, D_2}^D . Thus,

we have a parametrization of $(GL_n)^\wedge$ by pairs (D_1, D_2) of diagrams such that $\text{depth}(D_1) + \text{depth}(D_2) \leq n$. This is sometimes called the mixed tensor notation [Gup2], [Sty 1].

1.3: Multiplicity-free actions.

1.3.1: Let the (connected, reductive, algebraic) group G act rationally on the variety X . In particular, X could be a vector space on which G acts via a rational representation. Let $\mathcal{R}(X)$ be the ring of regular functions on X . We have a representation of G on $\mathcal{R}(X)$ by means of the usual formula:

$$g(f)(x) = f(g^{-1}(x)), \quad g \in G, x \in X, f \in \mathcal{R}(X).$$

It is obvious that this formula defines a representation of G on the space $C(X)$ of all complex-valued functions on X . The stipulation that G acts rationally on X amounts to requiring that $\mathcal{R}(X)$ be invariant for this action.

1.3.2: We will say that the action of G on X , or on $\mathcal{R}(X)$, is multiplicity-free if the multiplicity of each irreducible representation of G is at most one:

$$\dim \text{Hom}_G(\sigma, \mathcal{R}(X)) \leq 1, \quad \sigma \in \hat{G}.$$

We may also say that X is multiplicity-free. Multiplicity-free actions have been studied by numerous authors since the late 1970s. We mention the papers [Kr], [Br 1, 2], [BLV], [VK], [Kn], [GS] as a sample of the literature.

1.3.3: There is a very simple criterion for an action of G on X to be multiplicity-free, at least when X is affine.

PROPOSITION [Srv], [Vn]. *The action of G on the affine variety X is multiplicity-free if and only if a Borel subgroup of G has a (Zariski) dense and open orbit.*

PROOF. We will argue only that a dense B -orbit guarantees multiplicity-freeness of X . Given the highest weight theory, this is essentially an observation, which was made in [Srv]. That this condition is also necessary was shown in [Vn]. Although the argument for this is also quite short, we will not give it, because our main focus in this paper is on studying examples of multiplicity-free actions, not on classification.

Suppose that B has a dense orbit in X . We will show that, for any character ψ of B , there is at most one ψ -eigenvector in $\mathcal{R}(X)$. In particular, there is at most one highest weight vector for any given dominant weight, and so, by the highest weight theory, at most one copy of the irreducible representation with that highest weight. Let x_0 be a point in the dense B -orbit, and let Q be a ψ -eigenfunction in $\mathcal{R}(X)$. Then for any b in B we have

$$Q(b(x_0)) = (b^{-1}(Q))(x_0) = (\psi(b)^{-1})Q(x_0).$$

Thus, we see that Q is determined everywhere on the orbit $B(x_0)$ by its value at x_0 , so that $Q|_{B(x_0)}$ is unique up to multiples. Since $B(x_0)$ is dense, Q has a unique continuous (Zariski-continuous, i.e., regular) extension to all of X .

1.3.4: Remarks:

- a) In fact, for this argument, we do not need the full highest weight theory, only the fact that any representation of G contains B -eigenvectors.
- b) Also we do not need X to be affine. This is for the converse [Vn].
- c) Also we do not need to specify that the orbit $B(x_0)$ should be open, only that it should be dense. But if X is irreducible, denseness and openness of a subvariety are equivalent [Har]. Further, since G is connected, B is connected, and irreducibility of X is necessary for B to have a dense orbit. Hence we could specify that X be irreducible and B have an open orbit to reach the same conclusion.
- d) Since B must have a dense orbit for X to be multiplicity-free, *a fortiori* G must have a dense orbit. Thus, X is "almost" a homogeneous space. The multiplicity-free homogeneous spaces G/H for G simple and H reductive were classified in [Kr]. This paper also had the effect of focussing attention on the phenomenon of multiplicity-freeness. Arbitrary multiplicity-free homogeneous spaces for general reductive groups were classified in [Br1]. Irreducible representations which are multiplicity-free as actions are classified (up to some minor errors) in [Ka]. The problem of "compactifying" multiplicity-free homogeneous spaces has been dealt with quite effectively in [Kn], [LV], [Pa].

1.3.5: Example: We will call a homogeneous space $H \backslash G$ symmetric if H is the group of fixed points of an automorphism of order 2, i.e., an involution.

PROPOSITION. *A symmetric space $H \backslash G$ is multiplicity-free.*

PROOF. This may be thought of as a consequence of the Iwasawa Decomposition for real semisimple Lie groups [He 2], [Kna 2].

1.3.6: We can extend slightly the reasoning of Proposition 1.3.3 to give a necessary condition for the representation ρ_ψ with highest weight $\psi \in \hat{A}^+$ to occur in $\mathcal{R}(X)$. As we noted in the remark 1.3.3 d), if the B -orbit $B(x_0)$ is dense in X , then certainly the G -orbit $G(x_0)$ is also dense. Let $H_{x_0} = H$ denote the stabilizer of x_0 in G . Let $Q_\psi = Q$ be a B -eigenfunction with eigencharacter ψ . For h in $H \cap B$, we have

$$Q_\psi(x_0) = Q(h^{-1}(x_0)) = \psi(h)Q(x_0).$$

From this we conclude that $\psi(h) = 1$. Thus, we may make the following statement.

PROPOSITION. *Let the action of G on the variety X be multiplicity-free. Suppose x_0 in X belongs to the dense B -orbit. Let H be the stabilizer of x_0 in G . Then in order for the representation ρ_ψ of G with highest weight ψ to occur in $\mathcal{R}(X)$, it is necessary that ψ be trivial on $H \cap B$.*

1.3.7: As above, let X be a multiplicity-free G -variety. Observe that the B -eigenfunctions form a semigroup. More precisely, if Q_ψ is the B -eigenfunction with

eigencharacter ψ , then

$$Q_\psi Q_\phi = Q_{\psi\phi}.$$

Since Q_ψ is unique up to scalars, the mapping $Q_\psi \rightarrow \psi$ defines an injection of semigroups from the set of B -eigenvectors modulo scalars to $\hat{A}^+$. Let us denote the image of this map by $\hat{A}^+(X)$.

1.3.8: If the ring $\mathcal{R}(X)$ is a unique factorization domain [CLO], [La] which is the case for example if X is a vector space E , then we can describe the nature of the semigroup $\hat{A}^+(X)$ fairly precisely (cf. [Krf], [SK], [Srv]).

PROPOSITION. *If $\mathcal{R}(X)$ is a unique factorization domain, then $\hat{A}^+(X)$ is a free abelian semigroup, i.e., is isomorphic to the positive orthant $(\mathbb{Z}^+)^l$ in $\mathbb{Z}^l$ for some l . Such a semigroup has a canonical set of primitive generators. Let $\{\psi_j : 1 \leq j \leq l\}$ be the set of these generators. The eigenfunctions Q_{ψ_j} corresponding to the ψ_j are exactly the B -eigenfunctions which are prime, and every other V -eigenfunction is uniquely a product of these prime eigenfunctions.*

PROOF. Suppose $\mathcal{R}(X)$ is a unique factorization domain, and for some V -eigenfunction Q_ψ , let

$$Q_\psi = \prod_{j=1}^l p_j^{m_j}$$

be the factorization of Q_ψ in to powers of distinct prime factors. Unique factorization means that the p_j are unique as a set with multiplicities, up to multiplication by scalars. Apply an element b in B to this equation to get

$$\psi(b)Q_\psi = \prod_{j=1}^l b(p_j)^{m_j}.$$

Comparing these equations, we see that unique factorization implies

$$b(p_j) = c_j p_k$$

for some permutation $s(b) : j \rightarrow k$ and some scalars c_j . The map $b \rightarrow s(b)$ is easily checked to be a homomorphism from B to the symmetric group S_l . Since B is connected, this homomorphism must be trivial, so $s(b)$ is the identity map. Thus, this last equation says that all the prime factors of Q_ψ are also eigenfunctions for B . Thus, the semigroup of B eigenfunctions in $\mathcal{R}(X)$ is generated by its prime elements. Suppose ψ in $\hat{A}^+(X)$ is not primitive, i.e., suppose we can write $\psi = \psi_1 \psi_2$ with ψ_1, ψ_2 in $\hat{A}^+(X)$. The equation of §1.3.7 implies that Q_ψ is not prime. Hence prime eigenfunctions correspond to primitive elements of $\hat{A}^+(X)$. The factorization argument above shows that, conversely, primitive elements of $\hat{A}^+(X)$ must come from prime eigenfunctions. Finally, consider ψ in $\hat{A}^+(X)$, and suppose we have two distinct factorizations

$$\psi = \prod \psi_j = \prod \tilde{\psi}_k$$

of ψ in to primitive elements in $\hat{A}^+(X)$. Then following §1.3.7, we have two distinct factorizations of Q_ψ into prime elements, contradicting unique factorization in $\mathcal{R}(X)$. This shows there are no relations among the set $\{\psi_j : 1 \leq j \leq l\}$ of primitive elements, i.e., $\hat{A}^+(X)$ is the free semigroup generated by $\{\psi_j : 1 \leq j \leq l\}$.

2: Multiplicity-free Actions and Classical Invariant Theory: Case of GL_n .

In this section, we describe how to derive the Fundamental Theorems of Classical Invariant Theory for the general linear group using multiplicity-free actions. The central result in our approach is what we call (GL_n, GL_m) -duality, which we derive by observing the existence of a particular multiplicity-free action. We explain how to use it to derive Weyl's formulation of the First Fundamental Theorem (FFT) of Invariant Theory for GL_n , as well as some further, less classical but closely allied results. We will also compare our derivation with Weyl's treatment, and connect it with the celebrated Schur Duality. At the end we will have 6 results, each of which implies the others, more or less directly. Any one of these might be considered the FFT for GL_n . We then give a further result which combines several of these. We show that Weyl's Second Fundamental Theorem for GL_n is also easily derivable using multiplicity-free actions. We close with an explanation of how (GL_n, GL_m) -duality provides a transparent conceptual understanding of Weyl's finiteness theorem for typical basic invariants. Weyl proved these results using the Capelli Identity.

2.1: (GL_n, GL_m) -duality.

2.1.1: Let U and V be two vector spaces. Form their tensor product $U \otimes V$. The general linear group $GL(U)$ of invertible linear transformations on U acts on $U \otimes V$ via its action on the first factor:

$$g(u \otimes v) = g(u) \otimes v \quad u \in U, v \in V, g \in GL(U).$$

Similarly, $GL(V)$ acts on $U \otimes V$ via its action on the second factor. Combining these actions gives us a homomorphism

$$GL(U) \times GL(V) \rightarrow GL(U \otimes V).$$

2.1.2: The previous paragraph allows us to consider the structure of the symmetric algebra $S(U \otimes V)$ as a $GL(U) \times GL(V)$ module.

THEOREM. *The symmetric algebra $S(U \otimes V)$ is multiplicity-free as a $GL(U) \times GL(V)$ module. Precisely, we have a decomposition*

$$(2.1.2.1) \quad S(U \otimes V) = \sum_D \rho_U^D \otimes \rho_V^D$$

of $GL(U) \otimes GL(V)$ -modules. Here D varies over all diagrams of depth at most $\min(\dim U, \dim V)$ (see §1.2.6 for notation). The homogeneous component $S^d(U \otimes V)$ is the sum over all these diagrams which have size d .

2.1.3: Remark: The decomposition of §2.1.2 is significantly better than being just multiplicity-free. In fact, we see that each representation ρ_U^D of $GL(U)$ is paired with exactly one representation of $GL(V)$, namely the representation ρ_V^D parametrized by the same diagram. Hence the $GL(U)$ isotypic components of $S(U \otimes V)$ are

the same as the isotypic components for $GL(V)$, and are acted on irreducibly by $GL(U) \times GL(V)$. Thus, if we know the way a function transforms under $GL(U)$, we know how it transforms under $GL(V)$, and vice versa. It is for this reason that we refer to Theorem 2.1.2 as (GL_n, GL_m) -duality. (Here $n = \dim U$ and $m = \dim V$.)

(2.1.4) PROOF OF THEOREM 2.1.2. Without loss of generality, we may take $U = \mathbb{C}^n$ and $V = \mathbb{C}^m$. We may also identify $\mathbb{C}^n \otimes \mathbb{C}^m$ with the space $M_{n,m}$ of $n \times m$ matrices with GL_n acting by multiplication on the left and GL_m acting by multiplication on the right. Precisely, we have

$$(g, g')(T) = (g')^{-1} T g'^{-1} \quad g \in GL_n, g' \in GL_m, T \in M_{n,m}.$$

Here g' indicates the transpose of g . With these conventions, the algebra $\mathcal{S}(U \otimes V)$ is identified to the algebra $\mathcal{P}(M_{n,m})$ of polynomials on $M_{n,m}$. Let B be the upper triangular Borel subgroup in GL_n , and let B' be the upper triangular Borel in GL_m . The standard facts of elementary linear algebra (Gaussian elimination, LU decomposition [Stg]) imply that the $B \times B'$ -orbit of the matrix

$$J_{n,m} = \begin{vmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & \cdot & \cdot \\ 0 & 0 & 0 & \cdots & \cdot & \cdot \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \cdot & \cdot & \cdot & 0 & \cdot \\ 0 & \cdot & \cdot & \cdot & 0 & 1 \\ 0 & \cdot & \cdot & \cdot & 0 & 1 \\ 0 & \cdot & \cdot & \cdot & 0 & 1 \\ 0 & \cdot & \cdot & \cdot & 0 & 1 \\ 0 & \cdot & \cdot & \cdot & 0 & 1 \\ 0 & \cdot & \cdot & \cdot & 0 & 1 \end{vmatrix},$$

which has 1's on the diagonal starting from the upper left corner, and 0's elsewhere, is open and dense. According to the criterion of §1.3.3, this implies that the action of $GL(U) \times GL(V)$ is multiplicity-free. (It is implicit in our illustration of $J_{n,m}$ that $n < m$, and for convenience we will assume this for the rest of this proof.)

To compute which representations of $GL(U) \times GL(V)$ occur in $\mathcal{S}(U \otimes V)$, we use a two-part approach. We use the criterion of §1.3.6 to limit the possible highest weights which can occur. Then we produce all the possible highest weights by explicit construction. For the first part, consider the diagonal tori A_n and A_m in GL_n and GL_m respectively. Taking $n < m$, we may compute that the isotropy group $J_{n,m}$ in $A_n \times A_m$ consists of matrices of the form

$$\left\{ \begin{vmatrix} a_1^{-1} & & & & \\ & a_2^{-1} & & & \\ & & \ddots & & \\ & & & a_n^{-1} & \\ & & & & \times \end{vmatrix} \times \begin{vmatrix} a_1 & & & & \\ & a_2 & & & \\ & & \ddots & & \\ & & & a_n & \\ & & & & a_m \end{vmatrix} \right\}$$

According to §1.3.6, a highest weight of a representation which occurs in formula (2.1.2.1) must be trivial on this subgroup of $A_n \times A_m$. Consider a character $\varphi_l \varphi_{l'}$ of $A_n \times A_m$, with $l = (l_1, l_2, \dots, l_n)$ and $l' = (l'_1, l'_2, \dots, l'_m)$. It is easy to check that $\varphi_l \varphi_{l'}$ will be trivial on the stabilizer of $J_{n,m}$ if and only if $l_j = l'_j$ for $j \leq n$, and $l'_j = 0$ for $j > n$. If we refer to §§1.2.3 and 1.2.4, we see that the sum in

formula (2.1.2.1) runs over all highest weights satisfying these conditions. Hence to finish the proof, we only need to show that all the $GL(U) \times GL(V)$ modules in the right hand sum of formula (2.1.2.1) do occur in $\mathcal{S}(U \otimes V)$. We do this by producing elements of $\mathcal{S}(U \otimes V)$ which are the highest weight vectors of these modules. Guided by the discussion of §1.2.7, we see it suffices to exhibit the highest weight vectors for the modules $\rho_n^{1_k} \otimes \rho_m^{1_k}$, for $1 \leq k \leq \min\{n, m\}$. It is easy to check that these are given by the determinants of the leading minors of the typical element T of $M_{n,m}$. Thus, if $T = [t_{ij}]$, the highest weight vector of $\rho_n^{1_k} \otimes \rho_m^{1_k}$ equals

$$\delta_k = \det \begin{vmatrix} t_{11} & t_{12} & \cdots & t_{1k} \\ t_{21} & t_{22} & \cdots & \cdot \\ \vdots & \vdots & \ddots & \vdots \\ t_{k1} & \cdot & \cdot & t_{kk} \end{vmatrix}$$

This completes the proof of Theorem 2.1.2.

2.1.5: Remarks: a) A careful treatment of Gaussian elimination shows that a matrix T has an "LU-factorization" [Stg], i.e., can be written $T = LU$ where L is a lower triangular $n \times n$ matrix and U is an upper triangular $m \times m$ matrix, if and only if $\delta_k \neq 0$ for all $k \leq \min\{n, m\}$.

b) I do not know the earliest reference for Theorem 2.1.2. With our current understanding of the symmetric functions known as Schur functions as characters of the representations of GL_n , we can see this result as implicit in a combinatorial identity known to Cauchy [Ma]. However, the ingredients necessary for the interpretation of the identity as Theorem 2.1.2 were not available until Schur's thesis [Schu 1] in 1901, and I do not know when such an interpretation was first made, or whether it was the first proof of Theorem 2.1.2. As far as I can see from reading Weyl's book [Wy], he did not understand Cauchy's identity in this fashion, though he certainly knew both it and the character formula for GL_n . It is hard to imagine him not writing a very different book if he had understood this result. In any case, Theorem 2.1.2 seems not to have been recognized as common knowledge even in recent times, and to have been rediscovered by numerous authors, with various proofs, see for example [CGR], [GK], [Gup2], [Hw 1], etc.

2.2: First fundamental theorem for GL_n , standard polynomial form.

Let V be a vector space, and $G \subseteq GL(V)$ a group of linear transformations of V . We can let G act on V^* by the contragredient of its action on V , and we can let it act on $V^\ell \oplus (V^*)^m$, the sum of several copies of V and of V^* , by the direct sum action. By the First Fundamental Theorem of Invariants for G , Weyl meant the description of the algebra of G -invariants $\mathcal{P}(V^\ell \oplus (V^*)^m)^G$ for arbitrary ℓ and m . More precisely, he asked for a description of what he called "typical basic generators" for $\mathcal{P}(V^\ell \oplus (V^*)^m)^G$ and showed that the determination of these was in principle finite. To do this he used the Capelli identity. We will give later (see §2.7) a precise definition of typical basic invariants. Now we will show how to derive the FFT for $G = GL(V)$ using (GL_n, GL_m) -duality.

2.2.1: We can write a typical element of $V^\ell \oplus (V^*)^m$ as a tuple:

$$y = (v_1, v_2, \dots, v_\ell, \lambda_1, \lambda_2, \dots, \lambda_m) \quad v_j \in V, \lambda_k \in V^*.$$

We can see from the form of y that there is an obvious family of second order polynomials on $V^\ell \oplus (V^*)^m$ which are invariant under $GL(V)$. These are given by pairing copies of V with copies of V^* :

$$\xi_{jk}(y) = \lambda_k(v_j), \quad 1 \leq j \leq \ell, 1 \leq k \leq m.$$

2.2.2: THEOREM. *The polynomials ξ_{jk} , $1 \leq j \leq \ell, 1 \leq k \leq m$ generate the algebra $\mathcal{P}(V^\ell \oplus (V^*)^m)^{GL(V)}$.*

2.2.3:

PROOF. We may write

$$(2.2.3.1) \quad V^\ell \oplus (V^*)^m = V \otimes \mathbf{C}^\ell \oplus V^* \otimes \mathbf{C}^m$$

Doing this lets us see that inside $GL(V^\ell \oplus (V^*)^m)$ we have a subgroup $GL_\ell \times GL_m$ commuting with $GL(V) = G$. Thus, the action of $GL_\ell \times GL_m$ on $\mathcal{P}(V^\ell \oplus (V^*)^m)$ will preserve the algebra $\mathcal{P}(V^\ell \oplus (V^*)^m)^G$ of G -invariants, that is, $\mathcal{P}(V^\ell \oplus (V^*)^m)^G$ will be a $GL_\ell \times GL_m$ -module. We will investigate the structure of this module.

Theorem 2.1.2 tells us the structure of $\mathcal{P}(V^\ell)$ as a $GL(V) \times GL_\ell$ -module:

$$\mathcal{P}(V^\ell) = \sum_D (\rho_V^D)^* \otimes (\rho_\ell^D)^*$$

Here we have put $(\rho_V^D)^*$ instead of the ρ_V^D of Theorem 2.1.2 because we are dealing with the polynomial algebra rather than the symmetric algebra. We have an analogous description for $\mathcal{P}((V^*)^m)$:

$$\mathcal{P}((V^*)^m) = \sum_E \rho_V^E \otimes (\rho_m^E)^*.$$

Note that, because we are dealing with polynomials on V^* , the ρ_V^E are appropriate here. Using the natural isomorphism

$$\mathcal{P}(V^\ell \oplus (V^*)^m) = \mathcal{P}(V^\ell) \otimes \mathcal{P}((V^*)^m),$$

we can compute

$$\begin{aligned} (2.2.3.2) \quad \mathcal{P}(V^\ell \oplus (V^*)^m)^G &= (\mathcal{P}(V^\ell) \otimes \mathcal{P}((V^*)^m))^G \\ &= ((\sum_D (\rho_V^D)^* \otimes (\rho_\ell^D)^*) \otimes (\sum_E \rho_V^E \otimes (\rho_m^E)^*))^G \\ &= \sum_{D,E} ((\rho_V^D)^* \otimes \rho_V^E) \otimes (\rho_\ell^D)^* \otimes (\rho_m^E)^* \\ &= \sum_{D,E} ((\rho_V^D)^* \otimes \rho_V^E) \otimes (\rho_\ell^D)^G \otimes (\rho_\ell^D)^* \otimes \rho_m^E \\ &= \sum_D (\rho_\ell^D \otimes \rho_m^D)^*. \end{aligned}$$

Here the sum is over all diagrams D of depth at most $\min\{\ell, m, \dim V\}$. The last equality follows because, by Schur's Lemma [CR], [Ja 1], a tensor product of two representations can contain an invariant only when they are mutually contragredient.

Formula (2.2.3.2) describes $\mathcal{P}(V^\ell \oplus (V^*)^m)^G$ as a $GL_\ell \times GL_m$ -module. However, Theorem 2.2.2 describes $\mathcal{P}(V^\ell \oplus (V^*)^m)^G$ as an algebra with certain generators. We must deduce this from the description we have. We can observe that the final expression in formula (2.2.3.2) is reminiscent of formula (2.1.2.1) describing (the dual of) $S(\mathbf{C}^\ell \otimes \mathbf{C}^m)$. Reinforcing this, we note also that the proposed generators ξ_{jk} span the $GL_\ell \times GL_m$ -module $(\mathbf{C}^\ell \otimes \mathbf{C}^m)^*$, which indeed generates $S((\mathbf{C}^\ell \otimes \mathbf{C}^m)^*)$. To get from formula (2.2.3.2) to Theorem 2.2.2, all we need to do is show that the highest weight vectors of the modules in the final sum of formula (2.2.3.2) can be expressed in terms of the ξ_{jk} . We can do this just as in the proof of Theorem 2.1.2. Indeed, the ξ_{jk} transform under $GL_\ell \times GL_m$ just as do the usual entries of a matrix. Thus, the determinants

$$(2.2.3.3) \quad \Delta_b = \det \begin{vmatrix} \xi_{11} & \xi_{12} & \cdots & \xi_{1b} \\ \xi_{21} & \xi_{22} & \cdots & \xi_{2b} \\ \vdots & \vdots & \ddots & \vdots \\ \xi_{b1} & \cdots & \cdots & \xi_{bb} \end{vmatrix}, \quad 1 \leq b \leq \min\{\ell, m, \dim V\}$$

will transform in the desired way under $GL_\ell \times GL_m$, provided that they are non-zero. Thus, to finish the proof, we just need to check that the expressions Δ_b do not vanish identically. It is very easy to do this directly. However, we use it as the occasion to point to some pleasant geometry which makes contact with the Second Fundamental Theorem of invariants for $GL(V)$, to be discussed in §2.6.

Another way of thinking of $V^\ell \oplus (V^*)^m$ is in terms of spaces of matrices:

$$(2.2.3.4) \quad V^\ell \oplus (V^*)^m = \text{Hom}(\mathbf{C}^\ell, V) \oplus \text{Hom}(V, \mathbf{C}^m).$$

From this point of view, the action of $GL(V)$ is given by pre- and post-multiplication

$$(2.2.3.5) \quad g(T, S) = (gT, Sg^{-1}), \quad g \in GL(V), T \in \text{Hom}(\mathbf{C}^\ell, V), S \in \text{Hom}(V, \mathbf{C}^m).$$

If we form the map

$$(2.2.3.6) \quad \mu: \text{Hom}(\mathbf{C}^\ell, V) \oplus \text{Hom}(V, \mathbf{C}^m) \rightarrow M_{m,\ell}$$

given by composition:

$$\mu(T, S) = ST,$$

then we see that

$$\mu(g(T, S)) = \mu(T, S)$$

for all (T, S) in $\text{Hom}(\mathbf{C}^\ell, V) \oplus \text{Hom}(V, \mathbf{C}^m)$ and g in $GL(V)$. Therefore, the pullback map

$$\mathcal{P}(M_{m,\ell}) \rightarrow \mathcal{P}(V^\ell \oplus (V^*)^m)$$

given by composition by μ :

$$\mu^*(p) = p \circ \mu$$

maps $\mathcal{P}(M_{m,\ell})$ to $\mathcal{P}(V^\ell \oplus (V^*)^m)^G$. Furthermore, it is easy to verify that the proposed generators $\xi_{j,k}$ of $\mathcal{P}(V^\ell \oplus (V^*)^m)^G$ are just the pullbacks of the standard coordinate functions on $M_{m,\ell}$. Thus, Theorem 2.2.2 is equivalent to the statement

$$(2.2.3.7) \quad \mu^*(\mathcal{P}(M_{m,\ell})) = \mathcal{P}(V^\ell \oplus (V^*)^m)^G.$$

This formulation of Theorem 2.2.3 is given in [Pro]; I do not know if this is the first place it appears. Our discussion so far has reduced formula (2.2.3.7) to verifying that the polynomials Δ_b are non-zero. Comparison of formula (2.2.3.3) with the final formula in §2.1.4 tells us that

$$(2.2.3.8) \quad \Delta_b = \mu^*(\delta_b).$$

Elementary reduction theory for matrices [Stg] tells us that the image of μ consists of all matrices of rank at most $\min\{m, \ell, \dim V\}$. The basic theory of the determinant [Tur] tells us that the $k \times k$ minors of a matrix vanish on the variety of matrices of rank $k-1$, but not on the variety of matrices of rank k . Thus, these facts from elementary linear algebra complete the proof of Theorem 2.2.2.

2.3: Operator commutant version of the FFT.

2.3.1: Let V be a finite dimensional vector space, as above. Let $\mathcal{PD}(V)$ be the algebra of polynomial-coefficient differential operators on V . The group $GL(V)$ acts on $\mathcal{PD}(V)$ by conjugation. Thus, if $g \in GL(V)$, an operator $D \in \mathcal{PD}(V)$ is invariant under g acting $\mathcal{PD}(V)$ if and only if D commutes with g . Thus, for a subgroup $G \subseteq GL(V)$, the algebra $\mathcal{PD}(V)^G$ of G -invariant operators is the commuting algebra of G in $\mathcal{PD}(V)$.

2.3.2: As explained in §A2.1.1, the symbol mapping

$$\hat{s} : \mathcal{PD}(V) \rightarrow \mathcal{P}(V \oplus V^*)$$

is a natural isomorphism of $GL(V)$ -modules. Furthermore, for the subspace $V \otimes V^* \subseteq \mathcal{P}^2(V \oplus V^*)$, the corresponding subspace $\hat{g}\mathfrak{l}(V) = \hat{s}^{-1}(V \otimes V^*) \subseteq \mathcal{PD}(V)$ is a Lie algebra isomorphic to $\mathfrak{gl}(V)$, and taking commutators with $\hat{g}\mathfrak{l}$ produces infinitesimal generators of the conjugation action of $GL(V)$ on $\mathcal{PD}(V)$.

2.3.3: Now replace V with $V \otimes U$. As noted in §2.1.1, we can regard $GL(V)$ as a subgroup of $GL(V \otimes U)$. Consider the algebra $\mathcal{PD}(V \otimes U)^{GL(V)}$ of operators commuting with $GL(V)$. Theorem 2.2.2 says that $\hat{s}(\mathcal{PD}(V \otimes U)^{GL(V)}) = \mathcal{P}((V \otimes U) \oplus (V \otimes U)^*)^{GL(V)}$ is generated by the quadratic invariants $((V \otimes U) \otimes (V \otimes U)^*)^{GL(V)} = (V \otimes V^*)^{GL(V)} \otimes (U \otimes U^*) \cong (U \otimes U^*)$. It is clear that the space

$$\hat{s}^{-1}((V \otimes U) \otimes (V \otimes U)^*)^{GL(V)} \subseteq \hat{s}^{-1}((V \otimes U) \oplus (V \otimes U)^*) = \hat{g}\mathfrak{l}(U),$$

by which we mean here the Lie subalgebra of $\hat{g}\mathfrak{l}(V \otimes U)$ corresponding to the subgroup $GL(U)$, embedded in $GL(V \otimes U)$ in the obvious way (as described in §2.1.1). This observation, coupled with a small argument to compensate for the fact that $\mathcal{PD}(V \otimes U)^{GL(V)}$ is only filtered, not graded, allows us to convert Theorem 2.2.2 into a description of the commuting algebra of $GL(V)$ in $\mathcal{PD}(V \otimes U)$.

THEOREM. *The algebra $\mathcal{PD}(V \otimes U)^{GL(V)}$ of all polynomial-coefficient differential operators on $V \otimes U$ commuting with $GL(V)$ is generated (as associative algebra) by the Lie algebra $\hat{g}\mathfrak{l}(U)$ of infinitesimal generators of the group $GL(U)$ acting on $V \otimes U$. This Lie algebra is exactly the centralizer of $GL(V)$ in the Lie algebra $\mathfrak{gl}(V \otimes U)$.*

2.3.4: Remarks: a) We refer to [Hw1] for the details of the proof of this result. b) From this result, by the double commutant yoga used by Weyl in his treatment of Schur Duality ([W1]), one can deduce Theorem 2.1.2. Thus, Theorem 2.3.3 completes a circle of three results parallel to the tensor versions of the FFT given by Weyl. The tensor versions and the relations between the two sets of results are discussed below in §2.4.

2.4: Relation to Schur Duality.

The most famous version of the FFT for GL_n is Schur Duality [Schu 1,2]. Weyl gives it a prominent place in his book [W1]. Schur Duality does not explicitly involve polynomials, but instead describes the action of GL_n on tensor products of its standard representation on $\mathbb{C}^n$. We review these results, then discuss their relations to the versions of the FFT for GL_n given above.

2.4.1: For a vector space V , let $V^{\otimes k}$ denote the k -th tensor power of V . The group $GL(V)^k$ acts on $V^{\otimes k}$ by means of the standard action of $GL(V)$ on each factor:

$$(2.4.1.1) \quad (g_1, g_2, \dots, g_k)(v_1 \otimes v_2 \otimes \dots \otimes v_k) = g_1(v_1) \otimes g_2(v_2) \otimes \dots \otimes g_k(v_k)$$

for $g_i \in GL(V)$ and $v_i \in V$. By restricting to the diagonal subgroup, i.e., taking $g_1 = g_2 = \dots = g_k = g$, we obtain the standard tensor product action of $GL(V)$ on $V^{\otimes k}$. There is also a natural action of the symmetric group S_k on $V^{\otimes k}$, given on decomposable tensors by permutation of the factors:

$$(2.4.1.2) \quad \pi(v_1 \otimes v_2 \otimes v_3 \otimes \dots \otimes v_k) = v_{\pi^{-1}(1)} \otimes v_{\pi^{-1}(2)} \otimes v_{\pi^{-1}(3)} \otimes \dots \otimes v_{\pi^{-1}(k)}$$

for π in S_k and v_j in V . These two actions clearly commute with one another.

2.4.2: FFT for GL_n , operator commutant for tensors version

THEOREM. *The algebras spanned by the images of $GL(V)$ and of S_k , each acting on $V^{\otimes k}$ as described in the previous paragraph, are mutual commutants in $\text{End}(V^{\otimes k})$.*

Comments: Weyl's proof of this theorem uses some general nonsense of multilinear algebra (essentially the isomorphism $\text{End}(V^{\otimes k}) \cong (\text{End}(V))^{\otimes k}$), plus the polarization theorem, viz., that $S^k(V)$; the k -th symmetric power of V , is spanned by the diagonal elements $v \otimes v \otimes \dots \otimes v$, $v \in V$, to show that the algebra spanned by $GL(V)$ is the full commutant of S_k in $\text{End}(V^{\otimes k})$. Since S_k acts completely reducibly on $V^{\otimes k}$ by Maschke's theorem [CR], the double commutant yoga yields the theorem.

2.4.3: Schur Duality

THEOREM. Under the joint action of $GL(V) \times S_k$, the space $V^{\otimes^*}$ decomposes into a direct sum

$$V^{\otimes^*} \cong \sum_D \rho_V^D \otimes \sigma^D.$$

Here the index D runs over all diagrams of size k and depth at most $\dim V$.

Comments: As in the case of (GL_n, GL_m) -duality, the key fact here is that each $GL(V)$ -isotypic component is also an S_k -isotypic component and vice versa, so that the way a vector transforms under $GL(V)$ determines how it transforms under S_k and vice versa. This much follows immediately from Theorem 2.4.2 and the double commutant yoga. The explicit form of the correspondence between $(GL(V))^\wedge$ and $(S_k)^\wedge$, specifically that it is compatible with the independent parametrizations of $(GL(V))^\wedge$ and $(S_k)^\wedge$ by means of diagrams, is also very interesting, and can be verified by explicitly exhibiting vectors which transform appropriately, in analogy with the formulas of §2.1.4 for the joint highest weight vectors in (GL_n, GL_m) -duality.

2.4.4: FFT for GL_n , tensor invariants version

The final version of the FFT for the action of $GL(V)$ on tensors describes the invariants for the actions of $GL(V)$ on the "mixed tensors" $V^{\otimes^*} \otimes (V^*)^{\otimes \ell}$. Here "mixed" refers to the fact that both covariant variables (i.e., copies of V) and contravariant variables (i.e., copies of V^*) are involved. Since the center of $GL(V)$ acts by non-trivial scalars (by the character $z \rightarrow z^{k-\ell}$) if $k \neq \ell$, the only case in which any invariants can possibly exist is when k and ℓ are equal. In this case, we can describe a family of invariants as follows. Let

$$T = (v_1 \otimes v_2 \times \dots \otimes v_k) \otimes (\lambda_1 \otimes \lambda_2 \otimes \dots \otimes \lambda_k)$$

be a typical decomposable tensor in $V^{\otimes^*} \otimes (V^*)^{\otimes \ell}$. Choose a bijection β between the covariant factors of T and the contravariant factors of T . Since we have labeled each set of factors by the integers from 1 to k , we can identify β with a permutation of these integers. This is useful, but it should be kept in mind that it is not canonical, but depends on our labeling. Consider the function

$$J_\beta(T) = \prod_{j=1}^k \lambda_{\beta(j)}(v_j).$$

It is easy to check that J_β is multilinear in the v_j and λ_j , so it defines a linear functional on $V^{\otimes^*} \otimes (V^*)^{\otimes k}$. Since $V^{\otimes^*} \otimes (V^*)^{\otimes k}$ is self-dual, we may also regard J_β as a vector in $V^{\otimes^*} \otimes (V^*)^{\otimes k}$.

THEOREM. The elements J_β defined above span the space $(V^{\otimes^*} \otimes (V^*)^{\otimes^*})^{GL(V)}$ of $GL(V)$ invariants in $V^{\otimes^*} \otimes (V^*)^{\otimes^*}$.

This follows directly from the commutant version, Theorem 2.4.2, by using the inverse of the identification $V^{\otimes^*} \otimes (V^*)^{\otimes^*} \cong \text{End}(V^{\otimes^*})$ used in that result.

Specifically, it corresponds to the fact that S_k spans the commutant of $GL(V)$ in $\text{End}(V^{\otimes^*})$; this is exactly the input of the double commutant yoga to Theorem 2.4.2.

2.4.5: We now have three closely linked results, Theorems 2.4.2, 2.4.3 and 2.4.4, describing the action of $GL(V)$ on $V^{\otimes^*}$. Each of these implies the others. We have three parallel results, Theorems 2.3.3, 2.1.2, and 2.2.2, describing the action of $GL(V)$ on $P(V^m)$. In proving them, we found it easiest to establish (GL_n, GL_m) -duality first, using the fact that the joint $GL_n \times GL_m$ action on the $n \times m$ matrices is multiplicity-free. The operator commutant version came last. Indeed, I do not know if it appeared explicitly in the literature before [Hw1]. In contrast, Weyl's development of the results for tensors puts the operator commutant version first. However, once stated, each result in a given triple is easily seen to imply the others, as we have explained in the discussion above. In turn, these two triples of results are closely linked to each other: indeed, they are mutually equivalent. We will show how to deduce each of (GL_n, GL_m) -duality and Schur Duality from the other.

2.4.5.1 Schur Duality implies (GL_n, GL_m) -duality:

The argument for deriving (GL_n, GL_m) -duality from Schur Duality is well known [CGR], [Gup2]. The idea is to prove formula (2.1.2.1) for each homogeneous component $S^*(U \otimes V)$. We have

$$(2.4.5.1.1) \quad S^*(U \otimes V) \cong ((U \otimes V)^{\otimes^*})^{S_k} \cong (U^{\otimes^*} \otimes V^{\otimes^*})^{\Delta S_k}.$$

When we factor $(U \otimes V)^{\otimes^*}$ into the tensor product of $U^{\otimes^*}$ and $V^{\otimes^*}$, we see that the product $S_k \times S_k$ acts on this space. Our original copy of S_k , acting on $(U \otimes V)^{\otimes^*}$, is identified with the diagonal subgroup ΔS_k of $S_k \times S_k$, and this is the meaning of the notation of the last expression in formula (2.4.5.1.1).

We now use Schur Duality to describe the $GL(V) \times GL(U)$ -module structure of $(U \otimes V)^{\otimes^*}$.

$$(2.4.5.1.2) \quad \begin{aligned} (U^{\otimes^*} \otimes V^{\otimes^*})^{\Delta S_k} &\cong \left(\sum_E \rho_U^E \otimes \sigma^E \right) \otimes \left(\sum_D \rho_V^D \otimes \sigma^D \right)^{\Delta S_k} \\ &\cong \left(\sum_{E,D} \rho_U^E \otimes \rho_V^D \otimes (\sigma^E \otimes \sigma^D) \right)^{\Delta S_k} \\ &\cong \sum_{E,D} \rho_U^E \otimes \rho_V^D \otimes (\sigma^E \otimes \sigma^D)^{\Delta S_k}. \end{aligned}$$

Schur's Lemma [CR][Jal] says that a tensor product of representations contains an invariant for the diagonal subgroup if and only if the two representations are mutually contragredient, and in this case there is a unique invariant. For the symmetric group, it is known that all its irreducible representations are self-contragredient [JK], [Sag]. Thus, as a $GL(U) \times GL(V)$ -module, the final expression in formula (2.4.5.2) is equal to

$$\sum_D \rho_U^D \otimes \rho_V^D.$$

where the summation parameter ranges over all diagrams of size k and of depth at most $\min\{\dim U, \dim V\}$. This is exactly what is predicted by (GL_n, GL_m) -duality (formula (2.1.2.1)).

2.4.5.2: (GL_n, GL_m) -duality implies Schur Duality:

To derive the reverse implication, we begin with the homogeneous decomposition of $S(V)$:

$$S(V) = \sum_{k \geq 0} S^k(V).$$

The symmetric powers $S^k(V)$ are irreducible as $GL(V)$ -modules. For us it is important to note that the $S^k(V)$ are exactly the eigenspaces for the action of $C^\times$, the scalar matrices, on $S(V)$. (This is (GL_n, GL_1) -duality!) Now we observe that

$$\begin{aligned} (2.4.5.2.1) \quad S(V^m) &\cong S(V)^{\otimes m} \cong \left(\sum_{k \geq 0} S^k(V) \right)^{\otimes m} \\ &\cong \sum_{k_1, k_2, \dots, k_m \geq 0} S^{k_1}(V) \otimes S^{k_2}(V) \otimes \dots \otimes S^{k_m}(V). \end{aligned}$$

In this sum, the summands are the eigenspaces for $(C^\times)^m$, the product of the scalar matrices in each of m copies of $GL(V)$. Noting that $V^m \cong V \otimes C^m$, we see that this copy of $(C^\times)^m$ is just the diagonal torus A_m in $GL_m(C)$ acting on $V \otimes C^m$ via its action on the second factor. Thus, the tensor product $S^{k_1}(V) \otimes S^{k_2}(V) \otimes \dots \otimes S^{k_m}(V)$ is identified to $S(V \otimes C^m)^{A, \psi_k}$, the ψ_k -eigenspace for A_m , where $k = (k_1, k_2, \dots, k_m)$. In particular,

$$(2.4.5.2.2) \quad V^{\otimes m} \cong S(V \otimes C^m)^{A_m, \det},$$

where $\det = \psi_{1,m}$, the restriction to A_m of the determinant character of GL_m .

In the description of (2.4.5.2.2), there is no mention of the symmetric group S_m . It enters as the normalizer in GL_m of A_m . In any representation (τ, Y) of GL_m , the action of S_m will permute the A_m -eigenspaces:

$$(2.4.5.2.3) \quad \tau(s)(Y^{A_m, \psi}) = Y^{A_m, s(\psi)}, \quad s \in S_m, \psi \in (A_m)^\wedge$$

where $s(\psi)$ is the standard permutation action of s applied to ψ (see §1.2.3). In particular, if $s(\psi) = \psi$, then $\tau(s)$ defines an endomorphism of $Y^{A_m, \psi}$. Thus, since $\det$ is invariant under all of S_m , the space $Y^{A_m, \det}$ is an S_m -module. Since $\det$ is trivial on the diagonal matrices of determinant 1, we will call the $\det$ -eigenspace the zero-weight space.

If we combine formula (2.4.5.2.2) with (GL_n, GL_m) -duality, we obtain

$$\begin{aligned} (2.4.5.2.4) \quad V^{\otimes m} &\cong \left(\sum_D \rho_V^D \otimes \rho_m^D \right)^{A_m, \det} \\ &\cong \sum_D \rho_V^D \otimes (\rho_m^D)^{A_m, \det}. \end{aligned}$$

We note that $(\rho_m^D)^{A_m, \det} = \{0\}$ unless $|D| = m$, so the last sum is effectively over diagrams of size m .

Formula (2.4.5.2.4) describes $V^{\otimes m}$ as a $GL(V) \times S_m$ -module. To know that this description indeed amounts to Schur Duality, we need to know that the action of S_m on each space $(\rho_m^D)^{A_m, \det}$ is irreducible, and that for different D we get different representations. We formulate this as a separate result.

2.4.5.3: PROPOSITION. *If the diagram D has size m , the action of S_m on the space $(\rho_m^D)^{A_m, \det}$ is equivalent to the irreducible representation σ^D .*

Remark: This theorem has been observed by several authors in recent years [AMT], [Gut], [Kos3], but does not seem to be widely known. We will give a proof that is in the spirit of our overall discussion.

PROOF. Since we have not discussed in detail the nature of the parametrization of $(S_m)^\wedge$ by diagrams, we will content ourselves with showing that the $(\rho_m^D)^{A_m, \det}$ are irreducible and mutually inequivalent. In formula (2.4.5.2.4), let us take $V = C^m$, and consider the $\det$ -weight space for the torus A_m acting on the first factor of $C^m \otimes C^m$. From formula (2.4.5.2.4), we conclude that

$$(2.4.5.3.1) \quad (C^m)^{\otimes m})^{A_m, \det} \cong \sum_D (\rho_m^D)^{A_m, \det} \otimes (\rho_m^D)^{A_m, \det}.$$

Thus, if

$$(\rho_m^D)^{A_m, \det} \cong \sum_{\sigma \in (S_m)^\wedge} \mu_{D, \sigma} \sigma$$

is the decomposition of $(\rho_m^D)^{A_m, \det}$ as S_m -module, for appropriate multiplicities $\mu_{D, \sigma}$, the formula (2.4.5.3.1) implies that

$$\begin{aligned} (2.4.5.3.2) \quad ((C^m)^{\otimes m})^{A_m, \det} &\cong \sum_D \left(\sum_{\sigma \in (S_m)^\wedge} \mu_{D, \sigma} \sigma \right) \otimes \left(\sum_{\sigma \in (S_m)^\wedge} \mu_{D, \sigma} \sigma \right) \\ &\cong \sum_{\sigma, \tau \in (S_m)^\wedge} \left(\sum_{D, E} \mu_{D, \sigma} \mu_{E, \tau} \right) \sigma \otimes \tau \end{aligned}$$

We can also give an alternative, more elementary description of $((C^m)^{\otimes m})^{A_m, \det}$. Recall that the space $(C^m)^{\otimes m}$ has a basis consisting of elements

$$(2.4.5.3.3) \quad e_{j_1} \otimes e_{j_2} \otimes e_{j_3} \otimes \dots \otimes e_{j_m}$$

where the e_j are the standard basis for C^m . It is easy to check that the vectors (2.4.5.3.3) are eigenvectors for A_m , and further, that the A_m -eigencharacter of such a vector will be $\det$ if and only if the indices j_b include each integer from 1 to m exactly once. In other words, for the basis vector $e_{j_1} \otimes e_{j_2} \otimes e_{j_3} \otimes \dots \otimes e_{j_m}$ to belong to $((C^m)^{\otimes m})^{A_m, \det}$, the map $b \rightarrow j_b$ should be a permutation of the set $\{1, 2, 3, \dots, m\}$. Thus, the vectors $e_{j_1} \otimes e_{j_2} \otimes e_{j_3} \otimes \dots \otimes e_{j_m}$ which belong to

$((C^m)^{\otimes m})^{A_m, det}$ constitute for $((C^m)^{\otimes m})^{A_m, det}$ a basis which is in bijection with S_m . Furthermore, it is easy to check that the action of the two copies of S_m , the normalizer of A_m inside the two copies of GL_m , becomes the combined right and left regular action under this identification. Thus, in the decomposition (2.4.3.5.2) of $((C^m)^{\otimes m})^{A_m, det}$, the multiplicity of each $\sigma \otimes \tau \in (S_m \times S_m)^\wedge$ must be zero unless $\sigma = \tau$, and in this case must be 1. The only way for this to happen is for the multiplicities $\mu_{D, \sigma}$ to be zero except for a unique σ for each D , and for this $\mu_{D, \sigma}$ to equal 1. This means that representations $(\rho_m^D)^{A_m, det}$ are irreducible and distinct. Thus, the proposition is proved.

2.5: Full isotypic decomposition of $\mathcal{P}(V^\ell \otimes (V^*)^m)$.

Although we now have six versions of the FFT for GL_n , three in the context of polynomial algebras and three in the context of tensor powers of C^n , we can see that the story is still not complete. For example, although Theorem 2.1.2 describes the full isotypic decomposition for $\mathcal{P}(V^\ell)$, Theorem 2.2.2 describes only the invariants in $\mathcal{P}(V^\ell \otimes (V^*)^m)$. As observed in [Hw 1], the considerations which went into Theorems 2.1.2, 2.2.2, and 2.3.3 may be extended to yield a result which simultaneously includes Theorems 2.1.2 and 2.2.2. This development also involves the extension of Theorem 2.3.3 to a description of $\mathcal{P}(V^\ell \otimes (V^*)^m)^{GL(V)}$. This completion of the theory seems to have first been achieved by workers studying unitary representations [Gib], [GK], [KV]. Since the derivation of the isotypic decomposition is given in [Hw1] (as well as in [GK] and [KV] by very different methods), we will not give here a detailed proof, but will just review the main ingredients of the result.

2.5.1: First we have to recall that if

$$\tilde{s} : \mathcal{PD}(V) \rightarrow \mathcal{P}(V \oplus V^*)$$

is the isotropic symbol mapping (see §A.2.2.4), then $\tilde{s}\mathcal{P}(V \oplus V^*) = \tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*))$ is a Lie subalgebra of $\mathcal{PD}(V)$, isomorphic to the symplectic Lie algebra $\mathfrak{sp}_{2n}(C)$ where $n = \dim V$. The space $\mathcal{P}^2(V \oplus V^*)$ has a decomposition into 3 components: $\mathcal{P}^2(V)$, $V \otimes V^*$ and $\mathcal{P}^2(V^*)$. The corresponding summands of $\tilde{s}\mathcal{P}(V \oplus V^*)$:

$$\tilde{s}\mathcal{P}^{(2,0)} = \tilde{s}^{-1}(\mathcal{P}^2(V)), \quad \tilde{s}\mathcal{P}^{(1,1)} = \tilde{s}^{-1}(V \otimes V^*) \quad \text{and} \quad \tilde{s}\mathcal{P}^{(0,2)} = \tilde{s}^{-1}(\mathcal{P}^2(V^*))$$

define a three term grading of $\tilde{s}\mathcal{P}$. The zero level summand is isomorphic to $\mathfrak{gl}(V)$, and it normalizes the summands $\tilde{s}\mathcal{P}^{(2,0)}$ and $\tilde{s}\mathcal{P}^{(0,2)}$, acting on them by representations equivalent to those on $\mathcal{P}^2(V)$ and on $\mathcal{P}^2(V^*)$, respectively.

2.5.2: Consider now V of the form $V = Y \otimes U_1 \oplus Y^* \otimes U_2$, where Y is some other vector space. For this V , we have

$$V \oplus V^* = Y \otimes (U_1 \oplus U_2^*) \oplus Y^* \otimes (U_1^* \oplus U_2).$$

Thus,

$$(2.5.2.1) \quad \begin{aligned} \mathcal{P}^2(V \oplus V^*) &= \mathcal{P}^2(Y \otimes (U_1 \oplus U_2^*)) \oplus \mathcal{P}^2(Y^* \otimes (U_1^* \oplus U_2)) \\ &\quad \oplus (Y \otimes (U_1 \oplus U_2^*)) \otimes (Y^* \otimes (U_1^* \oplus U_2)). \end{aligned}$$

If we take the $GL(V)$ -invariants, we get

$$(2.5.2.2) \quad \begin{aligned} \mathcal{P}^2(V \oplus V^*)^{GL(V)} &= (Y \otimes (U_1 \oplus U_2^*)) \otimes (Y^* \otimes (U_1^* \oplus U_2))^{GL(Y)} \\ &= (Y \otimes Y^*)^{GL(Y)} \otimes ((U_1 \oplus U_2^*) \otimes (U_1^* \oplus U_2)) \\ &\cong (U_1 \oplus U_2^*) \otimes (U_1^* \oplus U_2). \end{aligned}$$

The image $\tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*)^{GL(Y)})$ is the centralizer of $\mathfrak{gl}(Y)$ in $\tilde{s}\mathcal{P}(V \oplus V^*)$, and this is a Lie algebra. It is isomorphic to $\mathfrak{gl}(U_1 \oplus U_2^*)$, but it does not consist entirely of linear coefficient vector fields. We have

$$(2.5.2.3) \quad \begin{aligned} \tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*)^{GL(V)}) \cap \tilde{s}\mathcal{P}^{(0,2)} &= (Y \otimes Y^*)^{GL(Y)} \otimes (U_1) \otimes (U_2) \\ \tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*)^{GL(V)}) \cap \tilde{s}\mathcal{P}^{(1,1)} &= (Y \otimes Y^*)^{GL(Y)} \otimes ((U_1 \oplus U_1^*) \otimes (U_2^* \otimes U_2)) \\ \tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*)^{GL(V)}) \cap \tilde{s}\mathcal{P}^{(2,0)} &= (Y \otimes Y^*)^{GL(Y)} \otimes (U_2^* \otimes U_2). \end{aligned}$$

Let us label these subspaces of $\mathfrak{gl}(U_1 \oplus U_2^*)$ with a^- , a^0 and a^+ , respectively. We see that $\mathfrak{gl}(U_1 \oplus U_2^*)^+$ consists of the quadratic generators of $\mathcal{P}(V)^{GL(Y)}$, i.e., the functions ξ_{jk} of Theorem 2.2.2. Also, $\mathfrak{gl}(U_1 \oplus U_2^*)^0$ consists of the infinitesimal generators of $\mathfrak{gl}(U_1) \oplus \mathfrak{gl}(U_2)$. Finally, the subspace $\mathfrak{gl}(U_1 \oplus U_2^*)^-$ consists of all second order constant coefficient differential operators which commute with $GL(Y)$. These are in various senses dual to the invariant polynomials, and are analogous to Laplacians. If we choose coordinates $\{(x_{jk}, \lambda_{j\ell}) : 1 \leq j \leq \dim V, 1 \leq k \leq \dim U_1, 1 \leq \ell \leq \dim U_2\}$ on Y , then the space $\mathfrak{gl}(U_1 \oplus U_2^*)^-$ is spanned by the operators

$$(2.5.2.4) \quad \Delta_{kl} = \sum_{j=1}^{\dim V} \frac{\partial^2}{\partial x_{jk} \partial \lambda_{j\ell}}$$

The same argument used to prove Theorem 2.3.3 can be used on this situation to make the same conclusion that: $\tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*)^{GL(Y)})$ generates the algebra of differential operators on V commuting with $GL(Y)$. From this, by an argument given in [Hw1], we can obtain a description of the full isotypic decomposition of $GL(Y)$ acting on $\mathcal{P}(V \oplus V^*)$. We give only the statement here, and refer to [Hw1] for the details of the argument.

2.5.3: Denote by $\mathcal{H}(V; GL(Y))$ the space of polynomials functions on V which are annihilated by all the operators Δ_{jk} of formula (2.5.2.4). We will refer to the functions in $\mathcal{H}(V; GL(Y))$ as the $GL(Y)$ -harmonics. Let (D_1, D_2) be a pair of diagrams with $\text{depth}(D_1) + \text{depth}(D_2) \leq \dim Y$, and let $\rho_V^{D_1, D_2}$ denote the irreducible representation of $GL(Y)$ parametrized by (D_1, D_2) (see §1.2.8). Let $\mathcal{P}(V; \rho_V^{D_1, D_2})$ denote the $\rho_V^{D_1, D_2}$ -isotypic component of $\mathcal{P}(V)$, and let $\mathcal{H}(V; \rho_V^{D_1, D_2})$ denote the subspace of $GL(Y)$ harmonics contained in $\mathcal{P}(V; \rho_V^{D_1, D_2})$. If $\mathcal{P}(V; \rho_V^{D_1, D_2})$ is non-zero, then so is $\mathcal{H}(V; \rho_V^{D_1, D_2})$. One can see this by noting that, since the operators Δ_{jk} reduce the degree of polynomials, the space of non-zero polynomials of lowest possible degree in $\mathcal{P}(V; \rho_V^{D_1, D_2})$ must consist of $GL(Y)$ -harmonics.

2.5.4: THEOREM. a) We have

$$\begin{aligned} \mathcal{P}(V) &= \mathcal{H}(V; GL(Y)) \times \mathcal{P}(V)^{GL(Y)} \\ &= \sum_{D_1, D_2} \mathcal{H}(V; \rho_V^{D_1, D_2}) \times \mathcal{P}(V)^{GL(Y)}. \end{aligned}$$

- b) The space $\mathcal{H}(V; \rho_V^{D_1, D_2})$ consists precisely of the elements of lowest degree in the isotypic component $\mathcal{P}(V; \rho_V^{D_1, D_2})$.
c) Furthermore, $\mathcal{H}(V; \rho_V^{D_1, D_2})$ is an irreducible module for the group $GL(Y) \times GL(U_1) \times GL(U_2)$, and consists of a copy of

$$\rho_V^{D_1, D_2} \otimes (\rho_{U_1}^{D_1})^* \otimes (\rho_{U_2}^{D_2})^*.$$

In the sum in part a), all pairs (D_1, D_2) of diagrams occur which satisfy $\text{depth}(D_1) \leq \dim U_2$, $\text{depth}(D_2) \leq \dim U_1$, and $\text{depth}(D_1) + \text{depth}(D_2) \leq \dim Y$.

2.5.5: Remarks: i) The dualization of the D_i in part c) is due to the dualization attendant on looking at the polynomial algebra rather than the symmetric algebra. The module in the analog of part c) for the symmetric algebra would be

$$\rho_V^{D_1, D_2} \otimes \rho_{U_1}^{D_1} \otimes \rho_{U_2}^{D_2}$$

- ii) As is explained in [Hw1], there is more structure in the situation than is revealed by Theorem 2.5.4. The isotypic components are irreducible modules for the joint action of $GL(Y)$ and $\tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*)^{GL(Y)})$, and the action of each of $GL(Y)$ and $\tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*)^{GL(Y)})$ determines the action of the other. We did not state this formally here because it involves infinite dimensional representations and other ideas somewhat alien to invariant theory as customarily understood. However, the action of $\tilde{s}^{-1}(\mathcal{P}^2(V \oplus V^*)^{GL(Y)})$ is significant even in very classical contexts - cf. [Hw 1].

- iii) Part a) holds in a quite general context [He3], [Kos2]. What is special about the current situation is the specific structure of the harmonics. Understanding of this seems to be fairly recent, and to be a by-product of the study of infinite dimensional representation theory. The analogous phenomenon for the orthogonal group described in [Gib] was helpful in suggesting the formulation of [Hw1]. See [GK], [KV], for similar results. In the context of invariant theory, a version of Theorem 2.5.4 was found by R. Brylinski [Bry].

- iv) One can again write down the joint $GL(Y) \times GL(U_1) \times GL(U_2)$ -highest weight vectors. We use the coordinates $\{x_{jk}, \lambda_{j\ell}\}$ introduced in §2.5.2. Denote by δ_a the determinant of the upper left hand $a \times a$ submatrix of the matrix $\{x_{jk}\}$. Thus, except for a change in notation from $\{t_{jk}\}$ to $\{x_{jk}\}$ as coordinates, these δ_a are the same functions as were defined at the end of §2.1.4. Define also determinants

$$(2.5.5.1) \quad \bar{\delta}_b = \det \begin{vmatrix} \lambda_{n,1} & \cdots & \lambda_{n,b} \\ \vdots & \ddots & \vdots \\ \lambda_{n-b+1,1} & \cdots & \lambda_{n-b+1,b} \end{vmatrix}$$

Thus, the $\bar{\delta}_b$ are the determinants of the lower left hand $b \times b$ submatrix of $\{\lambda_{ji}\}$. Given a diagram $D = (\tau_1, \tau_2, \tau_3, \dots)$ of depth not more than $\min\{\dim V, \dim U_1\}$, define a monomial

$$(2.5.5.2) \quad \delta^D = \prod_a \delta_a^{\tau_a - \tau_{a+1}}.$$

Similarly, given another diagram $E = (s_1, s_2, s_3, \dots)$, of depth not more than $\min\{\dim V, \dim U_2\}$, define

$$(2.5.5.3) \quad \bar{\delta}^E = \prod_b \bar{\delta}_b^{s_b - s_{b+1}}.$$

Consider the product $\delta^D \bar{\delta}^E$. Suppose that $\text{depth}(D) + \text{depth}(E) \leq \dim V$. It is then easy to compute that this is a highest weight vector of $GL(V) \times GL(U_1) \times GL(U_2)$, and that it generates a representation of type $\rho_V^{D_1, D_2} \otimes (\rho_{U_1}^{D_1})^* \otimes (\rho_{U_2}^{D_2})^*$. Here the main computation is or the case when $\delta^D \bar{\delta}^E$ is one of the δ_a or $\bar{\delta}_b$. The general case follows because $GL(V) \times GL(U_1) \times GL(U_2)$ acts by automorphisms in the polynomial algebra. The restriction on the depths of D and E is required to insure that there is no cancellation between the $GL(V)$ weights defined by δ^D and $\bar{\delta}^E$. This restriction also guarantees that $\delta^D \bar{\delta}^E$ is harmonic, because the set of j such that x_{jk} appears with positive degree in δ^D is disjoint from the set of j such that $\lambda_{j\ell}$ appears with positive degree in $\bar{\delta}^E$. Thus, the $\delta^D \bar{\delta}^E$ are exactly the set of harmonic joint $GL(V) \times GL(U_1) \times GL(U_2)$ highest weight vectors. As a final remark, we observe that if $\dim(U_1) + \dim(U_2) \leq \dim(V)$, then the condition on $\text{depth}(D) + \text{depth}(E)$ is automatically satisfied, and the set of harmonic highest weight vectors form a semigroup under multiplication. When $\dim(U_1) + \dim(U_2) > \dim(V)$, the set of highest weight vectors is a (non-disjoint) union of semigroups defined by requiring $\text{depth}(D) \leq d$ and $\text{depth}(E) \leq \dim(V) - d$ for some d between 0 and $\dim(V)$. These various semigroups correspond to the irreducible components of the null fiber of the quotient mapping (see §3.7 for a more complete discussion of these notions in the case of the orthogonal group).

2.6: The Second Fundamental Theorem.

Suppose for some group G acting on a vector space V , we know generators for the algebras $\mathcal{P}(V^m)^G$ for all m . By the Second Fundamental Theorem (SFT) of invariant theory for the action of G on V , Weyl means a description of the relations between the generators. We should understand this in the following sense: since $\mathcal{P}(V^m)^G$ is a GL_m -module, we should consider as sets of generators only sets which span GL_m -invariant subspaces of $\mathcal{P}(V^m)^G$. Weyl gives in [W] the SFT as well as the FFT for the standard actions of the classical groups. Here we will explain how the SFT can be understood in terms of multiplicity-free actions. The SFT is also discussed in [Vu]. To simplify the discussion, take $U_1 = C^\ell$ and $U_2 = C^m$. We have defined the quotient mapping $\mu : \text{Hom}(C^\ell, V) \oplus \text{Hom}(V, C^m) \rightarrow M_{m,\ell}$ in formula (2.2.3.6). The fundamental generators ξ_{jk} are the pullbacks by μ of the usual matrix entries on $M_{m,\ell}$. We have seen that the image of μ consists of the set R_d , where $d = \dim V$, of matrices of rank at most $\min\{d, \ell, m\}$.

Thus, the relations between the $\xi_{j,k}$ are the equations defining R_d . If $d \geq \min\{1, m\}$, then $R_d = M_{m,\ell}$ and there are no non-trivial relations. However, if $d < \min\{1, m\}$, it was well known classically that the determinant of any $(d+1) \times (d+1)$ submatrix of $T \in M_{m,\ell}$ must vanish if T belongs to R_d . It is also well known [Ru] [HU] that these determinants generate the ideal defining R_d . Thus, the SFT says that the vanishing of these determinants provide all the relations among the $\xi_{j,k}$. The point we wish to make here is that the result that the $(d+1) \times (d+1)$ minors generate the ideal defining R_d is a simple consequence of the theory of multiplicity-free actions. Indeed, we know from §2.1 that the action of $GL_m \times GL_\ell$ on $M_{m,\ell}$ is multiplicity-free. The variety R_d is invariant and irreducible (since it contains a dense orbit of $GL_m \times GL_\ell$), so the ideal defining R_d is $GL_m \times GL_\ell$ -invariant and prime. An elementary argument shows that, for a multiplicity-free action of a group G on a vector space V (or any variety whose coordinate ring has unique factorization), the prime G -invariant ideals must be generated by the G -modules generated by some subset of the fundamental highest weight vectors (cf. §1.3.8). The determinants of the $(d+1) \times (d+1)$ submatrices are a basis for the $GL_m \times GL_\ell$ -module generated by the fundamental highest weight vector δ_{d+1} (see §2.1.4 for notation). From this, and some simple calculations which show that for $s > d+1$ the determinants of $s \times s$ minors are in the ideal generated by the $(d+1) \times (d+1)$ minors, the SFT follows. For more details, we refer to [HU].

2.7: (GL_n, GL_m) -duality and the role of the Capelli identities.

Weyl [W] dubbed the “First Fundamental Theorem” of invariant theory as the description, for a given group $G \subseteq GL_n$, of the invariants of the action of G on $\mathcal{P}((\mathbb{C}^n)^m)$ for arbitrary m . These should be described in terms of “typical basic invariants”, which means a collection of generators for $\mathcal{P}((\mathbb{C}^n)^{m_0})^G$ for some fixed m_0 , such that when one acts on them by polarization operators, one obtains a set of generators for $\mathcal{P}((\mathbb{C}^n)^m)^G$, for any m . One of Weyl’s main general results in [W] is the solvability, in principle, of this problem. More precisely, he reduces it to the standard finiteness theorems by showing that $m_0 = n$ will do: if one has generators for $\mathcal{P}((\mathbb{C}^n)^n)^G$, then one can manufacture generators for $\mathcal{P}((\mathbb{C}^n)^m)^G$ for any $m > n$ by means of polarization operators applied to these generators. Weyl proved this result by means of the Capelli identities. Many have found this proof less than enlightening. In particular, the Capelli identities themselves have been called “mysterious” [ABP]. I and others have written elsewhere [Hw 1], [HU], [Ram], [KS] about the meaning of the Capelli identities themselves, and I will not repeat that story here. The goal of this section is to point out that Weyl’s finiteness theorem for the typical basic invariants can be understood conceptually and quite directly by means of (GL_n, GL_m) -duality, with no need for any formal tool such as the Capelli identities. This observation can be found in [Ram].

First we formalize the notion of “typical basic invariant” in the spirit of this paper. A collection $\{\gamma_j\}$ of elements of $\mathcal{P}(\mathbb{C}^n \otimes \mathbb{C}^{m_0})^G$ is a set of typical basic invariants if for any $m > m_0$, the images of the γ_j under the standard embedding $\mathcal{P}(\mathbb{C}^n \otimes \mathbb{C}^{m_0})^G \rightarrow \mathcal{P}(\mathbb{C}^n \otimes \mathbb{C}^m)^G$ generate a GL_m -module which generates $\mathcal{P}(\mathbb{C}^n \otimes \mathbb{C}^m)^G$. By the standard embedding, we mean that if we regard $\mathbb{C}^n \otimes \mathbb{C}^{m_0}$ as the $n \times m_0$ matrices, then elements of $\mathcal{P}(\mathbb{C}^n \otimes \mathbb{C}^{m_0})$ depend only on the first m_0

columns of these matrices.

2.7.1: THEOREM. For $G \subseteq GL_n$, a set of generators for $\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G)$ is a set of typical basic invariants. More precisely, for any $m > n$, the image of $\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G)$ under the standard embedding $\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G) \rightarrow \mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G)$ generates $\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G)$ as GL_m -module.

PROOF. Using (GL_n, GL_m) -duality Theorem, we can write

$$\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G) \cong \left(\sum_D \rho_n^D \otimes \rho_m^D \right)^G \cong \sum_D (\rho_n^D \otimes \rho_m^D)^G \cong \sum_D \rho_n^D \otimes \rho_m^D.$$

The summation is over diagrams D of depth at most n . For each such D , the GL_n -submodule $(\rho_n^D)^G \otimes (\rho_m^D)^G$ of $\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G)$ is embedded into $(\rho_n^D)^G \otimes \rho_m^D$ under the standard embedding of $\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G)$ in $\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^m)^G)$. Since ρ_m^D is irreducible as GL_m -module, it is generated by ρ_n^D . Hence also $(\rho_n^D)^G \otimes \rho_m^D$ is generated by $(\rho_n^D)^G \otimes \rho_n^D$.

Since the representation of GL_n corresponding to a diagram of depth n is a twist by some power of $\det$ of a representation of corresponding to a diagram of depth $n-1$, one can easily modify the above argument to obtain Weyl’s refinement, that if $G \subseteq SL_n$, then a set of generators for $\mathcal{P}((\mathbb{C}^n \otimes \mathbb{C}^{m-1})^G)$, together with $\det$, are a set of typical basic invariants.

In [Sch 2,3], Schwarz gave some improvements on Weyl’s results in case G preserves a bilinear form. These can be understood by the considerations of Theorem 2.6.1 applied to the isotypic decompositions for the orthogonal or symplectic groups. We will discuss the case of symplectic isometries, which is cleaner than the orthogonal group case.

2.7.2: THEOREM. For a group $G \subseteq Sp_{2n} \subseteq GL_{2n}$, a set of generators for $\mathcal{P}((\mathbb{C}^{2n} \otimes \mathbb{C}^n)^G)$ is a set of typical basic generators for G . More precisely, for any $m > n$, the image under the standard embedding $\mathcal{P}((\mathbb{C}^{2n} \otimes \mathbb{C}^n)^G) \rightarrow \mathcal{P}((\mathbb{C}^{2n} \otimes \mathbb{C}^m)^G)$ of the space $\mathcal{H}((\mathbb{C}^{2n})^n, Sp_{2n})^G$ of G -invariant harmonics in n variables, generates $\mathcal{P}((\mathbb{C}^{2n} \otimes \mathbb{C}^m)^G)$ as a joint module for GL_m and $\mathcal{P}((\mathbb{C}^{2n} \otimes \mathbb{C}^m)^{Sp_{2n}})$.

PROOF. This follows by the argument of Theorem 3.8.5.3 together with formulas (3.8.5.4) and (3.8.5.5).

3: The FFT for the Other Classical Groups.

The standard FFT for orthogonal groups O_n and for symplectic groups Sp_{2n} is quite analogous to that for GL_n and simpler to formulate since there is no need to explicitly deal with contragredient vectors, because the standard representations of these groups are self-contragredient. Here, we will show how to deduce the FFT for these groups from (GL_n, GL_m) -duality, as discussed in §2.1, and from the fact that the coset spaces GL_n/O_n and GL_{2n}/Sp_{2n} , being symmetric spaces, are multiplicity-free. Since the line of argument and the results are quite similar for the orthogonal and symplectic cases, we will discuss in detail only the orthogonal case and summarize briefly the results for the symplectic group.

3.1: The action of GL_n on symmetric matrices.

Let V be, as before, a complex finite-dimensional vector space.

THEOREM. *The natural action of $GL(V)$ on $S^2(V)$, the second symmetric power of V , is multiplicity-free. Precisely, we have the decomposition*

$$S(S^2(V)) \cong \sum_D \rho_V^D,$$

where D ranges over all diagrams of depth at most n . Thus, the representations appearing in this sum are all those corresponding to diagrams with rows of even length.

(3.2) PROOF. Without loss of generality, we can identify V with C^n in such a way that $S^2(V)$ becomes the symmetric $n \times n$ matrices, with GL_n acting by the recipe

$$(3.2.1) \quad g(T) = gTg^t \quad g \in GL_n, T \in S^2(C^n).$$

Here, g^t denotes the transpose of the matrix g . Also, we may replace the symmetric algebra by the polynomial algebra.

Consider the identity matrix I_n . A simple calculation shows that the orbit of I_n under the Borel subgroup of upper triangular matrices is open and dense in $S^2(C^n)$. More precisely, let δ_k denote the determinant of the $k \times k$ leading minor of T :

$$(3.2.2) \quad \delta_1 = t_{11}, \quad \delta_2 = \det \begin{vmatrix} t_{11} & t_{12} \\ t_{12} & t_{22} \end{vmatrix}, \quad \delta_3 = \det \begin{vmatrix} t_{11} & t_{12} & t_{13} \\ t_{12} & t_{22} & t_{23} \\ t_{13} & t_{23} & t_{33} \end{vmatrix},$$

etc. Here, $\{t_{jk} = t_{kj}\}$ are the entries of T . The basic theory of Gaussian elimination [Stg] tells us that if $\delta_k \neq 0$ for all k , then we can find a (unique) unipotent lower triangular matrix u such that uT is upper triangular (cf. Remark 2.1.5 a). Then the matrix uTu^t is still upper triangular and also symmetric; hence it is diagonal. Now acting by $(uTu^t)^{-1/2}$ will transform T to I_n .

It follows from Proposition 1.3.3 that the action of GL_n on $S^2(C^n)$ is multiplicity-free. Further, we can use the criterion of Proposition 1.3.6 to limit which representations of GL_n can occur in the decomposition of $\mathcal{P}(S^2(C^n))$. We see that the group E of diagonal matrices with entries equal to ± 1 stabilizes I_n . (This is exactly

the stabilizer of I_n in B , but we do not need to know that.) Proposition 1.3.6 says that in order for a character $\psi = \psi_\ell$ to be the highest weight of a representation in $S^2(C^n)$, ψ_ℓ should be trivial on the group E . It is straightforward to check that this holds if and only if the entries of ℓ are even. Furthermore, since the action of GL_n on $S^2(C^n)$ is clearly polynomial, all representations in $S(S^2(C^n))$ will also be polynomial. Thus, the sum on the right hand side of the formula in Theorem 3.1 contains all representations which could possibly occur in $S(S^2(C^n))$. To finish the proof, it remains only to show that all the stated representations do occur. According to the discussion of §1.3.8, the crucial point is to show that the highest weights corresponding to the generators of the semigroup $(A_n)^{++}(S^2(C^n))$ do occur. By inspection, we see that the highest weights in question are those corresponding to the vectors $2\mathbf{k}$, i.e., twice the fundamental generators of the cone C^+ (see §§1.2.5 and 1.2.6). But the reader may verify that the functions δ_k described in formula (3.2.2) have (the duals of) these highest weights.

3.3: Complements to Theorem 3.1.

The stabilizer of 1_n , the $n \times n$ identity matrix, in GL_n is the orthogonal group:

$$O_n = \{r \in GL_n : rr^t = 1_n\}.$$

Thus, the map $g \rightarrow gg^t$ defines an embedding of the coset space GL_n/O_n in $S^2(C^n)$. The image is the (open, dense) set of invertible symmetric matrices. Thus, the restriction map from $\mathcal{P}(S^2(C^n))$ to $\mathcal{R}(GL_n/O_n)$ is injective. We know that GL_n/O_n is multiplicity-free since it is a symmetric space (see §1.3.5; precisely, O_n is the set of fixed points for the involution $g \rightarrow (g^t)^{-1}$ of GL_n). This gives an alternate proof of the multiplicity-freeness of $S^2(C^n)$.

The representations occurring in $\mathcal{R}(GL_n/O_n)$ need not be polynomial. Indeed, the determinant function δ_n on $S^2(C^n)$ is invertible on GL_n/O_n . Note that δ_n is an eigenfunction for GL_n , with $\det^2$ as associated eigencharacter. If we multiply any GL_n highest weight vector in $\mathcal{R}(GL_n/O_n)$ by δ_n^{-1} , we get another highest weight vector, transforming by $\det^{-2}$ times the character associated to the original function. Thus, from the description in Theorem 3.1 of the GL_n -module structure of $S(S^2(C^n))$, we can conclude that every representation of the form $\rho_n^{2D, 2E}$ occurs in $\mathcal{R}(GL_n/O_n)$ (see §1.2.8 for notation). On the other hand, the argument in §3.2, involving the group E of diagonal matrices of order 2, remains valid and implies only representations of this form can occur. Thus, we have established the following variant of Theorem 3.1.

3.3.1: COROLLARY. *The coset space GL_n/O_n is multiplicity-free and we have the GL_n -module decomposition*

$$\mathcal{R}(GL_n/O_n) \cong \sum_{D, E} \rho_n^{2D, 2E}.$$

Here, the sum is over all pairs of diagrams, the sum of whose depths is not more than n .

Applying Frobenius Reciprocity (§1.1.17) to this description of $\mathcal{R}(GL_n/O_n)$, we get information about O_n -invariants in GL_n -modules.

3.3.2: COROLLARY. In any irreducible GL_n -module $\rho_n^{2D,2E}$, the space $(\rho_n^{2D,2E})^{O_n}$ of vectors invariant under the orthogonal group O_n is at most one dimensional. It has dimension 1 if and only if all the rows of D and E have even length.

Remark: The determinant of an orthogonal matrix is ± 1 . Let SO_n be the special orthogonal group - the subgroup of O_n whose elements have determinant one. It has index 2 in O_n . A vector invariant under SO_n is the sum of a vector invariant under O_n and a vector transforming by the determinant character of O_n . Using this observation, we can slightly extend Corollary 3.3.2 and state that a representation $\rho_n^{D,E}$ of GL_n has at most one SO_n -invariant vector and does have one if and only if the lengths of the rows of D and E are either all even or all odd. (If they are all odd, then we should have $\text{depth } D + \text{depth } E = n$.)

3.4: FFT for O_n , standard polynomial version.

We are now in a position to establish the FFT for O_n . We recall that O_n is the subgroup of GL_n which preserves the inner product

$$(3.4.1) \quad (u, v) = \sum_{j=1}^n u_j v_j$$

for n -tuples $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ of complex numbers. We consider the action of O_n on the direct sum $(\mathbb{C}^n)^m$ of m copies of $\mathbb{C}^n$. We may think of $(\mathbb{C}^n)^m$ as the space of $n \times m$ matrices, with O_n acting by multiplication on the left. Write a typical element of $(\mathbb{C}^n)^m$ as an m -tuple:

$$(3.4.2) \quad v = (v_1, v_2, v_3, \dots, v_m) \quad v_j \in \mathbb{C}^n.$$

On $(\mathbb{C}^n)^m$ we have a collection of second order polynomials $\tilde{\xi}_{ij} = \tilde{\xi}_{ji}$, $1 \leq i, j \leq n$, which are obviously invariant under O_n :

$$(3.4.3) \quad \tilde{\xi}_{ij}(v) = (v_i, v_j) = \sum_{a=1}^n v_{ai} v_{aj}.$$

3.4.4: THEOREM. The polynomials $\tilde{\xi}_{ij}$ of formula (3.4.3) generate the algebra $\mathcal{P}((\mathbb{C}^n)^m)^{O_n}$.

(3.4.5) PROOF. To begin the proof, we give a geometric interpretation of the theorem. The parallel between the discussion below and that of §2.2.3, from equation (2.2.3.4) through equation (2.2.3.8) should be clear to the reader. Think of $(\mathbb{C}^n)^m$ as $M_{n,m}$, the space of $n \times m$ matrices. Think of $S^2(\mathbb{C}^m)$ as the space of symmetric $m \times m$ matrices. Define a map

$$(3.4.5.1) \quad \begin{aligned} Q : (\mathbb{C}^n)^m &\rightarrow S^2(\mathbb{C}^m) \\ T &\rightarrow T^s T. \end{aligned}$$

By straightforward calculation, we see that

$$(3.4.5.2) \quad Q(gT) = Q(T) \quad Q(Th) = h^* Q(T) h, \quad T \in M_{n,m}, g \in O_n, h \in GL_m.$$

Thus, the pullback map

$$(3.4.5.3) \quad \begin{aligned} Q^* : \mathcal{P}(S^2(\mathbb{C}^m)) &\rightarrow \mathcal{P}(M_{n,m}) \\ Q^*(p)(T) &= p(Q(T)) \quad p \in \mathcal{P}(S^2(\mathbb{C}^m)), \end{aligned}$$

will, in fact, map to the algebra $\mathcal{P}(M_{n,m})^{O_n}$ of O_n -invariants. Furthermore, since GL_m commutes with O_n , the algebra $\mathcal{P}(M_{n,m})^{O_n}$ will be invariant under the action of GL_m on $\mathcal{P}(M_{n,m})$, and therefore $\mathcal{P}(M_{n,m})^{O_n}$ has the structure of GL_m -module. The second of equations (3.4.5.2) says that Q^* intertwines the actions of GL_m on $\mathcal{P}(S^2(\mathbb{C}^m))$ and $\mathcal{P}(M_{n,m})^{O_n}$. Finally, we observe that the functions of Theorem 3.4.4 are exactly the Q -pullbacks of the standard coordinate functions on $S^2(\mathbb{C}^m)$. Thus, Theorem 3.4.4 is equivalent to the following statement.

3.4.5.4: The map $Q^* : \mathcal{P}(S^2(\mathbb{C}^m)) \rightarrow \mathcal{P}(M_{n,m})^{O_n}$ is surjective.

We will prove this form of the theorem. According to Theorem 2.1.2, we have a $GL_n \times GL_m$ -module decomposition

$$\mathcal{P}(M_{n,m}) = \sum_D \rho_n^D \otimes \rho_m^D.$$

Thus, since O_n is a subgroup of GL_n , if we take the O_n -invariants, we find

$$(3.4.5.5) \quad \begin{aligned} \mathcal{P}(M_{n,m})^{O_n} &= \left(\sum_D \rho_n^D \otimes \rho_m^D \right)^{O_n} = \sum_D (\rho_n^D)^{O_n} \otimes \rho_m^D \\ &\cong \sum_E \rho_m^E. \end{aligned}$$

The last isomorphism follows from Corollary 3.3.2. The sum in the last expression is over all diagrams of depth at most $\min\{n, m\}$. We now understand $\mathcal{P}(M_{n,m})^{O_n}$ as a GL_m -module, and our task is to convert that knowledge into understanding of the algebra structure. What needs to be done is to show that all the irreducible GL_m -modules which appear in the final sum in equation (3.4.5.5) are in the image of Q^* . Theorem 3.1 tells us that copies of all the modules needed do appear in $\mathcal{P}(S^2(\mathbb{C}^m))$, so all we need to do is show that their images under Q^* are non-zero. To do this it suffices to check that Q^* does not kill the appropriate highest weight vectors. Because of the multiplicative structure of the set of highest weight vectors, this comes down to showing that Q^* applied to the δ_k of formula (3.2.2), for $1 \leq k \leq \min\{n, m\}$, is non-zero. By the relation between determinants and rank, this follows because of the fact from elementary linear algebra, that the image of the map Q will consist of all symmetric matrices of rank up to $\min\{n, m\}$. This finishes the proof of Theorem 3.4.4.

3.4.6: Remark: The strong parallels between this proof and the proof of Theorem 2.2.2 should be evident to the reader. In particular, this proof and that one each

produces a variety (namely the variety of matrices of appropriately bounded rank in the image space) whose ring of regular functions is isomorphic to the algebra of invariants. In the parlance of geometric invariant theory, such a variety is called a quotient for the action. The geometric viewpoint toward the FFTs for the classical groups was promoted in [Pro].

3.5: Operator commutant and full isotypic decomposition for O_n .

Using the yoga of the Weyl algebra (see Appendix 2), we can turn Theorem 3.4.4 into a description of the commuting algebra $\mathcal{PD}(M_{n,m})^{O_n}$. Since the invariants are generated by elements of degree two, just as in the case of the FFT for GL_n , we again obtain a description of $\mathcal{PD}(M_{n,m})^{O_n}$ as the associative algebra generated by the Lie algebra of the centralizer of O_n in the Lie algebra $\hat{\mathfrak{sp}}$. From this in turn, we can get a description of the full isotypic decomposition of $\mathcal{P}(M_{n,m})$ as an O_n -module. This line of argument is given in [Hw1], and we will not repeat it here, except to formulate the final result. We would emphasize, however, that the argument depends heavily on the action of the Lie algebra centralizing the classical group in question, which in this case is O_n .

As was the case for GL_n , the centralizing Lie algebra consists of three parts: i) a Lie algebra of normalized linear coefficient vector fields, essentially coming from the action of GL_m acting on $M_{n,m}$ by multiplication on the right; ii) the space of degree two invariant polynomials (this space is spanned by the ξ_{ij} of formula (3.4.3)); and iii) a space of second order constant coefficient differential operators dual to the invariant polynomials. These differential operators may be thought of as "partial Laplacians". A spanning set is given, in the standard coordinates on $M_{n,m}$, by the formulas

$$(3.5.1) \quad \Delta_{ij} = \sum_{a=1}^n \frac{\partial^2}{\partial v_{ai} \partial v_{aj}}$$

We denote by $\mathcal{H}(M_{n,m}; O_n)$ the space of polynomials on $M_{n,m}$ which are annihilated by all the operators Δ_{ij} . For an irreducible representation τ of O_n , we let $\mathcal{P}(M_{n,m}; \tau)$ denote the τ -isotypic component of $\mathcal{P}(M_{n,m})$, and $\mathcal{H}(M_{n,m}; \tau)$ denote the τ -isotypic component of $\mathcal{H}(M_{n,m})$.

3.5.2: THEOREM. a) We have

$$\mathcal{P}(M_{n,m}) = \mathcal{H}(M_{n,m}; O_n) \mathcal{P}(M_{n,m})^{O_n}.$$

That is,

$$\mathcal{P}(M_{n,m}; \tau) = \sum_{\tau' \in O_n} \mathcal{H}(M_{n,m}; \tau) \mathcal{P}(M_{n,m})^{O_n}$$

for each τ in $O_n^\wedge$.

- b) The space $\mathcal{H}(M_{n,m}; \tau)$ consists of precisely the elements of $\mathcal{P}(M_{n,m}; \tau)$ of lowest possible degree.
- c) The space $\mathcal{H}(M_{n,m}; \tau)$ is an irreducible module for the joint action of $O_n \times GL_m$, of the form $\tau \otimes \tau'$, where τ' is an appropriate irreducible module for GL_m . Furthermore, the correspondence $\tau \leftrightarrow \tau'$ is bijective (between the representations which occur - these may not exhaust either $O_n^\wedge$ or $GL_m^\wedge$).

3.6: Structure of the harmonics.

The reader will observe the virtually complete parallel between this result and Theorem 3.4. The main difference is that we do not describe explicitly the bijection (3.5.2)c). This was first done in [KV] (the case $n \geq 2m$ was treated in [Glb]). We are not at the moment in a position to do that, because we have not discussed how to parametrize $O_n^\wedge$. In fact, the correspondence gives us a parametrization of $O_n^\wedge$ by diagrams. We will describe this. Some care is required to take into account the non-connectedness of O_n .

3.6.1: Polarized coordinates for O_n

The standard diagonal form (3.4.1) for the inner product is not convenient for studying representations of O_n , because in these coordinates, it is hard to see what a Borel subgroup of O_n looks like. To simplify calculations of highest weight vectors, we should change coordinates. If $v = (v_1, v_2, v_3, \dots, v_n)^t$ are the usual coordinates on C^n , let us set

$$z_j = v_{2j-1} + v_{2j}, \quad 1 \leq j \leq [n/2]$$

and $z_{(n+1)/2} = \sqrt{2}v_n$ if n is odd.

Here, $i = \sqrt{-1}$. Then the z_j together with their "complex conjugates" $z_{n+1-j} = v_{2j-1} - v_{2j}$ define a coordinate system on C^n . In this new coordinate system, the inner product (3.4.1) takes the form

$$(z, z') = (1/2) \left(\sum_{j=1}^n z_j z'_{n+1-j} \right).$$

The associated Laplacian is

$$\tilde{\Delta} = 4 \sum_{j=1}^n \frac{\partial^2}{\partial z_j \partial z_{n+1-j}}.$$

In the coordinate system defined by the z_j , the Lie algebra $\mathfrak{so}_n$ of O_n is spanned by the linear vector fields

$$\tilde{e}_{jk} = z_j \frac{\partial}{\partial z_k} - z_{n+1-k} \frac{\partial}{\partial z_{n+1-j}}, \quad 1 \leq j, k \leq n.$$

Of these, the $\tilde{e}_{jk}$ with $j \leq k$ and $j \leq n/2$ span a Borel subalgebra $\mathfrak{b}_{\mathfrak{so}_n} = \mathfrak{b}$ of $\mathfrak{so}_n$. We note also that the $\tilde{e}_{jk}$ with $j, k \leq n/2$ span a Lie subalgebra of $\mathfrak{so}_n$ isomorphic to $\mathfrak{gl}_m$ with $m = [n/2]$. The intersection $\mathfrak{gl}_m \cap \mathfrak{b}$ is the standard upper triangular Borel subalgebra of $\mathfrak{gl}_m$ and the diagonal Cartan subalgebra $\mathfrak{a}_m$, spanned by the $\tilde{e}_{jj}$ with $j \leq n/2$, is also a Cartan subalgebra of $\mathfrak{so}_n$. Hence we can use the notation of §1.2 to denote the characters of $\mathfrak{a}_m$, and the highest weight of any representation of O_n will necessarily be a highest weight for $\mathfrak{gl}_m$, hence will belong to the cone described in §1.2.3.

3.6.2: Highest weight vectors and the diagrams for representations of O_n

On $(C^n)^m$ we take coordinates z_{ak} , which for each fixed k are a copy of the coordinates described in the preceding paragraph. The partial Laplacians are given by the formula

$$(3.6.2.1) \quad \tilde{\Delta}_{jk} = 4 \sum_{a=1}^n \frac{\partial^2}{\partial z_{aj} \partial z_{n+1-a,k}}.$$

Consider the functions

$$(3.6.2.2) \quad \delta_b = \det \begin{vmatrix} z_{11} & z_{12} & \cdots & z_{1b} \\ z_{21} & z_{22} & \cdots & z_{2b} \\ \vdots & \vdots & \ddots & \vdots \\ z_{b1} & z_{b2} & \cdots & z_{bb} \end{vmatrix}.$$

It is easily checked that the δ_b are all highest weight vectors for the Borel subgroup B described in the previous paragraph. Furthermore, the δ_b are harmonic, i.e., annihilated by the $\tilde{\Delta}_{jk}$. For $b \leq n/2$, this is quite obvious; in fact, every term of $\tilde{\Delta}_{jk}$ annihilates δ_b because it involves differentiating with respect to a variable z_{aj} or z_{ak} with $a > n/2$. For $b > n/2$, the verification that δ_b is harmonic depends on cancellation of the results of applying the a -th and $n+1-a$ -th terms of $\tilde{\Delta}_{jk}$ to δ_b . The weight of δ_b is $(1, 1, \dots, 1, 0, \dots, 0) = 1_m$, where the number of ones is $m = \min(b, n-b)$, and there are $[n/2] - m$ zeroes. Thus, if $b+b' = n$, both δ_b and $\delta_{b'}$ have the same weight for B , and therefore they generate equivalent representations of SO_n . However, we can see that the representation of O_n generated by $\delta_{b'}$ is the twist by determinant of the representation generated by δ_b (except, of course, when $b = b' = n/2$). To see this, consider the action of a suitable element from O_n on the δ_b . If n is odd, we can take for the generator of O_n modulo SO_n the transformation r_0 which sends $z_{(n+1)/2}$ to its negative and leaves all other coordinates fixed. Clearly, if $b < (n)/2$, then $r_0(\delta_b) = \delta_b$, while if $b > n/2$, then $r_0(\delta_b) = -\delta_b$. Similarly, if n is even, we can take r_0 to be the transformation which switches z_m and z_{m+1} , $m = n/2$, and leaves the other coordinates alone. We again find that the action of r_0 on $\delta_{b'}$ differs by sign from the action on δ_b , as long as $b \neq n/2$.

If we have a diagram $D = (a_1, a_2, a_3, \dots)$ of depth $d \leq n/2$, the polynomial

$$(3.6.2.3) \quad \delta_D = \prod_{j \leq d} \delta_j^{b_j},$$

where $b_j = a_j - a_{j+1}$, is easily checked to be a harmonic highest weight vector for B , of weight ψ_D . The polynomial

$$(3.6.2.4) \quad \tilde{\delta}_D = \left(\prod_{j < d} \delta_j^{b_j} \right) \delta_d^{a_d-1} \delta_d^{a'}$$

can be checked, not quite so easily as δ_D , also to be a harmonic highest weight vector for B and also of weight ψ_D . By our discussion of the relation between

δ_d and δ'_d , we know that the representation generated by $\tilde{\delta}_D$ will be the twist by determinant of the representation generated by δ_D .

It seems natural to associate to δ_D , and to the representation it generates, the diagram D . We do so and, in particular, denote with the symbol σ_n^D the (isomorphism class of the) representation generated by δ_D . Note that we must have $n \geq 2 \text{ depth } D$ for this definition to apply. It is less clear how to associate a diagram to $\tilde{\delta}_D$. However, if we consider the inclusion $O_n \subset GL_n$, and let $\tilde{B}_n$ be the Borel subgroup of upper triangular matrices with respect to the coordinates z_j , then we see by referring to §2.1 that $\tilde{\delta}_D$ is a joint $GL_n \times GL_m$ highest weight vector, with weight ψ_E where E is the diagram

$$(3.6.2.5) \quad E = t(D, n)$$

which results from D by changing the length of the first column of D from $d = \text{depth } D \leq n/2$ to $d' = n - d$. This consideration leads us to associate the diagram $E = t(D, n)$ to $\tilde{\delta}_D$. We will also use the notation $\sigma_n^{t(D,n)}$ to indicate the representation generated by $\tilde{\delta}_D$. Thus, for a diagram D of depth at most $n/2$, we have

$$(3.6.2.6) \quad \sigma_n^{t(D,n)} \cong \sigma_n^D \otimes \det.$$

By the depth of a representation of O_n , we will mean the depth of its associated diagram. This will never exceed n . If $\text{depth}(\sigma) \leq n/2$, then it equals the depth of the diagram associated to its highest weight; but if $\text{depth}(\sigma) > n/2$, then the diagram of its highest weight has depth $n - \text{depth}(\sigma)$. We will also write

$$(3.6.2.7) \quad \delta_{t(D,n)} = \tilde{\delta}_D.$$

Thus, we have found a parametrization of $(O_n)^\wedge$ by diagrams D such that

$$(3.6.2.8) \quad \begin{array}{ll} \text{i)} & \text{depth } D \leq n; \text{ and} \\ \text{ii)} & \text{if depth } D > n/2, \text{ then for } j > n - \text{depth } D, \\ & \text{the length of row } j \text{ is one.} \end{array}$$

Thus, the determinant character of O_n corresponds to the diagram consisting of one column of length n .

3.6.3 PROPOSITION. *No representation of O_n of depth greater than m occurs in $\mathcal{P}(M_{n,m})$. We have the decomposition of $O_n \times GL_m$ -modules*

$$(3.6.3.1) \quad \mathcal{H}(M_{n,m}; O_n) \cong \sum_D \sigma_n^D \otimes \rho_m^D,$$

where the summation index D ranges over all diagrams of representations of O_n of depth at most m .

PROOF. We have exhibited in §3.6.2 the highest weight vectors of the representations occurring in the summation on the right hand side of equation (3.6.3.1) and

have noted that they are harmonic. Hence all these representations do occur in $\mathcal{H}(M_{n,m}; O_n)$. Thus, our problem is to show that the sum on the right hand side of (3.6.3.1) exhausts $\mathcal{H}(M_{n,m}; O_n)$. For $m \leq n/2$, we can note that all possible representations of GL_m in $\mathcal{P}(M_{n,m})$ occur in this sum, so that Theorem 3.5.2 c) guarantees that the stated sum gives the full decomposition of $\mathcal{H}(M_{n,m}; O_n)$. For $n > n/2$, however, the sum does not contain all representations of GL_m , so a different argument must be given. Since the sum is taken over all representations of O_n of depth at most m , the first statement of the proposition combined with Theorem 3.5.2 c) will prove the completeness of the sum. Hence we consider the first statement. Suppose that the representation σ in $(O_n)^\wedge$ occurs in $\mathcal{P}(M_{n,m})$, and that $\sigma \otimes \det$, its twist by determinant, occurs in $\mathcal{P}(M_{n,\ell})$. Then if we keep in mind that $\mathcal{P}(M_{n,m+\ell}) \cong \mathcal{P}(M_{n,\ell}) \otimes \mathcal{P}(M_{n,m})$, we see that $\det$ must occur in $\mathcal{P}(M_{n,m+\ell})$. We know that σ_n^D does occur in $\mathcal{P}(M_{n,m})$ if $\text{depth } D \leq m$. We also know that if $\text{depth } E > n/2$, then $E = t(D, n)$ for some D , and $\sigma_n^E \cong \sigma_n^D \otimes \det$. Thus, we will be done if we show that the determinant character of O_n cannot occur in $\mathcal{P}(M_{n,m})$ if $m < n$. But this is easy to see, because when $m < n$, a generic m -tuple $(v_1, v_2, v_3, \dots, v_m)$ of points in $\mathbb{C}^n$ will span a subspace Y of dimension m , and Y will be non-degenerate with respect to the inner product on $\mathbb{C}^n$. Hence there are elements of O_n but not SO_n (for example, reflections in hyperplanes containing Y) which leave Y pointwise fixed, and which therefore preserve $(v_1, v_2, v_3, \dots, v_m)$. It follows that any function which transforms by $\det$ must vanish at $(v_1, v_2, v_3, \dots, v_m)$; and since $(v_1, v_2, v_3, \dots, v_m)$ is generic, there can be no non-zero eigenfunction for O_n , with eigencharacter $\det$, on $M_{n,m}$, for $m < n$. This concludes the proof of the proposition.

Acknowledgment: I learned this use of $\det$ from T. Przebinda.

3.7: Geometry of the quotient mapping.

It is illuminating to study the relation between the decompositions (3.5.2) and (3.6.3.1) and the geometry of the quotient map Q of formula (3.4.5.1). A basic tool for studying the geometry of Q is Witt's Theorem [Ja1],[La], which can be formulated as follows.

3.7.1: (WITT'S THEOREM). Given two $n \times m$ matrices T_1 and T_2 , there is an orthogonal $n \times n$ matrix g such that $gT_1 = T_2$ if and only if

- i) $T_1^t T_1 = T_2^t T_2$, and,
- ii) $\ker T_1 = \ker T_2$

3.7.2: Generic fibers

By the fibers of Q , we mean the inverse images of points: $Q^{-1}(A)$, for A in $S^2(\mathbb{C}^m)$. A first application of Witt's Theorem gives us a description of the nice fibers of Q . Let $M_{n,m} \max$ denote the subset of $M_{n,m}$ consisting of maps T of maximum possible rank i.e., $\min(m, n)$. Let $M_{n,m} \max^+ \subseteq M_{n,m} \max$ denote the set of T such that $T^t T$ is of rank $\min(m, n)$. Evidently $M_{n,m} \max^+ \subseteq M_{n,m} \max$. It is easy to convince oneself that both these sets are open and dense in $M_{n,m}$.

PROPOSITION. a) If T is in $M_{n,m} \max^+$, then the fiber of Q through T , i.e., the set $Q^{-1}(T^t T)$, consists of a single O_n -orbit.

b) If $n \geq m$, then Q is submersive at all points of $M_{n,m} \max$, that is, the derivative of Q at T in $M_{n,m} \max$ is surjective.

PROOF. a) We obviously have $\ker T \subseteq \ker T^t T$. Hence, if $T^t T$ has maximum rank, we must have $\ker T = \ker T^t T$. In other words, $\ker T$ is determined by $T^t T$. From this and Witt's Theorem, we see that $T^t T$ determines the O_n -orbit of T . This is equivalent to statement a) of the proposition. A simple calculation shows that the derivative of Q at T is

$$S \rightarrow S^t T + T^t S = S^t T + (S^t T)^t, \quad S \in M_{n,m}.$$

It is easy to see that if T has rank m this takes $M_{n,m}$ onto $S^2(\mathbb{C}^m)$. Indeed, we observe that $S^t T + T^t S$ is just the symmetrization of $S^t T$ and it is elementary that if T is injective, then $S^t T$ can be an arbitrary $m \times m$ matrix.

3.7.3 The null fiber

Of particular interest is the null fiber ("nullfaser" [Krf]), the fiber over 0. It is also sometimes called the null cone to reflect the fact that it is invariant under scalar dilations, and to recall the "light cone" of the special theory of relativity. We denote it by NCQ . Call an element T of NCQ a null mapping. Thus, T is a null mapping if and only if $T^t T = 0$, which means that the image of T in $\mathbb{C}^m$ is an isotropic subspace. According to Witt's Theorem, two null mappings define the same O_n -orbit if and only if they have the same kernel. Among the null mappings, an open dense set will have the maximum possible rank, which is $\min(m, [n/2])$. Call these generic null mappings. If $m \leq n/2$, then the generic null mappings are injective, so they all have kernel zero and all belong to the same O_n -orbit, which is dense in the null fiber. However, if $m > n/2$, the generic null-mappings will have a kernel of dimension $k = m - [n/2]$, so the O_n -orbits of generic null mappings are parametrized by a Grassmannian G_k^m . Also, when $m \leq n/2$, the generic null mappings are generic among all matrices, in the sense that they belong to $M_{n,m} \max$, and Proposition 3.7.2 b) applies to them. However, when $m > n/2$, no null mappings are generic among all matrices. We will see how these changes in the geometry of the null fiber are reflected in the expansions (3.5.2) and (3.6.3.1). First, we describe the ring $\mathcal{R}(NCQ)$ of functions on NCQ . We note that formulas (3.4.5.2) show that NCQ is invariant not only under O_n , but also under multiplication on the right by GL_m .

3.7.3.1: THEOREM. The action of $GL_m \times O_n$ on the null cone NCQ is multiplicity-free. Precisely, the $GL_m \times O_n$ -module structure of $\mathcal{R}(NCQ)$ is given by

$$(3.7.3.2) \quad \mathcal{R}(NCQ) \cong \sum_D \sigma_n^D \otimes \rho_m^D,$$

where the index D ranges over all diagrams of depth at most $\min(m, [n/2])$.

PROOF. We will use the criterion 1.3.3. Thus, we want to show that a Borel subgroup of $GL_m \times O_n$ has a dense orbit in NCQ . Consider a null mapping T .

The image of T in C^n will be an isotropic subspace of C^n , of dimension at most $d = \min(m, [n/2])$. In any case, we can find an isotropic subspace Y of dimension d , containing the image of T . Thus, if we fix an isotropic subspace Y of dimension d , every null mapping will have the form gT for some g in O_n and T in $\text{Hom}(C^m, Y)$. We can do better. Let Y be the third summand in a decomposition of the form (A.3.5). The first summand is X . Let $J_d(C^n; X)$ denote the subset of $J_d(C^n)$ whose elements are transverse to X . The set $J_d(C^n; X)$ is clearly open in $J_d(C^n)$. When $d < n/2$, this implies that $J_d(C^n; X)$ is also dense in $J_d(C^n)$, since for d in this range, $J_d(C^n)$ is irreducible. For $d = n/2$, $J_d(C^n)$ has two irreducible components. We can find an element τ_0 in $O_n - SO_n$ which normalizes U_X (See A.3.1.1 for notation). (If we use the coordinates z_j of §3.6.1, with X defined by the vanishing of the last m coordinates and Y by the vanishing of the first m coordinates, then we can take for τ_0 the map which switches z_m and z_{m+1} .) Let $\tilde{U}_X$ denote the extension of U_X by this element τ_0 when $d = m$, and otherwise $\tilde{U}_X = U_X$. From Proposition A.3.1.1 a), we know that $J_d(C^n; X) = U_X(Y)$. We can conclude that

$$(3.7.3.3) \quad J_d(C^n) = (\tilde{U}_X(\text{Hom}(C^m, Y)))^- ,$$

where the $-$ indicates closure. We can choose a Borel subgroup B_0 of O_n such that $\tilde{U}_X \subseteq B_0$ and $B_0 \cap SO_n \subseteq P_X$. The intersection of B_0 with $GL(Y) \cong GL(X) \subseteq P_X$ is Borel subgroup of $GL(Y)$.

By virtue of equation (3.7.3.3) and the form of $B_0 \cap GL(Y)$, the statement we wish to prove, namely that a Borel subgroup of $GL_m \times O_n$ acts on NCQ with a dense orbit, is reduced to the statement that a Borel subgroup of $GL_m \times GL(Y)$ acts on $\text{Hom}(C^m, Y)$ with a dense orbit. In other words, we reduced the multiplicity-freeness of NCQ under the action of $GL_m \times O_n$ to the multiplicity-freeness of $\text{Hom}(C^m, Y)$ under the action of $GL_m \times GL(Y)$. We know this to be true from our discussion of (GL_m, GL_n) -duality (§2.1), so multiplicity-freeness for NCQ follows. The same reduction allows us to establish the explicit decomposition (3.7.3.2). Ignoring for the moment the case $d = n/2$, we note that equation (3.7.3.3) implies that any U_X -invariant functions on NCQ , in particular any joint $GL_m \times O_n$ highest weight vectors, must be determined by their restrictions to $\text{Hom}(C^m, Y)$. Further, the restriction to $\text{Hom}(C^m, Y)$ of joint $GL_m \times O_n$ highest weight vectors must be joint $GL_m \times GL(Y)$ highest weight vectors. The possibilities have been described in formula (2.1.2.1). Note that the subgroup $O(V_1)$ (which is non-trivial under our current assumption of $d < n/2$) of P_X leaves $\text{Hom}(C^m, Y)$ pointwise fixed and, therefore, must leave invariant the restriction of any function to $\text{Hom}(C^m, Y)$. Hence all joint $GL_m \times O_n$ highest weight vectors on NCQ must be invariant under $O(V_1)$. From our discussion of representations of O_n in §3.7.6.2 and formula (2.1.2.1), we see that the right hand side of formula (3.7.3.2) contains all $GL_m \times O_n$ -modules which could possibly occur in $\mathcal{R}(NCQ)$. To know that they do occur, we just have to check that the highest weight vectors δ_D of formula (3.6.2.3) for diagrams D of no more than d rows do not vanish on NCQ . We leave this easy task to the reader.

The case when $d = n/2$ requires slightly more care, since highest weight vectors for O_n will not be invariant under $\tilde{N}_X$, only under the connected subgroup N_X . They may be eigenvectors with eigenvalues ± 1 under the supplementary element

τ_0 . For this case, the simplest procedure seems to be to observe that, since NCQ is by definition the zero-set of the generators ξ_{ij} of the O_n -invariants, the restriction map from $\mathcal{P}(C^m)$ to $\mathcal{R}(NCQ)$ must map the space $\mathcal{H}(M_{n,m}; O_n)$ of harmonic polynomials onto $\mathcal{R}(NCQ)$. But from our description (3.7.3.1) of the harmonics, we see that, for any joint $GL_m \times O_n$ highest weight vector in $\mathcal{H}(M_{n,m}; O_n)$, the GL_m weight determines the corresponding O_n representation. Using this, we see that the decomposition (3.7.3.2) must hold also when $d = n/2$. This concludes the proof of Theorem 3.7.3.1.

3.7.3.4 Remarks a) We could have used the obvious fact that the restriction mapping

$$\mathcal{H}(M_{n,m}; O_n) \rightarrow \mathcal{R}(NCQ)$$

must be surjective to conclude directly that NCQ is multiplicity-free for $GL_m \times O_n$, but the geometric analysis of the proof gives us the exact description (3.7.3.2) of the module structure of $\mathcal{R}(NCQ)$.

b) When $d = n/2$, we can refine formula (3.7.3.2) to give a description of the $GL_d \times SO_n$ -module structure of the two irreducible components of NCQ . From the discussion of §3.6.2, we see that the O_n -modules σ_n^D are irreducible on SO_n if the depth of the diagram D is less than $n/2$. But if depth $(D) = n/2$, then σ_n^D breaks up into two pieces on restriction to SO_n . Denote these pieces by σ_n^{D+} and σ_n^{D-} . Their highest weights are $\psi_{\ell+}$ and $\psi_{\ell-}$, where

$$\ell^+ = (a_1, a_2, a_3, \dots, a_d) \text{ and } \ell^- = (a_1, a_2, a_3, \dots, -a_d),$$

where the a_j are the row lengths of D .

3.7.3.5: LEMMA. When D is of depth $d = n/2$, the SO_n components σ_n^{D+} and σ_n^{D-} each vanish on one of the irreducible components of NCQ . The vanishing of the σ_n^{D-} on one component, call it NCQ^+ , of NCQ , are the defining equations of NCQ^+ as a subvariety of NCQ . Similar remarks apply to the other component.

PROOF. The first statement follows by taking note of the specific form of the fundamental highest weight vector δ_d , which is a determinant of size d , and using Proposition A.3.20. The second statement is clear from formula (3.7.3.2), since when the depth of D is less than d , the module σ_n^D is irreducible under SO_n , so the σ_n^D isotypic component of $\mathcal{R}(NCQ)$ cannot vanish on either component of NCQ .

According to this lemma, when $m \geq d = n/2$, we can write

$$(3.7.3.5.1) \quad \mathcal{R}(NCQ^+) \cong \sum_D (\sigma_n^D)^+ \otimes \rho_m^D,$$

where σ_n^{D+} is as just specified when $\text{depth}(D) = d$, and $\sigma_n^{D+} = \sigma_n^D$ when $\text{depth}(D) < d$. Here, the sum is over all diagrams of depth at most d .

3.7.3.6: COROLLARY. The restriction map from $\mathcal{H}(M_{n,m}; O_n)$ to $\mathcal{R}(NCQ)$ is an isomorphism if and only if the invariants ξ_{ij} generate the ideal of functions vanishing on NCQ if and only if $m \leq n/2$.

PROOF. Denote by $\mathcal{I}(\tilde{\xi}_{ij})$ the ideal in $\mathcal{P}(\mathbf{C}^{nm})$ generated by the basic invariants $\tilde{\xi}_{ij}$. It is immediate from Theorem 3.5.2 that

$$(3.7.3.6.1) \quad \mathcal{P}(\mathbf{C}^{nm}) = \mathcal{H}(M_{n,m}; O_n) \oplus \mathcal{I}(\tilde{\xi}_{ij}).$$

Let $\mathcal{I}(NCQ)$ be the ideal of functions vanishing on NCQ . From the definition of NCQ , we have $\mathcal{I}(\tilde{\xi}_{ij}) \subseteq \mathcal{I}(NCQ)$. Also by definition, $\mathcal{R}(NCQ) = \mathcal{P}(\mathbf{C}^{nm})/\mathcal{I}(NCQ)$. From these facts and equation (3.7.3.6), it is obvious that $\mathcal{I}(\tilde{\xi}_{ij}) = \mathcal{I}(NCQ)$ if and only if the restriction map from $\mathcal{H}(M_{n,m}; O_n)$ to $\mathcal{R}(NCQ)$ is an isomorphism. That the second condition holds exactly in the range $m \leq n/2$ follows by comparison of the formulas (3.6.3.1) and (3.7.3.2).

Remark: Recall that when $m \leq n/2$, the null fiber contains generic points to which Proposition 3.7.2 b) applies. This allows one to apply the criterion of [Kos 2] to conclude that $\mathcal{I}(\tilde{\xi}_{ij}) = \mathcal{I}(NCQ)$ for m in this range. This method is followed in [DT], which then deduces the isomorphism of $\mathcal{H}(M_{n,m}; O_n)$ and $\mathcal{R}(NCQ)$. Without explicitly computing $\mathcal{H}(M_{n,m}; O_n)$, [DT] also concludes that $\mathcal{I}(\tilde{\xi}_{ij}) \neq \mathcal{I}(NCQ)$ when $m > n/2$. Also, [DT] points out that the equality $\mathcal{I}(\tilde{\xi}_{ij}) = \mathcal{I}(NCQ)$ implies that NCQ is a complete intersection.

3.7.3.7: When $\mathcal{I}(\tilde{\xi}_{ij}) = \mathcal{I}(NCQ)$ and NCQ is irreducible, i.e., when $m < n/2$, the ideal $\mathcal{I}(\tilde{\xi}_{ij})$ is prime. This and the denseness of the single O_n -orbit of generic null mappings when $m \leq n/2$ allow one to apply [Kos 2], Proposition 2 to refine decomposition (3.7.3.6).

PROPOSITION. When $m < n/2$, multiplication inside $\mathcal{P}(\mathbf{C}^{nm})$ defines an isomorphism

$$(3.7.3.8) \quad \mathcal{P}(\mathbf{C}^{nm}) \cong \mathcal{H}(M_{n,m}; O_n) \otimes \mathcal{P}(\mathbf{C}^{nm})^{O_n}.$$

Remark: R. Brylinski [Bry 2] has analyzed the module structure of the null-cone for the action of $GL(V)$ on $V^\ell \oplus (V^+)^m$ (cf. §2.5).

3.8: Synopsis of results for Sp_{2n} .

3.8.1: THEOREM. The GL_n -module structure of $S(\Lambda^2(\mathbf{C}^m))$ is described by the formula

$$S(\Lambda^2(\mathbf{C}^n)) \cong \sum_D (2D)^t \rho_N^{(2D)}.$$

Here, the sum ranges over all diagrams D with row lengths not more than $n/2$, so that $(2D)^t$ ranges over all diagrams whose columns are of even lengths not exceeding n .

As with Theorem 3.1, this result has both a multiplicity-free proof and a Schur Duality proof based on the considerations of Appendix 1. If we think of $\Lambda^2(\mathbf{C}^n)$ as the skew-symmetric matrices, then the fundamental generators of the covariants in $S(\Lambda^2(\mathbf{C}^n))$ (or, taking duals, in $\mathcal{P}(\Lambda^2(\mathbf{C}^n)))$ are the Pfaffians [Ja1], [W1], of the leading $2k \times 2k$ minors of the typical element of $\Lambda^2(\mathbf{C}^n)$.

3.8.2: Let Sp_{2n} be the isometry group of the standard symplectic form $<, >$ on $\mathbf{C}^{2n}$. The formula for $<, >$ is

$$(3.8.2.1) \quad <v, v'> = \sum_{j=1}^n x_j y'_j - y_j x'_j.$$

Here, $v = (x, y)$ and $v' = (x', y')$ are elements of $\mathbf{C}^{2n}$, expressed as 2-tuples of elements of $\mathbf{C}^n$. Thus, $x = (x_1, x_2, x_3, \dots, x_n)^t$, etc. The t indicates transpose, so that x is a column vector of length n . The vectors v and v' are likewise column vectors, of length $2n$.

3.8.2.2: COROLLARY. An irreducible representation ρ_n^D of GL_{2n} contains at most one invariant vector (up to multiples) for Sp_{2n} . It does contain one if and only if the columns of D have even length.

As with Corollary 3.3.2, this is a case of the Cartan-Helgason Theorem [He3], and it was derived in some form in the invariant theory literature.

3.8.3: FFT for Sp_{2n} , polynomial invariants version

Let Sp_{2n} act on $(\mathbf{C}^{2n})^m$, which we may think of as $2n \times m$ matrices $M_{2n,m}$, or as m -tuples of elements $(v_1, v_2, v_3, \dots, v_m)$ of vectors in $\mathbf{C}^{2n}$. Using the m -tuple model, we can define polynomials $\tilde{\xi}_{ab}$ by the recipe

$$(3.8.3.1) \quad \tilde{\xi}_{ab}(v_1, v_2, v_3, \dots, v_m) = <v_a, v_b>, \quad 1 \leq a, b \leq m.$$

Here, $<, >$ is the symplectic form (3.8.2.1).

3.8.3.2: THEOREM. Consider the action of Sp_{2n} on the polynomial algebra $\mathcal{P}((\mathbf{C}^{2n})^m)$. The subalgebra $\mathcal{P}((\mathbf{C}^{2n})^m)^{Sp_{2n}}$ of polynomials invariant for Sp_{2n} is generated by the quadratic polynomials $\tilde{\xi}_{ab}$ of formula (3.8.3.1).

3.8.4: Commutant version of the FFT for Sp_{2n}

We may consider the polynomials $\tilde{\xi}_{ab}$ to be operators on $\mathcal{P}((\mathbf{C}^{2n})^m)$, acting by multiplication. We can define also some second order, constant-coefficient differential operators $\hat{\Delta}_{ab}$, symplectic analogs of Laplacians.

$$(3.8.4.1) \quad \hat{\Delta}_{ab} = \sum_{j=1}^n \frac{\partial^2}{\partial x_{ja} \partial y_{jb}} - \frac{\partial^2}{\partial y_{ja} \partial x_{jb}}$$

We also have the (normalized) polarization operators: normalized linear coefficient vector fields coming from the infinitesimal action of GL_m via the factorization $(\mathbf{C}^{2n})^m \cong \mathbf{C}^{2n} \otimes \mathbf{C}^m$.

$$(3.8.4.2) \quad \check{E}_{ab} = \left(\sum_{j=1}^n x_{ja} \frac{\partial}{\partial x_{jb}} + y_{ja} \frac{\partial}{\partial y_{jb}} \right) + n \delta_{ab}.$$

3.8.4.3: THEOREM. The operators $\check{\xi}_{ab}$, $\check{\Delta}_{ab}$ and $\check{E}_{ab}$ span a Lie algebra isomorphic to $\mathfrak{so}_{2m}$. They generate the full algebra of polynomial coefficient differential operators on $\mathcal{P}((\mathbb{C}^{2n})^m)$ commuting with Sp_{2n} .

3.8.5: Full isotypic decomposition

Denote by $\mathcal{H}((\mathbb{C}^{2n})^m, Sp_{2n})$ the polynomials on $(\mathbb{C}^{2n})^m$ which are annihilated by all the partial symplectic Laplacians $\check{\Delta}_{ab}$. For an irreducible representation τ , of Sp_{2n} , let $\mathcal{P}((\mathbb{C}^{2n})^m, \tau)$ denote the τ -isotypic component of $\mathcal{P}((\mathbb{C}^{2n})^m)$. Let us think of $(\mathbb{C}^{2n})^m$ as the space $M_{2n,m}$ of matrices. For $k \leq \min\{n, m\}$, let δ_k denote the determinant of the leading $k \times k$ submatrix of an element of $M_{2n,m}$:

$$(3.8.5.1) \quad \delta_k = \det \begin{vmatrix} x_{11} & x_{12} & \cdots & x_{1k} \\ x_{21} & x_{22} & \cdots & \cdot \\ \cdot & \cdot & \ddots & \cdot \\ x_{k1} & \cdot & \cdots & x_{kk} \end{vmatrix}$$

Notice that there are no y coordinates involved in the δ_k , no matter how large m is. It is easy to verify that, with the obvious choices of Borel subgroups in Sp_{2n} and GL_m , the δ_k are joint $Sp_{2n} \times GL_m$ highest weight vectors. Given a diagram $D = (a_1, a_2, a_3, \dots, a_1)$, with $1 \leq \min\{n, m\}$, define

$$(3.8.5.2) \quad \delta^D = \prod_{k=1}^1 \delta_k^{a_k - a_{k+1}}$$

The δ^D are all joint $Sp_{2n} \times GL_m$ highest weight vectors. The representation of GL_m generated by δ^D is ρ_m^D . Denote by τ_{2n}^D the representation of Sp_{2n} generated by δ^D .

3.8.5.3: THEOREM. a) We have

$$(3.8.5.4) \quad \mathcal{P}((\mathbb{C}^{2n})^m) = \mathcal{H}((\mathbb{C}^{2n})^m, Sp_{2n}) \cdot \mathcal{P}((\mathbb{C}^{2n})^m, Sp_{2n}).$$

In other words, for every irreducible representation τ of Sp_{2n} , we have

$$\mathcal{P}((\mathbb{C}^{2n})^m, \tau) = (\mathcal{H}((\mathbb{C}^{2n})^m, Sp_{2n}) \cap \mathcal{P}((\mathbb{C}^{2nm})^m, \tau)) \cdot \mathcal{P}((\mathbb{C}^{2n})^m, Sp_{2n})$$

b) The space $\mathcal{H}((\mathbb{C}^{2n})^m, Sp_{2n}) \cap \mathcal{P}((\mathbb{C}^{2n})^m, \tau)$ consists exactly of the elements of $\mathcal{P}((\mathbb{C}^{2n})^m, \tau)$ of lowest possible degree. It is an irreducible module under the joint action of $Sp_{2n} \times GL_m$.

c) The irreducible representations of Sp_{2n} such that $\mathcal{P}((\mathbb{C}^{2n})^m, \tau)$ is non-zero are exactly the τ_{2n}^D defined above, after formula (3.8.5.2). We have

$$(3.8.5.5) \quad \mathcal{H}((\mathbb{C}^{2n})^m, Sp_{2n}) \cap \mathcal{P}((\mathbb{C}^{2n})^m, \tau^D) \cong \tau_{2n}^D \otimes \rho_m^D.$$

The polynomials δ^D of formula (3.8.5.2) constitute all the harmonic joint $Sp_{2n} \times GL_m$ highest weight vectors.

3.8.6: Geometry of the quotient map

Note that $\check{\xi}_{ba} = -\check{\xi}_{ab}$. If we organize the $\check{\xi}_{ab}$ in to an $m \times m$ skew-symmetric matrix, we get a map

$$(3.8.6.1) \quad Q_{Sp} : M_{2n,m} \rightarrow \Lambda^2(\mathbb{C}^m).$$

The FFT for Sp_{2n} may be rephrased as saying that the pullback map $(Q_{Sp})^* : \mathcal{P}(\Lambda^2(\mathbb{C}^m)) \rightarrow \mathcal{P}(M_{2n,m})$ is onto the algebra $\mathcal{P}(M_{2n,m})^{Sp_{2n}}$ of Sp_{2n} -invariants. This means that the image $Q_{Sp}(M_{2n,m})$ is the quotient or orbit space $M_{2n,m}/Sp_{2n}$ in the sense of algebraic geometry [Krf]. As in the orthogonal case, we write $Q_{Sp}^{-1}(0) = \text{NCQ}$, or NCQ_{Sp} if we need to specify the symplectic case. We call NCQ the null fiber. The null fiber is invariant under $Sp_{2n} \times GL_m$, so the coordinate ring $\mathcal{R}(\text{NCQ})$ is a module for $Sp_{2n} \times GL_m$.

3.8.6.2: THEOREM. We have the $Sp_{2n} \times GL_m$ -module decomposition

$$\mathcal{R}(\text{NCQ}) \cong \sum_D \tau_{2n}^D \otimes \rho_m^D,$$

where the sum is over diagrams D of depth at most $\min\{n, m\}$. The restriction map

$$\tau : \mathcal{H}((\mathbb{C}^{2n})^m, Sp_{2n}) \rightarrow \mathcal{R}(\text{NCQ})$$

is an isomorphism. The ideal in $\mathcal{P}(M_{2n,m})$ generated by the $\check{\xi}_{ab}$ is the full ideal of functions vanishing on NCQ. In particular, it is prime.

(3.8.6.3) Remark: The symplectic case is thus somewhat simpler than the orthogonal case, in that the $\check{\xi}_{ab}$ always generate the ideal for NCQ, no matter how large m is. The non-primeness of the ideal generated by the basic invariants in the orthogonal case seems to be related to the disconnectedness of the orthogonal groups.

Study of the orbit structure of NCQ shows that when $m \leq n$, the Sp_{2n} -orbit of generic (maximal rank) elements in NCQ is dense in NCQ, while when $m > n$, the generic orbits are parametrized by a Grassmannian. Applying Kostant's criterion [Kos2], we get

3.8.6.4: THEOREM. When $m \leq n$, the multiplication map

$$\mu : \mathcal{H}((\mathbb{C}^{2n})^m, Sp_{2n}) \otimes \mathcal{P}((\mathbb{C}^{2n})^m, Sp_{2n}) \rightarrow \mathcal{P}((\mathbb{C}^{2n})^m)$$

is an isomorphism.

3.8.7: Skew symmetric FFT

This section and the following ones anticipate Chapter 4, on skew-symmetric multiplicity-free actions. For completeness and simplicity, we collect the skew symmetric results for Sp_{2n} together with the symmetric ones. However, we don't advise reading these sections before Chapter 4. Consider the action of Sp_{2n} on $\Lambda((\mathbb{C}^{2n})^m)$. We can create degree two invariants quite analogous to the $\check{\xi}_{ab}$. Let $\{e_j, f_j\}$ be the standard symplectic basis of $\mathbb{C}^{2n}$ with respect to which the coordinates x_j, y_j of

§3.8.2 are defined. Let $\{e_{ja}, f_{ja}\}$ be a basis for $(C^{2n})^m$ made of m copies of the basis $\{e_j, f_j\}$ for C^{2n} . Set

$$(3.8.7.1) \quad \widehat{\xi}_{a,b} = \sum_{j=1}^n \wedge e_{ja} \wedge f_{jb} - \wedge f_{ja} \wedge e_{jb}.$$

Note that because of the skew-commutativity of multiplication, $\widehat{\xi}_{b,a} = \widehat{\xi}_{a,b}$, and in particular $\widehat{\xi}_{a,a}$ is non-zero.

3.8.7.2: THEOREM. The algebra $\Lambda((C^{2n})^m)^{Sp_{2n}}$ of Sp_{2n} invariants in $\Lambda((C^{2n})^m)$ is generated by the degree two elements $\widehat{\xi}_{a,b}$.

(3.8.8) Skew-symmetric operator commutant version of the FFT (skew (Sp_{2n}, Sp_{2m}))-duality.

We can let the elements $\widehat{\xi}_{a,b}$ operate on $\Lambda((C^{2n})^m)$ by multiplication. In addition, we have "skew symplectic partial Laplacians"

$$(3.8.8.1) \quad \widehat{\Delta}_{a,b} = \sum_{j=1}^n \lfloor e_{ja} \rfloor f_{jb} - \lfloor f_{ja} \rfloor e_{jb}.$$

The commutators of these are the normalized skew polarization operators:

$$(3.8.8.2) \quad \widehat{E}_{a,b} = \sum_{j=1}^n (\wedge e_{ja} \rfloor e_{jb} + \wedge f_{ja} \rfloor f_{jb}) - n\delta_{a,b}.$$

3.8.8.3: THEOREM. The operators $\widehat{\xi}_{a,b}$, $\widehat{\Delta}_{a,b}$ and $\widehat{E}_{a,b}$ span a Lie algebra isomorphic to $\mathfrak{sp}_{2m}$. They generate the commutant of Sp_{2n} in $\text{End}(\Lambda((C^{2n})^m))$.

Note that since Sp_{2m} is connected and simply connected, a representation of $\mathfrak{sp}_{2m}$ gives rise to a unique representation of Sp_{2m} by exponentiation.

3.8.9: Complete isotypic decomposition of $\Lambda((C^{2n})^m)$
Theorem 3.8.8.3 tells us that we can decompose $\Lambda((C^{2n})^m)$ into a sum of representations of $Sp_{2n} \times Sp_{2m}$. These will have the form $\tau \otimes \tau'$, where $\tau \in (Sp_{2n})^\wedge$ and $\tau' \in (Sp_{2m})^\wedge$. The double commutant yoga tells us that τ determines τ' and vice versa, so that $\tau \otimes \tau'$ is the full τ isotypic component of $\Lambda((C^{2n})^m)$, and is likewise the full τ' isotypic component. We will describe the bijection $\tau \leftrightarrow \tau'$, and explicitly describe joint $Sp_{2n} \times \mathfrak{sp}_{2m}$ highest weight vectors. For Borel subgroup B_{Sp} of Sp_{2n} , choose the one which preserves the span of the basis vectors e_j (introduced in §3.8.7), and on this subspace acts as the standard upper triangular Borel. In other words, B_{Sp} preserves the span of the sets $\{e_j : 1 \leq j \leq c\}$ for any c between 1 and n . For Borel subalgebra $\mathfrak{g}'_{sp}$ of $\mathfrak{sp}_{2m}$, choose the one which contains the subalgebra $\mathfrak{sp}_{2m}^{0,2}$ (the span of the operators $\widehat{\Delta}_{a,b}$), and which intersects $\mathfrak{sp}_{2m}^{1,1}$ (the span of the $\widehat{E}_{a,b}$) in the standard Borel of $\mathfrak{gl}_m$.

For $a \leq n$ and $j \leq m$, define a decomposable a -vector

$$(3.8.9.1) \quad \omega_j(a) = e_{1j} \wedge e_{2j} \wedge e_{3j} \wedge \dots \wedge e_{aj}.$$

Given a diagram $D = (a_1, a_2, a_3, \dots, a_m)$, of depth at most m and row lengths at most n , define a vector

$$(3.8.9.2) \quad \omega^D = \omega_1(a_1) \wedge \omega_2(a_2) \wedge \omega_3(a_3) \wedge \dots \wedge \omega_m(a_m).$$

The weight of this vector with respect to the standard torus in Sp_{2n} is ψ_D . The weight with respect to the standard Cartan subalgebra of $\mathfrak{sp}_{2m}$ is $(a_1 - n, a_2 - n, a_3 - n, \dots, a_m - n)$. This is dominant, but it is not polynomial; rather it is "dual polynomial" - all the weights are negative. To obtain a polynomial weight to which we can associate a diagram, we should transform this to $(n - a_m, n - a_{m-1}, \dots, n - a_2, n - a_1)$. There is a simple diagrammatic interpretation of this operation. We should imagine our initial diagram D to be occupying the upper left corner of a rectangle of m rows and n columns (depth m , width n). If we take the complementary region inside the rectangle and rotate it 180° , we get the diagram $r_{n,m}(D) = (n - a_m, n - a_{m-1}, \dots, n - a_2, n - a_1)$ just described. Thus, we arrive at the following description of an $Sp_{2n} \times \mathfrak{sp}_{2m}$ -module.

3.8.9.3: THEOREM. We have the $Sp_{2n} \times \mathfrak{sp}_{2m}$ -module decomposition

$$\Lambda((C^{2n})^m) \cong \sum_D \tau_{2n}^D \otimes \tau_{2m}^{r_{n,m}(D')}.$$

Here, D runs through all diagrams which fit inside an $n \times m$ rectangle (depth at most n , width at most m). The labeling of the representations of Sp_{2n} (or $\mathfrak{sp}_{2m}$) in terms of diagrams is as defined in §3.8.5.

We point out that we have formulated this theorem in terms of the transpose of the diagram used in the discussion, in order to have D directly label one of the factors of the summand of formula (3.8.9.3).

4: Skew-symmetric Multiplicity-free Actions.

It has been gradually understood that skew-symmetric variables (Grassmann algebras) are just as important as the more commonly encountered symmetric variables (polynomial or symmetric algebras). Therefore, we consider the skew-symmetric analog of the theory developed in §2 and §3. Unfortunately, I do not know a simple geometric criterion for multiplicity-freeness in the skew-symmetric case. Also, the simple method of producing the highest weight vectors, as monomials in the fundamental (prime) highest weight vectors, fails completely since the product of highest weight vectors is typically zero. Nevertheless, there is a rich set of examples, and there are both a geometrically-based necessary condition for multiplicity-freeness and a method for constructing highest weight vectors. The skew-symmetric theory has many parallels to the symmetric case, and some interesting differences. It also offers some insights into the well-known results of Kostant [Kos] on the cohomology of nilradicals of parabolic subgroups. We begin with examples parallel to §2 and §3, and develop the general theory at the end of the section.

4.1: Skew (GL_n, GL_m) -duality.

4.1.1: Again, consider the action of $GL_n \times GL_m$ on $C^n \otimes C^m$. Consider further the resulting action by automorphisms of $GL_n \times GL_m$ on the exterior algebra $\Lambda(C^n \otimes C^m)$. This has a decomposition strongly analogous to that of the symmetric algebra $S(C^n \otimes C^m)$ described in §2.1.

THEOREM. We have the $GL_n \times GL_m$ -module decomposition

$$\Lambda(C^n \otimes C^m) \cong \sum_D \rho_n^D \otimes \rho_m^{D^t}$$

In this sum, D ranges over all diagrams with at most n rows and with rows lengths at most m . The diagram D^t is the transpose of D : the rows of D are the columns of D^t , and vice versa.

(4.1.2) PROOF. Probably the shortest proof of this is to appeal to Schur Duality. The argument is exactly parallel to that of §2.4.5.1, showing how to deduce the symmetric case of (GL_n, GL_m) -duality from Schur Duality. As far as the specific way in which representations of GL_n and GL_m are matched in the sum in §4.1.1, we should note the fact from the representation theory of the symmetric group [JK] that tensoring a representation of S_k with the sign character is accomplished by taking the transpose of the corresponding diagram.

4.1.3: Remarks: a) Because of the one-to-one matching of the relevant representations of GL_n and GL_m in Theorem 4.1.1, it follows that GL_n and GL_m generate mutual commutants in $\text{End}(\Lambda(C^n \otimes C^m))$. One may conclude this also from the Schur Duality argument, without having to know anything specific about $(S_k)^\wedge$. Then the specific form of the correspondence may be seen by explicitly writing down the joint highest weight vectors. Let $e_{jk} = e_j \otimes e_k$ be the basis of $C^n \otimes C^m$ consisting of the tensor products of the standard bases of C^n and C^m . Arrange

the e_{jk} in a rectangular array. Thus:

$$\begin{array}{ccccccc} e_{11} & e_{12} & e_{13} & \cdots & e_{1m} \\ e_{21} & e_{22} & e_{23} & \cdots & e_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ e_{n1} & e_{n2} & e_{n3} & \cdots & e_{nm} \end{array}$$

Given a diagram D , we fill its boxes with the corresponding elements from this array and take the wedge product of all the e_{jk} that fit in the diagram. We get the highest weight vector of the summand indexed by D in Theorem 4.1.1. For example, the highest weight vector corresponding to the diagram (5,3,2,2) is

$$\begin{aligned} & (e_{11} \wedge e_{12} \wedge e_{13} \wedge e_{14} \wedge e_{15}) \wedge (e_{21} \wedge e_{22} \wedge e_{23}) \wedge (e_{31} \wedge e_{32}) \wedge (e_{41} \wedge e_{42}) \\ & = (e_{11} \wedge e_{21} \wedge e_{31} \wedge e_{41}) \wedge (e_{12} \wedge e_{22} \wedge e_{31} \wedge e_{42}) \wedge (e_{13} \wedge e_{23}) \wedge e_{14} \wedge e_{15} \end{aligned}$$

This construction shows directly that all the summands claimed in Theorem 4.1.1 do indeed occur. One can also argue fairly easily that all highest weights of GL_n which occur in $\Lambda(C^n \otimes C^m)$ must yield diagrams with row length at most m . Hence, if we know that GL_n and GL_m generate mutual commutants in $\text{End}(\Lambda(C^n \otimes C^m))$ so that the decomposition of $\Lambda(C^n \otimes C^m)$ must yield a bijection between GL_n representations and GL_m -representations, we may conclude that Theorem 4.1.1 gives the correct correspondence and is exhaustive.

b) Just as the argument of §2.4.5.1, slightly modified, proves Theorem 4.1.1, we can deduce Schur Duality from skew (GL_n, GL_m) -duality by essentially the argument of §2.4.5.2. The difference is that I do not know for skew (GL_n, GL_m) -duality a simple independent proof, analogous to that of §2.1.4 for symmetric (GL_n, GL_m) -duality. Thus, at the moment the skew-symmetric theory seems to be logically subordinate to the symmetric theory or, more directly, to Schur Duality. It would be interesting to establish the skew-symmetric theory independently.

c) The module $C^n \otimes C^m$ for $GL_n \times GL_m$ occurs in the following way: regard $GL_n \times GL_m$ as a block diagonal subgroup of GL_{n+m} . Then the upper right $n \times m$ block is normalized by $GL_n \times GL_m$, and, under the action of $GL_n \times GL_m$ by conjugation, is isomorphic to $C^n \times C^m$. (Actually, it is isomorphic to $M_{n,m} \cong C^n \otimes C^{m*}$, but we continue our convention to ignore these distinctions.) With this observation, and taking into account that this realization of $C^n \otimes C^m$ gives it the structure of abelian Lie algebra, Theorem 4.1.1 is seen to be a special case of the description given by Kostant [Kos] of the cohomology of nilradicals of parabolic subalgebras. Several of the results below have similar interpretation. However, we will be concerned to give them proofs depending only on the principles under study here, which in these cases effectively means Schur Duality. We hope these proofs have some interest of their own. We will discuss further the connection with Kostant's results in §4.7 and §A4.3.

4.2: The skew FFT for GL_n .

Next consider the action of GL_n on $C^n \otimes C^\ell \oplus C^{n*} \otimes C^m$. We wish to describe the GL_n -invariants in $\Lambda(C^n \otimes C^\ell \oplus C^{n*} \otimes C^m)$. These will, of course, be a module for the group $GL_\ell \times GL_m$ which acts on $C^n \otimes C^\ell \oplus C^{n*} \otimes C^m$ and commutes with

GL_n . We can use Theorem 4.1.1 to give a description of $\Lambda(C^n \otimes C^l \oplus C^{n*} \otimes C^m)^{GL_n}$ as a $GL_\ell \times GL_m$ -module, in a fashion parallel to that of §2.2.3. We have

$$\begin{aligned}
 (4.2.1) \quad & \Lambda(C^n \otimes C^l \oplus C^{n*} \otimes C^m)^{GL_n} \cong \Lambda(C^n \otimes C^l) \otimes \Lambda(C^{n*} \otimes C^m)^{GL_n} \\
 & \cong ((\sum_D \rho_n^D \otimes \rho_l^{D'}) \otimes (\sum_E (\rho_n^{E*} \otimes \rho_m^{E'}))^{GL_n} \\
 & \cong (\sum_{D,E} (\rho_n^D \otimes (\rho_n^{E*}) \otimes \rho_l^{D'} \otimes \rho_m^{E'}))^{GL_n} \\
 & \cong \sum_D \rho_l^{D'} \otimes \rho_m^{D'}.
 \end{aligned}$$

In the last sum, D varies over diagrams with at most n rows of length at most $\min(l, m)$; or D^l varies over diagrams with at most $\min(l, m)$ rows of length at most n .

The last sum in formula (4.2.1) resembles the formula (2.1.2.1) for symmetric (GL_n, GL_m) -duality, except it is truncated after diagrams of length n . Moreover, in $\Lambda(C^n \otimes C^l \oplus C^{n*} \otimes C^m)$ we have the subspace $(C^n \otimes C^l) \otimes (C^{n*} \otimes C^m)$, and we have

$$(C^n \otimes C^l) \otimes (C^{n*} \otimes C^m)^{GL_n} \cong (C^n \otimes C^{n*})^{GL_n} \otimes (C^l \otimes C^m) \cong C^l \otimes C^m.$$

We may write a basis for this space as follows. Let $\{e_a : 1 \leq a \leq n\}$ be the standard basis for C^n , and let $\{\lambda_a : 1 \leq a \leq n\}$ be the dual basis for C^{n*} . Make a basis $\{e_{ab} : 1 \leq j \leq n, 1 \leq b \leq \ell\}$ for $C^n \otimes C^l$ by taking copies of $\{e_a : 1 \leq a \leq n\}$, and make a basis $\{\lambda_{ac} : 1 \leq a \leq n, 1 \leq c \leq m\}$ for $C^{n*} \otimes C^m$ similarly. Then the elements

$$(4.2.2) \quad \tilde{\xi}_{bc} = \sum_{a=1}^n e_{ab} \wedge \lambda_{ac}$$

are a basis for $(C^n \otimes C^l) \otimes (C^{n*} \otimes C^m)^{GL_n}$. They provide us with the skew-symmetric analog of Theorem 2.2.2.

4.2.3: THEOREM. The algebra $\Lambda(C^n \otimes C^l \oplus C^{n*} \otimes C^m)^{GL_n}$ is generated by the degree 2 elements $\tilde{\xi}_{bc}$ of formula (4.2.2).

(4.2.4) PROOF. The proof proceeds in direct analogy with the proof of Theorem 2.2.2: The identification $(C^n \otimes C^l) \otimes (C^{n*} \otimes C^m)^{GL_n} \cong C^l \otimes C^m$ leads to a homomorphism

$$S(C^l \otimes C^m) \rightarrow \Lambda(C^n \otimes C^l \oplus C^{n*} \otimes C^m)^{GL_n}.$$

(Note that since the $\tilde{\xi}_{bc}$ are of degree 2, their wedge multiplication is simply commutative.) This is a homomorphism of $GL_\ell \times GL_m$ -modules. To verify that it is surjective, we just need to express the highest weight vectors of the modules occurring in the final sum of formula (4.2.1) in terms of the $\tilde{\xi}_{bc}$. Monomials in determinants, parallel to formula (2.2.3.2) give us the answer, and our only job is to

show that these expressions do not vanish. Since we are working here in an exterior algebra, we should be particularly careful about this. Of course, the key fact must be that the elements $\tilde{\xi}_{bc}$ have rank n as elements of $\Lambda^2(C^n \otimes C^l \oplus C^{n*} \otimes C^m)$. For example, we have

$$\tilde{\xi}_{11}^n = n! \prod_{a=1}^n e_{a1} \wedge \lambda_{a1}.$$

With this hint, we will leave the general verification as an exercise to the reader.

4.2.5: Skew operator commutant theorem

We can use Theorem 4.2.3 to describe the commuting algebra of $GL(V)$ acting on $\Lambda(V \otimes U)$ for an auxiliary space U . According to §A2.3.1, we have an identification $\text{End}(\Lambda(V \otimes U)) \cong \Lambda((V \otimes U) \oplus (V \otimes U)^*)$ as $GL(V \otimes U)$ -modules. In this identification, the space $(V \otimes U) \oplus (V \otimes U)^* = \Lambda^1(V \otimes U) \otimes \Lambda^1((V \otimes U)^*) \subseteq \Lambda^2((V \otimes U) \oplus (V \otimes U)^*)$ is identified to the Lie algebra $\mathfrak{gl}(V \otimes U)$. Thus, the set of $GL(V)$ -invariants in $\Lambda^2((V \otimes U) \oplus (V \otimes U)^*)$ corresponds to the centralizer of $GL(V)$ inside $\mathfrak{gl}(V \otimes U)$, that is, to $\mathfrak{gl}(U)$. By a simple argument (see [Hw1]) comparing the algebra structures of the $GL(V)$ -invariants in the graded algebra $\Lambda((V \otimes U) \oplus (V \otimes U)^*)$ and the $GL(V)$ -invariants in the filtered algebra $\text{End}(\Lambda(V \otimes U))$, we conclude from Theorem 4.2.3 that $\mathfrak{gl}(U)$ generates the full commutant of $GL(V)$ inside $\text{End}(\Lambda(V \otimes U))$. From this, we can complete the circle and deduce skew (GL_n, GL_m) -duality, following Remark 4.1.3 a). As noted in that remark, we can pass directly from Schur Duality to the fact that $GL(U)$ spans the commutant of $GL(V)$ in $\text{End}(\Lambda(V \otimes U))$, and since $GL(U)$ is a connected group, its Lie algebra $\mathfrak{gl}(U)$ will generate the same subalgebra of $\text{End}(\Lambda(V \otimes U))$ as $GL(U)$ does. Hence, we can pass directly from Schur Duality to the operator commutant version of the skew FFT, and back again, without invoking the skew FFT, Theorem 4.2.3. However, in the case of symmetric variables, and for the other classical groups in both the symmetric and skew-symmetric situations, the FFT seems to be an essential part of the chain of reasoning, so we used it here also, to emphasize the parallels between all cases.

4.3: The FFT for isometry groups.

We can combine skew (GL_n, GL_m) -duality with the descriptions of the symmetric algebras of $\Lambda^2(V)$ or $S^2(V)$ as $GL(V)$ -modules to deduce skew FFTs for orthogonal or symplectic groups. For example, let Y be a vector space equipped with an inner product $(\cdot, \cdot)$. Consider the action of the orthogonal group $O(Y)$ on the sum $Y^m \cong Y \otimes C^m$ of m copies of Y . We would like to describe the $O(Y)$ -invariants in the exterior algebra $\Lambda(Y \otimes C^m)$. There are some obvious invariants of degree two. To describe them, introduce an orthonormal basis $\{b_j : 1 \leq j \leq \dim Y\}$ for Y , and let $\{b_{jk} : 1 \leq j \leq \dim Y, 1 \leq k \leq m\}$ be the basis of Y^m obtained by replicating $\{b_j\}$ m times. We can write down these degree two invariants

$$(4.3.1) \quad \tilde{\xi}_{ab} = \sum_{j=1}^{\dim Y} b_{ja} \wedge b_{jb} \quad 1 \leq a \neq b \leq m.$$

4.3.2: THEOREM. The elements $\tilde{\xi}_{ab}$ generate the algebra $\Lambda(Y \otimes C^m)^{O(Y)}$ of $O(Y)$ -invariant elements in $\Lambda(Y \otimes C^m)$.

(4.3.3) PROOF. The proof follows a pattern which should by now be familiar. We first describe $\Lambda(Y \otimes C^m)$ as a $GL(Y) \times GL_m$ -module, using (GL_n, GL_m) -duality (with $n = \dim Y$).

$$(4.3.3.1) \quad \Lambda(Y \otimes C^m) \cong \sum_D \rho_Y^D \otimes \rho_m^{D^t}.$$

Hence we have

$$\begin{aligned} \Lambda(Y \otimes C^m)^{O(Y)} & \left(\cong \sum_D \rho_Y^D \otimes \rho_m^{D^t} \right)^{O(Y)} \cong \sum_D (\rho_Y^D \otimes \rho_m^{D^t})^{O(Y)} \\ & \cong \sum_D (\rho_Y^D)^{O(Y)} \otimes \rho_m^{D^t} \cong \sum_E \rho_m^{(2E)^t}. \end{aligned}$$

The final equation follows from Theorem 4.1. In the final summation, E ranges over diagrams with at most $\dim Y$ rows and such that $2E$ has rows of length at most m . Except for the restriction on the length of the rows of E , this summation reminds us of the description (Theorem 3.8) of the GL_m -module structure for the symmetric algebra on the skew-symmetric matrices $\Lambda^2(C^m)$. Note that $\Lambda^2(C^m)$ does occur in the final sum, and it has the smallest degree among the non-trivial modules occurring there. In this context, we observe that the quadratic invariants $\tilde{\xi}_{ab}$ of formula (4.3.1) are a basis for the copy of $\Lambda^2(C^m)$ appearing in the final sum. Thus, the proof of the theorem again comes down to showing that the expressions in the $\tilde{\xi}_{ab}$ which should give the highest weights of the modules appearing in the final sum define non-zero elements of $\Lambda(Y \otimes C^m)$. We again leave this as an exercise for the reader.

4.3.4: Operator commutant version

As in previous situations, we may use the FFT as formulated in Theorem 4.3.2 to describe the commutant of $O(Y)$ in $\text{End}(\Lambda(Y \otimes C^m))$. We recall from equation (A2.3.2.1) that we have isomorphisms $\text{End}(\Lambda(Y \otimes C^m)) \cong \Lambda(Y \otimes C^m) \oplus (Y \otimes C^m)^* \cong \Lambda(Y \otimes (C^m \oplus C^{m*}))$. The space $C^m \oplus C^{m*}$ supports the canonical inner product obtained by symmetrizing the natural pairing between C^m and C^{m*} . Taking the tensor product of this inner product with the given inner product on Y , we get an inner product on $Y \otimes (C^m \oplus C^{m*})$. We may also recognize that by using the inner product on Y , we can identify $(Y \otimes C^m)^*$ with $Y \otimes (C^{m*})$; doing so, we see that the inner product just described may also be described as the canonical symmetrization of the pairing between $Y \otimes C^m$ and $(Y \otimes C^{m*})^*$. Thus, we may identify $\text{End}(\Lambda(Y \otimes C^m))$ with the Clifford algebra $C(Y \otimes (C^m \oplus C^{m*}))$ (see §A2.3.2). In this identification, the action of $O(Y)$ on $\text{End}(\Lambda(Y \otimes C^m))$ by conjugation becomes the natural action of $O(Y) \subseteq O(Y \otimes (C^m \oplus C^{m*}))$ on $C(Y \otimes (C^m \oplus C^{m*}))$. Thus the commutant of $O(Y)$ in $\text{End}(\Lambda(Y \otimes C^m))$ gets taken to the algebra of $O(Y)$ invariants in $C(Y \otimes (C^m \oplus C^{m*}))$;

this in turn is identified to the algebra of $O(Y)$ -invariants in $\Lambda(Y \otimes (C^m \oplus C^{m*}))$, which according to Theorem 4.3.2, is generated by the $O(Y)$ -invariants in $\Lambda^2(Y \otimes (C^m \oplus C^{m*}))$. But under the isomorphism of the Clifford and Grassmann algebras, (see §A2.3.5), $\Lambda^2(Y \otimes (C^m \oplus C^{m*}))$ becomes the Lie algebra $\mathfrak{so}(Y \otimes (C^m \oplus C^{m*}))$, so the $O(Y)$ -invariants are just the centralizer of $O(Y)$ in $\mathfrak{so}(Y \otimes (C^m \oplus C^{m*}))$. This is exactly the Lie algebra $\mathfrak{so}(C^m \oplus C^{m*})$. Thus, we have the following result.

4.3.4.1: THEOREM. The commutant of $O(Y)$ in

$\text{End}(\Lambda(Y \otimes C^m)) \cong C(Y \otimes (C^m \oplus C^{m*}))$ is generated as associative algebra by the Lie algebra $\mathfrak{so}(C^m \oplus C^{m*}) \subseteq C^2(Y \otimes (C^m \oplus C^{m*})) \cong \mathfrak{so}(Y \otimes (C^m \oplus C^{m*}))$.

(4.3.4.1.1) Remark: We should describe explicitly operators defining a basis for $\mathfrak{so}(C^m \oplus C^{m*})$. To this end, introduce a basis $\{e'_j : 1 \leq j \leq n\}$ for Y such that the inner product of these basis elements is $\langle e'_j, e'_k \rangle = \delta_{j,n+1-j}$ (Kronecker's delta). This is a basis corresponding to the polarized coordinates of §3.6.1. Make a basis $\{e'_{j,k}\}$, $1 \leq k \leq m$, for $Y \otimes (C^m \oplus C^{m*})$ by taking m copies of the chosen basis for V . Then $\mathfrak{so}(C^m \oplus C^{m*})$ has a three-step grading as described in §A2.3. We have

$$\begin{aligned} \mathfrak{so}^{(2,0)}(C^m \oplus C^{m*}) &= \text{span of } \{e_{ab}^+ = \sum_{j=1}^n \wedge_{e'_{j,a}} \wedge_{e'_{n+1-j,b}}\} \\ \mathfrak{so}^{(1,1)}(C^m \oplus C^{m*}) &= \text{span of } \{e_{a,b}^0 = (\sum_{j=1}^n \wedge_{e'_{j,a}} \wedge_{e'_{j,b}}) - \delta_{a,b} n/2\} \\ \mathfrak{so}^{(0,2)}(C^m \oplus C^{m*}) &= \text{span of } \{e_{a,b}^- = \sum_{j=1}^n \wedge_{e'_{j,a}} \wedge_{e'_{n+1-j,b}}\}. \end{aligned}$$

If we compare the reasoning behind Theorem 4.3.4.1 with the discussion in §A2.3, we see that the Clifford algebra formulation of this result can be refined, and we can replace $C^m \oplus C^{m*}$ with an arbitrary inner product space. Let Y and Z be vector spaces endowed with inner products. Let the tensor product $Y \otimes Z$ be given the inner product which is the tensor product of the inner products on the factors Y and Z . Observe that, under the standard embedding of $GL(Y)$ in $GL(Y \otimes Z)$, the orthogonal group $O(Y)$ embeds in $O(Y \otimes Z)$. Since $O(Y)$ acts irreducibly on Y , the centralizer of $O(Y)$ in $O(Y \otimes Z)$ is just $O(Z)$, embedded similarly.

4.3.4.2: THEOREM. Consider the action of $O(Y) \subseteq O(Y \otimes Z)$ on the Clifford algebra $C(Y \otimes Z)$. The algebra of invariants $C(Y \otimes Z)^{O(Y)}$ is generated as associative algebra by the Lie algebra $\mathfrak{so}(Z) \subseteq C^2(Y \otimes Z) \cong \mathfrak{so}(Y \otimes Z)$.

4.3.4.3: Remarks a) Theorem 4.3.4.1 is the case of Theorem 4.3.4.2 when $\dim Z$ is even. This follows because any even dimensional inner product on a complex vector space is isomorphic to $V \oplus V^*$ with its canonical inner product, where V is any maximal isotropic subspace of Z .

b) In §A2.3.5, it is explained that the Clifford algebra $C(Y)$ contains a copy of the spin cover of $SO(Y)$, and that $C^2(Y) \cong \mathfrak{so}(Y)$ is its Lie algebra. In the context of Theorem 4.3.4.2, the Lie algebra $\mathfrak{so}(Z)$ will generate the group $SO(Z)$, and the algebra spanned by $SO(Z)$ is the same as the algebra generated by $\mathfrak{so}(Z)$. As

explained in §A2.3.5, there is a cover of $O(Z)$ containing the cover of $SO(Z)$ commuting with $O(Y)$. However, the full group $O(Z)$ will not commute with $O(Y)$. Although Theorem 4.3.4.2 generalizes Theorem 4.3.4.1, it is not a double commutant theorem, because $O(Y)$ is acting on $C(Y \otimes Z)$ as automorphisms and is not actually embedded in $C(Y \otimes Z)$. Theorem 4.3.4.1 gives a double commutant version of Theorem 4.3.4.2 when $\dim Z$ is even. With some care, we may also formulate a double commutant version of Theorem 4.3.4.2 when $\dim Z$ is odd.

Consider the Clifford algebra $C(Y \otimes Z)$. The Lie algebras $\mathfrak{so}(Y)$ and $\mathfrak{so}(Z)$ are embedded in $\mathfrak{so}(Y \otimes Z)$ in the obvious way, and $\mathfrak{so}(Y \otimes Z)$ in turn is identified with $C^2(Y \otimes Z)$. Theorem 4.3.4.2 tells us that $\mathfrak{so}(Y)$ generates the subalgebra of $C(Y \otimes Z)$ invariant under the action by the automorphisms of $O(Z) \subseteq O(Y \otimes Z)$. We know from §A2.3.5 that the action of $SO(Z)$ is inner and that by exponentiating $\mathfrak{so}(Z)$ inside $C(Y \otimes Z)$, we obtain a covering group $Spin(Z)$ of $SO(Z)$.

The Lie algebra $C^2(Y \otimes Z)$ generates the even Clifford algebra $C^+(Y \otimes Z)$ (see §A2.3.8). In particular, both $\mathfrak{so}(Y)$ and $\mathfrak{so}(Z)$ are contained inside $C^+(Y \otimes Z)$.

4.3.4.4: THEOREM. *If $\dim Z$ is odd, the Lie algebras $\mathfrak{so}(Y)$ and $\mathfrak{so}(Z)$ generate mutual commutants inside $C^+(Y \otimes Z)$.*

PROOF. Let $\det_Z$ denote the determinant character of $O(Z)$. The commutant of $\mathfrak{so}(Y)$ in $C(Y \otimes Z)$ is the sum of the trivial eigenspace $C(Y \otimes Z)^{O(Z)}$ and the $\det_Z$ eigenspace for the action of $O(Z)$ on $C(Y \otimes Z)$. We know from Theorem 4.3.4.2 that $C(Y \otimes Z)^{O(Z)}$ is generated by $\mathfrak{so}(Y)$; in particular, $C(Y \otimes Z)^{O(Z)}$ is contained in $C^+(Y \otimes Z)$. We wish to show that the $\det_Z$ eigenspace for the action of $O(Z)$ on $C(Y \otimes Z)$ is contained in $C^-(Y \otimes Z)$. This will imply the theorem. Suppose we can construct an invertible element w of $C^-(Y \otimes Z)$ which is a $\det_Z$ eigenvector for O_Z . It is easy to check that multiplication by w will interchange $C(Y \otimes Z)^{O(Z)}$ and the $\det_Z$ eigenspace for $O(Z)$. In other words, the $\det_Z$ eigenspace is equal to $wC(Y \otimes Z)^{O(Z)}$. Since w is in $C^-(Y \otimes Z)$, we conclude that the whole $\det_Z$ eigenspace is contained in $C^-(Y \otimes Z)$, as desired.

Choose any unit vector y in Y . Let $\{e_j : 1 \leq j \leq 2m+1\}$ be an orthonormal basis for Z . Then the tensor products $y \otimes e_j$ are an orthonormal set in $Y \otimes Z$. It is evident that the mapping $e_j \rightarrow y \otimes e_j$ extends to a linear isometric embedding of Z into $Y \otimes Z$ that is equivariant for the natural actions of $O(Z)$ on the two spaces. It follows that the subalgebra C_y of $C(Y \otimes Z)$ generated by the $y \otimes e_j$ is preserved by the action of $O(Z)$, and the map $e_j \rightarrow y \otimes e_j$ defines an isomorphism of $C(Z)$ with C_y equivariant for the action of $O(Z)$ on each.

Consider the product $\tilde{\gamma}(y) = \prod_{j=1}^{2m+1} y \otimes e_j$. Up to sign, this is independent of the order of multiplication. We can see that it is the image under the isomorphism $C(Z) \cong C_y$ of the previous paragraph of the element γ of $C(Z)$. (See formula (A2.3.4.3).) It follows that it is an eigenvector with eigencharacter $\det_Z$ for the action of $O(Z)$ on C_y , or on $C(Y \otimes Z)$. Furthermore, from formula (A2.3.5.3), we see that conjugation by $\tilde{\gamma}(y)$ preserves $Y \otimes Z$, and acts on it by the element $-R_y$ of $O(Y)$. In particular, $\tilde{\gamma}(y)$ is invertible. This concludes the proof of Theorem 4.3.4.4.

4.3.4.5: Remarks: a) More details of the duality between orthogonal groups can be found in [Ada] and [Slu 1,2]. Especially, [Slu 2] contains a careful discussion of

what covering groups arise when the scalars are taken to be $\mathbf{R}$ and gives all possible duality relationships when nm is even.

b) It also follows from the proof of Theorem 4.3.4.4 that when $\dim Y$ is even, $\mathfrak{so}(Z)$ generates the full commutant of $Pin(Y) \subseteq Pin(Y \otimes Z)$.

4.3.5: $(O_n, \mathfrak{so}_{2m})$ -duality

The operator commutant version of the skew FFT immediately leads to a decomposition of $\Lambda(Y \otimes C^m)$ into joint $O(Y) \times \mathfrak{so}(C^m \oplus C^{m*})$ -isotypic components. Indeed, we have the $(GL(Y), GL_m)$ decomposition of $\Lambda(Y \otimes C^m)$, described in Theorem 4.1.1. The Lie algebra $\mathfrak{gl}_m$ is the subalgebra $\mathfrak{so}^{(1,1)}(C^n \oplus C^{m*})$ (see Remark 4.3.4.1.1 for notation) of $\mathfrak{so}(C^m \oplus C^{m*})$. This is the Levi component of a parabolic subalgebra of $\mathfrak{so}(C^m \oplus C^{m*})$, the nilradical of which is $\mathfrak{so}^{(0,2)}(C^m \oplus C^{m*})$. A standard extension of the highest weight theory tells us that, in any irreducible representation of $\mathfrak{so}(C^m \oplus C^{m*})$, the kernel of $\mathfrak{so}^{(0,2)}(C^m \oplus C^{m*})$ will be an irreducible representation of $\mathfrak{so}^{(1,1)}(C^m \oplus C^{m*})$, and this representation characterizes the representation of $\mathfrak{so}(C^m \oplus C^{m*})$ containing it. Since the summands in the decomposition of Theorem 4.1.1 are isotypic components for $\mathfrak{so}^{(1,1)}(C^m \oplus C^{m*})$, the intersection of one of them with $\ker \mathfrak{so}^{(0,2)}(C^m \oplus C^{m*})$ will be an irreducible representation for $\mathfrak{so}^{(1,1)}(C^m \oplus C^{m*})$.

Take for Borel subalgebra $\mathfrak{b}'$ in $\mathfrak{so}(C^m \oplus C^{m*})$ of the span of $\mathfrak{so}^{(0,2)}(C^m \oplus C^{m*})$ and the standard Borel in $\mathfrak{so}^{(1,1)}(C^m \oplus C^{m*})$. Consider the joint $GL(V) \times GL_m$ component of $\Lambda(Y \otimes C^m)$ labeled by the diagram D in the sum of Theorem 4.1.1. Denote it by $\Lambda(Y \otimes C^m)^D$. The conditions of dominance with respect to $\mathfrak{b}'$ imply that, in order for the intersection of $\ker \mathfrak{so}^{(0,2)}(C^m \oplus C^{m*})$ with $\Lambda(Y \otimes C^m)^D$ to be non-trivial, we must have

$$(4.3.5.1) \quad c_{j+1} \leq c_j \quad c_j + c_k \leq n, \text{ for } j \neq k$$

where c_j is the length of the j -th column of D .

On the other hand, if these conditions are satisfied, we can verify directly, using the formulas of Remarks 4.1.3 and 4.3.4.1.1, that the joint $GL(V) \times GL_m$ -highest weight vector of $\Lambda(Y \otimes C^m)^D$ is annihilated by $\mathfrak{so}^{(0,2)}(C^m \oplus C^{m*})$. (In writing these highest weight vectors, we are endowing V with a basis as in Remark 4.3.4.1.1.) Hence the conditions (4.3.5.1) exactly specify the joint $O(V) \times SO(C^m \oplus C^{m*})$ modules which occur in $\Lambda(Y \otimes C^m)$, and we have, moreover, explicit expressions for the joint $O(V) \times SO(C^m \oplus C^{m*})$ highest weight vectors.

Remark. The conditions (4.3.5.1) together imply that

$$c_j \leq \min\{(\dim V)/2, \dim V - c_1\}$$

for $j \geq 2$. Thus, if $\Lambda(Y \otimes C^m)^D$ contains a joint $O(V) \times SO(C^m \oplus C^{m*})$ highest weight vector, only the first column of D can be longer than $(\dim V)/2$. Thus, D is a diagram of the type described by conditions (3.6.2.8), and this association of diagrams of representations of $O(V)$ is the same as was made in the symmetric case (cf. §3.6.2).

The cases $m = 1$ and $m = 2$ are described in more detail in Appendix 4.

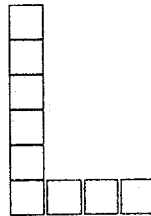
4.4: $\Lambda(S^2(C^n))$ and $\Lambda(\Lambda^2(C^n))$.

In this section we compute the GL_n module structures of the exterior algebras of $S^2(C^n)$, the symmetric two tensors, or symmetric matrices, and of $\Lambda^2(C^n)$, the skew-symmetric two tensors, or skew-symmetric matrices. These are the skew-symmetric analogs of Theorems 3.1 and 3.8.1. These results are special cases of calculations of Kostant [Kos1], and can also be found in [Iw]. They may be inferred from some formulas of Littlewood [Ltw]. As we did for skew (GL_n, GL_m) -duality, we give a proof based on Schur Duality. I find this proof attractive in itself, because it involves some nice representation theory of Weyl groups, the analog over the "field of one element" of Theorem 3.1 and these results. This approach also would give a proof of Theorems 3.1 and 3.8.1. Consider $S^2(C^n)$, the second symmetric power of C^n . This has a standard basis $\{e_{jk} : 1 \leq j \leq k \leq n\}$ consisting of the symmetrized tensor products of the standard basis $\{e_j\}$ of C^n . The weight or eigencharacter of e_{jk} is $a_j a_k$, where $\{a_j\}$ are the diagonal entries of the typical diagonal matrix in GL_n . Let us arrange the e_{jk} in a triangular array:

$$(4.4.1) \quad \begin{array}{cccccccc} & & & & e_{11} & e_{12} & e_{13} & e_{14} & e_{15} & e_{16} & \cdot & \cdot & \cdot \\ & & & & e_{22} & e_{23} & e_{24} & e_{25} & e_{26} & \cdot & \cdot & \cdot & \cdot \\ & & & & & e_{33} & e_{34} & e_{35} & e_{36} & \cdot & \cdot & \cdot & \cdot \\ & & & & & & e_{44} & e_{45} & e_{46} & \cdot & \cdot & \cdot & \cdot \\ & & & & & & & e_{55} & e_{56} & \cdot & \cdot & \cdot & \cdot \\ & & & & & & & & e_{66} & \cdot & \cdot & \cdot & \cdot \end{array}$$

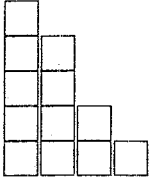
Let us say that $e_{\ell m} \geq e_{jk}$ if $j \geq \ell$ and $k \geq m$. In terms of the array (4.4.1), this says that $e_{\ell m} \geq e_{jk}$ if $e_{\ell m}$ appears above and to the left of e_{jk} . We can easily check that the action of U_n , the unipotent upper triangular matrices, on the e_{jk} transforms a given e_{jk} into linear combinations of $e_{\ell m}$ which are greater than or equal to e_{jk} . It follows that if we take a subset of the e_{jk} , which is increasing with respect to the specified order, and take the wedge product of all of them, we will get a B -eigenvector, that is, a highest weight vector, which will of course generate an irreducible GL_n -submodule of $\Lambda(S^2(C^n))$.

Let us investigate the highest weights of the modules constructed by the procedure just described. The diagram corresponding to the wedge product of the first r elements of the first row of array (4.4.1) has one row of length $r+1$ and $r-1$ rows of length 1. We will call it a $(r+1, r)$ -hook. Thus, the weight of $e_{11} \wedge e_{12} \wedge e_{13} \wedge e_{14}$ if $(5, 1, 1, 0, 0, \dots)$ and this corresponds to the diagram



If we now wedge this with the first s elements of the second row, the result is to nest a second hook, an $(s+1, s)$ -hook, inside the first hook. Of course, for this to be possible we must have $s < r$, and this is just the condition that the set of

e_{jk} be increasing. As an example, the diagram corresponding to the wedge product $e_{11} \wedge e_{12} \wedge e_{13} \wedge e_{14} \wedge e_{22} \wedge e_{23}$ is



Continuing to add elements from more rows, we find we can construct highest weight vectors whose diagrams are formed by nesting a sequence of $(r_i + 1, r_i)$ -hooks of decreasing size, each inside the previous one. The main result is that this process produces a complete decomposition of $\Lambda(S^2(C^n))$ as a GL_n -module.

4.4.2: THEOREM. The submodules of $\Lambda(S^2(C^n))$ generated by the highest weight vectors defined by taking wedge products of elements in an increasing subset of the array (4.4.1) give a complete decomposition of $\Lambda(S^2(C^n))$. The diagrams which parametrize these highest weights are formed by taking nested sequences of $\{k+1, k\}$ -hooks. In particular, the number of irreducible components of $\Lambda^m(S^2(C^n))$ is equal to the number of partitions of m into distinct parts of size no larger than n .

PROOF. Our proof is a typical application of the reciprocity laws implied by Schur Duality. More or less by definition, $S^2(C^n)$ is the space of S_2 -invariant vectors in $C^n \otimes C^n \cong C^{n \otimes 2}$. Similarly, $\Lambda^m(S^2(C^n))$ consists of the S_m -skew-invariant vectors in $(S^2(C^n))^{\otimes m}$. Combining these two descriptions, we can describe $\Lambda^m(S^2(C^n))$ as the space of vectors in $(C^n)^{\otimes 2m}$ which transform according to the character $\text{sgn}^\sim$ of the group D_m , where by D_m we mean the subgroup of S_{2m} which stabilizes the set Π_0 of pairs $\{\{1, 2\}, \{3, 4\}, \{5, 6\}, \dots, \{2m-1, 2m\}\}$ and by $\text{sgn}^\sim$ we mean the character which is pulled back from the sign character of S_m under the natural mapping from D_m to S_m defined by looking at the action of D_m on Π_0 . Combining this observation with Schur Duality, we find

$$(4.4.2.1) \quad \begin{aligned} \Lambda^m(S^2(C^n)) &\cong ((C^n)^{\otimes 2m})^{D_m, \text{sgn}^\sim} \cong \sum_{E'} \rho_n^E \otimes (\sigma^E)^{D_m, \text{sgn}^\sim} \\ &\cong \sum_E (\rho_n^E \otimes \sigma^E)^{D_m, \text{sgn}^\sim} \cong \sum_E \rho^E \otimes (\sigma^E)^{D_m, \text{sgn}^\sim} \end{aligned}$$

Here, E runs over diagrams of size $2m$ and at most n rows. We see from this formula that the multiplicity of ρ_n^E in $\Lambda^m(S^2(C^n))$ equals the multiplicity of the character $\text{sgn}^\sim$ in the restriction of σ^E to D_m . According to Theorem A1.4, the multiplicity of $\text{sgn}^\sim$ in any irreducible representation of S_{2m} is at most 1, and the number of representations for which it is one is at most the number of partitions of m into odd parts. We have produced a set of representations of GL_n which do occur in the final sum of formula (4.4.2.1), and if $n \geq m$, the number of them is equal to the number which can possibly occur. Thus, when $n \geq m$, we have produced the complete decomposition of $\Lambda^m(S^2(C^n))$. Since the dimension of $(\sigma^E)^{D_m, \text{sgn}^\sim}$ is independent of n , we may conclude that the representations we have produced decompose $\Lambda^m(S^2(C^n))$ also when $n < m$. This proves Theorem (4.4.2).

Similar considerations give the decomposition of $\Lambda(\Lambda^2(C^n))$. Let

$\{\tilde{e}_{jk} : 1 \leq k \leq n\}$ be the standard basis of $\Lambda^2(C^n)$. Arrange the $\tilde{e}_{jk}$ in an array like array (4.4.1):

$$(4.4.3) \quad \begin{array}{cccccccc} \tilde{e}_{12} & \tilde{e}_{13} & \tilde{e}_{14} & \tilde{e}_{15} & \tilde{e}_{16} & \tilde{e}_{17} & \cdots & \cdots \\ & \tilde{e}_{23} & \tilde{e}_{24} & \tilde{e}_{25} & \tilde{e}_{26} & \tilde{e}_{27} & \cdots & \cdots \\ & & \tilde{e}_{34} & \tilde{e}_{35} & \tilde{e}_{36} & \tilde{e}_{37} & \cdots & \cdots \\ & & & \tilde{e}_{45} & \tilde{e}_{46} & \tilde{e}_{47} & \cdots & \cdots \\ & & & & \tilde{e}_{56} & \tilde{e}_{57} & \cdots & \cdots \\ & & & & & \tilde{e}_{67} & \cdots & \cdots \end{array}$$

With the same order relation on the $\tilde{e}_{jk}$ as on the e_{jk} of (4.4.1), the U_n -invariant subspaces are again those spanned by increasing subsets, and an increasing subset is one which contains all $\tilde{e}_{jk}$ above and to the left of each of its members. We can form highest weight vectors by taking the wedge product of all members in any increasing set of $\tilde{e}_{jk}$. The wedge product of the first r elements of the first row produces a highest weight vector whose corresponding diagram has a first row of length r and r more rows of length 1. We call this an $(r, r+1)$ -hook. The wedge product of an increasing set with elements from several rows yields a diagram formed by nesting several (r_j, r_j+1) -hooks of decreasing sizes.

4.4.4: THEOREM. The submodules of $\Lambda(\Lambda^2(C^n))$ generated by highest weight vectors formed by taking wedge products of elements of an increasing subset of the array (4.4.3) exhaust $\Lambda(\Lambda^2(C^n))$. The diagrams of the modules which occur in $\Lambda(\Lambda^2(C^n))$ are formed by taking nested sequences of (r_j, r_j+1) -hooks. In particular, the number or components of $\Lambda^m(\Lambda^2(C^n))$ is the number of partitions of m into distinct parts of size at most $n-1$.

Remarks on the proof. The argument is quite parallel to the proof of Theorem 4.4.2, so we will omit the details. We remark that the reason the parts of the partitions parametrizing the components of $\Lambda^m(\Lambda^2(C^n))$ are bounded by $n-1$ is that an $(r, r+1)$ -hook has $r+1$ rows.

4.5: Skew-multiplicity-free actions, generalities.

The decompositions studied in §4.4 of $\Lambda(S^2(C^n))$ and $\Lambda(\Lambda^2(C^n))$, as well as the decomposition of $\Lambda(M_n(C^n))$ described by the skew- (GL_n, GL_m) -duality, are, like their symmetric counterparts, all multiplicity-free. Thus, the cleanness of the decompositions and the imperative to put skew-symmetric theory on the same footing as the symmetric theory, lead us to ask for a theory of skew-multiplicity-free modules, by which we mean a module V for a reductive group G such that the natural action of G on $\Lambda(V)$ is multiplicity-free. In this section, we will make some general observations about the shape of such a theory, and in the next few sections we will articulate the theory in more detail.

A major technique for constructing highest weight vectors in a polynomial algebra is to form monomials in known highest weight vectors. We have noted in §1.3.8 that this procedure is extremely effective for multiplicity-free actions: the semigroup of highest weight vectors is a free semigroup on the prime highest weight vectors, which are finite in number, and the highest weights of the modules which

appear is the semigroup generated by the weights of the prime highest weight-vectors. Review of the examples of §4.4 shows that this technique fails completely in these cases: the product of any two highest weight vectors is zero and the collection of highest weights which occur has no semigroup structure.

However, the examples of §4.4 also suggest an alternative approach which meshes well with our understanding of the geometric significance of the Grassmann algebra. To describe this construction, let V be a module for the connected reductive group G , and let B be a Borel subgroup of G . Suppose Y is a B -invariant subspace of V . Let $\{b_j : 1 \leq j \leq \dim Y\}$ be a basis for Y . Let

$$(4.5.1) \quad \Lambda^{\text{top}}(Y) = \prod_j \wedge b_j$$

be the wedge product of the b_j . Another choice of basis will yield the same $\Lambda^{\text{top}}(Y)$, up to a scalar multiple. From the basic properties of the Grassmann algebra [Ja1], [La], we know that for b in B we have

$$(4.5.2) \quad b(\Lambda^{\text{top}}(Y)) = \det(b|_Y) \Lambda^{\text{top}}(Y),$$

where $b|_Y$ denotes the restriction of b to Y . Thus $\Lambda^{\text{top}}(Y)$ is a B -eigenvector, i.e., a highest weight vector for G ; therefore, $\Lambda^{\text{top}}(Y)$ generates an irreducible G -module. We will call highest weight vectors in $\Lambda(V)$ of the form $\Lambda^{\text{top}}(Y)$ for B -invariant subspaces Y of V , geometric highest weight vectors. In the examples of §4.4, the geometric highest weight vectors gave us a full decomposition of $\Lambda(V)$. When this happens, we will call $\Lambda(V)$, or sometimes just V , geometrically decomposable.

In general, the construction of geometric highest weight vectors allows us to give a readily checked necessary condition on V in order for it to be skew-multiplicity-free. Let G be a reductive group, and let A be a Cartan subgroup of G . We will say that a G -module V is weight multiplicity-free (WMF) if the A -eigenspaces (the weight spaces) are all one-dimensional.

4.5.3: PROPOSITION. If a G -module V is skew-multiplicity-free, it is weight-multiplicity-free.

(4.5.4) PROOF. Select a Borel subgroup B of G , with B containing A . Write $B = AU$, where U is the unipotent radical of B . Recall that the positive roots are the weights of A acting on u , the Lie algebra of U , by conjugation. Recall also the standard order on the weights of A : we say φ in $\hat{A}$ is positive (not to be confused with dominant) if it is expressible as a product (with positive exponents) of positive roots, and we say that $\psi \geq \varphi$ if $\psi\varphi^{-1}$ is positive. If v in V is an A -eigenvector of weight φ , let $Y(v, B)$ denote the B -invariant subspace generated by v . Since v is an A -eigenvector, the space $Y(v, B)$ will be spanned by v together with vectors obtained by applying products of elements of u to v . It follows that the weights of A on $Y(v, B)$ consist of one copy of φ , and other weights which are larger than φ with respect to the order just described. Let $Y^+(\varphi)$ be the span of all weight spaces for weights (strictly) larger than φ . Then according to what we have just said, the space $Y^+(\varphi)$ is invariant by B , and if v is any A -eigenvector of weight φ , then the space $Cv \oplus Y^+(\varphi)$ is also B -invariant.

Now, suppose the φ -weight space has dimension larger than one. Select two linearly independent φ -eigenvectors for A , say v_1 and v_2 . Consider the spaces $Y_1 = \mathbb{C}v_1 \oplus Y^+(\varphi)$ and $Y_2 = \mathbb{C}v_2 \oplus Y^+(\varphi)$. From our discussion just above, we know that these spaces are both B -invariant. Further, it is easy to check that the $\det(b \mid Y_1)$ is the same as $\det(b \mid Y_2)$. It follows that $\Lambda^{\text{top}}(Y_1)$ and $\Lambda^{\text{top}}(Y_2)$ are two distinct B -eigenvectors in $\Lambda^m(V)$, of the same weight, where $m = \dim Y_j$, $j = 1, 2$. Hence, V is not skew-multiplicity-free.

4.5.5: Remark. The WMF condition on a module brings us in contact with basic structure theory of reductive Lie algebras. Let G be a reductive group with parabolic subgroup P . Let $P \cong M_P U_P$ be a Levi decomposition of P , with M_P the Levi component and U_P the unipotent radical of P . Let $A \subseteq M_P$ be a Cartan subgroup. Then A is also a Cartan subgroup of G . Let u_P be the Lie algebra of U_P . Then considered as an M_P -module by means of the adjoint action, u_P is WMF.

4.5.6: Remark. There is a natural ordering on the weight spaces of a WMF module. If ψ and φ are two weights occurring in the WMF module V , we will say that $\psi \geq \varphi$ if $V(\psi)$, the ψ -weight space (which is a line) is contained in $Y^+(\varphi)$ (notation as in theorem 4.5.4). This gives us a partially ordered set (poset) which we can represent as a graph (its Hasse diagram [Sty 2]). Some examples are given in Appendix 5.

4.6: Classification of weight multiplicity-free modules.

In this section, we will classify the irreducible modules of simple groups which are weight multiplicity-free (see the definition preceding Proposition 4.5.3). As with all classifications, this involves some case-by-case checking, but the amount of checking is greatly reduced by a number of results which show that weight multiplicities have certain monotonicity properties.

As above, let G be a connected reductive algebraic group, with Borel subgroup B containing a maximal torus A and unipotent radical U . If φ is a dominant weight of A , let V_φ be the irreducible G -module with highest weight φ .

For any weight χ of A , let $V_\varphi(\chi)$ be the χ -weight space in V_φ .

4.6.1: Lemma. If φ and ψ are dominant weights of A , then $\dim V_{\psi\varphi}(\psi\chi) \geq \dim V_\varphi(\chi)$ for any weight χ of A .

Proof. According to §1.1.19, we may consider V_φ as a subspace of the coordinate ring $\mathcal{R}(G/U)$. Let v_ψ denote the function on G/U which is a highest weight vector with weight ψ . Then multiplication by v_ψ defines an injective (since G/U is connected) map from V_φ to $V_{\psi\varphi}$. It is easy to check that this map maps $V_\varphi(\chi)$ into $V_{\psi\varphi}(\psi\chi)$, whence the desired inequality in dimensions.

Next, let $P \supseteq B$ be a parabolic subgroup of G , and let $P \cong M_P U_P$, where M_P is a Levi component of P , and U_P is the unipotent radical. Note that we may assume that M_P contains the torus A , and this uniquely determines M_P . Let φ be a dominant weight for A , considered as a torus in B . Then φ is also dominant as a weight of A considered as a torus in $B \cap M$. Let $V_{M,\varphi}$ denote the irreducible module of M with highest weight φ ; we denote the previous V_φ by $V_{G,\varphi}$ to be certain of what group is acting.

4.6.2: Lemma. For any character χ of A , we have $\dim V_{M,\varphi}(\chi) \leq \dim V_{G,\varphi}(\chi)$.

Proof. We simply observe that $V_{M,\varphi}$ may be regarded as the subspace of $V_{G,\varphi}$ generated by M acting on the highest weight vector.

4.6.3: Theorem. The irreducible modules of simple groups which are weight-multiplicity-free, are those on the following list.

A: SL_n : i) The fundamental representations of $\Lambda^m(\mathbb{C}^n)$, and
ii) the symmetric powers $S^m(\mathbb{C}^n)$ and $S^m(\mathbb{C}^{n*})$ of the standard representation and its dual.

B: $Spin_{2n+1}$: i) The standard representation on $\mathbb{C}^{2n+1}$, and
ii) the spin representation.

C: Sp_{2n} : i) The standard representation on $\mathbb{C}^{2n}$, and

ii) if $n = 2$ or 3 , the last fundamental representation, on $\Lambda^2_{\text{prim}}(\mathbb{C}^4)$ and $\Lambda^3_{\text{prim}}(\mathbb{C}^6)$ respectively.

D: $Spin_{2n}$: i) The standard representation on $\mathbb{C}^{2n}$, and

ii) the two half-spin representations.

E: i) The two 27-dimensional representations of E_6 , and

ii) The 56-dimensional representation of E_7 .

G: The 7-dimensional representation of G_2 .

There are no WMF modules for E_8 or F_4 .

(4.6.4) Proof. First consider the case of SL_n . The diagram $(2, 1, 1, \dots, 1)$, with $k-1$ rows all together (i.e., one row of length 2 and $k-2$ rows of length $(k-1)$) corresponds to the adjoint representation of SL_k , and the zero weight space of this has dimension $k-1$, and so is not WMF. From this, using Lemma 4.6.1, we can conclude that any representation of SL_k given by a diagram in which the first row is larger than the second row, and the $(k-1)$ -th row is larger than the k -th row, is not WMF. From this, using Lemma 4.6.2 applied to an upper triangular parabolic in SL_n which has one diagonal block of size k and $n-k$ blocks of size one, we conclude that any representations whose diagram has rows of three different lengths cannot be WMF. This leaves only representations whose diagrams are rectangles - the Cartan powers of the fundamental representations - as candidates to be WMF.

Consider the representation of SL_4 whose diagram is $(2, 2)$, i.e., $\rho_4^{(2,2)}$. By Proposition 2.4.5.3, the dimension of the zero weight space of $\rho_4^{(2,2)}$ is the dimension of the corresponding representation of the symmetric group S_4 . This dimension is 2, hence $\rho_4^{(2,2)}$ is not WMF. Using Lemma 4.6.1, we conclude that $\rho_4^{(k,k)}$ is never multiplicity-free for any $k > 1$. By applying Lemma 4.6.2 to an upper block triangular parabolic subgroup having one 4×4 block at the appropriate place and 1×1 blocks otherwise, we can conclude that the representations whose diagrams are rectangular blocks with r rows of length ℓ , for $2 \leq r \leq n-2$ and $\ell > 1$, are not WMF. This leaves only the representations listed in part A of the theorem as potentially WMF. It is easy to see by inspection that these are indeed WMF. The space $S^m(\mathbb{C}^n)$ is spanned by monomials $e_1^{i_1} e_2^{i_2} \dots e_n^{i_n}$, each of which is an eigenvector for the diagonal torus A_n with weight ψ_ℓ , with $\ell = (i_1, i_2, \dots, i_n)$. We see that knowledge of the eigencharacter allows us to reconstruct the eigenvector, so that we are in a WMF situation. Similarly, the fundamental representations $\Lambda^m(\mathbb{C}^n)$ have

vectors in $\Lambda(V)$ by a well-defined procedure: one determines the weight vectors in V , then determines the order relation between them following remark 4.5.6, then for each increasing set, forms the wedge product of its elements. By means of the Weyl dimension formula, one can then see how much of $\Lambda(V)$ is accounted for. For a substantial fraction of the modules on the list of the theorem, one finds one is already done, that is, many of the modules on the list are not merely skew-multiplicity-free, but also geometrically decomposable.

Among the modules which are not geometrically decomposable, the standard action of Sp_{2n} on C^{2n} is well understood; it is described in §A4.3. We have the Sp_{2n} -invariant element ω in $\Lambda^2(C^{2n})$. It is not decomposable; on the contrary, its n -th wedge power gives a non-zero element of $\Lambda^{2n}(C^{2n})$. However, every highest weight vector is the product of a decomposable highest weight vector with some power of ω . Thus, if we take into account the inevitability of ω due to the existence of an invariant symplectic form, everything else is given by geometric highest weight vectors. This suggests we might define a symplectic geometrically decomposable module to be one which admits an invariant symplectic form, and for which the geometric highest weight vectors give a complete decomposition of the primitive vectors $\Lambda_{\text{prim}}^m(C^{2n})$ for $1 \leq m \leq n$. One can check for symplectic geometric decomposability by the same procedure suggested above for geometric decomposability. Symplectic geometric decomposability accounts for a substantial fraction of the remaining modules in the list of the Theorem 4.7.1. In fact, after the decomposable and symplectically decomposable modules are crossed off, there are only 7 low dimensional modules left to investigate. These can be handled directly.

This discussion suggests it would be valuable to recast Theorem 4.7.1 to point out the three categories of modules distinguished above. Here is the result.

4.7.3: THEOREM. *The list in Theorem 4.7.1 can be broken up according to the following classes of modules.*

<i>Geometrically Decomposable</i>	<i>Symplectic Geometrically Decomposable</i>	<i>Other Skew- multiplicity-free</i>
A: $C^n, S^2(C^n), \Lambda^2(C^n)$, all n	$S^3(C^2); \Lambda^3(C^6)$	$S^m(C^2)$, $m = 4, 5, 6; S^3(C^3)$
B: C^{2n+1}		$spin_{2n+1}$, $n = 3, 4$
C: $\Lambda_{\text{prim}}^2(C^4)$	$C^{2n}, \Lambda_{\text{prim}}^3(C^6)$	
D: $C^{2n}, spin_8^\pm, spin_{10}^\pm$	$spin_{12}^\pm$	
E: V_{27}, V_{27}^* for E_6	V_{56} for E_7	
G:		V_7 for G_2

In this list, the subscripts on the modules of the exceptional groups indicate their dimensions.

4.7.4: REMARKS: With the list in this form, several observations become possible.

a) The geometrically decomposable modules all occur as abelian nilradicals of parabolic subgroups. Indeed, comparing with the list in §5.5, we see that our modules

are exactly the abelian nilradicals which are irreducible modules for simple groups. Having noted this, we see that the decomposition of $\Lambda(V)$ for these V is a part of Kostant's calculation of the cohomology of nilradicals [Kos1], since the cohomology of an abelian Lie algebra is just the exterior algebra of the Lie algebra. On the other hand, in §4.4 we have given the decomposition of $\Lambda(S^2(C^n))$ and $\Lambda(\Lambda^2(C^n))$, and the decomposition of $\Lambda(C^n \otimes C^m)$ is in §4.1, and of $\Lambda(C^n)$ as a module for O_n , in §A4.5. These are all the infinite classes of modules in the first column of Theorem 4.7.3. The remaining finite number of modules can be checked by the mechanical process described in the discussion of §4.7.2. This gives an alternative and very elementary if not elegant alternative demonstration of Kostant's result in case of abelian nilradicals.

b) As a corollary of the observation of remark a), we obtain an elegant characterization of the abelian nilradicals among irreducible modules for simple groups: an irreducible module V for a simple group appears as an abelian nilradical if and only if $\Lambda(V)$ is multiplicity-free and geometrically decomposable. It would be interesting to extend this characterization to general modules (not necessarily irreducible modules for not necessarily simple groups).

c) We may make a similar observation about the symplectically decomposable modules. Comparing with the list in §5.5 of Heisenberg nilradicals, we see that this part of our list consists exactly of the irreducible modules for simple groups which appear as the commutator quotient of nilradicals isomorphic to the Heisenberg Lie algebra. From our discussion in §A4.3 of the cohomology of the Heisenberg Lie algebra, we see that the decomposition of $\Lambda(V)$ for these V also follows from Kostant's calculation in [Kos1]; and conversely, our method of calculation provides an elementary approach to this case of Kostant's result.

d) Observation c) leads us to a characterization of the irreducible symplectic modules for simple groups which appear as the commutator quotient of a Heisenberg nilradical: they are precisely the (irreducible) symplectic modules (of simple groups) which are skew-multiplicity-free and symplectic geometrically decomposable. Again, it is desirable to lift the restrictions of irreducibility of V and simplicity on G .

e) I do not know any simple principle to explain the modules in the third column of Theorem 4.7.3. However, it is not difficult to deal with these modules on a case-by-case basis. Their dimensions are not large; the largest is $spin_9$, of dimension 16. The others have dimensions 10, 8, 7, 6 and 5. The two spin representations can be decomposed by using the branching law from O_{2n} to O_{2n-1} , as described in §5.4.2. The SL_2 -modules may be described by methods going back to the 19th century. The 7-dimensional representation of G_2 can be dealt with essentially by dimension counting. The descriptions of $\Lambda(V)$ for these V is given in Appendix 5.

4.7.5: Exhaustion of skew-multiplicity-free modules

We consider the list of Theorem 4.7.3 together with the remarks of 4.7.4 to be a sufficient demonstration that all the modules listed do belong there. To prove theorems 4.7.1 and 4.7.3, it remains to show that these lists are exhaustive. This involves some computations. Fortunately, monotonicity properties like those of Lemma 4.6.1 and 4.6.2 greatly reduce the number of explicit verifications we must perform. In particular, we have the following labor-saving observation.

basis $e_{j_1} \wedge e_{j_2} \wedge \dots \wedge e_{j_r}$, with $j_1 < j_2 < \dots < j_r$, which are again A_n -eigenvectors with distinct weights, so the $\Lambda^m(\mathbb{C}^n)$ are also WMF. This establishes part A of the theorem.

Part A puts strong restrictions on the WMF representations of the other types of groups, since GL_n appears as a Levi component in all the other groups. For $Spin_{2n+1}$, the Levi component of the stabilizer of a maximal (n -dimensional) isotropic subspace of $\mathbb{C}^{2n+1}$ is isomorphic to a twofold cover of GL_n . The first $n-1$ fundamental representations of $Spin_{2n+1}$ are the actions of O_{2n+1} on the exterior powers $\Lambda^m(\mathbb{C}^{2n+1})$ for $1 \leq m \leq n-1$. The restriction of this to GL_n has the form

$$\sum_{m-1 \leq a+b \leq m} \Lambda^a(\mathbb{C}^n) \otimes \Lambda^b((\mathbb{C}^n)^*).$$

The decomposition of $\Lambda^a(\mathbb{C}^n) \otimes \Lambda^b((\mathbb{C}^n)^*)$, $1 \leq a, b \leq n-1$ contains representations which are not on the list of part A, except when one of a and b equals 1 and the other equals $n-1$; and in this exceptional case, which is a twist by $\det^{-1}$ of $\mathbb{C}^n \otimes \mathbb{C}^n$, the representation as a whole is not WMF, although the irreducible summands, viz., the twists by $\det^{-1}$ of $S^2(\mathbb{C}^n)$ and $\Lambda^2(\mathbb{C}^n)$, are WMF. Hence $\Lambda^m(\mathbb{C}^{2n+1})$ is not WMF except for $m = 0, 1, 2n, 2n+1$.

The last fundamental representation of $Spin_{2n+1}$ is the spin representation. The restriction of this to GL_n is well known to be a twist by a power of $\det$ of the action on the exterior algebra (see §A4.4), which is WMF. Hence this is WMF. The second Cartan power of the spin representations is $\Lambda^n(\mathbb{C}^{2n+1})$, which we have seen is not WMF. By an extension of the argument of Lemma 4.6.2, we can see that the restriction to GL_n of the Cartan product of $\mathbb{C}^{2n+1}$ with the spin representation contains the Cartan product of $\mathbb{C}^n$ with $\Lambda^m(\mathbb{C}^n)$ for $0 \leq m \leq n$. Hence this representation is also clearly not WMF. Finally, the second Cartan power of $\mathbb{C}^{2n+1}$ is just the action of $Spin_{2n+1}$ on the harmonic polynomials of degree 2 on $\mathbb{C}^{2n+1}$, and this contains a zero weight space of dimension n , so it is not WMF for $n \geq 2$. This concludes case B of the theorem.

Case D of the theorem is quite similar to case B, and we will not give the complete details. We note that the first $n-2$ fundamental representations are realized on the modules $\Lambda^m(\mathbb{C}^{2n})$ for $1 \leq m \leq n-2$ and that the last two fundamental representations are the two half spin representations. The Cartan square of each of the half spin representations is one of the two SO_{2n} -invariant subspaces of $\Lambda^n(\mathbb{C}^{2n})$, and $\Lambda^{n-1}(\mathbb{C}^{2n})$ is the Cartan product of the two half spin representations. $Spin_{2n}$ contains (a twofold cover of) GL_n as the Levi component of the stabilizer of a maximal isotropic subspace in $\mathbb{C}^{2n}$.

Consider now the symplectic group Sp_{2n} . It is similar in some respects to the orthogonal groups. It contains GL_n as the Levi component of the stabilizer of a maximal isotropic subspace. Its fundamental representations are realized on the spaces $\Lambda_{\text{prim}}^m(\mathbb{C}^{2n})$ for $1 \leq m \leq n$; see §A4.3. As in the case of $Spin_{2n}$, we know that the restriction of $\Lambda^m(\mathbb{C}^{2n})$ to GL_n is a sum

$$\sum_{a+b=m} \Lambda^a(\mathbb{C}^n) \otimes \Lambda^b((\mathbb{C}^n)^*).$$

Out of a given tensor product $\Lambda^a(\mathbb{C}^n) \otimes \Lambda^b((\mathbb{C}^n)^*)$, it is clear that at least the Cartan product of $\Lambda^a(\mathbb{C}^n)$ and $\Lambda^b((\mathbb{C}^n)^*)$ must belong to $\Lambda_{\text{prim}}^m(\mathbb{C}^{2n})$. This Cartan

product is the twist by $\det^{-1}$ of $\Lambda^a(\mathbb{C}^n) \otimes \Lambda^{n-b}(\mathbb{C}^n)$, which is not on the list of part A unless one of a or $n-b$ is zero or n , or $a = n-b$ and both are equal to 1 or $n-1$. If $2 \leq m \leq n-1$, these conditions are violated for at least one pair (a, b) with $a+b=m$. If $m=1$, they hold and we can see directly that $\mathbb{C}^{2n}$ is WMF. If $m=n$, then we obtain the diagrams $(2, 2, 2, \dots, 2)$ with a $2s$, for $1 \leq a \leq n$. The choice of $a=2$ will be on the list of part A only when $n=2$ or $n=3$. The Cartan product of $\mathbb{C}^{2n}$ with itself is realized as $S^2(\mathbb{C}^2)$, which is isomorphic to the adjoint representation of Sp_{2n} , and so has a zero weight space of dimension n . It is also easy to see that when $n=2$ or 3 , the Cartan product of $\mathbb{C}^{2n}$ with $\Lambda_{\text{prim}}^n(\mathbb{C}^{2n})$ is not WMF. This concludes part C of the theorem.

The proofs of the assertions of parts E, F and G of the theorem can be verified by various special arguments. Rather than draw out the details, we simply refer to [BMP], from the tables of which the necessary facts can be gleaned.

4.7: Classification of skew-multiplicity-free modules.

In this section, we use the results of the previous two sections to classify the simple modules for simple groups which are skew-multiplicity-free. By Proposition 4.5.3, we know that these modules are a subset of the list of Theorem 4.6.3. Our problem, then, is to eliminate the non-SMF members from that list and to verify that the remaining ones are indeed SMF. The result is as follows.

4.7.1: THEOREM. *The irreducible modules of simple groups (possibly augmented by $\mathbb{C}^x$, the scalar matrices) which are skew-multiplicity-free are those on the following list.*

A: SL_n : i) $S^m(\mathbb{C}^n)$ and $S^m(\mathbb{C}^{n*})$

a) for $m=1$ and all $n > 0$

b) for $m=2$ and all $n > 0$

c) for $m=3$ and $n=2$ or 3

d) for $m=4, 5$ or 6 and $n=2$

ii) $\Lambda^m(\mathbb{C}^n)$ and $\Lambda^m(\mathbb{C}^{n*})$

a) for $m=1$ or $n-1$ and all $n > 0$

b) for $m=2$ or $n-2$ and all $n > 0$

c) for $m=3$ and $n=6$

B: $Spin_{2n+1}$: i) $\mathbb{C}^{2n+1}$ for all $n \geq 3$

ii) the spin representation, for $n=3$ or 4

C: Sp_{2n} : i) $\mathbb{C}^{2n}$ for all $n > 0$

ii) $\Lambda^n(\mathbb{C}^{2n})$ for $n=2$ or 3

D: $Spin_{2n}$: i) $\mathbb{C}^{2n}$ for all $n \geq 4$

ii) the half spin representations for $n=4, 5$ or 6

E: i) the two 27-dimensional representations of E_6

ii) the 56-dimensional representation of E_7

G: The 7-dimensional representation of G_2 .

4.7.2: Beginning of proof

Our proof of Theorem 4.7.1 is unfortunately fairly computational. We begin by worrying how to show that all the modules listed are indeed SMF. To begin, choose one of these modules and call it V . One can construct the geometric highest weight

4.7.5.1: LEMMA. Let V be a module for the reductive group G . Let $P = M_P U_P = M U_P$ be a parabolic subgroup of G . Then if V is SMF for G , any irreducible summand of $V|_M$ must be SMF for M .

PROOF. Let $Y \subseteq V$ be an irreducible summand for M . Let $\tilde{Y}$ denote the P -invariant subspace generated by Y . We can write

$$\tilde{Y} = \tilde{Y}^+ \oplus Y,$$

where $\tilde{Y}^+$ is a P -invariant complement to Y in $\tilde{Y}$. If ν is any highest weight vector for M in $\Lambda^r(Y)$, then $\Lambda^{\text{top}}(\tilde{Y}^+) \wedge \nu$ is a highest weight vector for G in $\Lambda^s(V)$, with $s = r + \dim \tilde{Y}^+$. It is clear that if ν_1 and ν_2 are two highest weights of $\Lambda^r(Y)$ with the same weight, then likewise the vectors $\Lambda^{\text{top}}(\tilde{Y}^+) \wedge \nu_1$ and $\Lambda^{\text{top}}(\tilde{Y}^+) \wedge \nu_2$ have the same weight. The conclusion of the lemma follows.

4.7.5.2. Remark: A more elegant way of formulating the above result would be to say that there is a character ψ of M such that the M -module $\psi \otimes \Lambda(Y)$, considered as a P -module on which U_P acts trivially, embeds (as a P -module) in $\Lambda(V)$. (The character ψ is $\det(M|_{\tilde{Y}^+})$.)

By means of this lemma, we see that to verify the completeness of part A of the list in Theorem 4.6.1, it suffices to check that the modules $S^7(C^2)$, $S^4(C^3)$, $S^3(C^4)$, and $\Lambda^3(C^7)$ are not SMF.

Outside of case A, the only modules in the list of Theorem 4.6.3 which are missing from Theorem 4.7.1 are the spin modules spin_{2n+1} for $n \geq 5$ and half-spin representations $\text{spin}_{2n}^\pm$ for $n \geq 7$. Since Spin_n contains $\text{Spin}_{n-2} \cdot C^x$ as a Levi component of a parabolic (the stabilizer of an isotropic line) and since a spin representation of Spin_n restricted to Spin_{n-2} is the sum of two spin representations, Lemma 4.5.7.1 tells us that if spin_{2n+1} nor is $\text{spin}_{2n}^\pm$ is not SMF, then neither is spin_{2n+3} or $\text{spin}_{2n+2}^\pm$. Thus, to show that only the specified spin representations are SMF, it will suffice to show that spin_{11} and $\text{spin}_{14}^\pm$ are not SMF.

Thus, to complete the proof of Theorem 4.7.1 and 4.7.3, we just need to show that the modules $S^7(C^2)$, $S^4(C^3)$, $S^3(C^4)$, $\Lambda^3(C^7)$, spin_{11} and $\text{spin}_{14}^\pm$ are not SMF. The module $S^7(C^2)$ is easily handled by the classical methods of invariant theory of binary forms. The result is given in Appendix 5. Since Spin_{14} contains (a two-fold cover of) GL_7 as a Levi component and since, up to twisting with a (half-integral) power of det, we have

$$\text{spin}_{14}|_{GL_7} \cong \Lambda^{\text{odd}}(C^7) \supseteq \Lambda^3(C^7),$$

we see Lemma 4.5.7.1 says that if $\Lambda^3(C^7)$ is not SMF, neither is spin_{14} . For spin_{11} , we can use our knowledge of $\text{spin}_{12}^\pm$ and of the branching law from Spin_{12} to Spin_{11} (see §5.4.2) to describe spin_{11} . The result is described in A5. The cases $S^4(C^3)$, $S^3(C^4)$, and $\Lambda^3(C^7)$ are handled by direct computation. Fortunately, they are well within the scope of existing technology. For example, one can read off from the tables of [ER] that $\Lambda^5(S^4(C^3))$ contains the representations ρ_3^D with multiplicity 2 for $D = (10, 7, 3)$ and $D = (11, 6, 3)$, and with multiplicity 3 for $D = (10, 6, 4)$. Also, $\Lambda^6(S^3(C^4))$ contains ρ_4^D with multiplicity 2 for $D = (6, 6, 3, 3)$, $(8, 6, 3, 1)$ and $(9, 4, 4, 1)$. Finally, $\Lambda^6(\Lambda^3(C^7))$ contains ρ_7^D with multiplicity 2 for $D = (5, 4, 3, 3, 1, 1, 1)$ and $(4, 3, 3, 3, 1, 1)$. This completes our argument for Theorems 4.7.1 and 4.7.3.

5: Examples.

5.1: Classical symmetric spaces: cohomology and invariant differential operators.

The classical compact symmetric spaces can be grouped 10 infinite families, which can be further grouped into 4 classes [He2], [FJ].

5.1.1: List of classical compact symmetric spaces

- | | | | |
|-------------------|-----------------------------|---------------------------------|-------------------------------|
| a) Groups: | i) U_n | ii) Sp_n | iii) O_n |
| b) Grassmannians | i) $U_{m+n}/U_m \times U_n$ | ii) $Sp_{m+n}/Sp_m \times Sp_n$ | iii) $O_{m+n}/O_m \times O_n$ |
| c) Polarizations | i) Sp_n/U_n | ii) O_{2n}/U_n | |
| d) Bilinear forms | i) U_{2n}/Sp_n | ii) U_n/O_n | |

The next list describes for each of these spaces the action of the complexified isotropy group on the complexified tangent space to the identity coset.

5.1.2: List of complexified tangent actions for the classical symmetric spaces

- | | | |
|---|--|-----------------------------|
| a) i) GL_n , adjoint action | ii) Sp_{2n} , adjoint action | iii) O_n , adjoint action |
| b) i) $GL_n \times GL_m$, | ii) $Sp_{2n} \times Sp_{2m}$, | iii) $O_n \times O_m$ |
| on $C^n \otimes C^m \oplus C^{n*} \otimes C^{m*}$ | on $C^{2n} \otimes C^{2m}$ | on $C^n \otimes C^m$ |
| c) i) GL_n , | ii) GL_n , | |
| on $S^2(C^n) \oplus S^2(C^{n*})$ | on $\Lambda^2(C^n) \oplus \Lambda^2(C^{n*})$ | |
| d) i) Sp_{2n} , on $\Lambda^2(C^{2n})$ | ii) O_n , on $S^2(C^n)$ | |

The invariants in the polynomial or symmetric algebras of these modules are of interest in connection with analysis on (both compact and non-compact) symmetric spaces, because these invariants determine the invariant differential operators on the associated symmetric space [He3], [FJ]. (They are more or less the symbols of these differential operators.) These invariants have been systematically described for general symmetric spaces, not just classical ones, in [KR]. The invariants in the exterior algebras of these modules are of interest because they describe the DeRham cohomology of the compact symmetric spaces [He2], [W]. This has been computed by Brauer [Bra], [W] in the case of groups, by Ehresmann [Ehr] for Hermitian symmetric spaces (cases bi), ci) and cii), and biii) when n or m equals 2), and in general by Iwamoto [Iw]. Iwamoto's results seems to be relatively little known, even among experts. Perhaps this is partially due to the sad event that he died before his paper was published. As Brauer did for the classical groups themselves, Iwamoto's method produces an explicit basis for the cohomology. It is based on the skew-symmetric part of the following remarkable fact.

5.1.3: For every action (G, V) in the list 5.1.2, there is a group H acting on V and containing G , such that

- the action of H on V is multiplicity-free and skew-multiplicity-free; and
- H/G is a symmetric space.

Fact a) allows us to give the H -isotypic decomposition of the polynomial or exterior algebra of V , and we can then easily determine which of the H -modules occurring contain invariants for G , by fact b). Thus, these facts together give us a painless way

4.7.5.1: LEMMA. Let V be a module for the reductive group G . Let $P = M_P U_P = MU_P$ be a parabolic subgroup of G . Then if V is SMF for G , any irreducible summand of $V|_M$ must be SMF for M .

PROOF. Let $Y \subseteq V$ be an irreducible summand for M . Let $\tilde{Y}$ denote the P -invariant subspace generated by Y . We can write

$$\tilde{Y} = \tilde{Y}^+ \oplus Y,$$

where $\tilde{Y}^+$ is a P -invariant complement to Y in $\tilde{Y}$. If ν is any highest weight vector for M in $\Lambda^r(Y)$, then $\Lambda^{\text{top}}(\tilde{Y}^+) \wedge \nu$ is a highest weight vector for G in $\Lambda^s(V)$, with $s = r + \dim \tilde{Y}^+$. It is clear that if ν_1 and ν_2 are two highest weights of $\Lambda^r(Y)$ with the same weight, then likewise the vectors $\Lambda^{\text{top}}(\tilde{Y}^+) \wedge \nu_1$ and $\Lambda^{\text{top}}(\tilde{Y}^+) \wedge \nu_2$ have the same weight. The conclusion of the lemma follows.

4.7.5.2. Remark: A more elegant way of formulating the above result would be to say that there is a character ψ of M such that the M -module $\psi \otimes \Lambda(Y)$, considered as a P -module on which U_P acts trivially, embeds (as a P -module) in $\Lambda(V)$. (The character ψ is $\det(M|_{\tilde{Y}^+})$.)

By means of this lemma, we see that to verify the completeness of part A of the list in Theorem 4.6.1, it suffices to check that the modules $S^7(C^2)$, $S^4(C^3)$, $S^3(C^4)$, and $\Lambda^3(C^7)$ are not SMF.

Outside of case A, the only modules in the list of Theorem 4.6.3 which are missing from Theorem 4.7.1 are the spin modules spin_{2n+1} for $n \geq 5$ and half-spin representations $\text{spin}_{2n}^\pm$ for $n \geq 7$. Since Spin_n contains $\text{Spin}_{n-2} \cdot C^\pi$ as a Levi component of a parabolic (the stabilizer of an isotropic line) and since a spin representation of Spin_n restricted to Spin_{n-2} is the sum of two spin representations, Lemma 4.5.7.1 tells us that if spin_{2n+1} nor is $\text{spin}_{2n}^\pm$ is not SMF, then neither is spin_{2n+3} or $\text{spin}_{2n+2}^\pm$. Thus, to show that only the specified spin representations are SMF, it will suffice to show that spin_{11} and $\text{spin}_{14}^\pm$ are not SMF.

Thus, to complete the proof of Theorem 4.7.1 and 4.7.3, we just need to show that the modules $S^7(C^2)$, $S^4(C^3)$, $S^3(C^4)$, $\Lambda^3(C^7)$, spin_{11} and $\text{spin}_{14}^\pm$ are not SMF. The module $S^7(C^2)$ is easily handled by the classical methods of invariant theory of binary forms. The result is given in Appendix 5. Since Spin_{14} contains (a two-fold cover of) GL_7 as a Levi component and since, up to twisting with a (half-integral) power of $\det$, we have

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we see Lemma 4.5.7.1 says that if $\Lambda^3(C^7)$ is not SMF, neither is spin_{14} . For spin_{11} , we can use our knowledge of $\text{spin}_{12}^\pm$ and of the branching law from Spin_{12} to Spin_{11} (see §5.4.2) to describe spin_{11} . The result is described in A5. The cases $S^4(C^3)$, $S^3(C^4)$, and $\Lambda^3(C^7)$ are handled by direct computation. Fortunately, they are well within the scope of existing technology. For example, one can read off from the tables of [ER] that $\Lambda^5(S^4(C^3))$ contains the representations ρ_3^D with multiplicity 2 for $D = (10, 7, 3)$ and $D = (11, 6, 3)$, and with multiplicity 3 for $D = (10, 6, 4)$. Also, $\Lambda^6(S^3(C^4))$ contains ρ_4^D with multiplicity 2 for $D = (6, 6, 3, 3)$, $(8, 6, 3, 1)$ and $(9, 4, 4, 1)$. Finally, $\Lambda^6(\Lambda^3(C^7))$ contains ρ_7^D with multiplicity 2 for $D = (5, 4, 3, 3, 1, 1)$ and $(4, 3, 3, 3, 1, 1)$. This completes our argument for Theorems 4.7.1 and 4.7.3.

5: Examples.

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The classical compact symmetric spaces can be grouped into 10 infinite families, which can be further grouped into 4 classes [He2], [FIJ].

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The next list describes for each of these spaces the action of the complexified isotropy group on the complexified tangent space to the identity coset.

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| on $C^n \otimes C^m \oplus C^{n*} \otimes C^{m*}$ | on $C^{2n} \otimes C^{2m}$ | on $C^n \otimes C^m$ |
| c) i) GL_n , | ii) GL_n , | |
| on $S^2(C^n) \oplus S^2(C^{n*})$ | on $\Lambda^2(C^n) \oplus \Lambda^2(C^{n*})$ | |
| d) i) Sp_{2n} , on $\Lambda^2(C^{2n})$ | ii) O_n , on $S^2(C^n)$ | |

The invariants in the polynomial or symmetric algebras of these modules are of interest in connection with analysis on (both compact and non-compact) symmetric spaces, because these invariants determine the invariant differential operators on the associated symmetric space [He3], [FIJ]. (They are more or less the symbols of these differential operators.) These invariants have been systematically described for general symmetric spaces, not just classical ones, in [KR]. The invariants in the exterior algebras of these modules are of interest because they describe the DeRham cohomology of the compact symmetric spaces [He2], [W1]. This has been computed by Brauer [Bra], [W1] in the case of groups, by Ehresmann [Ehr] for Hermitian symmetric spaces (cases bi), ci) and cii), and biii) when n or m equals 2), and in general by Iwamoto [Iw]. Iwamoto's results seems to be relatively little known, even among experts. Perhaps this is partially due to the sad event that he died before his paper was published. As Brauer did for the classical groups themselves, Iwamoto's method produces an explicit basis for the cohomology. It is based on the skew-symmetric part of the following remarkable fact.

5.1.3: For every action (G, V) in the list 5.1.2, there is a group H acting on V and containing G , such that

- the action of H on V is multiplicity-free and skew-multiplicity-free; and
- H/G is a symmetric space.

Fact a) allows us to give the H -isotypic decomposition of the polynomial or exterior algebra of V , and we can then easily determine which of the H -modules occurring contain invariants for G , by fact b). Thus, these facts together give us a painless way

to determine the polynomial and exterior invariants for all the classical symmetric spaces.

We exhibit below the list of the groups H announced by statement 5.1.3.

5.1.4: List of multiplicity-free actions associated to the classical symmetric spaces

- | | | |
|---|--|--|
| a) i) $GL_n \times GL_n$,
on $C^n \otimes C^n$ | ii) GL_n ,
on $S^2(C^n)$ | iii) GL_n ,
on $\Lambda^2(C^n)$ |
| b) i) $(GL_n \times GL_m) \times (GL_n \times GL_m)$,
on $C^n \otimes C^m \oplus C^{n*} \otimes C^{m*}$ | ii) $GL_{2n} \times GL_{2m}$,
on $C^{2n} \otimes C^{2m}$ | iii) $GL_n \times GL_m$,
on $C^n \times C^m$ |
| c) i) $GL_n \times GL_n$,
on $S^2(C^n) \oplus S^2(C^{n*})$ | ii) $GL_n \times GL_n$,
on $\Lambda^2(C^n) \oplus \Lambda^2(C^{n*})$ | |
| d) i) GL_{2n} ,
on $\Lambda^2(C^{2n})$ | ii) GL_n ,
on $S^2(C^n)$ | |

All of these actions are known to be multiplicity-free by the results of §§2,3 and to be skew-multiplicity-free by §§4.1, 4.4.

5.1.5: Remarks: a) The alert reader may have noted that the list 5.1.1 does not always give quite the same description of the symmetric spaces as is given, for example, in [He2]. Thus, rather than take SU_n for the simple group of type A , we take U_n . These small modifications are made in order to make statement 5.1.3 valid. Thus, the complexified Lie algebra of SU_n is the traceless $n \times n$ matrices, and there is no action of $GL_n \times GL_n$ on this space. These small modifications, and the differences between the actions of O_n and of SO_n , are easily accounted for if one wishes to deal with the standard symmetric spaces rather than our slightly deviant versions.

b) Examples d) and e) of §4 of [Hw1] implicitly invoke the principle of statement 5.1.3. They are the skew-symmetric case a) and the symmetric case aii), respectively.

c) Some, though evidently not all, of the non-classical symmetric spaces can be treated by considerations similar to the principle of statement 5.1.3.

5.1.6: We illustrate the type of computation involved in determining the invariants of actions 5.1.2 by the procedure made possible by statement 5.1.3. We take case di) of U_{2n}/Sp_n . For this we want to analyze the invariants of Sp_{2n} acting on the polynomial algebra and on the exterior algebra of the skew-symmetric matrices. We know from §3.8, that as a module for GL_{2n} , $\mathcal{P}(\Lambda^2(C^{2n}))$ breaks up into a sum of one copy each of the modules ρ_{2n}^D such that the diagram D has columns which are all of even length. From this and Frobenius reciprocity (see §1.1.17), we also know that a representation ρ_{2n}^D of GL_{2n} contains at most one invariant for Sp_{2n} and does contain one if and only if the columns of D all have even length. Thus, there is one Sp_{2n} -invariant contained in each GL_{2n} -module appearing in the decomposition of $\mathcal{P}(\Lambda^2(C^{2n}))$. In particular, for each k , $1 \leq k \leq n$, there is an Sp_{2n} -invariant J_k in the copy of $\Lambda^{2k}(C^{2n})$ appearing in $\mathcal{P}^k(\Lambda^2(C^{2n}))$. An argument using the filtration on $\mathcal{P}(\Lambda^2(C^{2n}))$ imposed by the GL_{2n} -module structure (cf. [HU] or [Pop2]) shows that $\mathcal{P}(\Lambda^2(C^{2n}))^{Sp_{2n}}$ is a polynomial algebra in the generators $\{J_k\}$.

Remark: We can give explicit formulas for the invariants if we appeal to the Hodge theory of §A4.3. From there, we know that the invariant of Sp_{2n} contained

in $\Lambda^{2k}(C^{2n})$ is the k -th exterior power of the invariant 2-vector in $\Lambda^2(C^{2n})$.

Turning to the exterior algebra $\Lambda(\Lambda^2(C^{2n}))$, we know from §4.4 that, as a GL_{2n} -module, it is a multiplicity-free sum of representations ρ_{2n}^D where D is formed by nesting a set of $(r+1, r)$ -hooks for different values of r . In order that the first column of such a diagram should have even length, it is necessary that r_1 , the size of the largest hook, be odd. The diagram will have the same number, r_1 , of columns as the largest hook. The first column will be contained in the first hook, but this hook will contribute only the first box to the $r_1 - 1$ remaining columns. In order that these have even length, the second hook must contribute a second element to them. This requires that r_2 , the size of the second hook, equal $r_1 - 1$. These first two hooks will then fit together to make a diagram of the required type, with the first two columns of length $r_1 + 1$, and the remaining $r_1 - 2$ columns of length 2. We will call this type of diagram an $(r+1, r)$ -double hook, where r here is r_1 . We can now see that the general diagram of the type required will be composed by nesting a sequence of $(r+1, r)$ -double hooks of different sizes r , all of which should be odd. The number of boxes in an $(r+1, r)$ -double hook is $2(2r-1)$. If r is odd, say $r = 2s+1$, then $2r-1 = 4s+1$ is congruent to 1 modulo 4. Conversely, if t is congruent to 1 modulo 4, then $t = 4s+1 = 2(2s+1) - 1$; that is, t can be written as $2r-1$ with r odd. For a representation ρ_{2n}^D appearing in $\Lambda^k(\Lambda^2(C^{2n}))$, the diagram D will have $2k$ boxes. If D is made of (r_j+1, r_j) -double hooks for certain r_j , then the number of boxes in d is $\sum_j 2(2r_j-1)$. Hence $k = \sum_j (2r_j-1)$, and D defines a partition of k by distinct numbers congruent to 1 modulo 4. Thus, we obtain a basis for the cohomology parametrized by such partitions. This analysis also suggests, but does not establish, that, as a ring, $\Lambda(\Lambda^2(C^{2n}))^{Sp_{2n}}$ is an exterior algebra on the elements corresponding to the $(r+1, r)$ -double hooks for odd r .

5.2: Other First Fundamental Theorems.

5.2.1: Schwarz's FFTs

Besides the First Fundamental Theorems of invariant theory for the standard actions of GL_n, O_n , and Sp_{2n} , and variants of these [Pri], [W1], very few other cases of the FFT are known. In the 1980s, Schwarz [Schw 1,2,3], proved some new cases of the FFT, for the 7-dimensional representation of G_2 , the spin representation of $Spin_7$, and the cubic (four-dimensional) representation of SL_2 . In [Hw4], it is explained how the FFTs for G_2 and $Spin_7$ can be derived using ideas of multiplicity-free actions. We will not repeat the arguments of [Hw4] here, but we will briefly describe the strategy, which is quite similar to the proofs given in §2.2.3 and §3.4.5 of the classical FFTs. We have shown in [Hw1] how a description of the invariant differential operators allows one to describe the full isotypic decomposition of the polynomials on multiples of the standard module of a classical group. From Krämer's list [Krm], we find that G_2 is a spherical subgroup of O_7 . Thus, if we have a good description of $\mathcal{P}((C^7)^m)$ a O_7 -module, we can describe the G_2 -invariants by the same technique as in the argument of §2.2.3 for the FFT for GL_n or §3.4.5 for the FFT for O_n . As with those examples, this description is not a description of $\mathcal{P}((C^7)^m)^{G_2}$ as an algebra, but it has a lot of structure coming from the O_7 -module structure. This structure suggests certain invariants as likely

generators for the algebra, and allows one to prove they in fact are generators. One obtains in this way two complementary descriptions of the G_2 -invariants: the conventional one as an algebra with certain generators, and the description relative to the O_7 -module structure. Both descriptions can be useful.

5.2.2: Huang's FFT for G_4^4

Here, we would like to describe another result, very much like an FFT, due to R. Huang [Hn]. This describes the invariants for GL_4 of an arbitrary collection of lines in projective 3-space $P^3(C)$. If we think of $P^3(C)$ in the usual model, as the set of lines through the origin in C^4 , then a line in $P^3(C)$ is identified to a 2-dimensional plane in C^4 , so the variety of lines in $P^3(C)$ is the same as the Grassmann variety G_2^4 of 2-planes in 4-space. To associate to G_2^4 a ring of functions, we identify G_2^4 with the set of decomposable 2-vectors in $\Lambda^2(C^4)$, modulo scalar multiplication. The set of all decomposable vectors, with scalar multiples not identified, is a cone cover G_2^4 . We will denote it by $C(G_2^4)$. Denote by $\mathcal{R}(C(G_2^4))$ the ring of restrictions to $C(G_2^4)$ of all polynomial functions on $\Lambda^2(C^4)$. We call this the coördinate ring of $C(G_2^4)$, and it is the ring we attach to G_2^4 . Sometimes we may abuse the language and call $\mathcal{R}(C(G_2^4))$ the coördinate ring of G_2^4 . Elements of $\mathcal{R}(C(G_2^4))$ are not actually functions on G_2^4 , but the restrictions of functions on $\Lambda^2(C^4)$ which are homogeneous of some fixed degree are sections of a line bundle over G_2^4 . This is quite analogous to associating to $P^{n-1}(C)$ the polynomial ring $C(z_1, z_2, \dots, z_n) = P(C^n)$, whose k -th homogeneous piece consists of the sections of the k -th power of the hyperplane section bundle over $P^{n-1}(C)$. The problem of describing the invariants of an arbitrary collection of lines in $P^3(C)$ is to describe the relative invariants for the action of GL_3 (i.e., the invariants for SL_3) on the coördinate ring of $C(G_2^4)^k$, which is the tensor product $\mathcal{R}(C(G_2^4))^{\otimes k}$. This is like an FFT, but with the variety $C(G_2^4)$ taking the place of a vector space. Clearly, we could envision an expansion of the notion of FFT to encompass actions on arbitrary varieties as well as linear actions on vector spaces.

Wedge product defines a bilinear mapping

$$(5.2.2.1) \quad B : \Lambda^2(C^4) \times \Lambda^2(C^4) \rightarrow \Lambda^4(C^4)$$

Since $\dim \Lambda^4(C^4) = 1$, if we choose a basis element for $\Lambda^4(C^4)$, we can think of B as a symmetric bilinear form on $\Lambda^2(C^4)$. The mapping B is equivariant with respect to the action of GL_4 on the relevant $\Lambda^2(C^4)$. Since SL_4 acts trivially on $\Lambda^4(C^4)$, the bilinear form B will be invariant under SL_4 . Thus, the natural action of SL_4 on $\Lambda^2(C^4)$ defines a homomorphism

$$(5.2.2.2) \quad c : SL_4 \rightarrow O(\Lambda^2(C^4), B).$$

It is well known and not hard to check that the map c is two to one. Since $\dim SL_4 = \dim SO_6$, the mapping will be onto, so c exhibits SL_4 as $Spin_6$, the two-fold cover of SO_6 , here realized concretely on $\Lambda^2(C^4)$.

In this context, we can further note that the variety $C(G_2^4)$ consists of vectors which are isotropic with respect to the inner product B . Since the variety $X_0(\Lambda^2(C^4)) = X_0$ of non-zero isotropic vectors consists of a single orbit for

$O(\Lambda^2(C^4), B)$, (by Witt's Theorem, [Ja1] for example), we see that $C(G_2^4)$ is exactly the same as X_0 . This connection between lines in P^3 and the quadric in P^5 was an early discovery of S. Lie [Lie], [Hel].

This lets us give a concrete description of $\mathcal{R}(C(G_2^4))$. Introduce the notation

$$(5.2.2.3) \quad r^2(v) = B(v, v).$$

Then

$$(5.2.2.4) \quad \mathcal{R}(C(G_2^4)) = P(\Lambda^2(C^4))/I(r^2),$$

where $I(r^2)$ denotes the ideal in $P(\Lambda^2(C^4))$ generated by r^2 . We can use equation (5.2.2.4) to give a precise description of $\mathcal{R}(C(G_2^4))^k$. Consider the sum of k copies of $\Lambda^2(C^4)$. It consists of k -tuples $v = (v_1, v_2, \dots, v_k)$ of vectors in $\Lambda^2(C^4)$. Denote by r_{ab}^2 the degree two polynomial defined by pairing the a -th and b -th components of v by means of B :

$$(5.2.2.5) \quad r_{ab}^2(v) = B(v_a, v_b).$$

The variety $(C(G_2^4))^k$ sits in $\Lambda^2(C^4)^k$ as the product of the copies of $C(G_2^4)$ in each copy of $\Lambda^2(C^4)$; it is not hard to see that it is the set of simultaneous zeroes of the r_{aa}^2 , for $1 \leq a \leq k$. Thus, we arrive at the equation

$$(5.2.2.6) \quad \mathcal{R}((C(G_2^4))^k) = P(\Lambda^2(C^4)^k)/I(r_{11}^2, r_{22}^2, \dots, r_{kk}^2).$$

Further, the ring which we are seeking to describe, the ring $\mathcal{R}(C(G_2^4)^k)^{SL_4}$ of SL_4 -invariants, is identified with the ring

$$(5.2.2.7) \quad \begin{aligned} & (P(\Lambda^2(C^4)^k)/I(r_{11}^2, r_{22}^2, \dots, r_{kk}^2))^{SO_6} \\ &= P(\Lambda^2(C^4)^k)^{SO_6} / (I(r_{11}^2, r_{22}^2, \dots, r_{kk}^2) \cap P(\Lambda^2(C^4)^k)^{SO_6}) \\ &= P(\Lambda^2(C^4)^k)^{SO_6} / \mathcal{J}(r_{11}^2, r_{22}^2, \dots, r_{kk}^2), \end{aligned}$$

where $\mathcal{J}$ is the ideal generated inside $P(\Lambda^2(C^4)^k)^{SO_6}$ by the r_{ij}^2 . In this equation, we are using SO_6 as an abbreviation for $SO(\Lambda^2(C^4), B)$. The equation (5.2.2.7) is valid because of the complete reducibility of the action of SO_6 .

The structure of the ring $P(\Lambda^2(C^4)^k)^{SO_6}$ is easily deduced from §§3.4 and 3.6. The ring $P(\Lambda^2(C^4)^k)^{O_6}$ of invariants for the full orthogonal group is just generated by the inner products r_{ab}^2 . Inspecting the isotypic decomposition described by Theorem 3.5.2 and Proposition 3.6.3, we see that to get all the invariants for SO_6 , we must also include the determinants

$$(5.2.2.8) \quad \Delta(c_1, c_2, c_3, c_4, c_5, c_6) = \det \begin{array}{c|cccccc} & x_{1c_1} & x_{1c_2} & x_{1c_3} & x_{1c_4} & x_{1c_5} & x_{1c_6} \\ \hline x_{2c_1} & & x_{2c_2} & x_{2c_3} & x_{2c_4} & x_{2c_5} & x_{2c_6} \\ x_{3c_1} & & x_{3c_2} & x_{3c_3} & x_{3c_4} & x_{3c_5} & x_{3c_6} \\ x_{4c_1} & & x_{4c_2} & x_{4c_3} & x_{4c_4} & x_{4c_5} & x_{4c_6} \\ x_{5c_1} & & x_{5c_2} & x_{5c_3} & x_{5c_4} & x_{5c_5} & x_{5c_6} \\ x_{6c_1} & & x_{6c_2} & x_{6c_3} & x_{6c_4} & x_{6c_5} & x_{6c_6} \end{array}$$

Here the x_{ac} , $1 \leq a \leq 6$, are the coordinates on the c -th copy of $\Lambda^2(\mathbb{C}^4) \cong \mathbb{C}^6$. Since these determinants generate $\mathcal{P}(\Lambda^2(\mathbb{C}^4)^k)^{SO_6}$ as a module over $\mathcal{P}(\Lambda^2(\mathbb{C}^4)^k)^{O_6}$, they together with the r_{ab}^2 certainly generate $\mathcal{P}(\Lambda^2(\mathbb{C}^4)^k)^{SO_6}$ as an algebra, and so, therefore, will their images in $\mathcal{R}((\mathbb{C}(\mathbb{G}_2^4))^k)$ generate the algebra $\mathcal{R}((\mathbb{C}(\mathbb{G}_2^4))^k)^{SL_4}$. In fact this approach also lets us describe relations between the generators r_{ab}^2 and $\Delta(c_1, c_2, c_3, c_4, c_5, c_6)$. Indeed, we know from the analog for O_n that the relations between the r_{ab}^2 are defined by requiring all 7×7 minors of the matrix $[r_{ab}^2]$ to vanish. Further, the product of any two of the $\Delta(c_1, c_2, c_3, c_4, c_5, c_6)$ is a 6×6 determinant in the r_{ab}^2 . In addition to these relations and the defining equations $r_{aa}^2 = 0$, there are some subtler relations between the r_{ij}^2 and the Δ 's. These are described in [DES].

5.3: Covariants of subspaces.

In this section, we will describe the covariants of two arbitrary linear subspaces of projective space and, using this, show how to determine the invariants of four subspaces.

5.3.1: One subspace

Begin with one subspace. We can parametrize subspaces by decomposable vectors in exterior powers. Precisely, if Y is a k -dimensional subspace of $\mathbb{C}^n$, and $\{b_j : 1 \leq j \leq k\}$ is a basis for Y , then the wedge product

$$(5.3.1.1) \quad \Lambda^{\text{top}}(Y) = b_1 \wedge b_2 \wedge b_3 \wedge \dots \wedge b_k$$

belongs to $\Lambda^k(\mathbb{C}^n)$, and it determines Y in the following way: a vector z in $\mathbb{C}^n$ belongs to Y if and only if $z \wedge \Lambda^{\text{top}}(Y) = 0$. If we choose a different basis for Y , then we will get the same vector $\Lambda^{\text{top}}(Y)$ up to a multiple. Thus, we should think of $\Lambda^{\text{top}}(Y)$ not as a specific vector, but as a line in $\Lambda^k(\mathbb{C}^n)$. We will denote the set of vectors in $\Lambda^k(\mathbb{C}^n)$ of the form (5.3.1.1) by $\mathcal{C}(\mathbb{G}_k^n)$, and the set of lines through them by $\mathbb{G}_k^n$. Note that the notation of §5.2 is consistent with this.

Let $\mathcal{R}(\mathcal{C}(\mathbb{G}_k^n))$ be the algebra of restrictions to $\mathcal{C}(\mathbb{G}_k^n)$ of polynomial functions on $\Lambda^k(\mathbb{C}^n)$. This is what we mean by the coordinate ring of $\mathbb{G}_k^n$. The action of GL_n on $\Lambda^k(\mathbb{C}^n)$ stabilizes $\mathcal{C}(\mathbb{G}_k^n)$, so $\mathcal{R}(\mathcal{C}(\mathbb{G}_k^n))$ is a module for GL_n .

Like all flag manifolds, $\mathbb{G}_k^n$ is a multiplicity-free space for GL_n . We can analyze it using the general principles of §1.3. Let $\{e_b : 1 \leq b \leq n\}$ be the standard basis for $\mathbb{C}^n$. The standard upper triangular Borel subgroup B_n stabilizes the span of $\{e_b : 1 \leq b \leq m\}$ for any $m \leq n$. A generic k -plane with respect to B_n is the space W spanned by the last k of the e_b , namely the set $\{e_b : n - k + 1 \leq b \leq n\}$. Its stabilizer in B_n is the group Γ_k of matrices of the form

$$g = \begin{vmatrix} T_1 & O \\ O & T_2 \end{vmatrix}$$

where T_1 and T_2 are upper triangular matrices of sizes $n - k$ and k , respectively. If we require that g stabilize individual vectors $\Lambda^{\text{top}}(W)$ and not just the line through them, we should demand that $\det T_2 = 1$. Call the group of matrices satisfying this additional restriction Γ_k^1 . The dominant characters which are trivial on Γ_k^1 are those of the form ψ_ℓ where notation as in §1.2.4.

$$(5.3.1.2) \quad \ell = c(1_{n-k} - 1_n), \quad c \in \mathbb{Z}^+.$$

The representations with these highest weights are the duals of the representations corresponding to diagrams $R(k, c)$ which are rectangles with k rows of length c . This is consistent with what we know about $\mathcal{R}(\mathcal{C}(\mathbb{G}_k^n))$ from its definition. The linear functions (homogeneous first order polynomials) on $\Lambda^k(\mathbb{C}^n)$ define the representation $(\Lambda^k(\mathbb{C}^n))^*$ of GL_n , and the c -th power of the highest weight vector in $(\Lambda^k(\mathbb{C}^n))^*$ will generate the dual of the representation corresponding to the diagram $R(k, c)$. Combining the restriction imposed by equation (5.3.1.2) with the existence of the obvious representations, we have the following description of $\mathcal{R}(\mathcal{C}(\mathbb{G}_k^n))$ as a GL_n -module.

$$(5.3.1.3) \quad \mathcal{R}(\mathcal{C}(\mathbb{G}_k^n)) \cong \sum_{c \geq 0} (\rho_n^{R(k, c)})^*.$$

5.3.2: Two subspaces

To describe the covariants of two subspaces means to describe the structure of the ring

$$\mathcal{R}(\mathcal{C}(\mathbb{G}_k^n) \times \mathcal{C}(\mathbb{G}_\ell^n)) \cong \mathcal{R}(\mathcal{C}(\mathbb{G}_k^n)) \otimes \mathcal{R}(\mathcal{C}(\mathbb{G}_\ell^n))$$

as a GL_n -module. The key observation here is that the product variety $\mathcal{C}(\mathbb{G}_k^n) \times \mathcal{C}(\mathbb{G}_\ell^n)$ is multiplicity-free. Before stating the precise result, we should note that, in addition to the action of GL_n on $\mathcal{C}(\mathbb{G}_k^n)$, the group $\mathbb{C}^\times$ of scalar matrices also acts, by dilations. Of course, this action coincides with the action of the center of GL_n , but if we have two Grassmannians, we have an action of $\mathbb{C}^\times \times \mathbb{C}^\times$ on $\mathcal{C}(\mathbb{G}_k^n) \times \mathcal{C}(\mathbb{G}_\ell^n)$, one $\mathbb{C}^\times$ on each factor, and the center of GL_n only accounts for a one-parameter subgroup in $\mathbb{C}^\times \times \mathbb{C}^\times$. (Precisely, this is the subgroup of elements of the form (z^k, z^ℓ) , $z \in \mathbb{C}^\times$.) This extra action by copies of $\mathbb{C}^\times$ is significant at several points. In particular, we can observe at the moment that it provides gradings for our rings. Thus $\mathcal{R}(\mathcal{C}(\mathbb{G}_k^n))$ is graded by $\mathbb{Z}^+$ (with the summands in formula (5.3.1.2) being the homogeneous components), and $\mathcal{R}(\mathcal{C}(\mathbb{G}_k^n) \times \mathcal{C}(\mathbb{G}_\ell^n))$ is graded by $\mathbb{Z}^{+2}$.

5.3.2.1: THEOREM. The action of $GL_n \times (\mathbb{C}^\times \times \mathbb{C}^\times)$ on $\mathcal{R}(\mathcal{C}(\mathbb{G}_k^n) \times \mathcal{C}(\mathbb{G}_\ell^n))$ is multiplicity-free. Except when $k = \ell$, the action of GL_n alone is multiplicity-free.

Remark: This result is the A_n case of a result of Littelman [Ltm], §5.5.3. It has also been observed by Panyushev [Pan].

PROOF. We will show that a Borel subgroup of $GL_n \times (\mathbb{C}^\times \times \mathbb{C}^\times)$ acts with a dense orbit on $\mathcal{C}(\mathbb{G}_k^n) \times \mathcal{C}(\mathbb{G}_\ell^n)$. Since $\mathcal{C}(\mathbb{G}_k^n) \times \mathcal{C}(\mathbb{G}_\ell^n)$ fibers over $\mathbb{G}_k^n \times \mathbb{G}_\ell^n$ and $\mathbb{C}^\times \times \mathbb{C}^\times$ acts transitively on the fibers, it is enough to show that a Borel subgroup of GL_n acts with a dense orbit in $\mathbb{G}_k^n \times \mathbb{G}_\ell^n$. This is what we will do. For simplicity, we treat the case $k + \ell \leq n$. In fact, this is no loss of generality, because duality, which assigns to a given subspace of a vector space its annihilator in the dual vector space, defines a GL_n -equivariant bijection between $\mathbb{G}_k^n$ and $\mathbb{G}_{n-k}^n$. If $k + \ell > n$, then $(n - k) + (n - \ell) < n$. We may also assume without loss of generality that $k \leq \ell$. Let $\{e_b : 1 \leq b \leq n\}$ be the standard basis for $\mathbb{C}^n$. The standard upper triangular Borel B_n preserves the span of the first m of the e_b , for any $m \leq n$. Let W be the ℓ -dimensional space spanned by the vectors e_b for $n - \ell + 1 \leq b \leq n$. Let Z be the k -dimensional space spanned by the vectors $e_a + e_{n+1-a}$ for $n - k - \ell + 1 \leq a \leq n - \ell$.

* An élément g of the group B_n will take a given e_a to a linear combination of e_b with $b \leq a$. In order to preserve W , the element g will take e_{n+1-a} to a sum of e_{n+1-b} , for $a \leq b \leq \ell$, whence $n+1-b > n-\ell$. It follows that the only way that $g(e_a + e_{n+1-a})$ can be a sum of the $e_b + e_{n+1-b}$ for b in the correct range, is if $g(e_a) = c_a e_a$ and $g(e_{n+1-a}) = c_a e_{n+1-a}$, for some common scalar c_a . Thus, the subgroup $\Gamma_{k,\ell}$ of B_n which preserves W and Z consists of matrices of the form

$$(5.3.2.2) \quad \begin{vmatrix} B_1 & O & O & O \\ O & A_1 & O & O \\ O & O & B_2 & O \\ O & O & O & A_2 \end{vmatrix}$$

where B_1 is an $(n-k-\ell) \times (n-k-\ell)$ upper triangular matrix, B_2 is an $(\ell-k) \times (\ell-k)$ upper triangular matrix, and A_1 and A_2 are $k \times k$ diagonal matrices, with the same entries, but in reverse order. Counting dimensions, we find that the codimension in B_n of the stabilizer of W and Z is $k(n-k) + \ell(n-\ell)$, which equals the sum of the dimensions of G_k^n and G_ℓ^n . It follows that the B_n -orbit of the pair (Z, W) is open, hence dense, in $G_k^n \times G_\ell^n$. This proves the first statement of the theorem. As to the second, we can observe that the matrices A_1 and A_2 will multiply $\Lambda^{\text{top}}(W)$ and $\Lambda^{\text{top}}(Z)$ by the same arbitrary scalar, while if $k \neq \ell$, the matrix B_2 will multiply $\Lambda^{\text{top}}(W)$ by an independent arbitrary scalar, showing that B_n acts transitively on the fibers of the projection $\mathcal{C}(G_k^n) \times \mathcal{C}(G_\ell^n) \rightarrow G_k^n \times G_\ell^n$. Thus, if $k \neq \ell$, GL_n by itself acts multiplicity-free on $\mathcal{C}(G_k^n) \times \mathcal{C}(G_\ell^n)$.

We, of course, want to describe the generators of the covariants for two subspaces. Since we have described the stabilizer in B_n of a generic point of $G_k^n \times G_\ell^n$, we can use criterion 1.3.6 to restrict the possible highest weights which can occur in $\mathcal{R}(\mathcal{C}(G_k^n) \times \mathcal{C}(G_\ell^n))$. In applying criterion 1.3.6, we should use not the full group $\Gamma_{k,\ell}$ but the subgroup $\Gamma_{k,\ell}^1$ of matrices of the form (5.3.2.2) such that $\det A_1 = \det A_2 = 1$ and $\det B_2 = 1$. These conditions guarantee that $\Gamma_{k,\ell}^1$ preserves the individual vectors $\Lambda^{\text{top}}(W)$ and $\Lambda^{\text{top}}(Z)$, not just the lines through them. The dominant weights which are trivial on $\Gamma_{k,\ell}^1$ are generated by the characters ψ_ℓ where ℓ is one of the vectors

$$(5.3.2.3) \quad 1_{n-\ell} - 1_n, \quad 1_{n-k} - 1_n, \quad 1_{n-b} + 1_{n-k+\ell+b} - 21_n, \quad \text{for } 0 \leq b \leq k.$$

These are the highest weights of representations dual to $\Lambda^\ell(C^n)$ to $\Lambda^k(C^n)$, and to the representations associated to diagrams with two columns, of lengths $k + \ell - b$ and b , with $0 \leq b \leq k$. From §A4.2, we know that these are the constituents of the tensor product $\Lambda^\ell(C^n) \otimes \Lambda^k(C^n)$. Thus, these representations all occur in $\mathcal{R}(\mathcal{C}(G_k^n) \times \mathcal{C}(G_\ell^n))$. In summary, we have the following description of the covariants of two subspaces.

5.3.2.4: PROPOSITION. The algebra $\mathcal{R}(\mathcal{C}(G_k^n) \times \mathcal{C}(G_\ell^n))^{U_n}$ of covariants of two subspaces of dimensions k and ℓ , with $k + \ell \leq n$, is a polynomial algebra on $2 + \min(k, \ell)$ generators, which are, respectively, the highest weight vectors in $(\Lambda^\ell(C^n))^*$ and $(\Lambda^1(C^n))^*$, and the highest weight vectors, other than the product of the first two generators, of the constituents of $(\Lambda^\ell(C^n) \otimes \Lambda^k(C^n))^*$. In terms of the $(Z)^{+2}$

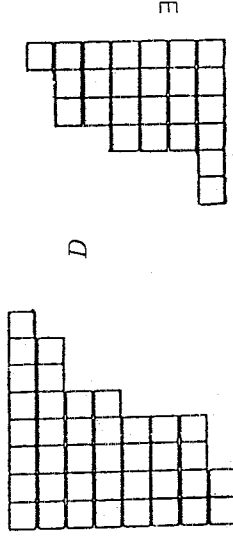
grading on $\mathcal{R}(\mathcal{C}(G_k^n) \times \mathcal{C}(G_\ell^n))$, the first two generators have degree $(1, 0)$ and $(0, 1)$ and the remaining generators have degree $(1, 1)$.

5.3.3 Invariants of four subspaces

We now turn to the problem of describing the (relative or SL_n) invariants of four subspaces. If the dimensions of the subspaces are a, b, c and d , this means computing the algebra $\mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n) \times \mathcal{C}(G_c^n) \times \mathcal{C}(G_d^n))^{SL_n}$. Because of our firm grasp on the structure of the covariants of two subspaces, this problem becomes an exercise in the combinatorics of diagrams. Precisely, we can write

$$(5.3.3.1) \quad \begin{aligned} \mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n) \times \mathcal{C}(G_c^n) \times \mathcal{C}(G_d^n)) \\ \simeq \mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n)) \otimes \mathcal{R}(\mathcal{C}(G_c^n) \times \mathcal{C}(G_d^n)). \end{aligned}$$

Thus, we have a decomposition of $\mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n) \times \mathcal{C}(G_c^n) \times \mathcal{C}(G_d^n))$ into a sum of tensor products $U \otimes V$, where U is an irreducible subspace of $\mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n))$, and V is an irreducible subspace of $\mathcal{R}(\mathcal{C}(G_c^n) \times \mathcal{C}(G_d^n))$. We know that a tensor product $\rho_n^D \otimes \rho_n^E$, of representations corresponding to diagrams D and E , will contain an SL_n -invariant if and only if ρ_n^D and ρ_n^E are mutually contragredient up to twisting by some power of $\det$. This is equivalent to the following condition: if one of the diagrams D or E is rotated 180° , then the two should fit exactly into an n -rowed rectangle $R(n, \ell)$ for some ℓ . Here is an example with $n = 8$.



Since we know very precisely the diagrams associated to irreducible subspaces of $\mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n))$, we can determine all possibilities for combining two diagrams, one from $\mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n))$ and one from $\mathcal{R}(\mathcal{C}(G_c^n) \times \mathcal{C}(G_d^n))$, to get an n by 1 rectangle for some 1. The details are slightly involved; we refer to [HH] for the full story. We will state here the result in the case when the invariants are most abundant.

To avoid trivial cases, we assume that $0 < a, b, c, d < n$. By taking duals if necessary, we can always assume that $a + b + c + d \leq 2n$. The situation in which invariants are easiest to find is when $a + b + c + d = 2n$. The space $\mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n) \times \mathcal{C}(G_c^n) \times \mathcal{C}(G_d^n))$ has a grading by $(Z)^{+4}$, from the Z^+ gradings on each of the factors $\mathcal{R}(\mathcal{C}(G_a^n))$, etc. The total degree is the sum of the degrees in the four factors.

5.3.3.2: THEOREM. Assume that $a + b + c + d = 2n$. Then the algebra $\mathcal{R}(\mathcal{C}(G_a^n) \times \mathcal{C}(G_b^n) \times \mathcal{C}(G_c^n) \times \mathcal{C}(G_d^n))^{SL_n}$ is generated by invariants of the following two types.

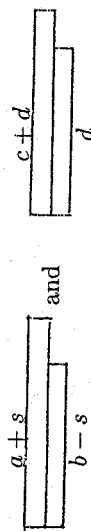
Type I): For any pair of a, b, c, d which sum to n , there is an invariant of total degree two. Suppose $a + b = n$. Then the tensor product $(\Lambda^a(C^n))^* \otimes (\Lambda^b(C^n))^*$ contains one of these generators. If a, b, c, d are all distinct, there are either zero or two of these generators. If exactly two of a, b, c, d are equal to $n/2$, there are two. If a, b, c, d are equal in pairs, there are four. If $a = b = c = d = n/2$, there are six.

Type II): Suppose without loss of generality that $a \geq b \geq c \geq d$. Then there are $\min(n-a, d) + 1$ generators of degree $(1, 1, 1, 1)$. The generators of type II arise by combining diagrams whose shapes are the transposes of



where r and s are chosen so that $a + d + r - s = n$.

If $a + b + c + d < 2n$, it is much harder to find invariants. Nevertheless, some may exist which are not of types I or II. We will call these remaining invariants type III. There are at most four generators for the type III invariants. Most typically there are zero or one. However, these generators may be of arbitrarily high degree. We refer to [HH] for the complete story. Here we indicate one of the simplest of type III generators. Suppose $a + b + c + 2d = 2n$, and $n - (a + d) = s \geq 0$. Then we can find an invariant by combining the transposes of the diagrams



5.4: Branching laws.

In this section, we consider the restriction of representations from GL_n to GL_{n-1} , and from O_n to O_{n-1} . The classical terminology for describing such restrictions is “branching rules”, and the branching rules we are considering are very well known [Prt 1], [Zb]. The restriction of an irreducible representation of GL_n to GL_{n-1} is multiplicity-free, and likewise for O_n . We will show that this is a consequence of the existence of certain multiplicity-free actions, and that the precise combinatorics of the branching follows directly from the structure of the generators of the semigroup of B -eigenvectors.

5.4.1: Branching from GL_n to GL_{n-1}

Consider the action of $GL_n \times GL_{n-1}$ on the space $M_{n,n-1}$ of n by $n-1$ matrices. This is the context of (GL_n, GL_{n-1}) -duality, and the module structure of $\mathcal{P}(M_{n,n-1})$ is described in §2.1. We are going to want to restrict the representations of GL_n appearing in $\mathcal{P}(M_{n,n-1})$ to the subgroup GL_{n-1} . Thus, we will be working with two copies of GL_{n-1} , one inside GL_n and the other commuting with GL_n . To distinguish them, we will label the copy of GL_{n-1} acting on $M_{n,n-1}$ by right multiplication with a, ι , and all objects associated to it (subgroups, etc.) will also

be marked with a prime. Unprimed GL_{n-1} and associated objects will indicate the subgroup of GL_n . Thus, U'_{n-1} will denote the group of unipotent upper triangular matrices in GL'_{n-1} acting on $M_{n,n-1}$ by multiplication on the right. We can see from formula (2.1.2.1) and the highest weight theory that the space of U_{n-1} invariants in $\mathcal{P}(M_{n,n-1})$ consists of one copy of (the dual of) each irreducible of the form ρ_n^D where D is a diagram with at most $n-1$ rows. Each irreducible representation of GL_n is isomorphic to the twist of a unique representation of this form by a unique power of $\det$. Furthermore, the representations are labeled by the eigenvalues of A'_{n-1} . If we restrict this realization of ρ_n^D to GL_{n-1} , then the copies of the various GL_{n-1} representations that appear as summands will all be eigenspaces of A'_{n-1} . Their highest weights will thus be eigenvectors for A'_{n-1} and for A_{n-1} . The A_{n-1} -eigencharacter determines what GL_{n-1} module we have, and the A'_{n-1} -eigencharacter determines in which GL_n -module it occurs. Thus, the joint $A'_{n-1} \times A_{n-1}$ eigenvectors determine the branching law from GL_n to GL_{n-1} . Therefore, we want to describe $\mathcal{P}(M_{n,n-1})^{U'_{n-1} \times U_{n-1}}$ as an $A'_{n-1} \times A_{n-1}$ -module.

5.4.1.1: THEOREM. a) The action of $GL_{n-1} \times GL'_{n-1}$ on $\mathcal{P}(M_{n,n-1})$ is multiplicity-free.

b) There are $2(n-1)$ prime generators for $\mathcal{P}(M_{n,n-1})^{U'_{n-1} \times U_{n-1}}$. They are of two types:

$$\begin{aligned} \text{i) } \quad & \delta_k = \delta_{k,O} = \det \begin{vmatrix} v_{11} & v_{12} & \cdots & v_{1k} \\ v_{21} & v_{22} & \cdots & v_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ v_{k1} & v_{k2} & \cdots & v_{kk} \end{vmatrix}, \text{ for } 1 \leq k \leq n-1; \\ \\ \text{ii) } \quad & \delta_{k,1} = \det \begin{vmatrix} v_{11} & v_{12} & \cdots & v_{1,k+1} \\ v_{21} & v_{22} & \cdots & v_{2,k+1} \\ \vdots & \vdots & \ddots & \vdots \\ v_{k1} & v_{k2} & \cdots & v_{k,k+1} \\ y_1 & y_2 & \cdots & y_{k+1} \end{vmatrix}, \text{ for } 0 \leq k \leq n-2. \end{aligned}$$

In the formula for $\delta_{k,1}$, we have written y_j for $v_{n,j}$, to emphasize the different role played by the last variable.

(5.4.1.2) PROOF. We use the criterion of §1.3.3 to show multiplicity-freeness. The action of $GL_n \times GL'_{n-1}$ on $M_{n,n-1}$ is given explicitly by

$$(g, g')(T) = (g^*)^{-1} T g'^{-1} \quad g \in GL_n, g' \in GL'_{n-1}, T \in M_{n,n-1}.$$

Consider the matrix

$$C = \begin{vmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \\ 1 & 1 & \cdots & 1 & 1 \end{vmatrix}$$

Notice that, under the action of $GL_{n-1} \times GL'_{n-1}$, the space $M_{n,n-1}$ splits as a direct sum $M_{n,n-1} = M_{n-1,n-1} \oplus M'_{1,n-1}$, one summand consisting of the first $n-1$ rows and the other summand of the last row, respectively. As we have seen in §2.1.4, under the action of $B_{n-1} \times B'_{n-1}$ on $M_{n-1,n-1}$, the matrix $J_{n-1,n-1}$ (which is the component of C in $M_{n-1,n-1}$) sweeps out an open orbit, and the stabilizer H_B of $J_{n-1,n-1}$ is a subgroup of the product $A_{n-1} \times A'_{n-1}$ of diagonal subgroups, described explicitly in §2.1.4. From that description, it is easy to check that H_B acting on the final row R_n of C sweeps out an open orbit in $M_{1,n-1}$, and that the isotropy group of R_n in H_B is trivial. It follows that C sweeps out an open orbit for $B_{n-1} \times B'_{n-1}$ acting on $M_{n,n-1}$, and that the isotropy group of C in $B_{n-1} \times B'_{n-1}$ is trivial. Hence the criterion of §1.3.3 shows that the action of $GL_{n-1} \times GL'_{n-1}$ on $M_{n,n-1}$ is multiplicity-free.

Since the isotropy group of C in $B_{n-1} \times B'_{n-1}$ is trivial, the criterion of §1.3.6 puts no restriction on the highest weights of $GL_{n-1} \times GL'_{n-1}$ which can occur in $\mathcal{P}(M_{n,n-1})$. However, it is clear that not all joint highest weights do occur, since a given representation of GL_n cannot contain all representations of GL_{n-1} . The set of joint highest weights which do occur is determined not by the general criterion of §1.3.6, but by the form of the prime generators which occur. We know from §1.3.8 that there can be at most $2(n-1) = \dim(A_{n-1} \times A'_{n-1})$ of these. In part b) of the theorem, we proposed a list with the correct number of elements. To know that this is the list of generators, we only have to show that they are primitive. We can do this by appealing to the classical elementary result [Tur] that the determinant is a prime polynomial in the entries of a matrix, or we can just observe that the weights of the listed polynomials must be primitive in the semigroup $(A_{n-1} \times A'_{n-1})^\wedge(M_{n,n-1})$, since for each one, the $(A'_{n-1})^\wedge$ -component is primitive in $(A'_{n-1})^{\wedge+}$, and any non-zero element of $(A_{n-1} \times A'_{n-1})^{\wedge+}(M_{n,n-1})$ must have non-zero projection onto $(A'_{n-1})^\wedge$ since the trivial representation can contain only the trivial representation.

5.4.1.3: The explicit branching rule

We will discuss the explicit form of the branching rule from GL_n to GL_{n-1} implied by Theorem 5.4.1.2. In the notation of §1.2.7, the weight of the polynomial δ_k of Theorem 4.4.1.2, part b), as an element of $(A_{n-1} \times A'_{n-1})^\wedge$, is $(1_k, 1_k)$, and the weight of $\delta_{k,1}$ is $(1_k, 1_{k+1})$. The elements δ_k are joint highest weight vectors for $GL_n \times GL_{n-1}$, while the elements $\delta_{k,1}$ are highest weight vectors only for GL_{n-1} , not GL_n . Consider a monomial

$$\mu(a_1, a_2, \dots, a_{n-1}; b_1, b_2, \dots, b_{n-1}) = \prod_{j=1}^{n-1} \delta_j^{a_j - a_{j-1}} \delta_{j-1,1}^{b_j - b_{j-1}}.$$

Here $\mathbf{a} = (a_1, a_2, a_3, \dots, a_{n-1})$ and $\mathbf{b} = (b_1, b_2, b_3, \dots, b_{n-1})$ define dominant weights of A_{n-1} , i.e., $a_j \geq a_{j+1}$; and likewise for the b_j . The A'_{n-1} weight of μ is

$$(a_1, a_2, a_3, \dots, a_{n-1}) + (b_1, b_2, b_3, \dots, b_{n-1}) = (c'_1, c'_2, c'_3, \dots, c'_{n-1}).$$

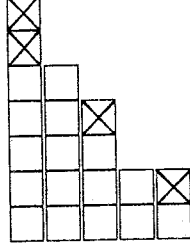
$$(a_1, a_2, a_3, \dots, a_{n-1}) + (b_2, b_3, b_4, \dots, b_{n-1}, 0) = (c_1, c_2, c_3, \dots, c_{n-1}).$$

The A_{n-1} weight of μ is

The difference in these weights is

$$\mathbf{c}' - \mathbf{c} = (b_1 - b_2, b_2 - b_3, b_3 - b_4, \dots, b_{n-1}) = \mathbf{d}.$$

We may think of this as follows. For monomials μ with a given A'_{n-1} -weight $\mathbf{c}'$, the largest possible A_{n-1} -weight is $\mathbf{c}'$ and can only be achieved by taking $\mathbf{a} = \mathbf{c}'$ and $\mathbf{b} = 0$. To create monomials with the same A'_{n-1} -weight but smaller A_{n-1} -weight, we substitute copies of $\delta_{k-1,1}$ for copies of δ_k . When we replace d_k copies of δ_k with copies of $\delta_{k-1,1}$, we reduce $\mathbf{c}'_k$ by d_k . The total number of copies of δ_k available to be replaced by $\delta_{k-1,1}$ is $\mathbf{c}'_k - \mathbf{c}'_{k+1}$. If we visualize the process of replacing copies of δ_k with copies of $\delta_{k-1,1}$, in terms of the diagrams of $\mathbf{c}$ and $\mathbf{c}'$, replacing d_k copies of δ_k amounts to erasing d_k boxes from the end of the k -th row of $\mathbf{c}'$ to get the k -th row of $\mathbf{c}$. We are allowed to erase any number of boxes from each row, as long as we do not make it shorter than the original length of the next row. We illustrate this procedure for $\mathbf{c} = (7, 5, 4, 2, 2)$ and $\mathbf{d} = (2, 0, 1, 0, 1)$:



This is the common description of the branching rule [Zb].

5.4.1.4: Gelfand-Tsetlin basis

Another way of formulating the branching rule from GL_n to GL_{n-1} is as follows: a weight $\mathbf{b} = (b_1, b_2, b_3, \dots, b_{n-1})$ of A_{n-1} can be a highest weight of a GL_{n-1} -submodule of the GL_n -module with highest weight $\mathbf{a} = (a_1, a_2, a_3, \dots, a_n)$ if and only if the b_j fit in between the a_j , in the sense that $a_j \geq b_j \geq a_{j+1}$. A pleasing way of exhibiting this relation between the a_j and the b_j is to arrange the two sequences in two successive rows, with the b_j between a_j and a_{j+1} :

$$\begin{array}{cccccccc} a_1 & & a_2 & & a_3 & & a_4 & & a_n \\ & b_1 & & b_2 & & b_3 & & b_4 & & b_{n-1} \end{array}$$

The order relations between the a_j and b_k may be indicated by requiring that, for two numbers which are adjacent in adjacent rows, the leftmost one is larger.

This scheme has an obvious continuation. If we wish to describe which GL_{n-2} -modules will appear inside a given GL_{n-1} -module inside a given GL_n -module, we can arrange a third row below the first two:

$$\begin{array}{cccccccccccc} a_1 & & a_2 & & a_3 & & a_4 & & & & & & a_n \\ & b_1 & & b_2 & & b_3 & & b_4 & & & & & b_{n-1} \\ & & c_1 & & c_2 & & c_3 & & & & & & c_{n-2} \end{array}$$

Again, the order relations between these elements should be that, for two elements which are adjacent in adjacent rows, the leftmost one should be larger. To continue this indefinitely, we should change to a more systematic notation, and write

$a_j = a_{1j}, b_j = a_{2j}$, and $c_j = a_{3j}$. Then we can continue, and obtain an array $\{a_{ij} : 1 \leq i \leq n, 1 \leq j \leq n+1-i\}$ of non-negative integers such that

$$a_{ij} \geq a_{(i+1)j} \geq a_{i(j+1)}.$$

We call such an array a Gelfand-Tsetlin pattern [BL], [NT], [Zb], or GT pattern for short. The set of all possible GT patterns with given top row label all possible containments of a GL_1 -module in a GL_2 -module in a $\dots$ in a GL_{n-1} -module in the GL_n -module V_{a_1} whose highest weight is the first row. Since GL_1 is abelian, an irreducible GL_1 -module is just a line. Thus, the set of GT-patterns describe a decomposition of an irreducible GL_n -module V_{a_1} into lines. If we choose a vector in each line, we get a basis for V_{a_1} . (There is a relatively canonical choice for such vectors [NT], but we will not worry about this issue.) Such a basis is called a Gelfand-Tsetlin basis. If the space V_{a_1} is given a Hermitian inner product invariant by the subgroup $U_n \subset GL_n$ of unitary matrices, a GT basis will be orthogonal, and, of course, may be taken to be orthonormal. It may also be characterized as the eigenbasis for a certain maximal abelian subalgebra of $\mathbf{U}(\mathfrak{gl}_n)$, the universal enveloping algebra of GL_n [Hw4].

5.4.1.5: Tableaux and standard monomials

The Gelfand-Tsetlin basis of the previous paragraph was somewhat abstract, being obtained by our knowledge of the structure of GL_n modules deriving from the branching law. We did not give formulas for the basis elements as polynomials on some space of matrices, and indeed, such formulas are rather complicated (see [BL], [NT]). However, guided by the combinatorial scheme of GT patterns, one can produce explicit bases for GL_n -modules consisting of functions of a rather simple form. To prepare for this, let us note another way to express the data encoded in a GT pattern. The point is to exhibit the highest weights, or equivalently the diagrams, of a nested family of GL_k -modules, $1 \leq k \leq n$. We can do this by starting with the diagram of the original GL_n -module, then indicating which subdiagram gives the highest weight of each GL_k module specified by the GT pattern. As we have explained, the GL_{n-1} diagram is obtained from the GL_n diagram by striking out a certain number of boxes at the end of each row of the GL_n diagram. We can identify which boxes we strike out by filling them with the number n . To obtain the GL_{n-2} diagram, we then strike out further boxes from the ends of the rows of the GL_{n-1} diagram. We can identify these boxes by filling them with the letter $n-1$. Continuing in this fashion, when we are done we will have a diagram whose boxes are filled with the numbers from 1 to n . The numbers in each row increase from left to right, and the numbers in each column strictly increase from top to bottom. The strict increase along columns reflects the condition that, in a GT pattern $\{a_{ij}\}$, we must have $a_{(i+1)j} \geq a_{i(j-1)}$. A diagram with its boxes filled with the numbers from 1 to n is called a (Young, or Weyl) tableaux [JK], [Sag]. The diagram D is called the shape of the tableaux. When the numbers satisfy the conditions just noted, i.e., increasing along rows and strict increasing down columns, the tableaux is called standard. It is not hard to convince oneself that the procedure described above for producing a standard tableaux from a GT pattern establishes a bijection between the two kinds of objects.

For each set $R = \{\tau_1, \tau_2, \tau_3, \dots, \tau_k\} \subseteq \{1, 2, 3, \dots, n\}$, consider the determinant

$$(5.4.1.5.1) \quad \delta_{(R)} = \delta_{(\tau_1, \tau_2, \dots, \tau_k)} = \det \begin{vmatrix} v_{\tau_1 1} & v_{\tau_1 2} & \cdots & v_{\tau_1 k} \\ v_{\tau_2 1} & v_{\tau_2 2} & \cdots & v_{\tau_2 k} \\ \vdots & \vdots & \ddots & \vdots \\ v_{\tau_k 1} & v_{\tau_k 2} & \cdots & v_{\tau_k k} \end{vmatrix}$$

For each R of cardinality k , $\delta_{(R)}$ is a highest weight vector for GL'_{n-1} , of highest weight 1_k , and the $\delta_{(R)}$, for $\#(R) = k$, form a basis of a copy of the GL_n -module $\Lambda^k(\mathbf{C}^n)$. Consider a product

$$(5.4.1.5.2) \quad M = \prod_{c=1}^a \delta_{(R_c)}$$

of the $\delta_{(R)}$, for various R , not necessarily of the same size. Since M is a product of highest weight vectors for GL'_{n-1} , it will be one also. The weight of M will be the product of the weights of the $\delta_{(R)}$; thus, the diagram corresponding to the weight of M will have one column of length ℓ for each set R_c of cardinality ℓ . We will call this diagram the shape of M . Fix a diagram D , and consider the set $M(D)$ of all functions of the form (5.4.1.5.2) with shape D . Since the $\delta_{(R)}$ with $\#(R) = k$ span a GL_n -invariant space, the span of $M(D)$ will likewise be GL_n -invariant. From (GL_n, GL_m) -duality (with $m = n-1$), we know that this GL_n -module must be a copy of the irreducible representation ρ_n^D . Thus, the collection $M(D)$ of monomials M in the functions $\delta_{(R)}$ provide a spanning set of ρ_n^D . We will see how the yoga of tableaux instructs us how to refine this spanning set to get a basis of ρ_n^D .

Given a monomial M of shape D , we attach to M a tableaux as follows. Assume that in the product (5.4.1.5.2) defining M , the sets R_c decrease in size as c increases. Then the cardinality $\#(R_c)$ will be the same as the length of the c -th column of D , by definition of shape. In the c^{th} column of D , insert the elements of R_c in increasing order from top to bottom. It is clear that this gives a tableaux of shape D . If D has several columns of a given length ℓ_0 , then the order in which the R_c of cardinality ℓ_0 appear in the product (5.4.1.5.2) does not affect the polynomial M , but does affect the tableaux assigned to M by the procedure just outlined. Thus, the procedure gives a mapping between the set of M of the form (5.4.1.5.2) and the set of tableaux of shape D , modulo the permutations of the equal-length columns of D . If a tableaux T is standard, then any other standard tableaux equivalent to T under permutations of equal length columns will in fact be equal to T . This is because, if T is standard, and we reverse the order of two columns of T , resulting in a tableaux T' , which is again standard, then the two columns must be identical. This is immediate from the definition of standard tableaux. If M as in (5.4.1.5.2) gives rise to a standard tableaux for some choice of ordering of the factors $\delta_{(R_c)}$ of M , then we will call the M standard. Denote the set of standard M by $M_{\text{stan}}(D)$. It is clear that the procedure just described gives a bijection between $M_{\text{stan}}(D)$ and the set of standard tableaux of shape D .

5.4.1.6: THEOREM. The set $M_{\text{stan}}(D)$ is a basis for ρ_n^D .

(5.4.1.7) PROOF. Part I, the straightening rules: To prove Theorem 5.4.1.6, we only have to show that $M_{\text{stan}}(D)$ spans ρ_n^D , because we already know from our

... discussion of GT patterns that the cardinality of $M_{\text{stan}}(D)$ equals the dimension of ρ_n^D .

The proof that $M_{\text{stan}}(D)$ spans ρ_n^D involves relations between the polynomials of formula (5.4.1.5.2). These relations are often called straightening rules, because they provide an algorithm for writing any M as a linear combination of standard M . The straightening rules have a clear representation-theoretic interpretation; see §A4.2 for an explanation.

The straightening rules are quadratic relations among the polynomials $\delta_{(R)}$. To derive them, let $W = \{w_{jk}\}$, $X = \{x_{jk}\}$, $Y = \{y_{jk}\}$ and $Z = \{z_{jk}\}$ be generic $n \times n$ matrices. For subsets $A, B \subseteq \{1, 2, 3, \dots, n\}$, let $W_{A,B}$ denote the submatrix of W defined by taking the rows from A and the columns from B . Denote by $|W_{A,B} X_{A,C}|$ the matrix gotten by juxtaposing the two matrices $W_{A,B}$ and $X_{A,C}$. We similarly juxtapose vertically matrices with the same column entries, and will use similar notation for more complicated constructions. Let A, B, C, D, E, F and G be sets in $\{1, 2, 3, \dots, n\}$ such that $\#(B) + \#(C) + \#(D) = \#(A)$, and $\#(G) + \#(F) = \#(E)$, where $\#()$ indicates cardinality. Assume also that $A \subseteq E$ and that $F \subseteq B$. The straightening rules are the following identities between determinants.

$$\begin{aligned}
 & \det |X_{A,B} Y_{A,C} Z_{A,D}| \det |X_{E,F} W_{E,G}| \\
 &= \det \begin{vmatrix} X_{A,B} Y_{A,C} Z_{A,D} & O \\ O & X_{E,F} W_{E,G} \end{vmatrix} \\
 &= \det \begin{vmatrix} X_{A,B} & Y_{A,C} & Z_{A,D} & O \\ O_{E,B} & O_{E,C} & -Z_{E,D} & X_{E,F} W_{E,G} \end{vmatrix} \\
 &= \det \begin{vmatrix} X_{A,B} & Y_{A,C} & O_{A,D} & X_{A,F} W_{A,G} \\ O_{E,B} & O_{E,C} & -Z_{E,D} & X_{E,F} W_{E,G} \end{vmatrix} \\
 &= \det \begin{vmatrix} X_{A,B} & Y_{A,C} & O_{A,D} & O_{A,F} W_{A,G} \\ O_{E,B} & O_{E,C} & -Z_{E,D} & X_{E,F} W_{E,G} \end{vmatrix} \\
 &= \sum_{\substack{H \subseteq G \\ \#(H) = \#(D)}} \pm \det |X_{A,B} Y_{A,C} W_{A,H}| \det |X_{E,F} Z_{E,D} W_{E,G-H}|.
 \end{aligned}$$

Here $O_{E,B}$ simply indicates the zero matrix of the appropriate size. The assumptions about inclusions of the sets is made to permit the row operations which account for the third and fourth equations. This identity says that the product of two determinants may be computed as a signed sum of other products of determinants, where these other determinants are gotten by selecting a fixed set of columns from the first determinant and inserting them in the second, and compensating by in turn selecting any equal number of columns from the second and putting them in the first, in place of the originals; except that, should the second determinant have some columns in common with the first, these columns can be left out of the reinsertion process, as they would just yield a zero determinant.

(5.4.1.8) PROOF. Part II, the quadratic case.

Let us see how the straightening rules allow us to express the product of two of the $\delta_{(R)}$ in terms of standard products. Thus, take sets $R = \{r_1 < r_2 < r_3 < \dots < r_k\}$ and $S = \{s_1 < s_2 < s_3 < \dots < s_\ell\}$, and consider the product $\delta_{(R)}\delta_{(S)}$. Assume for convenience that $k \geq \ell$. This product will be standard if and only if $r_a \leq s_a$ for $a \leq 1$. Suppose this fails, and let a be the largest index such that $r_a > s_a$. Apply the straightening rule of §5.4.1.7 (actually, apply its transpose, since we need to move rows here instead of columns) where the specified set of rows to be moved from $\delta_{(S)}$ to $\delta_{(R)}$ is the first a rows. This will replace $\delta_{(R)}\delta_{(S)}$ by a sum of products $\delta_{(R')}\delta_{(S')}$ where R' is gotten from R by having the first a rows of S inserted in it, and a compensating set of a rows removed, and inserted in S' to replace the first a rows which were transferred to R' . Of course, after the transfer, it will be necessary to rearrange the elements of both R' and S' to put them in increasing order. The crucial observations to make about this process are:

- i) if $b \geq a$, then $r'_b \leq r_b$;
- ii) if $b \geq a$, then $s'_b \geq s_b$; and
- iii) that $r'_a \leq s_a < r_a \leq s'_a$.

To see i), consider r'_b with $b \geq a$. If $r'_b = s_c$ or if $r'_b = r_c$ for $c \leq a$, then $r'_b \leq r_a \leq r_b$. If $r'_b = r_{b'}$ with $b' > a$, then $b' \leq b$ since all of the a elements of S which were inserted in R' must all precede $r_{b'}$ in R' (since they do not exceed r_a). To see ii) note that a given s_b with $b > a$ must become an $s'_{b'}$ with $b' \leq b$, since S' contains at least the $\ell - b$ elements s_c , $b < c \leq \ell$, larger than s_b . Thus, $s'_b \geq s'_{b'} = s_b$. To see iii), first note that $s_a = r'_{a'}$ with $a' \geq a$, since s_a is larger than the $a - 1$ elements s_c , $c < a$, of R' . Thus, $r'_a \leq r'_{a'} = s_a$. Further, the only way that we can avoid having $s'_a \geq s_{a+1}$ is if the a elements of S' which were transferred from R are exactly the first a elements, in which case, $s'_a = r_a$. Since $s_{a+1} \geq r_{a+1} > r_a$ (by definition of a), we get $s_a \geq r_a$ in all cases. This proves the observations. It follows from the observations that, in any pair of sets R', S' resulting from our substitution procedure, the last index a' for which $r'_{a'} > s'_{a'}$ must be strictly smaller than a . Hence by a finite number of such operations, we will arrive at a collection of pairs R'', S'' such that $\delta_{(R'')}\delta_{(S'')}$ is always standard.

Thus, Theorem 5.4.1.6 is proved for products of two $\delta_{(R)}$.

5.4.1.9: Conclusion of proof

To conclude the proof, consider an arbitrary monomial M as in formula (5.4.1.5.2). Create the tableaux with the entries from the factors of M , taken in the order they are written in the product defining M . This tableaux will be standard if and only if any subtableaux formed by a pair of adjacent columns is standard. If this fails, choose a pair of adjacent columns and perform the operations of (5.4.1.8) in order to bring this pair into standard form. When this is done, if any one of the resulting tableaux is not standard, choose from it a new pair of adjacent columns and operate to make this pair standard. Continue doing this. Since operating on one pair of columns to make them standard could destroy standardness between the pairs adjacent to the chosen pair, we must explain why this procedure will eventually terminate in a collection of standard tableaux. Consider the boxes which contain the index n , it is easy to convince oneself that in this "straightening procedure", these boxes can only move to the right. Thus, after a finite number

of steps, they will end up in one of a finite number of possible final configurations. Once the positions of the boxes labeled "n" are fixed, the procedure is effectively dealing with a smaller diagram with boxes only labeled from 1 to $n-1$. Thus, by an obvious induction, the process will indeed terminate. This concludes the proof of Theorem 5.4.1.6.

5.4.1.10: Remarks: a) We have effectively shown that the coordinate ring $\mathcal{R}(GL_n/U_n)$ is an algebra generated by the $\delta_{(R)}$, with relations defined by the straightening rules of 5.4.1.7. Since the straightening rules only involve coefficients of ± 1 , this algebra can be given a structure over $\mathbb{Z}$. This is the point of view taken in [Tow], where $\mathcal{R}(GL_n/U_n)$, or its $\mathbb{Z}$ -form, is called the "shape algebra".

b) One can extend the analysis given here to get a basis of standard monomials for each $GL_n \times GL_m$ module in $\mathcal{P}(\mathbb{C}^m) [\text{GKR}]$, [Tur].

5.4.2: Branching from O_n to O_{n-1}

In this section, we will discuss the analogous branching for orthogonal groups. This is very strongly influenced by the branching law for GL_n , because either of O_{2n} or O_{2n+1} contains a copy of GL_n , and the diagonal torus in GL_n remains a Cartan subgroup in the orthogonal group. We must, however, discuss separately the restriction from O_{2n+1} to O_{2n} , and from O_{2n} to O_{2n-1} . Also, since O_n is disconnected, we should work with SO_n instead.

5.4.2.1: Let V be a vector space equipped with an inner product $\langle, \rangle$. Set $m = [n/2]$. Consider the space $V^m \cong \text{Hom}(\mathbb{C}^m, V)$. Inside V^m we have the null fiber NCQ , consisting of mappings T in $\text{Hom}(\mathbb{C}^m, V)$ such that the image of T is an isotropic subspace of V . The variety NCQ is invariant under premultiplication by GL_m and by postmultiplication by $G(V)$, and the algebra $\mathcal{R}(NCQ)$ of functions on NCQ is described as a $GL_m \times G(V)$ -module by Theorem 3.7.3.1. We can improve the qualitative statement in that result. Let $\tilde{V}$ be a non-degenerate hyperplane in V . Denote by $G(V)$ and $G(\tilde{V})$ the orthogonal groups of V and $\tilde{V}$. Regard $G(\tilde{V})$ as a subgroup of V in the usual way.

5.4.2.2: THEOREM. *The joint action of $GL_m \times G(\tilde{V})$ on NCQ is multiplicity-free.*

PROOF. We again appeal to criterion 1.3.3 and show that a Borel subgroup of $GL_m \times G(\tilde{V})$ has a dense orbit. We treat separately the cases of $\dim V$ even and $\dim V$ odd.

Begin with $\dim V$ even; thus, $\dim V = 2m$. We have seen during the proof of Theorem 3.7.3.1 that we can describe NCQ as a union of spaces $\text{Hom}(\mathbb{C}^m, Y)$ where Y is a maximal isotropic subspace of V . Fix one such space Y and find another space X such that (X, Y) form a complete polarization of V (see Proposition A3.8). According to Proposition A3.26, the group $U_{X \cap \tilde{V}}$ acts transitively on the set of maximal isotropic subspaces of V which are transverse to X . If we extend $U_{X \cap \tilde{V}}$ by an element which is in $G(V) - SG(V)$, the result will be a group $\tilde{U}_{X \cap \tilde{V}}$ such that $\tilde{U}_{X \cap \tilde{V}}(Y)$ is dense in the variety $P_m(V)$ of maximal isotropic subspaces of V . It follows that $\tilde{U}_{X \cap \tilde{V}}(\text{Hom}(\mathbb{C}^m, Y))$ is dense in NCQ .

According to Proposition A3.9, the stabilizer of $Y \cap \tilde{V}$ in $P_{X \cap \tilde{V}}$ maps onto all of $GL(Y \cap \tilde{V})$ via restriction to $Y \cap \tilde{V}$. Thus, the stabilizer of $\text{Hom}(\mathbb{C}^m, Y)$ in

$GL_m \times P_{X \cap \tilde{V}}$ contains $GL_m \times GL(Y \cap \tilde{V})$. Since the space $X \cap \tilde{V}$ is an isotropic subspace of both V and $\tilde{V}$, the notation $P_{X \cap \tilde{V}}$ is ambiguous. Here, by $P_{X \cap \tilde{V}}$ we mean the stabilizer of $X \cap \tilde{V}$ in $G(\tilde{V})$. Since $Y \cap \tilde{V}$ has codimension one in Y , an argument like that in §5.4.1.2 tells us that the Borel subgroups of GL_m and $GL(Y \cap \tilde{V})$ together have a dense orbit in $\text{Hom}(\mathbb{C}^m, Y)$. This proves the theorem for the case of $\dim V$ even.

Now consider $\dim V$ odd: $\dim V = 2m + 1$. Again, fix a maximal isotropic subspace Y and find a decomposition of type (A3.5) containing Y . In this case, the space V_1 is one dimensional. We also require that Y not be contained in $\tilde{V}$. In this case, Proposition A3.32 guarantees us that the group $SG(V_2) \cdot U_{X \cap \tilde{V}}$ acts transitively on the set of all maximal isotropic subspaces of V which are transverse to $X^\perp$ and not contained in $\tilde{V}$. Furthermore, the subgroup of $P_{X \cap \tilde{V}}$ which stabilizes $Y \cap \tilde{V}$ maps onto $GL(Y \cap \tilde{V})$ under restriction. The space $Y \cap \tilde{V}$ is a hyperplane in Y , so we are again in a situation essentially the same as in Theorem 5.4.1.1. We conclude that the Borel of $GL_m \times G(\tilde{V})$ has a dense orbit, and the theorem follows in this second case also.

5.4.2.3: Explicit description of the branching Theorem 5.4.2.2 implies that the irreducible representations of O_n decompose in a multiplicity-free fashion on restriction to O_{n-1} . We should sharpen it to give an explicit description of the restrictions.

5.4.2.3.1: Consider first the case when $n = 2m$ is even. Consider the irreducible component of NCQ^+ of NCQ . The coordinate ring $\mathcal{R}(NCQ^+)$ is described by formula (3.7.3.4.2). Let U_m be the standard upper triangular unipotent subgroup of GL_m . Consider the algebra $\mathcal{R}(NCQ^+)^{U_m}$ of GL_m highest weight vectors in $\mathcal{R}(NCQ^+)$. From formula (3.7.3.4.2), we conclude that $\mathcal{R}(NCQ^+)^{U_m}$ consists of one copy of each representation $\sigma_n^{D_+}$ of SO_n . Every irreducible representation of SO_n is either one of the $\sigma_n^{D_+}$, or its conjugate by O_n . Thus, if we understand how the representations $\sigma_n^{D_+}$ decompose when restricted to SO_{n-1} , we will understand how all irreducible representations of SO_n behave on restriction.

Further, the algebra $\mathcal{R}(NCQ^+)^{U_m}$ is graded by the characters of A_m , the diagonal torus of GL_m : the ψ_D -eigenspace is just the irreducible representation $\sigma_n^{D_+}$ as an SO_{n-1} -module. Let $B'_{n-1} = A'_{m-1} U'_{n-1}$ be a suitably chosen Borel subgroup of SO_{n-1} , with maximal unipotent subgroup U'_{n-1} and torus A'_{m-1} . (We use $m-1$ rather than $n-1$ to label A' because $m-1$ is the dimension of A'_{m-1} .) If we look at the algebra $(\mathcal{R}(NCQ^+)^{U_m})^{U'_{n-1}} = \mathcal{R}(NCQ^+)^{U_m \times U'_{n-1}}$, we see that this is doubly graded by $(A_m)^{\wedge^+} \times (A'_{m-1})^{\wedge^+}$. Knowledge of the ψ_D -eigenspace for A_m tells us the SO_{n-1} highest weight vectors in $\sigma_n^{D_+}$. Hence knowledge of $\mathcal{R}(NCQ^+)^{U_m \times U'_{n-1}}$ as an $(A_m)^{\wedge^+} \times (A'_{m-1})^{\wedge^+}$ graded algebra tells us how to decompose representations of SO_n to SO_{n-1} .

The argument of Theorem 5.4.2.2 allows us to give a set of $A_m \times A'_{m-1}$ -homogeneous generators for $\mathcal{R}(NCQ^+)^{U_m \times U'_{n-1}}$. It will be easy to see that these generators are algebraically independent, so $\mathcal{R}(NCQ^+)^{U_m \times U'_{n-1}}$ is a polynomial algebra. Consider the polarization (X, Y) as in the proof of Theorem 5.4.2.2. We can introduce coordinates as in §3.6.1 in such a way that X is the subspace on which the coordinates z_1 through z_m vanish, and Y is the subspace on which z_{m+1} through z_n vanish. Furthermore, we can arrange that the hyperplane $Y \cap \tilde{V}$ of Y

' is defined by the condition $z_m = O$. We may assume that the hyperplane $\tilde{V}$ is defined by $z_m + z_{m+1} = O$. In other words, $z_m + z_{m+1}$ is the coordinate on the line invariant by O_{n-1} . Let us write $z_m + z_{m+1} = w$. Recall from formula (3.6.2.2) the functions δ_b for $1 \leq b < m$. Let us now denote these as $\delta_{b,0}$, for $b < m$, and define, in addition,

$$(5.4.2.3.1.1) \quad \delta_{b,1} = \begin{vmatrix} z_{11} & z_{12} & \cdots & z_{1,b+1} \\ z_{21} & z_{22} & \cdots & z_{2,b+1} \\ \vdots & \vdots & \ddots & \vdots \\ z_{b,1} & z_{b,2} & \cdots & z_{b,b+1} \\ w_1 & w_2 & \cdots & w_{b+1} \end{vmatrix} \quad \text{for } 1 \leq b < m.$$

Here the column $(z_{1a}, z_{2a}, z_{3a}, \dots, z_{m-1,a}, w_a)^t$ are the coordinates on the a -th copy of $Y \cap \tilde{V} \oplus \tilde{V}^\perp$ in $(Y \cap \tilde{V} \oplus \tilde{V}^\perp)^m \subset V^m$. However, since the restriction of w to Y is a multiple of z_m , they also define coordinates on the a -th copy of Y in $Y^m \subset V^m$. According to §5.4.1, the restrictions of the $\delta_{b,0}$ and $\delta_{b,1}$ to Y^m form an algebraically independent generating set for the $GL_m \times GL(Y \cap \tilde{V})$ highest weight vectors in $P(Y^m)$.

We have a factorization ${}^aU'_{n-1} = U_{X \cap \tilde{V}} \cdot \tilde{U}_{m-1}$, where $U_{X \cap \tilde{V}}$ is as in Proposition A3.26, and $\tilde{U}_{m-1}$ is a maximal unipotent subgroup of $GL(Y \cap \tilde{V}) \subset SO_{n-1}$. As argued in the proof of Theorem 5.4.1, a $U_{X \cap \tilde{V}}$ -invariant function on NCQ^+ is determined by its restriction to $Y^m \cong Hom(C^m, Y)$. Thus, any U'_{n-1} -invariant vector will be determined by its restriction to Y^m , and, moreover, this restriction will be invariant for $\tilde{U}_{m-1}$. If it is also invariant for U_m , its restriction to Y^m must be a polynomial in the $\delta_{b,0}$ and $\delta_{b,1}$. Thus, we conclude:

5.4.2.3.1.2: PROPOSITION. The (restriction to NCQ^+ of) $\delta_{b,0}$ and $\delta_{b,1}$ form an algebraically independent set of generators for the algebra $\mathcal{R}(NCQ^+)^{U_m \times U'_{n-1}}$.

5.4.2.3.1.3: Remark: Let us label the representations σ_n^+ of SO_{2m} with the diagram D . We have already labeled representations of O_{2m-1} with diagrams. Up to tensoring with $\det$, these diagrams can be taken to be of depth at most m . Since $O_{2m-1} = (\pm 1) \times SO_{2m-1}$, where ± 1 here denotes plus or minus the $(2m-1) \times (2m-1)$ identity matrix, an irreducible representation of O_{2m-1} remains irreducible when restricted to SO_{2m-1} . We will label an irreducible representation of SO_{2m-1} with the diagram of depth at most $m-1$ which labels one of the possible extensions to O_{2m-1} . With these conventions, the calculation just above shows that, when expressed in terms of diagrams, the branching law from SO_{2m} to SO_{2m-1} looks exactly like the branching law from GL_m to GL_{m-1} . However, we should keep in mind several points that are disguised by our pleasant notation.

i) The σ_n^+ do not constitute all irreducible representations of SO_{2m} —one must also include the σ_n^- . (However, the restrictions of the σ_n^- follow the same diagrammatic yoga as the σ_n^+ .) There is also the (here unaddressed) issue of the spin coverings of the orthogonal groups.

ii) The representation σ_n^+ , where D is the diagram with one column of length m , is not one dimensional, but is the Cartan square of one of the spin representations.

(Likewise, its restriction to SO_{2m-1} is the Cartan square of the spin representations for $Spin_{2m-1}$. It follows from the fact that O_{2m-1} acts transitively on the Grassmannian of maximal isotropic subspaces in C^{2n} (see Corollary A3.24) that the Cartan powers of the $spin_{2m}^\pm$ remain irreducible when restricted to $Spin_{2m-1}$.)

iii) The representation σ_n^+ , where D is the diagram with one column of length $m-1$, is not a fundamental representation of SO_{2m} : it is the Cartan product of the two half-spin representations.

5.4.2.3.2: Now consider the branching from SO_{2m+1} to SO_{2m} . Retain the notation of this case of the proof of Theorem 5.4.2.2. We introduce coordinates z_j , $1 \leq j \leq 2m+1$ on V such that $\tilde{V}$ is the hyperplane defined by $z_{m+1} = 0$. We abbreviate $z_{m+1} = w$. We can take the space $Y \cap \tilde{V}$ to be defined by the vanishing of the z_j for $j \geq m$, and the space $X \cap \tilde{V}$ to be defined by $z_j = 0$ for $j < m$ or $j > m+2$, and then $\tilde{V}_3$ (see formula (A3.31)) is defined by $z_j = 0$ for $j < m$ or $j > m+2$, and the space $\tilde{V}_2 = V_3 \cap \tilde{V}$ satisfies the further condition $w = 0$. Then the space Y will be spanned by $Y \cap \tilde{V}$ together with one more vector, which should be isotropic and contained in V_3 , but not contained in $\tilde{V}_2$. The vector defined by $z_m = z_{m+1} = 2$, $z_{m+2} = -1$, and other coordinates zero is a possible choice. All others arise from this one by scaling and action of $SG(V_2)$.

Since a joint $GL_m \times SO_{2m}$ highest weight vector in $\mathcal{R}(NCQ(V))$ is, in particular, an eigenvector for $SG(V_2) \cdot U_X \cap \tilde{V}$, it must be determined by its restriction to $Hom(C^m, Y)$. Furthermore, this restriction must be a joint highest weight vector for $GL_m \times GL(Y \cap \tilde{V})$. We know from the discussion in §5.4.1.1 of the branching rule for GL_n that a polynomial on $Hom(C^m, Y)$ which is a joint highest weight vector for $GL_m \times GL(Y \cap \tilde{V})$ is determined by its weight. Thus, a joint $GL_m \times SO_{2m}$ highest weight vector is determined by its weight under $SG(V_2)$ and the weight of its restriction to $Hom(C^m, Y)$. The polynomials $\delta_{b,0}$ and $\delta_{b,1}$ used above in §5.4.2.3.1 are again joint $GL_m \times SO_{n-1}$ highest weight vectors in this case. They are all invariant by $SG(V_2)$. In addition, we have the polynomials

$$(5.4.2.3.2.1) \quad \delta_{m-1,1}^+ = \begin{vmatrix} z_{11} & z_{12} & \cdots & z_{1,m} \\ z_{21} & z_{22} & \cdots & z_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ z_{m-1,1} & z_{m-1,2} & \cdots & z_{m-1,m} \\ z_{m,1} & z_{m,2} & \cdots & z_{m,m} \end{vmatrix} \quad \text{and}$$

$$(5.4.2.3.2.1) \quad \delta_{m-1,1}^- = \det \begin{vmatrix} z_{11} & z_{12} & \cdots & z_{1m} \\ z_{21} & z_{22} & \cdots & z_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ z_{m-1,1} & z_{m-1,2} & \cdots & z_{m-1,m} \\ z_{m+2,1} & z_{m+2,2} & \cdots & z_{m+2,m} \end{vmatrix}$$

These have the same restriction to Y^m as does $\delta_{m-1,1}$, but they have different weights for $SG(V_2)$. They have the same weights for $SG(V_2)$ as do z_m and z_{m+2} , respectively. We will call these weights $+1$ and -1 .

Since the product $(\delta_{m-1,1}^+)(\delta_{m-1,1}^-)$ is invariant under $SG(V_2)$, and has the same restriction to Y^m as does $(\delta_{m-1,1}^+)^2$, it must equal $(\delta_{m-1,1}^+)^2$ up to multiples, by multiplicity-freeness. Indeed, by expanding the determinants involved along their last rows, we can verify that the isotropy of the line $Y \cap V_3$ implies that

$$(5.4.2.3.2.2) \quad (\delta_{m-1,1}^+)(\delta_{m-1,1}^-) + (\delta_{m-1,1}^+)^2 = 0.$$

5.4.2.3.2.3: PROPOSITION. When $n = 2m + 1$ is even, the polynomials $\delta_{b,0}$ and $\delta_{b,1}$, for $0 \leq b \leq m - 1$, together with $\delta_{m-1,1}^+$ and $\delta_{m-1,1}^-$, generate the algebra $\mathcal{R}(NCQ(V))^{U_m \times U_{n-1}^+}$, subject to the relation (5.4.2.3.2.2).

PROOF. Since the $\delta_{b,0}$ and $\delta_{b,1}$ are already algebraically independent on Y^m , and the restriction to Y^m of $\delta_{m-1,1}^+$ and $\delta_{m-1,1}^-$ equals the restriction of $\delta_{m-1,1}^+$, any relations between the proposed generators must be between $\delta_{m-1,1}^+$, $\delta_{m-1,1}^-$ and $\delta_{m-1,1}$. We can use the relation (5.4.2.3.2.2) to reduce any monomial in these three functions to one involving only one of $\delta_{m-1,1}^+$ and $\delta_{m-1,1}^-$. Then keeping in mind the weights with respect to $SG(V_2)$, we see these monomials must be linearly independent. Hence (5.4.2.3.2.2) is a complete set of relations for our proposed generating set.

Now we will show that the listed elements do generate $\mathcal{R}(NCQ(V))^{U_m \times U_{n-1}^+}$. Let A_m be the diagonal torus in GL_m and A'_m the diagonal torus (in the z_j coordinate system) of O_{n-1} . Consider the semigroup of weights in $(A_m \times A'_m)^\wedge$ generated by the weights of the proposed generators. It is easy to check (cf. §5.4.1.3) that the weights $((a_1, a_2, a_3, \dots, a_m)(b_1, b_2, b_3, \dots, b_m))$ of this semigroup are exactly those satisfying

$$(5.4.2.3.2.4) \quad a_1 \geq b_1 \geq a_2 \geq b_2 \geq a_3 \geq \dots \geq a_{m-1} \geq b_{m-1} \geq a_m \geq |b_m|.$$

If we can show that any $GL_m \times O_{n-1}$ constituent of $\mathcal{R}(NCQ(V))$ must have its highest weight in this semigroup, then we will know its highest weight vector must coincide with appropriate monomial in our generators, by multiplicity-freeness of $\mathcal{R}(NCQ(V))$.

Consider the structure of the Lie algebra $\mathfrak{so}_{2m+1}$. We have a decomposition

$$(5.4.2.3.2.5) \quad \mathfrak{so}_{2m+1} \cong \mathfrak{so}_{2m} \oplus \mathbb{C}^{2m}.$$

Inside $\mathfrak{so}_{2m+1}$ we have a copy of $\mathfrak{gl}_m$, and we have

$$\mathfrak{so}_{2m} \cong \mathfrak{gl}_m \oplus \mathfrak{p}^+ \oplus \mathfrak{p}^-,$$

where $\mathfrak{p}^+$ and $\mathfrak{p}^-$ are abelian subalgebras normalized by $\mathfrak{gl}_m$. (They are isomorphic to $\Lambda^2(\mathbb{C}^m)$ and $\Lambda^2(\mathbb{C}^m)^*$ as $\mathfrak{gl}_m$ modules.) As a module for $\mathfrak{gl}_m$, the space $\mathbb{C}^{2m}$ breaks up as $\mathbb{C}^m \oplus \mathbb{C}^m$. The subspaces $\mathfrak{gl}_m \oplus \mathfrak{p}^+ \oplus \mathbb{C}^m$ and $\mathfrak{gl}_m \oplus \mathfrak{p}^- \oplus \mathbb{C}^m$ are parabolic subalgebras of $\mathfrak{so}_{2m+1}$. Their nilradicals are $\mathfrak{p}^+ \oplus \mathbb{C}^m$ and $\mathfrak{p}^- \oplus \mathbb{C}^m$, respectively. We may choose for Borel subalgebra of $\mathfrak{so}_{2m+1}$ a subalgebra of $\mathfrak{gl}_m \oplus \mathfrak{p} \oplus \mathbb{C}^m$. Then if w_0 is highest weight vector for an irreducible-module W , the Lie algebra $\mathfrak{p}^+ \oplus \mathbb{C}^m$ will annihilate w_0 . Under the action of $\mathfrak{gl}_m$, the vector w_0 will generate an irreducible module W_0 . The $\mathfrak{so}_{2n}$ -submodule of W generated by w_0

may be obtained by applying the universal enveloping algebra $U(\mathfrak{p}^-)$ to W_0 , and the full module W is obtained by applying the enveloping algebra $U(\mathfrak{p}^- \oplus \mathbb{C}^m)$ to W_0 , corresponding to the decomposition $\mathfrak{p}^- \oplus \mathbb{C}^m$, we have a decomposition

$$U(\mathfrak{p}^- \oplus \mathbb{C}^m) \cong S(\mathbb{C}^m) \oplus \mathfrak{p}^- U(\mathfrak{p}^- \oplus \mathbb{C}^m).$$

The space $S(\mathbb{C}^m)$ is the span of the powers of the elements of $\mathbb{C}^m$. It is not a subalgebra of $U(\mathfrak{p}^- \oplus \mathbb{C}^m)$, because $\mathbb{C}^m$ is not a Lie subalgebra of $\mathfrak{p}^- \oplus \mathbb{C}^m$. (The commutators of elements of $\mathbb{C}^m$ land in $\mathfrak{p}^-$.) However, it is a $\mathfrak{gl}_m$ -module (the complement to $\mathfrak{p}^- U(\mathfrak{p}^- \oplus \mathbb{C}^m)$ in $U(\mathfrak{p}^- \oplus \mathbb{C}^m)$). Since $\mathfrak{p}^- \subset \mathfrak{so}_{2m}$, we see that the possible $\mathfrak{so}_{2m}$ highest weight vectors will be contained in the space generated by applying $S(\mathbb{C}^m)$ to W_0 . Thus, these $\mathfrak{so}_{2m}$ highest weight vectors will, in particular, be $\mathfrak{gl}_m$ highest weight vectors for $W_0 \times S(\mathbb{C}^m)$. By the contragredient Pieri Rule (see formula (5.6.4.2)), we see that if the weight of w_0 is $(a_1, a_2, a_3, \dots, a_m)$, then the $\mathfrak{gl}_m$ highest weights $(b_1, b_2, b_3, \dots, b_m)$ of $W \otimes S(\mathbb{C}^m)$ must be of the form specified in formula (5.4.2.3.2.4), except that b_m can be arbitrarily large negative. However, a highest weight of $W_0 \otimes S(\mathbb{C}^m)$ with $b_m < -a_m$ cannot be a weight of W , because in $\mathfrak{so}_{2m+1}$ there is an element which normalizes $\mathfrak{so}_{2n}$ and which switches b_m to $-b_m$. Hence the upper bound on b_m coming from the Pieri Rule implies a bound on the absolute value of b_m in the highest weights of $\mathfrak{so}_{2m}$ submodules of W . This concludes the proof of Theorem 5.4.2.3.2.3.

5.5: Abelian and Heisenberg nilradicals.

A naturally arising family of linear actions are those defined by the adjoint action of the Levi component M of a parabolic subgroup P on the Lie algebra $\mathfrak{u}$ of the unipotent radical of the parabolic subgroup. Among these, the simplest examples come from maximal parabolic subgroups. Among maximal parabolics, we can in turn distinguish level of difficulty, and the simplest cases are when M acts irreducibly on $\mathfrak{u}$, which is equivalent to $\mathfrak{u}$ being abelian. These cases are quite restricted, and the list of them is well-known.

5.5.1: List of parabolic subgroups with abelian nilradical

Group	Levi component	nilradical
GL_n	$GL_k \times GL_{n-k}$	$\mathbb{C}^k \otimes \mathbb{C}^{n-k}$
SO_{2n+1}	$O_{2n-1} \times GL_1$	$\mathbb{C}^{2n-1}$
Sp_{2n}	GL_n	$S^2(\mathbb{C}^n)$
SO_{2n}	$O_{2n-1} \times GL_1$	$\mathbb{C}^{2(n-1)}$
SO_{2n}	GL_n	$\Lambda^2(\mathbb{C}^n)$
E_6	$Spin_1 O \times GL_1$	$spin_1 O^+$
E_7	$E_6 \times GL_1$	V_{27}

The list appears somewhat haphazard: a given simple group may have several (SL_n) , two (SO_{2n}) , one $(SO_{2n+1}, Sp_{2n}, E_6, E_7)$ or none (E_8, F_4, G_2) . Interestingly, the groups which contain no abelian nilradicals are the "last of the line" - the natural endpoints of the list of simple groups. In any case, the main point we wish to make about this list is that it consists of the nicest actions known. All of them are both (symmetric) multiplicity-free and skew multiplicity-free. In fact, their

exterior algebras are geometrically decomposable in the sense of §4.7, and we have shown there, at least for the cases in which the commutator subgroup of the Levi component is simple, that this characterizes these modules among simple modules for simple groups. The decomposition of their exterior algebras is described in Appendix 5, and was computed in a uniform way in terms of the Weyl group by Kostant [Kos 1]. The classical cases were known well before. The symmetric algebras are described in various places [Jon], [Hu], [Schm]. Again, the classical ones have been known for some time. Some early references are [Th], [Tur]. The first uniform treatment is in [Schm], (which, however, asserts that the results were already known to Kostant).

After abelian Lie algebras, the next simplest are the Heisenberg Lie algebras: two-step nilpotent Lie algebras with one dimensional center. See Appendix 2. The commutator quotient of a Heisenberg Lie algebra carries a symplectic form defined in terms of the Lie bracket. This form will be preserved by the action of the Levi component. Thus, from parabolics with Heisenberg nilradicals, we get a family of symplectic representations. A remarkable fact, first communicated to me by G. Savin, is that each simple Lie algebra, except for $\mathfrak{sl}_2$, contains a unique parabolic with Heisenberg nilradical. (It is the stabilizer of the highest root, and so forms the isotropy group of the minimal nilpotent orbit for the coadjoint action.) Here is a list of the associated representations.

5.5.2: List of parabolics with Heisenberg nilradical

Group	Levi component	Commutator Quotient
GL_n	$GL_1 \times GL_{n-2} \times GL_1$	$C^{n-2} \oplus C^{n-2*}$
O_n	$O_{n-4} \times GL_2$	$(C^{n-4})^2$
Sp_{2n}	$Sp_{2(n-1)} \times GL_1$	$C^{2(n-1)}$
E_6	GL_6	$\Lambda^3(C^6)$
E_7	$Spin_{12} \times GL_1$	$spin_{12}$
E_8	$E_7 \times GL_1$	V_{56}
F_4	$Sp_6 \times GL_1$	$\Lambda^3_{prim}(C^6)$
G_2	GL_2	$S^3(C^2)$

These actions are all skew-multiplicity-free, in fact symplectic-geometrically decomposable in the sense of §4.7, and again, the set of those which are simple modules for simple groups is characterized by the property of symplectic-geometric decomposability. However, these actions are not all (symmetric) multiplicity-free. More precisely, the ones for GL_n and Sp_{2n} are multiplicity-free. The fact that they are, and their explicit decompositions, are fairly direct consequences of the theory of dual pairs ([Hw1], see also Theorems 2.5.4 and 3.5.2) and the Branching Laws (see §5.4). The one for O_n is not multiplicity-free, but it may also be analyzed using the theory of dual pairs [ATZ]. The ones for the exceptional groups are also not multiplicity-free. Nevertheless, except for the smallest, they satisfy a strong regularity property: they are *exceptional*, in the sense of [Br3], which is to say, the algebra of their covariants is a polynomial algebra. From [Br3] also, their explicit decompositions may be read off. Here we would like to point out the very strong family resemblance between these five modules, and to relate their common, pleasant structure to multiplicity-free actions. We will refer to them as the "exceptional

nilradicals", even though this terminology is not strictly accurate.

First, we arrange the 5 exceptional nilradicals in order of increasing dimension.

5.5.3: The exceptional Heisenberg nilradicals

Group:	SL_2	Sp_6	GL_6	$Spin_{12}$	E_7
Module	$S^3(C^2)$	$\Lambda^3_{prim}(C^6)$	$\Lambda^3(C^6)$	$spin_{12}^+$	V_{56}
Dimension	4	14	20	32	56

Note that the group listed is the group which acts on the nilradical, i.e., the Levi component of the parabolic subgroup containing the nilradical. Next, we remark that each group in this list embeds in the next in such a way that its associated module occurs in the restriction of the module associated to the next group. For the last three groups, the embedding is as a nilradical of a parabolic. For $Sp_6 \rightarrow GL_6$, it is the standard action on C^6 . For $SL_2 \rightarrow Sp_6$, it is the diagonal action of SL_2 on three copies of C^2 . Now consider the algebra of covariants $\mathcal{P}(V)^U$, where V denotes one of the five modules and U a maximal unipotent subgroup of the appropriate group. The next table, derived from [Br3], describes the degrees and the representation type of the fundamental generators of $S(V)^U$ for each of the five V 's.

5.5.4: Generators for the symmetric covariants

Module:	Degree 1	Degree 2	Degree 3	Degree 4	Total
$S^3(C^2)$	same	adjoint	same	1	4
$\Lambda^3_{prim}(C^6)$	same	adjoint	same	$1, 2\omega_2$	5
$\Lambda^3(C^6)$	same	adjoint	same	$1, \omega_2 + \omega_4$	5
$spin_{12}^+$	same	adjoint	same	$1, \omega_4$	5
V_{56}	same	adjoint	same	$1, \omega_6$	5

In this table, "same" means the same representation as the given module, as listed in the first column; "adjoint" means the adjoint representation of G on its Lie algebra; "1" denotes the trivial representation, and the ω 's in the degree 4 column indicate the highest weight of the representation of the non-invariant fundamental generator of degree 4, following the notation of [Br3].

The similarity structure in these actions is evident from Table 5.5.4. Allowing for the inevitable difference in notations for the noninvariant degree 4 fundamental covariant, the algebras $S(V)^U$ for the four largest modules are isomorphic as $\mathbb{Z}^{+4}$ graded modules. The case of $S^3(C^2)$ is slightly different, in that the non-invariant degree four covariant is missing, and also there is an algebraic relation between the generators, but the similarity is still strong, and one is tempted to attribute the difference to a degeneration due to small dimensionality. However, for simplicity, we delete it from the discussion below.

We propose an explanation of the coincidences in Table 5.5.4 in terms of their varieties of highest weight vectors. By this we mean the collection of all vectors v in V such that there is a Borel subgroup of G for which v is an eigenvector. Denote this variety by $HW(V)$. Fix a vector v_0 in $HW(V)$. Then the stabilizer of the line Cv_0 must be a parabolic subgroup of G , and in fact, it is a maximal

parabolic, with abelian nilradical. Denote it by P_V . Then the dimension of $HW(V)$ is $\dim G - \dim P_V + 1$. Here is a table of these dimensions together with a listing of the Levi component of P_V .

5.5.5: Dimensions of the highest weight varieties for the exceptional nilradicals

Module:	$S^3(\mathbf{C}^2)$	$\Lambda^3_{prim}(\mathbf{C}^6)$	$\Lambda^3(\mathbf{C}^6)$	$spin_{12}^+$	V_{56}
Levi factor of P_V	GL_1	GL_3	GL_3	GL_6	E_6
$\dim HW(V)$:	2	7	10	16	28

Comparing Tables 5.5.5 and 5.5.3, we see that in all cases, the dimension of $HW(V)$ is exactly half the dimension of V . We can define a map

$$(5.5.6) \quad \alpha : HW(V) \times HW(V) \rightarrow V$$

by addition:

$$\alpha(v_1, v_2) = v_1 + v_2, \quad v_1, v_2 \in HW(V).$$

Since the dimension of $HW(V) \times HW(V)$ is the same as $\dim V$, we might guess that the map α is generically a diffeomorphism. This is correct. Thus, the pullback map

$$(5.5.7) \quad \alpha^* : \mathcal{P}(V) \rightarrow \mathcal{R}(HW(V) \times HW(V))$$

is injective. We note that the scalar dilations on $HW(V)$ define a $\mathbf{Z}^+$ grading on $\mathcal{R}(HW(V))$. Hence $\mathcal{R}(HW(V) \times HW(V))$ has a $(\mathbf{Z})^{+2}$ grading, and we can sum the two partial degrees to get a total degree. Because of the compatibility of the degrees on $\mathcal{R}(HW(V))$ and $\mathcal{P}(V)$, the map α^* is degree preserving when $\mathcal{R}(HW(V) \times HW(V))$ is graded by total degree (and $\mathcal{P}(V)$ has its usual degree).

According to [Ltm 1], the varieties $HW(V) \times HW(V)$ are among the multiplicity-free "double cones". Each of the covariant algebras $\mathcal{R}(HW(V) \times HW(V))^U$ is a polynomial algebra on five generators (excluding $S^3(\mathbf{C}^2)$ as announced above). Two of the generators belong to $V^* \otimes 1$ and $1 \otimes V^*$, the generating spaces of the factors $\mathcal{R}(HW(V)) \otimes 1$ and $1 \otimes \mathcal{R}(HW(V))$ of $\mathcal{R}(HW(V) \times HW(V))$. The other three generators are the highest weights of the non-Cartan components in $V^* \otimes V^*$. The first two generators have bidegree $(1, 0)$ and $(0, 1)$, and the last three have bidegree $(1, 1)$. Let us call these generators $\beta_{(1,0)}$, $\beta_{(0,1)}$, γ and ε . We call one of them ad because the adjoint representation always occurs as a component of the tensor square of any self-dual representation; ad will denote the highest weight of this factor. Also, the trivial representation clearly occurs in $V^* \otimes V^*$. The G -invariant generator is ε .

The degrees of the generators of $\mathcal{R}(HW(V) \times HW(V))^U$ do not match well with the Table 5.5.4. However, we should observe that the ring $\mathcal{R}(HW(V) \times HW(V))$ has a $\mathbf{Z}/2$ symmetry, from exchanging the two factors. Since addition is commutative, the map α clearly commutes with this involution of $\mathcal{R}(HW(V) \times HW(V))$, so it actually defines a map

$$(5.5.8) \quad \alpha^* : \mathcal{P}(V) \rightarrow \mathcal{R}(HW(V) \times HW(V))^{\mathbf{Z}/2}.$$

Of the generators of $\mathcal{R}(HW(V) \times HW(V))^U$, the degree one generators, $\beta_{(1,0)}$ and $\beta_{(0,1)}$, are exchanged by the involution. Thus, we can form

$$\beta_+ = (\beta_{(1,0)} + \beta_{(0,1)})/2 \quad \text{and} \quad \beta_- = (\beta_{(1,0)} - \beta_{(0,1)})/2,$$

which will be eigenvectors for the involution. Of the degree two generators, the one denoted ad will be invariant under the involution because the adjoint representation of Sp_{2n} is realized on $S^2(\mathbf{C}^{2n})$. The other two generators, γ and ε , will be sent to their negatives by the involution. Thus, as generators for the algebra $\mathcal{R}(HW(V) \times HW(V))^{\mathbf{Z}/2}$ of involution-invariants, we find the set

$$(5.5.9) \quad \beta_+, \quad ad, \quad (\beta_-)^2, \quad (\gamma)^2, \quad (\varepsilon)^2, \quad (\beta_-)\gamma, \quad (\beta_-)\varepsilon, \quad \text{and} \quad \gamma\varepsilon.$$

These generators are clearly not algebraically independent; they satisfy the relations

$$(5.5.10) \quad (\beta_-)^2(\gamma)^2 = (\beta_- \gamma)^2, \quad (\beta_-)^2(\varepsilon)^2 = (\beta_- \varepsilon)^2, \quad (\gamma)^2(\varepsilon)^2 = (\gamma \varepsilon)^2,$$

$$(\beta_- \gamma)(\beta_- \varepsilon) = (\gamma \varepsilon)(\beta_-)^2, \quad (\beta_- \gamma)(\gamma \varepsilon) = (\beta_- \varepsilon)(\gamma)^2,$$

$$(\beta_- \varepsilon)(\gamma \varepsilon) = (\beta_- \gamma)(\varepsilon)^2.$$

Thus, we have in lists (5.5.9) and (5.5.10) an explicit description of an algebra which contains an isomorphic copy of $\mathcal{P}(V)^U$ as a subalgebra. It remains to find this subalgebra. Computation using Weyl's Dimension Formula shows that in $\alpha^*(\mathcal{P}(V))$ one has the five generators

$$(5.5.11) \quad \beta_+, \quad ad, \quad (\beta_-)\varepsilon, \quad \gamma\varepsilon, \quad \text{and} \quad (\varepsilon)^2.$$

If one accepts from [Br3] that the algebra $\mathcal{P}(V)^U$ must be a polynomial algebra, then since it is contained in $\mathcal{R}(HW(V) \times HW(V))^U$ and must have the same dimension (since α^* is generically a local isomorphism), the list (5.5.11) must constitute the generators of $\mathcal{P}(V)^U$. Thus, this analysis finds an explanation for the uniformity in Table 5.5.4 in terms of a uniform relation of V with a multiplicity-free double cone. However, we have not found an independent explanation of why $\mathcal{P}(V)^U$ should be a polynomial algebra, since although $\mathcal{R}(HW(V) \times HW(V))^U$ is polynomial, the subalgebra $(\mathcal{R}(HW(V) \times HW(V))^U)^{\mathbf{Z}/2}$ is not. We also have not explained why the particular generators (5.5.11) are the ones which come from α^* (although the first two are obvious). Perhaps a deeper understanding of the map α would clarify these points.

5.6: The tensor product problem.

In many situations, one would like to be able to decompose tensor products $U \otimes V$ of irreducible modules U, V , for a reductive group G . One can derive a formula for multiplicities of irreducible representations in tensor products using Weyl's Character Formula [Hmp], [Ja3]. However, because the formula involves a large summation with much cancellation, it is hard to derive specific information from it. Although there has recently been some striking progress on this problem from the unexpected direction of quantum groups [Kas], [Lus], as well as more

direct attacks [BL], [BF], [Ltm 2], there are still many aspects of tensor products which we do not understand. The major successes to date have been in giving combinatorial formulas for the multiplicities of irreducible constituents in a tensor product. Combinatorial formulas have the virtue that every term contributes positively to the final answer. The result for GL_n is called the Littlewood-Richardson Rule [BK], [Mac], [Sag], and has been known for quite a while, though established rigorously more recently. The quantum group yoga as well as more conventional work has furnished analogous formulas for other groups [Ltm 3], [Nak].

Combinatorial formulas have the disadvantage that they are sometimes unwieldy to compute with and may give little insight into structural questions. Further, one sometimes likes to know more than just the multiplicities of representations: one sometimes would like to be able to identify specific irreducible subspaces of $U \otimes V$. Physicists have been the most demanding in the information they seek about a tensor product. They would like to have explicit standard bases for each irreducible module and to have expressions for a basis of $U \otimes V$ which consists of standard bases of irreducible subspaces in terms of the tensor products of the standard bases of U and V . Results at this level of explicitness exist for SL_2 (or GL_2), which, in its compact real form $SO_3(\mathbb{R})$, has been a mainstay of quantum mechanics [BL], [Ham], [Jns]. The primary investigator of this kind of problem for groups beyond SL_2 has been Biedenharn who has developed over a period of decades the study of "tensor operators" [BL]. This theory was brought to some sort of completion for SL_3 in the beautiful paper [BF]. The theory of canonical bases for quantum groups in principle answers this question [Lus 2].

In this section, we formulate a problem whose solution would give information intermediate between the various multiplicity formulas and a full blown theory of tensor operators. The problem is geometrically based and has the virtue that it contains many subproblems, some of which may have accessible and useful solutions. We will offer solutions to some subproblems and will, in particular, give a fairly explicit description of tensor products for GL_3 and GL_4 .

5.6.1: The tensor product problem

As was discussed in §§1.1.18 and 1.1.19, if U is a maximal unipotent subgroup of a reductive group G , the coset space $U \backslash G$ supports one copy of each irreducible representation of G :

$$(5.6.1.1) \quad \mathcal{R}(U \backslash G) \cong \sum_{\psi \in A^+} V_\psi,$$

where V_ψ has a double characterization:

(5.6.1.2)

- i) V_ψ is an irreducible G submodule of $\mathcal{R}(U \backslash G)$, and gives a realization of the representation of G with highest weight ψ .
- ii) V_ψ is the ψ^* -eigenspace for A , the maximal torus of G which normalizes U , acting on $\mathcal{R}(U \backslash G)$ by left translations. Here ψ^* denotes the highest weight of the representation of G contragredient to V_ψ .

In particular, description i) of V_ψ tells us that $\mathcal{R}(U \backslash G)$ has the structure of

graded algebra, where the grading is provided by the semigroup A^{++} of dominant characters of A .

Consider now the product of $U \backslash G$ with itself. We have

$$(5.6.1.3) \quad \mathcal{R}((U \backslash G) \times (U \backslash G)) \cong \mathcal{R}(U \backslash G) \otimes \mathcal{R}(U \backslash G) \cong \sum_{\psi, \varphi \in A^+} V_\psi \otimes V_\varphi.$$

Thus, $\mathcal{R}((U \backslash G) \times (U \backslash G))$ is a sum of the tensor product of every pair of irreducible modules for G , and these summands are the homogeneous components of a graded algebra structure graded by $(A^{++})^2$. To understand how these tensor products decompose, we look at the covariants - the subalgebra of highest weight vectors, $\mathcal{R}((U \backslash G) \times (U \backslash G))^U$. This algebra is triply generated by A^{++} : in addition to the $(A^{++})^2$ -grading of the homogeneous components $V_\psi \otimes V_\varphi$, the space $(V_\psi \otimes V_\varphi)^U$ is naturally an A -module, and so can be decomposed into A -eigenvectors, which will be exactly the highest weight vectors in $V_\psi \otimes V_\varphi$. Knowledge of these is knowledge about the irreducible components of $V_\psi \otimes V_\varphi$. Hence the algebra $\mathcal{R}((U \backslash G) \times (U \backslash G))^U$, with its $(A^{++})^3$ -grading, gives us an explicit picture of the submodule structure of each $V_\psi \otimes V_\varphi$, without also incorporating details of the internal structure of the irreducible submodules. A description of $\mathcal{R}((U \backslash G) \times (U \backslash G))^U$ via $(A^{++})^3$ -homogeneous generators and relations would give us considerable insight into tensor products.

5.6.2: Triple multiplicities

An alternative formulation, in the spirit of [BZ], of this approach to tensor products would be to consider the algebra $\mathcal{R}((U \backslash G) \times (U \backslash G) \times (U \backslash G))^G$. This also is clearly triply graded by A^{++} . Its homogeneous components are the G -invariants in triple tensor products $V_\psi \otimes V_\varphi \otimes V_\chi$ of representations of G . Since the χ -highest weight vectors in $V_\psi \otimes V_\varphi$ are proxies for G -intertwining operators from V_χ to $V_\psi \otimes V_\varphi$, the homogeneous components of $\mathcal{R}((U \backslash G) \times (U \backslash G))^U$ essentially describe $\text{Hom}_G(V_\chi, V_\psi \otimes V_\varphi) \cong (V_\psi \otimes V_\varphi \otimes (V_\chi)^*)^U$. Thus there is a simple dictionary for translating between the information provided by the two algebras. An advantage of $\mathcal{R}((U \backslash G) \times (U \backslash G) \times (U \backslash G))^G$ is that the three factors of each triple tensor product enter in an equivalent way, so the algebra has an S_3 symmetry obtained by permuting the factors in the triple tensor product. On the other hand, the algebra $\mathcal{R}((U \backslash G) \times (U \backslash G))^U$ involves fewer variables. Knowledge of either one is highly desirable.

5.6.3: Specializations

Although the relatively complete and explicit information about tensor products afforded by a description of $\mathcal{R}((U \backslash G) \times (U \backslash G))^U$ will probably be difficult to obtain, a pleasant aspect of this formulation of the problem of decomposing tensor products is that it affords many subproblems which may be substantially simpler than the full problem, and still interesting and useful. The general scheme for describing such subproblems is to replace the Borel subgroup with general parabolic subgroups. Thus, instead of $U \backslash G$, we can consider spaces $P^1 \backslash G$, where P is a parabolic subgroup and P^1 is the commutator subgroup of P . Similarly, instead of trying to find all irreducible components, we may only try to find certain ones, by

computing the P^1 -invariants instead of the U -invariants. Thus, we obtain a family of subproblems of the problem formulated in the last section: given parabolic subgroups P_a, P_b and P_c , describe the algebra $\mathcal{R}((P_a^1 \setminus G) \times (P_b^1 \setminus G))^{P_c^1}$. The description should include the triple grading on $\mathcal{R}((P_a^1 \setminus G) \times (P_b^1 \setminus G))^{P_c^1}$, which is a grading by $(A^{\wedge+})_{P_c^1} \times (A^{\wedge+})_{P_b^1} \times (A^{\wedge+})_{P_a^1}$, where $(A^{\wedge+})_{P_i}$ indicates the dominant characters of A which are trivial on $A \cap P_i$.

An example of this kind of problem was given in §5.4, in the description of the covariants of two subspaces. Here G was GL_n or SL_n , and the parabolics were the stabilizer of an ℓ -dimensional subspace, the stabilizer of a k -dimensional subspace and the Borel. In this case, the algebra was multiplicity-free as a graded algebra, and that is, multiplicity-free when the group acting was enlarged to include the action of $(P_a/P_a^1) \times (P_b/P_b^1)$ on the left. Littelmann [Ltm 1] calls a space of the form $(P_a^1 \setminus G) \times (P_b^1 \setminus G)$ where P_a and P_b are maximal parabolics, a double cone. He classifies all the multiplicity-free double cones in [Ltm 1]. Multiplicity-free double cones also play a role in §5.

5.6.4: Tensoring with modules of limited depth

In this section, we will study tensor products of representations of GL_n from the following point of view. We fix a depth d , and consider tensor products $\rho_n^E \otimes \rho_n^D$, where E is an arbitrary diagram, but D is limited to have depth at most d . Of course, if $d \geq n$, in fact, if $d \geq n-1$, this describes all tensor products for GL_n , so if we can solve this problem for any depth d , we will understand tensor products for all GL_n . We will actually find solutions for $d = 1, 2, 3$. This will provide descriptions of arbitrary tensor products for GL_n for $n \leq 4$.

We will study the indicated tensor products as an example of the class of problems described in the previous section. Let Q_d be the block upper triangular subgroup of GL_n which starts with d blocks of size one and then has one block of size $n-d$. The typical element of Q_d looks like this:

$$(5.6.4.1) \quad \begin{vmatrix} T & X \\ 0 & M \end{vmatrix}$$

where T is a $d \times d$ upper triangular matrix and M is an invertible $(n-d) \times (n-d)$ matrix. To understand the indicated tensor products, we seek to understand the algebra $\mathcal{R}((U \setminus GL_n) \times (Q_d^1 \setminus GL_n))^U$, where U here denotes the unipotent upper triangular matrices.

To attack this problem, we use the reciprocity between tensor products and restrictions to Levi components (see §5.7). By means of this reciprocity, the limited depth tensor product problem we wish to study is equivalent to describing the restriction of representations of GL_{n+d} to $GL_n \times GL_d$ (or since n is arbitrary here, of restricting from GL_n to $GL_{n-d} \times GL_d$). This equivalence holds not simply on the level of multiplicities: the algebras describing the two situations are naturally isomorphic. Begin with the case $d = 1$. This is reciprocal to the problem of restricting to $GL_{n-1} \times GL_1$, whose solution has been described in Theorem 4.4.1.1. According to that result, the algebra of $GL_{n-1} \times GL_1$ highest weight vectors in $\mathcal{R}(U \setminus GL_n)$ is a polynomial algebra generated by the elements $\delta_{k,0}$ and $\delta_{k,1}$ described there. We will not rewrite the generators here, but remark that they are determinants in the

variables x_{jk} associated to GL_{n-1} and variables y_j associated to GL_1 . The triple weight attached to $\delta_{a,b}$ is $(1_{a+b}, 1_a, 1_b)$. Here the first weight is the GL_n -weight, the second weight is the GL_{n-1} -weight, and the third weight is the GL_1 -weight. Of course, $b \leq 1$ in the current situation. The notation 1_a comes from §1.2.7. If we interpret this algebra in terms of tensor products and only worry about multiplicities, it says that the tensor product $\rho_n^E \otimes S^m(C^n)$ decomposes as a multiplicity-free sum of representations ρ_n^D where the diagrams D and E are related as follows. If $\{a_j\}$ are the row lengths of E and $\{b_j\}$ are the row lengths of D , then:

$$(5.6.4.2) \quad \begin{array}{ll} i) & a_j \leq b_j \leq a_{j-1}, \\ ii) & \sum_j b_a - a_j = m. \end{array}$$

This is the classical Pieri Rule [Wem]. Of course, our algebraic formulation of the Pieri Rule does more than simply identify the submodules which occur in the tensor product, it gives explicit highest weight vectors for the submodules. In this situation, when the decomposition is multiplicity-free, this may not seem important, but it becomes more significant in non-multiplicity-free situations.

The discussion in §§5.4.1.4 - 5.4.1.9 provides an extension of the Branching Rule, to a description not of the restriction to $GL_{n-d} \times GL_d$, but to $GL_{n-d} \times (GL_1)^d$. Let us formulate this precisely. Introduce coordinates on the $n \times n$ matrices $M_{n,n}$ by letting $\{x_{jk} : 1 \leq j \leq n, 1 \leq k \leq n-d\}$ be the coordinates for the first $n-d$ columns, and letting $\{y_{jl} : 1 \leq j \leq n, 1 \leq l \leq d\}$ be the coordinates for the last d columns. For a set $S = \{s_1 < s_2 < \dots < s_b\} \subseteq \{1, 2, 3, \dots, d\}$ and an integer $c, 1 \leq c \leq n-d$, let $\tilde{\delta}_{(c,S)}$ denote the determinant

$$(5.6.4.3) \quad \tilde{\delta}_{(c,S)} = \det \begin{vmatrix} x_{11} & x_{12} & \dots & x_{1c} & y_{1s_1} & y_{1s_2} & \dots & y_{1s_b} \\ x_{21} & x_{22} & \dots & x_{2c} & y_{2s_1} & y_{2s_2} & \dots & y_{2s_b} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{c+b,1} & x_{c+b,2} & \dots & x_{c+b,c} & y_{c+b,s_1} & y_{c+b,s_2} & \dots & y_{c+b,s_b} \end{vmatrix}$$

These polynomials $\tilde{\delta}_{(c,S)}$ are a subset of the polynomials $\delta_{(R)}$ of formula (5.4.1.5.1). The precise connection is

$$(5.6.4.4) \quad \tilde{\delta}_{(c,S)} = \delta_{(R(c,S))},$$

where

$$R(c,S) = \{1, 2, 3, \dots, c, n-d+s_1, n-d+s_2, \dots, n-d+s_b\}.$$

In other words, the $\tilde{\delta}_{(c,S)}$ are precisely those $\delta_{(R)}$ for which the set $R \cap \{1, 2, 3, \dots, n-d\}$ contains exactly the initial c indices.

From their definition, it is clear that the $\tilde{\delta}_{(c,S)}$ are highest weight vectors for GL_{n-d} , and since they are weight vectors, they are highest weight vectors for

of GL_{n-d+1} which stabilizes a flag of subspaces of dimensions $1, 2, 3, \dots, d$. (See formula (5.6.4.1).) This algebra is also describable as

$$P(M_{n-d+1,d})^{U_d}.$$

To get the full ring of $GL_{n-d+1} \times GL_d$ highest weight vectors, we must take into account the contribution of the elements $\tilde{\delta}_{(c,S)}$ for sets S of cardinality larger than 1. As a start, we note the following special case of the straightening rules.

$$\begin{aligned} (5.6.4.7) \quad \tilde{\delta}_{(c+1)} \tilde{\delta}_{(c,\{a,b\})} &= \tilde{\delta}_{(c,a)} \tilde{\delta}_{(c+1,b)} - \tilde{\delta}_{(c,b)} \tilde{\delta}_{(c+1,a)} \\ &= \det \begin{vmatrix} \tilde{\delta}_{(c,a)} & \tilde{\delta}_{(c+1,a)} \\ \tilde{\delta}_{(c,b)} & \tilde{\delta}_{(c+1,b)} \end{vmatrix} \end{aligned}$$

Thus, up to multiplication by the function $\tilde{\delta}_{(c+1)}$, which is independent of the y_{jk} , we can express the $\tilde{\delta}_{(c,S)}$ for $\#(S) = 2$ in terms of the $\tilde{\delta}_{(c,a)}$. This holds for all the $\tilde{\delta}_{(c,S)}$. To see this in a systematic way, we adopt notation similar to that used in §5.4.1.7. Let X denote the $n \times (n-d)$ matrix whose entries are the $\{x_{jk}\}$, and let Y denote the $n \times d$ matrix whose entries are the $\{y_{jk}\}$. For any sets $S \subseteq \{1, 2, 3, \dots, n\}$ and $T \subseteq \{1, 2, 3, \dots, d\}$, let $Y_{S,T}$ denote the submatrix of Y gotten by selecting the rows of Y with entries in S and the columns of Y with entries in T . We will let $a = \{1, 2, 3, \dots, a\}$ denote the set consisting of the first a numbers. Thus, $Y_{a,b}$ is the leading $a \times b$ submatrix of Y . We will use similar notation for submatrices of X , but we will usually only use the leading submatrices of X . By $|X_{S,T} Y_{S',T'}|$ we mean the matrix whose first $\#(T')$ columns consists of $X_{S,T}$ and whose last $\#(T')$ columns consists of $Y_{S',T'}$. For this to make sense, we should have $\#(S) = \#(S')$. We will usually have $S = S'$. Thus, $|X_{S,T} Y_{S',T'}|$ is an $\#(S) \times (\#(T) + \#(T'))$ matrix. If it is square, we can take its determinant. We can also use the matrices $X_{S,T}$ and $Y_{S',T'}$ as blocks in larger matrices. With this notation, we can write

$$(5.6.4.8) \quad \tilde{\delta}_{(c,S)} = \det |X_{c+d,c} Y_{c+d,a} S|$$

where $d = \#(S)$. Thus, we may rewrite the identity (5.6.4.7) as follows:

$$(5.6.4.9) \quad \det \begin{vmatrix} X_{a+1,a} & 0 & Y_{a+1,\{r,s\}} \\ 0 & X_{a+2,a+1} & Y_{a+2,\{r,s\}} \end{vmatrix} = \det |X_{a+2,a} Y_{a+2,\{r,s\}}|$$

Identity (5.6.4.9) can be established by simple, standard manipulations of determinants. First subtract the first $a+1$ rows of the second row of blocks from the rows of the first row of blocks. This eliminates the block $Y_{a+1,\{r,s\}}$ and puts a block $X_{a+1,a+1}$ next to $X_{a+1,a}$. Now add the first a columns of the second block of columns to the columns of the first block of columns. This eliminates $X_{a+1,a}$ leaving only the square block $-X_{a+1,a+1}$ in the first $a+1$ rows, and two blocks making $|X_{a+2,a} Y_{a+2,\{r,s\}}|$ in the last $a+2$ rows, whence the identity. Repeated application of this sort of procedure leads to an extension of identity (5.6.4.9) for

$GL_{n-d} \times (GL_1)^d$. If we take all standard monomials, in the sense of §5.4.1.6, formed from the $\tilde{\delta}_{(c,S)}$, and take the corresponding tableaux, and convert these to Gelfand-Tsetlin patterns, we see that we get all Gelfand-Tsetlin patterns such that the bottom $n-d$ rows are obtained by taking the $(d+1)$ -th row and truncating it, one entry at a time, from the right. These are clearly in bijection with truncated Gelfand-Tsetlin patterns which only go down to the $(d+1)$ -th row. From the discussion of §5.4.1.4, we see that these truncated Gelfand-Tsetlin patterns parametrize pairs of GL_{n-d} submodules ρ_{n-d}^E of a GL_n -module ρ_n^D , where the diagram D corresponds to the first row of the pattern and the diagram E corresponds to the $(d+1)$ -th row. Thus, if we fix the shape of the diagram D , we have a collection of GL_{n-d} highest weight vectors, in bijection with the collection of GL_{n-d} -submodules of ρ_n^D . Since our GL_{n-d} highest weight vectors are all standard monomials, and the standard monomials are a basis, our highest weight vectors are a linearly independent set. Hence, they are a basis for the set of GL_{n-d} highest weight vectors in ρ_n^D . To summarize, we have proved the following statement.

5.6.4.5: THEOREM. *The polynomials $\tilde{\delta}_{(c,S)}$ generate the algebra of simultaneous $(GL_{n-d} \times (GL_1)^d, GL_n)$ highest weight vectors in $\mathcal{P}(M_{n,n})$. The relations between the $\tilde{\delta}_{(c,S)}$ are a consequence of the straightening rules of §5.4.1.7. The monomials in the $\tilde{\delta}_{(c,S)}$ which are standard monomials in the sense of §5.4.1.5 form a basis for the simultaneous $(GL_{n-d} \times (GL_1)^d, GL_n)$ highest weight vectors in $\mathcal{P}(M_{n,n})$.*

When $d = 1$, this result is exactly what we want and gives the Branching Law from GL_n to $GL_{n-1} \times GL_1$, and reciprocally, the Pieri Rule. When $d > 1$, to pass from this result to a description of the $GL_{n-d} \times GL_d$ highest weight vectors, we must reconstitute the GL_d highest weight vectors out of the set of all $(GL_1)^d$ weight vectors obtained as monomials in the $\tilde{\delta}_{(c,S)}$. To see how this might be done, consider first the case when S contains only one element. This gives us the elements $\tilde{\delta}_{(c,\{s_1\})} = \tilde{\delta}_{(c,s_1)}$ to work with. If we fix c and let s_1 vary, the $\tilde{\delta}_{(c,s_1)}$ span a GL_d -module isomorphic to $\mathbb{C}^d$. The span of all of these, as c varies from 0 to $n-d$, gives us a GL_d -module isomorphic to $(\mathbb{C}^d)^{n-d+1}$. By virtue of (GL_n, GL_m) -duality, we know that the GL_d highest weight vectors in the symmetric algebra on $(\mathbb{C}^d)^{n-d+1}$ look like. They are generated as algebra by the sum of GL_{n-d+1} -modules isomorphic to $\Lambda^k(\mathbb{C}^{n-d+1})$, for $k \leq d$. The basis for the copy of $\Lambda^k(\mathbb{C}^{n-d+1})$ in $S((\mathbb{C}^d)^{n-d+1})$ consists of the (determinants of the) $k \times k$ minors of the matrix

$$(5.6.4.6) \quad \begin{vmatrix} \tilde{\delta}_{(0,1)} & \tilde{\delta}_{(1,1)} & \tilde{\delta}_{(2,1)} & \cdots & \tilde{\delta}_{(n-d+1,1)} \\ \tilde{\delta}_{(0,2)} & \tilde{\delta}_{(1,2)} & \tilde{\delta}_{(2,2)} & \cdots & \tilde{\delta}_{(n-d+1,2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \tilde{\delta}_{(0,k)} & \tilde{\delta}_{(1,k)} & \tilde{\delta}_{(2,k)} & \cdots & \tilde{\delta}_{(n-d+1,k)} \end{vmatrix}$$

The theory of standard monomials and the straightening rules give a basis for the algebra generated by these generators and the relations between them. Thus, this part of the algebra of $GL_{n-d+1} \times GL_d$ highest weight vectors looks just like the coordinate ring of the flag manifold $Q_d \backslash GL_{n-d+1}$, where Q_d here is the parabolic

a set $S = \{s_1, s_2, \dots, s_k\}$ of cardinality $\#(S) = k$.

$$\begin{aligned}
 (5.6.4.10) \quad & \det \begin{vmatrix} \delta_{(a,s_1)} & \delta_{(a,s_2)} & \delta_{(a,s_k)} \\ \delta_{(a+1,s_1)} & \delta_{(a+2,s_2)} & \cdot \\ \delta_{(a+k-1,s_1)} & \cdot & \delta_{(a+k-1,s_k)} \end{vmatrix} \\
 &= \det \begin{vmatrix} X_{a+1,a} & 0 & \cdot & \cdot & 0 & Y_{a+1,s} \\ 0 & X_{a+2,a+1} & 0 & \cdot & 0 & Y_{a+2,s} \\ \cdot & 0 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & \cdot & 0 & X_{a+k,a+k-1} & Y_{a+k,s} \end{vmatrix} \\
 &= \left(\prod_{j=1}^k \det |X_{a+j,a+j-1}| \right) \det |X_{a+k,a} Y_{a+k,s}|.
 \end{aligned}$$

Identity (5.6.4.10) supports the following statement about the structure of $\mathcal{P}(M_{n,n})^{U_{n-d} \times U_d \times U_n}$. We recall that if R is a ring and $x \in R$ is not a zero divisor, then one can embed R in a larger ring in $R_{(x)}$ which x is invertible. This larger ring is called the localization of R at x [Ja 2], [La]. One can likewise localize at any set L of elements which are not zero divisors.

5.6.4.11: THEOREM. The algebra $\mathcal{P}(M_{n,n})^{U_{n-d} \times U_d \times U_n} \{ \det |X_{a,a}| \}$ of simultaneous $GL_{n-d} \times GL_d \times GL_n$ highest weight vectors in $\mathcal{P}(M_{n,n})$, localized at the elements $\det |X_{a,a}|$, is generated by the functions $\delta_{(a,1)} = \det |X_{a+1,a} Y_{a+1,1}|$ (in addition to the $\det |X_{a,a}|$ and their inverses). The localized algebra is isomorphic to the tensor product of the algebras $\mathcal{P}(M_{n-d,n-d})^{U_{n-d} \times U_d \times U_n} \{ \det |X_{a,a}|^{\pm 1} \}$ and $\mathcal{P}(M_{d,n-d+1})^{U_d}$. The isomorphism is realized by taking the functions $\delta_{(a,b)}$, for $0 \leq a \leq n-d$ and $1 \leq b \leq d$, to be the coordinates of $M_{d,n-d+1}$. Concretely, the copy of $\mathcal{P}(M_{d,n-d+1})^{U_d}$ inside $\mathcal{P}(M_{n,n})^{U_{n-d} \times U_d \times U_n}$ is generated by the maximal minors of the matrices (5.6.4.6), for $1 \leq k \leq d$.

5.6.4.12: COROLLARY. The algebra $\mathcal{P}(M_{n,n})^{U_{n-d} \times U_d \times U_n} \{ \det |X_{a,a}| \}$ has a basis over $\mathcal{P}(M_{n-d,n-d})^{U_{n-d} \times U_d \times U_n} \{ \det |X_{a,a}| \}$ consisting of the standard monomials in the maximal minors of matrices (5.6.4.6), for $1 \leq k \leq d$.

The corollary follows because, as emphasized by Towber [Tow], Kung-Rota [KuRo], and others, the fact that the straightening rules of §5.4.1.7 have only ± 1 for coefficients implies that the standard monomials are actually a basis for the algebra over $\mathbb{Z}$ generated by the $\delta_{(R)}$ of §5.4.1, and therefore they will be a basis when any other ring is used for coefficients. See also Remark 5.4.1.10.

5.6.4.13: Remarks: a) These results are the algebraic counterpart of (and they imply) the well-known fact that, in a tensor product $U \otimes V$, if the highest weight of U is sufficiently non-degenerate relative to the highest weight of V , then the

irreducible components of $U \otimes V$ are in one-to-one correspondence with the weight vectors of V .

b) The statement of these results are easily translated into statements about tensor products of general reductive groups. It is natural to wonder whether they might not hold in that context. P. Littelmann has sketched to me orally a geometric argument that this is so.

Theorem 5.6.4.11 implies, in particular, that the maximal minors of the matrices (5.6.4.6), together with the $\det |X_{a,a}|$, generate the field of quotients of $\mathcal{P}(M_{n,n})^{U_{n-d} \times U_d \times U_n}$. However, they do so rather inefficiently, for the dimension of $U_n \setminus GL_n / (U_{n-d} \times U_d)$ is only $d(n-d+1)$, and the number of minors of matrices (5.6.4.6) is far larger than that. With the help of theorem 5.6.4.11, we can find certain of the generators which themselves generate a polynomial algebra, but whose field of quotients is the full field of quotients of $\mathcal{P}(M_{n,n})^{U_{n-d} \times U_d \times U_n}$. Producing such a set exhibits the variety $U_n \setminus GL_n / (U_{n-d} \times U_d)$ as a rational variety, but the fact that it is a rational variety can be seen directly using the Bruhat decomposition.

5.6.4.14: THEOREM. The polynomials $\det |X_{a+b,a} Y_{a+b,b}|$ for $0 \leq a \leq n-d$ and $0 \leq b \leq d-1$, are algebraically independent, and they generate the field of quotients of $\mathcal{P}(M_{n,n})^{U_{n-d} \times U_d \times U_n}$.

PROOF. Since we have the $\det |X_{a,a}|$ in our set, we can use Theorem 5.6.4.11, which means we only have to show we can express the maximal minors of the matrices (5.6.4.6) as rational functions in the given set. We can use Theorem 5.6.4.11 to translate the desired result to a statement about the coordinate ring of the flag manifold. The translated result is the following, which may have independent interest (and may be classical).

5.6.4.15: THEOREM. Consider the coordinate ring of the flag manifold, generated by the minors $\det |X_{a,s}|$ of an $n \times n$ matrix $X = [x_{jk}]$, where S runs over all subsets of $\{1, 2, 3, \dots, n\}$ of cardinality a . The field of quotients of this ring is generated by the minors $\det |X_{a,S_b}|$ where $S_b = \{b, b+1, b+2, \dots, b+a-1\}$, for $1 \leq b \leq n-a+1$, is a set of consecutive indices.

PROOF. We prove this by induction on a . We sketch the argument in the case of $a=3$, which should suffice to enable the reader to fill in the rest.

We suppose that we have expressed all 2×2 minors $\det \begin{vmatrix} x_{1,c} & x_{1,d} \\ x_{2,c} & x_{2,d} \end{vmatrix}$ as rational functions of the $\det \begin{vmatrix} x_{1,b} & x_{1,b+1} \\ x_{2,b} & x_{2,b+1} \end{vmatrix}$. Consider the 4×4 determinant

$$\det \begin{vmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \end{vmatrix}$$

This clearly equals zero. Expanding the determinant along the first row, we get the equation

$$x_{11}Z_{234} + x_{12}Z_{134} + x_{13}Z_{124} + x_{14}Z_{123} = 0,$$

where, up to sign, Z_{abc} is the determinant of the 3×3 submatrix made of the last 3 rows and the indicated columns. Replacing the first row with $\{x_{21}, x_{22}, x_{23}, x_{24}\}$ gives us a similar equation. We are given the determinants Z_{234} and Z_{123} in our original list of generators. Solving for the other two determinants gives the equations

$$\begin{aligned} x_{12}Z_{134} + x_{13}Z_{124} &= -x_{11}Z_{234} - x_{14}Z_{123} \\ x_{22}Z_{134} + x_{23}Z_{124} &= -x_{21}Z_{234} - x_{24}Z_{123}. \end{aligned}$$

Solving these equations by Cramer's Rule gives the formula

$$\begin{aligned} Z_{134} &= - \left(\frac{\det \begin{vmatrix} x_{11} & x_{13} & x_{14} \\ x_{21} & x_{23} & x_{24} \end{vmatrix}}{\det \begin{vmatrix} x_{12} & x_{13} \\ x_{22} & x_{23} \end{vmatrix}} Z_{234} - \det \begin{vmatrix} x_{13} & x_{14} \\ x_{23} & x_{24} \end{vmatrix} Z_{123} \right) \\ Z_{124} &= - \left(-\det \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix} Z_{234} + \det \begin{vmatrix} x_{12} & x_{14} \\ x_{22} & x_{24} \end{vmatrix} Z_{123} \right) \end{aligned}$$

These are formulas of the desired type.

5.6.5: Tensoring with modules of depth two

In this section, we specialize the considerations of §5.6.4 to the case when $d = 2$. For this case, the only generators besides the $\delta_{(c,a)}$, $a = 1$ or 2 , are the $\delta_{(c,\{1,2\})}$, which are eigenvectors of GL_2 . As we have noted, the $\delta_{(c,a)}$ for fixed c and $a = 1, 2$, span a GL_2 -module isomorphic to C^2 .

Thus, the algebra generated by the $\delta_{(c,a)}$ is some quotient of $\mathcal{P}((C^2)^m)$ for suitable m . From (GL_2, GL_m) -duality, we know that the GL_2 highest weight vectors in $\mathcal{P}((C^2)^m)$ are generated as algebra by the $\delta_{(c,1)}$ and by the determinants

$$(5.6.5.1) \quad \Delta(c_1, c_2) = \det \begin{vmatrix} \delta_{(c_1,1)} & \delta_{(c_1,2)} \\ \delta_{(c_2,1)} & \delta_{(c_2,2)} \end{vmatrix} = \det \begin{vmatrix} X_{c_1+1,c_1} & 0 \\ 0 & X_{c_2+1,c_2} \end{vmatrix} \begin{vmatrix} Y_{c_1+1,2} \\ Y_{c_2+1,2} \end{vmatrix}$$

Without loss of generality, we may take $c_2 \geq c_1$. If $c_2 = c_1$, then this determinant is equal to zero. If $c_2 = c_1 + 1$, then we have equation (5.6.4.9) which expresses $\Delta(c_1, c_2)$ in terms of $\delta_{(c_1+1)}$ and $\delta_{(c_1,\{1,2\})}$. So we need only consider the cases when $c_2 \geq c_1 + 2$.

Thus, we have the following generators for the algebra of $GL_{n-2} \times GL_2$ highest weight vectors in $\mathcal{P}((C^n)^n)$.

$$(5.6.5.2) \quad \tilde{\delta}_c = \tilde{\delta}_{(c,0)}, \tilde{\delta}_{(c,1)}, \tilde{\delta}_{(c,2)}, \text{ for } 0 \leq c \leq n-2$$

and

$$\Delta(c_1, c_2), \text{ for } 0 \leq c_1 \leq c_2 - 2 \leq n-4.$$

We can also describe the relations between these generators. We know from Theorem 5.6.4.14 that the $\tilde{\delta}_{(c,a)}$, for $a = 0, 1, 2$, are algebraically independent and that the $\Delta(c_1, c_2)$ are rational functions in the $\tilde{\delta}_{(c,a)}$. Thus, the relations for this algebra consist in expressing the $\Delta(c_1, c_2)$ as rational functions in the $\tilde{\delta}_{(c,a)}$, or in relations which permit such expressions. We can find such relations by using the "Lewis Carroll Identity" [Dod], [RR]. This is a formula for the determinant of an $n \times n$ matrix A in terms of the four $(n-1) \times (n-1)$ matrices occupying the "corners" of A , and the $(n-2) \times (n-2)$ matrix in the center of A . Specifically, let A_{11} be the submatrix of A formed by the first $n-1$ rows and $n-1$ columns of A . Let A_{12} be the submatrix formed by the first $n-1$ rows and the last $n-1$ columns. Define A_{21} and A_{22} similarly. Let $\tilde{A}$ denote the matrix formed by rows 2 through $n-2$, and columns 2 through $n-2$. Then the Lewis Carroll Identity says

$$(5.6.5.3) \quad (\det A)(\det \tilde{A}) = (\det A_{11})(\det A_{22}) - (\det A_{12})(\det A_{21}).$$

To use this identity, we rearrange the matrix whose determinant is $\Delta(c_1, c_2)$. We reverse the order of the first $c_1 + 1$ columns and the first c_1 rows. If we apply the Lewis Carroll Identity to the resulting matrix, we find the following relation

$$\begin{aligned} \Delta(c_1, c_2) \tilde{\delta}_{(c_2,0)} \tilde{\delta}_{(c_1-1,1)} &= \tilde{\delta}_{(c_2,0)} \tilde{\delta}_{(c_1,1)} \Delta(c_1 - 1, c_2) - \tilde{\delta}_{(c_2,0)} \tilde{\delta}_{(c_1-1,2)} \tilde{\delta}_{(c_1,0)} \tilde{\delta}_{(c_2,1)}. \end{aligned}$$

Factoring out of this equation the common term, we arrive at a relation

$$(5.6.5.4) \quad \Delta(c_1, c_2) \tilde{\delta}_{(c_1-1,1)} = \tilde{\delta}_{(c_1,1)} \Delta(c_1 - 1, c_2) - \tilde{\delta}_{(c_1-1,2)} \tilde{\delta}_{(c_1,0)} \tilde{\delta}_{(c_2,1)}.$$

We can also write this relation in the form

$$(5.6.5.5) \quad \frac{\Delta(c_1, c_2)}{\tilde{\delta}_{(c_1,1)}} = \frac{\Delta(c_1 - 1, c_2)}{\tilde{\delta}_{(c_1 - 1,1)}} - \frac{\tilde{\delta}_{(c_1,0)} \tilde{\delta}_{(c_1 - 1,2)} \tilde{\delta}_{(c_2,1)}}{\tilde{\delta}_{(c_1,1)} \tilde{\delta}_{(c_1 - 1,1)}}.$$

From this, one sees easily how to write $\Delta(c_1, c_2)$ as a rational function in the $\tilde{\delta}_{(c,a)}$. We summarize the preceding discussion.

5.6.5.6: PROPOSITION. *The algebra $\mathcal{P}(M_{n,n})^{U_{n-1} \times U_2 \times U_n}$ is generated by the $(n+2)(n-1)/2$ elements (5.6.5.2). Of these, the $3n-4$ elements $\tilde{\delta}_{(c,a)}$ for $0 \leq c \leq n-2$ and $0 \leq a \leq 2$ (and $ac > 0$) are algebraically independent and generate the fields of quotients of $\mathcal{P}(M_{n,n})^{U_{n-1} \times U_2 \times U_n}$. The remaining elements $\Delta(c_1, c_2)$ are rational expressions in the $\tilde{\delta}_{(c,a)}$, and are explicitly described by equations (5.6.5.4) or ((5.6.5.5)).*

If we take the case $n = 4$, we obtain an algebra with 9 generators:

$$(5.6.5.6) \quad \begin{aligned} \tilde{\delta}_{(1,0)} &= y_{11}, & \tilde{\delta}_{(1,1)} &= \begin{vmatrix} x_{11} & y_{11} \\ x_{21} & y_{21} \end{vmatrix}, & \tilde{\delta}_{(2,1)} &= \begin{vmatrix} x_{11} & x_{12} & y_{11} \\ x_{21} & x_{22} & y_{21} \\ x_{31} & x_{32} & y_{31} \end{vmatrix}, \\ \tilde{\delta}_{(2,0)} &= \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix}, \end{aligned}$$

$$\tilde{\delta}_{(0,2)} = \begin{vmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{vmatrix}, \quad \tilde{\delta}_{(1,2)} = \begin{vmatrix} x_{11} & y_{11} & y_{12} \\ x_{21} & y_{21} & y_{22} \\ x_{31} & y_{31} & y_{32} \end{vmatrix}, \quad \tilde{\delta}_{(2,2)} = \begin{vmatrix} x_{11} & x_{12} & y_{11} & y_{12} \\ x_{21} & x_{22} & y_{21} & y_{22} \\ x_{31} & x_{32} & y_{31} & y_{32} \\ x_{41} & x_{42} & y_{41} & y_{42} \end{vmatrix},$$

$$\Delta(0, 2) = \begin{vmatrix} 0 & 0 & y_{11} & y_{12} \\ x_{11} & x_{12} & y_{11} & y_{12} \\ x_{21} & x_{22} & y_{21} & y_{22} \\ x_{31} & x_{32} & y_{31} & y_{32} \end{vmatrix}.$$

Among these generators, there is one relation:

$$(5.6.5.7) \quad \Delta(0, 2)\tilde{\delta}_{(1,1)} = \tilde{\delta}_{(0,1)}\tilde{\delta}_{(2,0)}\tilde{\delta}_{(1,2)} + \tilde{\delta}_{(1,0)}\tilde{\delta}_{(0,2)}\tilde{\delta}_{(2,1)}.$$

This algebra describes the highest weight vectors in the tensor product of two representations whose diagrams have at most two rows. This covers all representations of GL_3 , up to twisting by determinant. However, for GL_3 , the row index should only go up to three, so the generator $\tilde{\delta}_{(2,2)}$ is excluded. Thus, the algebra $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ for $G = SL_3$ is described by the eight polynomials of (5.6.5.6), leaving out $\tilde{\delta}_{(2,2)}$ and subject to the relations (5.6.5.7). This description of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ is used in Appendix 6 to give an explicit description of the decomposition of tensor products of representations of SL_3 (or GL_3).

5.6.6: Tensoring with modules of depth 3

We can perform an analysis like that of the previous section for the case of $d = 3$. In this case, the algebra of simultaneous $(GL_n, GL_{n-3} \times (GL_1)^3)$ highest weight vectors is generated by the elements $\tilde{\delta}_{(c,S)}$, where $S \subseteq \{1, 2, 3\}$. If $\#(S) = 0$ or 3, then $\tilde{\delta}_{(c,S)}$ is already a $GL_{n-3} \times GL_3$ highest weight vector; indeed, it is a GL_3 -eigenvector. We have noted that the collection of $\tilde{\delta}_{(c,S)}$ with $\#(S) = 1$ span a GL_3 -module isomorphic to $n - 2$ copies of $\mathbb{C}^3$. Similarly, the $\tilde{\delta}_{(c,S)}$ with $\#(S) = 2$ span a GL_3 -module isomorphic to $n - 2$ copies of $\Lambda^2(\mathbb{C}^3)$, which is isomorphic to the twist by determinant of $(\mathbb{C}^3)^*$. Thus, the $\tilde{\delta}_{(c,S)}$ with $\#(S) = 1$ or 2 span a space which as SL_3 -module (i.e., ignoring twists by det) is isomorphic to $(\mathbb{C}^3 \oplus (\mathbb{C}^3)^*)^{n-2}$. The way to construct the SL_3 , or GL_3 , highest weight vectors in the symmetric algebra on this module is described in §2.5. We describe the results of this construction in the current situation.

The elements $\Delta(c_1, c_2)$ constructed in §5.6.5 may be generalized in various ways. We will need several of these extensions to discuss tensoring with three-rowed diagrams. The first family of extensions are

$$(5.6.6.1) \quad \Delta(c_1, c_2', k_1, k_2) = \det \begin{vmatrix} X_{c_1+k_1, c_1} & 0 & Y_{c_1+k_1, k_1+k_2} \\ 0 & X_{c_2+k_2, c_2} & Y_{c_2+k_2, k_1+k_2} \end{vmatrix}.$$

Evidently the elements here denoted $\Delta(c_1, c_2, 1, 1)$ coincide with the elements $\Delta(c_1, c_2)$ of §5.6.5. We can further define elements

$$(5.6.6.2) \quad \Delta(c_1, c_2, c_3, k_1, k_2, k_3) = \det \begin{vmatrix} X_{c_1+k_1, k_1} & 0 & 0 & Y_{c_1+k_1, \ell} \\ 0 & X_{c_2+k_2, k_2} & 0 & Y_{c_2+k_2, \ell} \\ 0 & 0 & X_{c_3+k_3, k_3} & Y_{c_3+k_3, \ell} \end{vmatrix},$$

where $\ell = k_1 + k_2 + k_3$. There is evidently a pattern here which could be continued, but for our current purpose it will suffice to stop here. Another generalization of the $\Delta(c_1, c_2, k_1, k_2)$ are the elements

$$(5.6.6.3) \quad \Delta(c_1, c_2, d; k_1, k_2) = \det \begin{vmatrix} X_{e_1, c_1} Y_{e_1, d} & 0 & Y_{e_1, \{d'\}} \\ 0 & X_{e_2, c_2} Y_{e_2, d} & Y_{e_2, \{d'\}} \end{vmatrix}$$

Here $e_1 = c_1 + d + k_1$, and $e_2 = c_2 + d + k_2$, and $d' = k_1 + k_2$. The matrix $Y_{e_i, \{d'\}}$ means the e_i by d' matrix of ys , made from the first e_i rows and the d' columns from $d+1$ to $d+d'$. We could construct a common generalization of the $\Delta(c_1, c_2, d; k_1, k_2)$ and the $\Delta(c_1, c_2, c_3, k_1, k_2, k_3)$, but we do not need this construction for the present purpose.

We need one other type of generator. It is similar to the ones already described, but has a slightly different flavor. It is given by the formula

$$(5.6.6.4) \quad E(\ell_1, \ell_2, \ell_3) = \det \begin{vmatrix} X_{\ell_1+2, \ell_1} & 0 & 0 & Y_{\ell_1+2, \{1,2\}} & Y_{\ell_1+2, \{3,1\}} & 0 \\ 0 & X_{\ell_2+2, \ell_2} & 0 & Y_{\ell_2+2, \{1,2\}} & 0 & Y_{\ell_2+2, \{2,3\}} \\ 0 & 0 & X_{\ell_3+2, \ell_3} & 0 & Y_{\ell_3+2, \{3,1\}} & Y_{\ell_3+2, \{2,3\}} \end{vmatrix} = \det \begin{vmatrix} X_{\ell_1+2, \ell_1} & 0 & 0 & Y_{\ell_1+2, 3} & 0 & 0 \\ 0 & X_{\ell_2+2, \ell_2} & 0 & 0 & Y_{\ell_2+2, 3} & 0 \\ 0 & 0 & X_{\ell_3+2, \ell_3} & Y_{\ell_3+2, 3} & Y_{\ell_3+2, 3} & 0 \end{vmatrix}.$$

According to §2.5.5, a collection of generators for the GL_3 highest weight vectors in $\mathcal{P}((\mathbb{C}^3 \oplus \mathbb{C}^{3*})^n)$ is constituted by the sum of the $GL_n \times GL_n$ modules generated by the functions $\delta_1, \delta_2, \delta_3$, and $\tilde{\delta}_1, \tilde{\delta}_2, \tilde{\delta}_3$ of the formula at the end of §2.1.4 and formula (2.5.5.1), together with the generators for the GL_3 invariants, i.e., the inner products between the various copies of $\mathbb{C}^3$ and $\mathbb{C}^{3*}$. We have noted in §5.6.4 that the polynomials which we have denoted there by $\tilde{\delta}_{(c,1)}$ correspond to the elements in the GL_n module generated by δ_1 . Likewise, the elements of the current set of generators corresponding to the elements of the GL_n module generated by δ_2 are the $\Delta(c_1, c_2, 1, 1)$ of formula (5.6.6.1). Similarly, it is not hard to convince oneself that, of the current generators, the ones corresponding to elements in the GL_n module generated by δ_3 are the $\Delta(c_1, c_2, c_3, 1, 1, 1)$ of formula (4.6.6.2). The elements associated to $\tilde{\delta}_1$ are the $\tilde{\delta}_{(c,2)}$. Those associated to $\tilde{\delta}_2$ are the elements $\Delta(c_1, c_2, 1, 1)$. Those associated to $\tilde{\delta}_3$ are the elements $E(\ell_1, \ell_2, \ell_3)$. Finally, there are elements corresponding to the GL_n -invariants. These are the $\Delta(c_1, c_2, 1, 2)$. To all these generators we should add the $\tilde{\delta}_{(c,0)}$ and the $\tilde{\delta}_{(c,3)}$ to get a complete set of generators for $\mathcal{R}((U \setminus GL_n \times (Q_3^1 \setminus GL_n))^U)$.

The elements $\tilde{\delta}_{(c,j)}$, for $j = 0, 1, 2, 3$, generate the field of fractions of our ring $\mathcal{R}((U \setminus GL_n \times (Q_3^1 \setminus GL_n))^U)$. The relations in the ring can therefore be described by formulas giving the remaining generators as rational functions in the $\tilde{\delta}_{(c,j)}$ or permitting reduction to such formulas. We have seen in formulas (5.6.5.3) through (5.6.5.5) how to express the $\Delta(c_1, c_2, 1, 1)$ as rational functions in the $\tilde{\delta}_{(c,j)}$. A similar procedure, again relying on the Lewis Carroll Identity (5.6.5.3), also gives the

desired expressions for the other generators. Two types of identities are particularly worth noting, because they allow us to cut down on the number of generators. One, which we have already taken advantage of in §5.6.5, is that if $c_1 = c_2$, then $\Delta(c_1, c_2; 1, 1) = 0$. Essentially the same fact is that $\delta(c_1, c_2; 1, 1) = 0$ if $c_1 = c_2$. Also, $\Delta(c_1, c_2; 1, 2) = 0$ if $c_2 = c_1$ or $c_2 = c_1 - 1$. (Note that while for $\Delta(c_1, c_2; 1, 1)$ we can assume $c_1 \leq c_2$, since we have $\Delta(c_1, c_2; 1, 1) = \pm \Delta(c_2, c_1; 1, 1)$.) The same symmetry between c_1 and c_2 holds for $\Delta(c_1, c_2; 1, 1)$, but clearly does not hold for $\Delta(c_2, c_1; 1, 2) = \Delta(c_1, c_2; 2, 1)$. With the $E(\ell_1, \ell_2, \ell_3)$, one can assume that $\ell_1 \leq \ell_2 \leq \ell_3$. Further, if any two of them coincide, then $E(\ell_1, \ell_2, \ell_3) = 0$, so one may as well take $\ell_1 < \ell_2 < \ell_3$. Further, equation (5.6.4.9) shows that if $c_2 = c_1 + 1$, then $\Delta(c_1, c_2; 1, 1)$ is a product of the $\delta_{(c,j)}$. Hence we may as well take $c_2 \geq c_1 + 2$ in our list of generators. Similar reasoning applies to the $\Delta(c_1, c_2; 1, 1)$, $\Delta(c_1, c_2; 1, 2)$, and $\Delta(c_1, c_2; 1, 1)$. For the $E(\ell_1, \ell_2, \ell_3)$, we may assume that $\ell_1 < \ell_2 < \ell_3$, and further that $\ell_3 \geq \ell_1 + 3$.

We will not exhibit explicitly the full list of generators and relations for $\mathcal{R}((U \setminus GL_n) \times (Q_3^1 \setminus GL_n))^U$. However, by specializing to $c_j \leq 3$ and $\ell_j \leq 3$, as well as limiting ourselves to use of variables x_{jk} or y_{jk} with $j \leq 4$, we can obtain the generators of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ for $g = SL_4$. This is an algebra of dimension 12, with 18 generators, which we exhibit below.

5.6.6.5: Generators for the algebra $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ for $G = SL_4$,

$\delta_{(c,a)}$ for $0 \leq c, a \leq 3$, and $1 \leq c + a \leq 4$

$$\Delta(0, 2; 1, 1) = \det \begin{vmatrix} 0 & 0 & y_{11} & y_{12} \\ x_{11} & x_{12} & y_{11} & y_{12} \\ x_{21} & x_{22} & y_{21} & y_{22} \\ x_{31} & x_{32} & y_{31} & y_{32} \end{vmatrix};$$

$$\Delta(0, 3; 2, 1) = \det \begin{vmatrix} 0 & 0 & 0 & y_{11} & y_{12} & y_{13} \\ 0 & 0 & 0 & y_{21} & y_{22} & y_{23} \\ x_{11} & x_{12} & x_{13} & y_{11} & y_{12} & y_{13} \\ x_{21} & x_{22} & x_{23} & y_{21} & y_{22} & y_{23} \\ x_{31} & x_{32} & x_{33} & y_{31} & y_{32} & y_{33} \\ x_{41} & x_{42} & x_{43} & y_{41} & y_{42} & y_{43} \end{vmatrix};$$

$$\Delta(0, 3; 1, 1) = \det \begin{vmatrix} 0 & 0 & 0 & y_{11} & y_{12} \\ x_{11} & x_{12} & x_{13} & y_{11} & y_{12} \\ x_{21} & x_{22} & x_{23} & y_{21} & y_{22} \\ x_{31} & x_{32} & x_{33} & y_{31} & y_{32} \\ x_{41} & x_{42} & x_{43} & y_{41} & y_{42} \end{vmatrix};$$

$$\Delta(0, 2; 1, 2) = \det \begin{vmatrix} 0 & 0 & y_{11} & y_{12} & y_{13} \\ x_{11} & x_{12} & y_{11} & y_{12} & y_{13} \\ x_{21} & x_{22} & y_{21} & y_{22} & y_{23} \\ x_{31} & x_{32} & y_{31} & y_{32} & y_{33} \\ x_{41} & x_{42} & y_{41} & y_{42} & y_{43} \end{vmatrix};$$

$$\Delta(1, 2; 1, 1) = \det \begin{vmatrix} x_{11} & 0 & 0 & 0 & y_{11} & y_{12} \\ x_{21} & 0 & 0 & 0 & y_{21} & y_{22} \\ 0 & x_{11} & x_{12} & x_{13} & y_{11} & y_{12} \\ 0 & x_{21} & x_{22} & x_{23} & y_{21} & y_{22} \\ 0 & x_{31} & x_{32} & x_{33} & y_{31} & y_{32} \\ 0 & x_{41} & x_{42} & x_{43} & y_{41} & y_{42} \end{vmatrix}; \text{ and}$$

$$\Delta(0, 2; 1, 1) = \det \begin{vmatrix} x_{11} & x_{12} & 0 & y_{11} & y_{12} & y_{13} \\ x_{21} & x_{22} & 0 & y_{21} & y_{22} & y_{23} \\ x_{31} & x_{32} & 0 & y_{31} & y_{32} & y_{33} \\ x_{41} & x_{42} & 0 & y_{41} & y_{42} & y_{43} \\ x_{11} & x_{12} & 0 & 0 & y_{11} & y_{12} \\ x_{21} & x_{22} & 0 & 0 & y_{21} & y_{22} \\ x_{11} & x_{12} & y_{11} & y_{12} & y_{13} & 0 \\ x_{21} & x_{22} & y_{21} & y_{22} & y_{23} & 0 \\ x_{31} & x_{32} & y_{31} & y_{32} & y_{33} & 0 \\ x_{41} & x_{42} & y_{41} & y_{42} & y_{43} & 0 \end{vmatrix} = \det$$

Note that this set of generators is symmetric under interchange of x and y , as it should be, although this symmetry is not completely obvious from the form in which we write our generators, and is not reflected in our notation. The 12 generators $\delta_{(c,a)}$ generate the field of fractions of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$, and the other six generators are rational functions in these 12. The expressions of the last six generators in terms of the first 12 may be found by repeated use of the Lewis Carroll Identity (5.6.5.3). We give these expressions below. Note that the first equation of (5.6.6.6) is the same as formula (5.6.5.7).

$$\Delta(0, 2; 1, 1)\delta_{(1,1)} = \delta_{(0,1)}\delta_{(2,0)}\delta_{(1,2)} + \delta_{(1,0)}\delta_{(0,2)}\delta_{(2,1)}$$

$$\Delta(1, 2; 1, 1)\delta_{(2,1)} = \delta_{(1,1)}\delta_{(3,0)}\delta_{(2,2)} + \delta_{(2,0)}\delta_{(1,2)}\delta_{(3,1)}$$

$$\Delta(0, 2; 1, 1)\delta_{(1,2)} = \delta_{(0,2)}\delta_{(2,1)}\delta_{(1,3)} + \delta_{(1,1)}\delta_{(0,3)}\delta_{(2,2)}$$

$$\begin{aligned} \Delta(0, 3; 1, 1)\delta_{(2,1)}\delta_{(1,1)} &= \delta_{(0,1)}\delta_{(1,1)}\delta_{(3,0)}\delta_{(2,2)} \\ &+ \delta_{(0,1)}\delta_{(2,0)}\delta_{(1,2)}\delta_{(3,1)} \\ &+ \delta_{(1,0)}\delta_{(0,2)}\delta_{(2,1)}\delta_{(3,1)} \end{aligned}$$

$$\begin{aligned} \Delta(0, 2; 1, 2)\delta_{(1,2)}\delta_{(1,1)} &= \delta_{(0,1)}\delta_{(2,0)}\delta_{(1,2)}\delta_{(1,3)} \\ &+ \delta_{(1,0)}\delta_{(0,2)}\delta_{(2,1)}\delta_{(1,3)} \\ &+ \delta_{(1,0)}\delta_{(1,1)}\delta_{(0,3)}\delta_{(2,2)} \end{aligned}$$

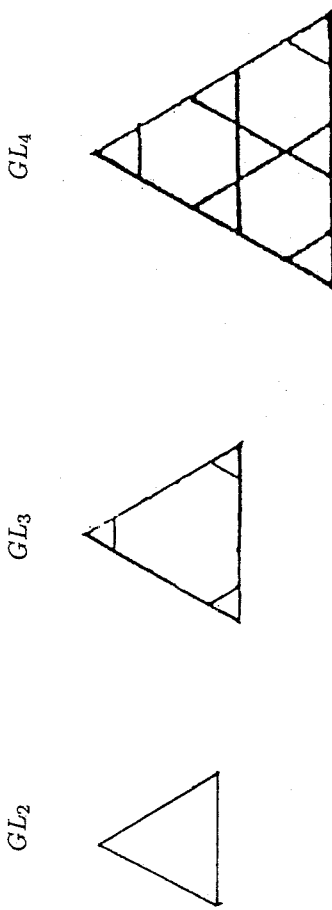
$$\begin{aligned}\Delta(0, 3, 2, 1)\tilde{\delta}_{(2,1)}\tilde{\delta}_{(1,2)} &= \tilde{\delta}_{(0,2)}\tilde{\delta}_{(2,1)}\tilde{\delta}_{(3,0)}\tilde{\delta}_{(1,3)} \\ &+ \tilde{\delta}_{(2,0)}\tilde{\delta}_{(1,2)}\tilde{\delta}_{(0,3)}\tilde{\delta}_{(3,1)} \\ &+ \tilde{\delta}_{(1,1)}\tilde{\delta}_{(3,0)}\tilde{\delta}_{(0,2)}\tilde{\delta}_{(2,2)}.\end{aligned}$$

It would be interesting to use this collection of generators and relations to perform the same kind of calculation for GL_4 as is done in Appendix 6 for GL_3 .

5.6.7: Comparison with Berenstein-Zelevinsky

The most familiar combinatorial description of multiplicities in tensor products of representations of GL_n is the Littlewood-Richardson Rule [JK], [Ma], [Sag]. In [BZ], A. Berenstein (as distinguished from J. Bernstein) and A. Zelevinsky give an alternative description. They formulate it in terms of "triple multiplicities", as described in §5.6.2. The multiplicities are described as the number of integral points in certain convex sets. The convex sets are defined by certain linear inequalities and equations derived from some planar arrays of hexagons and triangles. For example, for GL_2 , GL_3 and GL_4 , one considers the configurations illustrated in Figure 5.6.7.1.

Figure 5.6.7.1: BZ diagrams for $GL_{2,3,4}$



configuration of integers a triple of highest weights of SL_n . The triple multiplicity attached to a given triple of highest weights is the number of BZ configurations whose associated triple is the given one. The set of BZ configurations is clearly the set of integral points in a convex cone, and the set of BZ configurations with given associated triple is clearly a convex subset of this cone.

The integral cone of BZ configurations is a finitely generated semigroup, which we will call the *BZ cone*. It is generated by its so-called *indecomposable* or *primitive* elements, the elements which are not expressible as the sum of other BZ configurations. For the cases $n = 2, 3, 4$, we exhibit the primitive BZ configurations on Figure 5.6.7.4. To aid in the visualization of these configurations, we associate to them paths. The restriction of a BZ configuration to any hexagon will be a sum of the five arrays illustrated in Figure 5.6.7.2. To each of these arrays we can attach a path, illustrated in Figure 5.6.7.3. By joining the paths for the hexagons in the relevant array, we attach paths to the primitive BZ configurations for $n = 2, 3, 4$. These paths are what is illustrated in Figure 5.6.7.4.

Figure 5.6.7.2: Primitive BZ configurations on one hexagon

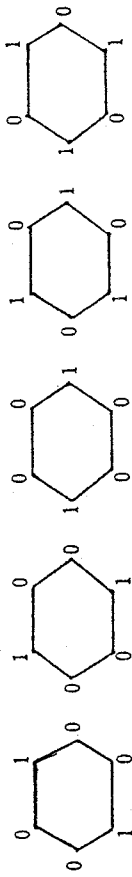
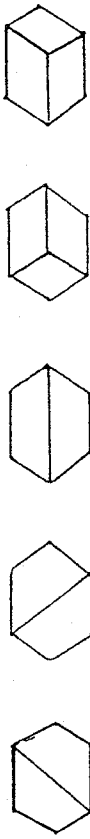


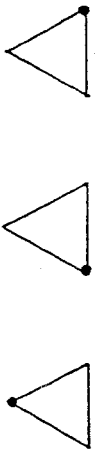
Figure 5.6.7.3 Paths associated to primitive hexagonal BZ configurations



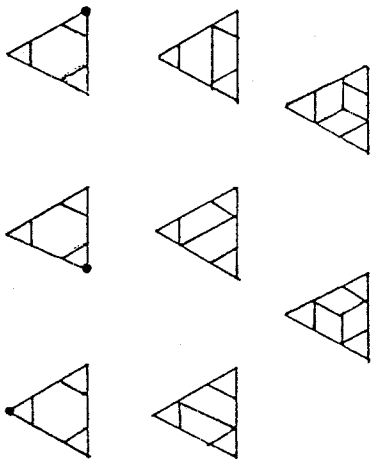
Using these configurations, one constructs certain arrays of integers: one puts a non-negative integer at each intersection of lines in the configuration, requiring that each pair of integers attached to the ends of a side of a hexagon should have the same sum as the pair attached to the opposite side. Call these *BZ configurations*. The way a highest weight is attached to a BZ configuration is as follows. One chooses an orientation for the triangle, say clockwise. One labels the small triangles on each side of the boundary of the big triangle from one to $n - 1$, following the orientation of the big triangle. Attach to each side of the triangle the highest weight of SL_n whose coefficients with respect to the k -th fundamental weight is the sum of the two integers at the vertices of the k -th triangle. In this way, we attach to each BZ

Figure 5.6.7.4: Primitive BZ configurations for $n = 2, 3, 4$

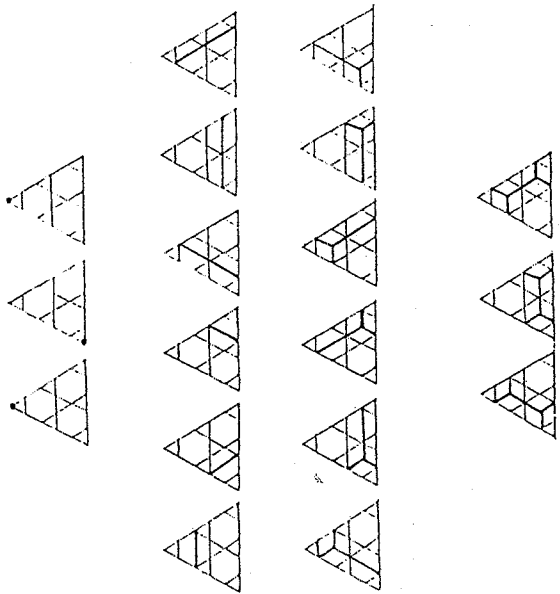
$n = 2$:



$n = 3$:



$n = 4$:

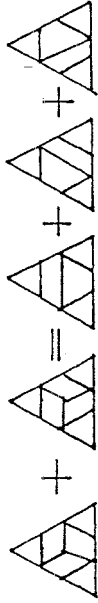


If we compare the paths of Figure 5.6.7.4 with the generators of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ for $G = SL_2, SL_3, SL_4$, we see there is a fairly evident one-to-one correspondence, which is consistent with the triple $(\hat{A}_N)^+$ grading on each set. We see further that the correspondence between generators of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ and generators of the BZ cone has some pleasant features. First, we can attach to each primitive BZ configuration illustrated in Figure 5.6.7.4 a number, which is the sum of the integers attached to the triangles containing the ends of the path of the configuration. It is easy to check that this number must always be divisible by n (as in SL_n). We call the multiple of n , the degree of the configuration. We may observe that the configurations of degree one correspond exactly to the generators $\delta_{(a,e)}$ of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$; we recall that we know these elements in fact generate the field of fractions of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$. Further, the relations between the generators of the algebras $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ seem to parallel the relations between the generators of the BZ cones for $G = SL_n$, $n = 2, 3, 4$. Figure 5.6.7.5 illustrates this for the single relation in each case when $n = 3$.

These parallels in structure between the BZ cone and the algebras $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$, when $n = 2, 3$ and 4 , suggest the possibility that the BZ cone might tell us the structure of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ for any SL_n . However, this hope seems to be too naïve. We can argue that if the analogy noted for $n = 2, 3, 4$ holds for all n , then the algebras $\mathcal{R}((U \setminus GL_n) \times (Q_d^1 \setminus GL_n))^U$, discussed in §5.6.6.4 would correspond to BZ configurations based in strips of width $d - 1$, as illustrated in Figure 5.6.7.6. For $d = 1$ and $d = 2$, there is a very good correspondence between generators of $\mathcal{R}((U \setminus GL_n) \times (Q_d^1 \setminus GL_n))^U$ and generators of the BZ cone. Examples of primitive generators of the BZ cone for a strip of width 1 or 2 are illustrated in Figure 5.6.7.6. For $d = 2$ (i.e., a strip of width 1) can be matched very nicely with the generators of $\mathcal{R}((U \setminus GL_n) \times (Q_2^1 \setminus GL_n))^U$, and we have written beside each diagram its associated generator.

However, for $d = 3$, the BZ cone requires many more generators than does $\mathcal{R}((U \setminus GL_n) \times (Q_3^1 \setminus GL_n))^U$. In fact, the number of generators of the BZ cone is exponential in n . It may be described as a solution of a difference equation whose characteristic polynomial is $(\lambda(\lambda - 1)^2 - 1)(\lambda - 1)$. The largest root of this polynomial is approximately 1.755. A typical primitive generator of the BZ cone is illustrated in Figure 5.6.7.7. For $n = 4$, the situation is worse. The structure of the primitive generators of the BZ cone appears to be very complicated. At any rate, I do not understand it. It seems, however, that the number of generators is superexponential in n . Thus, it seems that for large n , the BZ cone provides an unnecessarily complicated description of the multiplicities of tensor products.

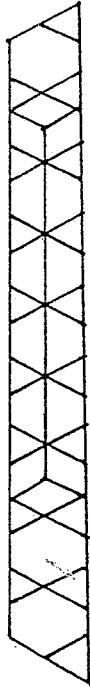
Figure 5.6.7.5: Relations among BZ configurations, and among generators of $\mathcal{R}((U \setminus G) \times (U \setminus G))^U$ for $n = 3$



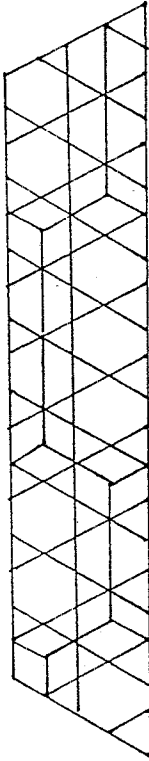
$$\Delta(0, 2)\tilde{\delta}_{(1,1)} = \tilde{\delta}_{(0,1)}\tilde{\delta}_{(2,0)}\tilde{\delta}_{(1,2)} + \tilde{\delta}_{(1,0)}\tilde{\delta}_{(0,2)}\tilde{\delta}_{(2,1)}$$

Figure 5.6.7.6: Primitive BZ configurations in strips (representative examples)

$d = 2$:



$d = 3$:



5.7: Reciprocity laws and stability properties.

Duality results such as Theorems 2.1.2, 2.4.3, 2.5.4, 3.5.2, 3.8.5.3, 3.8.9.3, 4.1.1, 4.3.4.2 and 4.3.4.4 give rise to a variety of reciprocity laws, of the same general nature as Frobenius Reciprocity for induced representations of groups. We will formulate some of these reciprocity laws here. As an application, we will discuss some of the “stability” results in representation theory which have been treated by a number of authors since the late 1970’s [BBL], [BKW], [Bry 1], [Han 1,2], [KE], [KP], [KT], [Wyb] (see [BBL] for more references.)

5.7.1: Reciprocity laws attached to see-saw pairs

One family of reciprocity laws arises by considering “see-saw” pairs [Hw 3], [Kud]. To formulate this idea, first consider the general shape of the duality theorems mentioned above. They involve a group G and a Lie algebra $\mathfrak{g}'$, both acting on a vector space Y in such a way that G and $\mathfrak{g}'$ generate each other’s commutants in a large subalgebra of $\text{End}(Y)$ (all of $\text{End}(Y)$ if Y is finite dimensional). (The Lie algebra can be exponentiated to a group action in many cases, most transparently in the case of (GL_n, GL_m) -duality, but for uniformity of discussion, we ignore this possibility.) Let us call a pair $(G, \mathfrak{g}')$ in this relation to each other a dual pair.

A major property of dual pairs, provided by the double commutant yoga, is that if we decompose Y as a sum of representations $\rho \otimes \rho'$, where ρ is an irreducible representation of G and ρ' is an irreducible representation of $\mathfrak{g}'$, then ρ determines ρ' and vice versa. Thus, $\rho \otimes \rho'$ is the full ρ -isotypic component for G and is also the full ρ' -isotypic component for $\mathfrak{g}'$. It is this over determination of $\rho \otimes \rho'$ that gives rise to the reciprocity laws. Suppose we have a pair $(G, \mathfrak{g}')$ and $(K, \mathfrak{k}')$ of dual pairs, such that $K \subset G$, and (hence) $\mathfrak{k}' \subset \mathfrak{g}'$. We write these inclusions one below the other:

$$K \subset G$$

$$(5.7.1.1)$$

$$\mathfrak{k}' \subset \mathfrak{g}'$$

In this figure, we find that the objects in the opposite corners form dual pairs. Meekly following [Kud], we will call this configuration a see-saw pair (although it is actually a pair of pairs).

The representations of G decompose into direct sums of K -submodules:

$$(5.7.1.2) \quad \rho \mid K \cong \sum_{\sigma} m_{\rho\sigma} \sigma,$$

where the sum is over the irreducible representations σ of K , and $m_{\rho\sigma}$ is the multiplicity with which σ occurs in (the restriction to K of) ρ . Similarly, the representations of $\mathfrak{k}'$ decompose into $\mathfrak{g}'$ submodules:

$$(5.7.1.3) \quad \sigma' \mid \mathfrak{g}' \cong \sum_{\rho'} m_{\sigma'\rho'} \rho'.$$

These latter sums may be infinite.

Consider the decomposition of Y as a $K \times g'$ -module. If $\sigma \in K^\wedge$ and $\rho' \in g'^\wedge$, we can compute the $\sigma \otimes \rho'$ -isotypic component $Y^{\sigma \otimes \rho'}$ of Y in two ways. We have

$$(5.7.1.4) \quad Y^{\sigma \otimes \rho'} = (Y^{\sigma'})^{\rho'} \cong (\sigma \otimes \sigma')^{\rho'} \cong \sigma \otimes (\sigma')^{\rho'} \cong \text{Hom}_{g'}(\rho', \sigma') \otimes (\sigma \otimes \rho')$$

and

$$Y^{\sigma \otimes \rho'} = (Y^{\rho'})^{\sigma} \cong (\rho \otimes \rho')^{\sigma} \cong \rho^{\sigma} \otimes \rho' \cong \text{Hom}_K(\sigma, \rho) \otimes (\sigma \otimes \rho').$$

Thus, we find

$$(5.7.1.5) \quad \text{Hom}_K(\sigma, \rho) \cong \text{Hom}_{K \times g'}(\sigma \otimes \rho', Y) \cong \text{Hom}_{g'}(\rho', \sigma').$$

In particular,

$$(5.7.1.6) \quad m'_{\sigma', \rho'} = m_{\rho\sigma'}$$

Frobenius Reciprocity can be proved by essentially the same argument.

5.7.2: Tensor product, restriction reciprocity and stability for tensor products

While the argument for the reciprocity law (5.7.1.5) is of a very general nature, some of its consequences for the dual pairs we are studying depend also on the structure of the system of all possible such pairs. This overall structure provides examples of several classes of reciprocity laws, in particular, some involving tensor products.

The simplest examples come from the pairs (GL_n, GL_m) , or (GL_n, gl_m) . These occur in both the symmetric and the skew-symmetric theory. Consider the symmetric case. The pair (GL_n, gl_m) acts on $\mathcal{P}(M_{n,m})$. The representations which occur are the $\rho_n^D \otimes \rho_m^D$, for diagrams D of depth at most $\min\{n, m\}$. On $\mathcal{P}(M_{n,m+\ell})$, we have the pairs $(GL_n, gl_{m+\ell})$ and $(GL_n \times GL_n, gl_m \times gl_\ell)$. They form a see-saw pair in the sense of §5.7.1. The second pair is the product of the pairs (GL_n, gl_m) and (GL_n, gl_ℓ) and depends on the isomorphism $\mathcal{P}(M_{n,m+\ell}) \cong \mathcal{P}(M_{n,m}) \otimes \mathcal{P}(M_{n,\ell})$. The embedding of GL_n in $GL_n \times GL_n$ is the diagonal embedding. The product algebra $gl_m \times gl_\ell$ embeds in $gl_{m+\ell}$ as a block diagonal algebra (a Levi component). Thus, the restriction multiplicities for the inclusion $GL_n \subset GL_n \times GL_n$ describe the decomposition of tensor products while the restriction multiplicities for $gl_m \times gl_\ell \subset gl_{m+\ell}$ describe restriction to Levi components. The reciprocity law (5.7.1.6) specializes to

$$(5.7.2.1) \quad m(\rho_n^D \otimes \rho_n^E, \rho_n^F) = m(\rho_{m+\ell}^F | GL_m \times GL_\ell, \rho_m^D \otimes \rho_\ell^E).$$

This is valid when D has depth at most $\min\{n, m\}$, E has depth at most $\min\{n, \ell\}$, and F has depth at most $\min\{n, m+\ell\}$. If n is large, then it poses no restrictions. We then see that the right hand side of this equation depends only on m and ℓ , not at all on n . Thus, for large n , the decomposition of the tensor product $\rho_m^D \otimes \rho_n^E$ is independent of n . This is the first stability result. The multiplicities on the left hand side of (5.7.2.1) are computed in terms of D, E and F by the combinatorial recipe known as *Littlewood-Richardson Rule* [JK], [Mac], [Sag]. Working with the exterior algebra instead of the symmetric algebra results in a formula like (5.7.2.1),

except that on the right hand side the transposed diagrams D^t, E^t and F^t appear in the place of D, E and F . Combining the two gives the invariance of tensor product multiplicities under transpose, corresponding to the well-known symmetry connecting the complete symmetric functions and the elementary symmetric functions [Mac].

As a second example, consider the dual pairs (Sp_{2n}, so_{2m}) acting on $\mathcal{P}(M_{2n,m})$. We have seen (cf. §3.8.5) that this situation gives rise to the labeling of representations of Sp_{2n} by diagrams. For a given τ_{2n}^D in $(Sp_{2n})^\wedge$ there is a unique (up to multiples) harmonic joint highest weight vector δ^D for $Sp_{2n} \times so_{2m}^{(1,1)}$ in the $\tau_{2n}^D \otimes (\tau_{2n}^D)'$ isotypic component in $\mathcal{P}(M_{2n,m})$. The notation $so_{2m}^{(1,1)}$ used here is consistent with §A2.2.6. Since $so_{2m}^{(1,1)}$ is normalized gl_m acting by its direct sum action on $\mathcal{P}(M_{2n,m})$, the representation of $so_{2m}^{(1,1)}$ generated by δ^D is $(det_m)^n \otimes \rho_m^D$. Thus, in contrast to the situation for the pairs (GL_n, gl_m) , the representation of so_{2m} corresponding to τ_{2n}^D is not independent of n , owing to the need to work with normalized gl_m . However, it varies in a relatively simple way as n varies. Now consider the dual pairs $(Sp_{2n}, so_{2(m+\ell)})$ and $(Sp_{2n} \times Sp_{2m}, so_{2m} \times so_{2\ell})$ acting on $\mathcal{P}(M_{2n,m+\ell})$. They form a see-saw pair. Again, the restriction multiplicities from $Sp_{2n} \times Sp_{2m}$ to Sp_{2n} describe the decomposition of tensor products for representations of Sp_{2n} . The product Lie algebra $so_{2m} \times so_{2\ell}$ embeds in $so_{2(m+\ell)}$ in the obvious way. We remark that the quotient space $SO_{a+b}/SO_a \times SO_b$ is a symmetric space.

The representations $(\tau_{2n}^D)'$ of so_{2m} which appear in $\mathcal{P}(M_{2n,m})$ are infinite dimensional. However, they are quite similar to finite-dimensional representations in many ways. In particular, they have vectors which are annihilated by the abelian subalgebra $so_{2m}^{(0,2)}$, and this space $\ker so_{2m}^{(0,2)}$ is an irreducible module for $so_{2m}^{(1,1)}$. It is called the lowest K -type. The lowest K -type determines the full representation of so_{2m} . These facts are all part of the (most elementary part of the) theory of Verma modules, which is now a popular approach to representation theory of semisimple Lie algebras [Hmp], [Jan]. We will label such a representation of so_{2m} by its lowest K -type. Thus, following the discussion above, we write

$$(5.7.2.2) \quad (\tau_{2n}^D)' = \tau'((det_m)^n \otimes \rho_m^D).$$

With this notation, we may write reciprocity law (5.7.1.6) for this case in the form

$$(5.7.2.3)$$

$$\begin{aligned} m(\tau_{2n}^D \otimes \tau_{2n}^E, \tau_{2n}^F) = \\ m(\tau'((det_{m+1})^n \otimes \rho_{m+\ell}^F | so_{2m} \times so_{2\ell}, \tau'((det_m)^n \otimes \rho_m^D) \otimes \tau'((det_\ell)^n \otimes \rho_\ell^E)). \end{aligned}$$

Now suppose n is large. The groups involved on the right are independent of n , but their representations are not. Nevertheless, these representations vary in a mild way. The separation of variables theorem, Theorem 3.8.6.4, says that when $n \geq m$, we have

$$(5.7.2.4) \quad \tau'((det_m)^n \otimes \rho_m^D | so_{2m}^{(1,1)}) \cong (det_m)^n \otimes \rho_m^D \otimes S(\Lambda^2(C^m)).$$

Given this and the knowledge that $\tau'((det_m)^n \otimes \rho_m^D)$ is determined by its restriction to $so_{2m}^{(1,1)}$, indeed by its lowest K -type, we see that the multiplicity on the right hand

side of equation (5.7.2.4) is independent of n . Thus, we have stability for tensor products for Sp_{2n} also:

$$(5.7.2.5) \quad \tau_{2n}^D \otimes \tau_{2n}^E \cong \sum_F \tilde{m}_{DE}^F \tau_{2n}^F,$$

where the coefficients are independent of n for large n . A similar result holds for the orthogonal groups, and also for the "mixed tensor models" of the representations of GL_n [BBL], [Bry 1].

There is also a reciprocity law for tensor products for Sp_{2n} in the skew-symmetric case. Here we have the see-saw pair consisting of the dual pairs $(Sp_{2n}, sp_{2(m+\ell)})$ and $(Sp_{2n} \times Sp_{2n}, sp_{2m} \times sp_{2\ell})$ acting on $\Lambda(C^{2n(m+\ell)})$. The representations of sp_{2m} , etc. corresponding to representations τ_{2n}^D of Sp_{2n} again vary in a regular way as n varies, but we are now dealing with ordinary finite dimensional representations. In the notation of §2.6.8.9, we have

$$(5.7.2.6) \quad \begin{aligned} m(\tau_{2n}^D \otimes \tau_{2n}^E, \tau_{2n}^F) &= \\ m(\tau_{2(m+\ell)}^{r_{n,m+\ell}(F^t)} | \mathfrak{so}_{2m}^{r_{n,m}(D^t)} \otimes \tau_{2\ell}^{r_{n,\ell}(E^t)}). \end{aligned}$$

As n increases, the highest weights $r_{n,m}(D^t)$, etc., vary in a regular way:

$$(5.7.2.7) \quad r_{n+1,m}(D^t) = r_{n,m}(D^t) + 1_m,$$

where 1_m is as in §1.2.7. Since we know that the left hand side of formula (5.7.2.6) is independent of n , the right hand side must be also. Thus, formula (5.7.2.7) is a statement about stability of certain restrictions as the highest weights go to infinity in a regular way. Results of this flavor are in the literature; I do not know a reference for this particular one.

5.7.2.8: Remark: Because for the dual pair (GL_n, GL_m) both members act via automorphisms on $\mathcal{P}(M_{n,m})$ (or if we take $(GL_n, \mathfrak{gl}_m)$), then $\mathfrak{gl}_m$ acts via derivations, the reciprocity laws take a particularly sharp form. In all cases, the reciprocity law (5.7.1.5) is "spatial" in the sense that it does not simply assert an equality of multiplicities, but an isomorphism of intertwining spaces. The theory of the highest weight lets us identify a space such as $Hom_K(\sigma, \rho)$ with the subspace of (the space of) ρ of highest weight vectors for a pre-selected Borel subgroup of K . When a group G acts on a variety X , then B -eigenvectors in $\mathcal{R}(X)$ form an algebra. As is explained in §5.6.1, we can describe the structure of tensor products by means of an appropriate graded algebra. We can describe restrictions similarly. The reciprocity law for the see-saw pair $(GL_n, \mathfrak{gl}_{m+\ell})$ and $(GL_n \times GL_n, \mathfrak{gl}_m \times \mathfrak{gl}_\ell)$ establishes not simply an equality of multiplicities, but an isomorphism of algebras between the algebra which describes the decomposition of tensor products $\rho_n^D \times \rho_n^E$, where D has depth at most m and E has depth at most ℓ , and the algebra which describes restrictions from $GL_{m+\ell}$ to $GL_m \times GL_\ell$.

5.7.3: Stability for restrictions and for multiplicities

By considering other see-saw pairs, we obtain reciprocity laws which describe restrictions from a classical group to certain subgroups. These also imply stability

laws for such restrictions. The simplest such stability law is the branching law from GL_n to GL_{n-1} which is discussed in detail in §5.4. A major feature of that discussion is the fact that the restriction can be described completely in terms of diagrams, with no mention of the size of n . Iteration of that branching law leads to the conclusion that if we fix diagrams D and E , then the multiplicity with which the weight ψ_E of the torus A_n occurs in ρ_n^D is independent of n (as long as n is large). As is shown in [BBL], for other series of classical groups, the weight multiplicities are not independent of the size of the group, but they depend polynomially on the rank. We can understand this also in terms of reciprocity laws for see-saw pairs, combined with a generalization of the stability result for multiplicities for GL_n from polynomial representations to representations described by the "mixed tensor" notation of §1.2.8.

Consider the inclusion $Sp_{2n} \subset GL_{2n}$. If we let these groups act on $M_{2n,m}$, they form part of the see-saw pair $(GL_{2n}, \mathfrak{gl}_m)$ and $(Sp_{2n}, \mathfrak{so}_{2m})$. By the same reasoning as in §5.7.2 (and keeping in mind that (GL_n, GL_m) -duality is usually stated in terms of unnormalized GL_m), we find that in the restriction

$$(5.7.3.1) \quad \rho_{2n}^D | Sp_{2n} \cong \sum_E \tilde{m}_{DE}^E \tau_{2n}^E,$$

the multiplicities $\tilde{m}_{D,E}$ are independent of n for n large ($n > 2m$). This allows one to create a theory of "stable characters" for Sp_{2n} , analogous to that for GL_n . A similar procedure works for orthogonal groups. This has been carried out in detail by Koike and Terada [KT], [Ko].

We can reverse the situation just considered and study the inclusion $GL_n \subset Sp_{2n}$. Here, GL_n is the subgroup of Sp_{2n} which preserves the coordinates $\{x_j\}$ described in §2.6.8.2, and also the coordinates $\{y_j\}$. As a module for GL_n , C^{2n} decomposes into $C^n \oplus C^{n*}$. Thus $M_{2n,m}$ can be viewed as $M_{n,m} \oplus M_{n,n}$ as GL_n -module, and its structure is described in §2.5. In particular, we find that we are dealing with a see-saw pair $(Sp_{2n}, \mathfrak{so}_{2m})$ and $(GL_n, \mathfrak{u}_{m,m})$, where the Lie algebra $\mathfrak{u}_{m,m}$ is described in §2.5.2. As abstract Lie algebra, $\mathfrak{u}_{m,m}$ is isomorphic to $\mathfrak{gl}_{2m}$, but it does not consist of linear vector fields. It has a three-term grading, of which the $(1,1)$ term consists of (normalized) linear vector fields, and is isomorphic to $\mathfrak{gl}_m \oplus \mathfrak{gl}_m$. We use the notation to $\mathfrak{u}_{m,m}$ to remind the reader of this structure.

At any rate, from the reciprocity law for this see-saw pair, together with the observations of §5.7.2 on the harmlessness of the variation caused by using normalized $\mathfrak{gl}_m$, tells us that we have the restriction law

$$(5.7.3.2) \quad \tau_{2n}^D | GL_n \cong \sum_{E,F} \tilde{m}_D^{(E,F)} \rho_n^{(E,F)},$$

where the multiplicities $\tilde{m}_D^{(E,F)}$ are independent of n for large n . Here, the representations $\rho_n^{(E,F)}$ of GL_n are described by the mixed tensor or double diagram notation (see §1.2.8 and §§2.5.3-2.5.4).

Since GL_n contains the standard Cartan subgroup of Sp_{2n} , we can compute weight multiplicities using the restriction law (5.7.3.2). This formula tells us that a description of weight multiplicities for Sp_{2n} as $n \rightarrow \infty$ involves studying weight multiplicities of mixed tensor representations $\rho_n^{(E,F)}$ of GL_n as $n \rightarrow \infty$.

Appendix 1: The Symmetric Space S_{2n}/D_n .

Let S_n be the symmetric group on n letters. Let D_n denote the subgroup of S_{2n} consisting of elements which stabilize the partition $\Pi_0 = \{\{1, 2\}, \{3, 4\}, \{5, 6\}, \dots, \{2n-1, 2n\}\}$ of $2n = \{1, 2, 3, \dots, 2n\}$. Let $C(S_{2n}/D_n)$ be the convolution algebra of D_n bi-invariant functions on S_{2n} . We want to analyze the structure of this convolution algebra, and to specify which representations of S_{2n} it "sees", i.e., which representations of S_{2n} contain fixed vectors for D_n (cf. [Gro]).

A1.1: THEOREM.

- a) The algebra $C(S_{2n}/D_n)$ is abelian, of dimension equal to the number of partitions of n .
- b) A representation σ^E of S_{2n} contains a non-zero invariant vector for D_n if and only if the diagram E has rows of even length.

PROOF. The dimension of $C(S_{2n}/D_n)$ equals the number of D_n double cosets on S_{2n} . Thus, we focus our attention on describing these double cosets. We can think of the left coset space S_{2n}/D_n as the set of all possible partitions of $2n$ into n pairs and the double coset space then becomes the set of D_n orbits on this set of partitions. Thus, let us study the relation between two such partitions, P_1 and P_2 . Let us say a set $X \subseteq 2n$ is P_1, P_2 stable if X is a union of pairs from P_1 and a union of pairs from P_2 . Clearly, the set $\pi(P_1, P_2)$ of P_1, P_2 stable subsets of $2n$ is closed under unions and intersections, and the minimal non-empty P_1, P_2 stable sets will define a partition of $2n$, and also partitions $\Pi_{P_2}(P_1)$ of P_1 and $\Pi_{P_1}(P_2)$ of P_2 respectively.

A1.2: PROPOSITION.

- a) The numerical partition D of $2n$ defined by $\Pi(P_1, P_2)$ has all of its parts even.
- b) The numerical partitions of n defined by $\Pi_{P_2}(P_1)$ and $\Pi_{P_1}(P_2)$ are both equal to $D/2$, the partition of n whose parts are equal to the parts of D divided by 2.
- c) P_2 is determined by P_1 and $\Pi(P_1, P_2)$, equivalently, by P_1 and $\Pi_{P_2}(P_1)$, up to permutations which preserve both P_1 and $\Pi_{P_2}(P_1)$.
- d) Any partition of P_1 arises as $\Pi_{P_2}(P_1)$ for some P_2 .

PROOF. Statement a) is clear, since each set X in $\Pi(P_1, P_2)$ is the union of disjoint pairs, from P_1 or from P_2 . Since the number of pairs from P_1 , or the number of pairs from P_2 , in any X in $\Pi(P_1, P_2)$, is one half the cardinality of X , statement b) is likewise clear.

The main thing to show is statement c). Fix X in $\Pi_{P_2}(P_1)$, and let $\{a_1, b_1\}$ be a pair from P_1 contained in X . If X is not equal to $\{a_1, b_1\}$, then we can find a pair $\{w, z\}$ from P_2 contained in X , and which is neither equal to nor disjoint from $\{a_1, b_1\}$. After switching a_1 and b_1 if necessary, we may assume that $\{w, z\} = \{b_1, a_2\}$ where a_2 is our new name for the element of $\{w, z\}$ not contained in $\{a_1, b_1\}$. Let the pair in P_1 containing a_2 be $\{a_2, b_2\}$. If $X = \{a_1, b_1, a_2, b_2\}$, then the pair of P_2 containing b_2 is $\{b_2, a_1\}$; otherwise it is $\{b_2, a_3\}$ where a_3 is some new element. Let $\{a_3, b_3\}$ be the pair in P_1 containing a_3 . Continue in this fashion, to obtain a

sequence of pairs

$$(A1.2.1) \quad \{a_1, b_1\} \quad \{a_2, b_2\} \quad \{a_3, b_3\} \quad \{a_4, b_4\} \dots$$

where the first row consists of pairs from P_1 and the second row consists of pairs from P_2 . By permuting the members of X suitably, we may arrange the pairs of P_1 in any order we wish and also order the elements of each pair as desired. Thus, if $X = 2k = \{1, 2, 3, \dots, 2k\}$ and $P_1 = \{\{1, 2\}, \{3, 4\}, \dots, \{2k-1, 2k\}\}$, we can arrange for the top row in (A1.2.1) to be $\{1, 2\}, \{3, 4\}$, etc. Then the pairs of the second row are $\{2, 3\}, \{4, 5\}, \{6, 7\}$, etc. It is clear that they are completely determined by the top row. This proves statement c). It also makes clear how, given a partition Π of $2n$ such that each member of Π is a union of pairs of P_1 , we can construct a partition P_2 such that $\Pi(P_1, P_2) = \Pi$. In fact, we can do this in a systematic way. If the set X belongs to Π , and if we arrange the pairs of P_1 in some desired way, as in the first row of (A1.2.1), then the cycle $a_1 \rightarrow a_2 \rightarrow a_3 \rightarrow a_4 \rightarrow \dots \rightarrow a_1$ will convert the pairs of P_1 contained in X to the pairs of a suitable P_2 . This finishes the proof of the proposition and gives us a more refined picture of the D_n double cosets. We state this more formally.

A1.3: PROPOSITION. The correspondence which takes a permutation s of the set $\{1, 3, 5, 7, \dots, 2n-1\}$ to the partition $s(\Pi_0)$ defines a bijection between conjugacy classes in S_n and D_n orbits in the set of partitions of $2n$ into pairs, i.e., on D_n double cosets in S_{2n} .

We now return to the proof of Theorem A1.1. The correctness of the dimension of $C(S_{2n}/D_n)$ stated in part a) follows from Proposition A1.3 just above. That $C(S_{2n}/D_n)$ is abelian also follows from this proposition. We know that every conjugacy class of S_n is self-inverse, i.e., every element of S_n is conjugate to its inverse. It follows from the Proposition that every D_n -double coset in S_{2n} is self-inverse. From this, the commutativity of $C(S_{2n}/D_n)$ follows by the Gelfand Lemma [Gro]. Since $C(S_{2n}/D_n)$ cannot have more representations than its dimension, statement b) will follow provided we know that all the representations of the form mentioned there do have D_n -fixed vectors. But this follows from the explicit highest weight vectors for the $GL(V)$ action on $S(S^2(V))$, as discussed in §3.1, plus Schur Duality, as described in §2.4.

We observe that

$$(A1.3.1) \quad D_n = S_n \times (Z/2)^n,$$

where $(Z/2)^n$ preserves the pairs $\{2j-1, 2j\}$, and s in S_n sends $\{2j-1, 2j\}$ to $\{2s(j)-1, 2s(j)\}$. Thus, there is a character on D_n which is trivial on $(Z/2)^n$ and which is the sign character on S_n . Note that this is *not* the restriction of the sign character of S_{2n} , because all elements of $S_n \subset D_n$ act as even permutations of $2n$. Call this character $\text{sgn} \sim$. We also want to describe the convolution algebra $C(S_{2n}/D_n; \text{sgn} \sim)$, consisting of functions on S_{2n} , which transform by $\text{sgn} \sim$ under right and left translations by D_n .

A1.4: THEOREM.

- a) The algebra $C(S_{2n}/D_n; \text{sgn} \sim)$ is abelian, of dimension equal to the number of partitions of n into only odd parts.
- b) The representations of S_{2n} which contain a vector which transforms by the character $\text{sgn} \sim$ of D_n are exactly those which can be constructed as nested systems of $(h, h+1)$ hooks.

PROOF. The same argument which proves that $C(S_{2n}/D_n)$ is abelian proves that $C(S_{2n}/D_n; \text{sgn} \sim)$ is abelian also, since $C(S_{2n}/D_n; \text{sgn} \sim)$ is supported on fewer D_n double cosets than is $C(S_{2n}/D_n)$. Assume we know the dimension of $C(S_{2n}/D_n; \text{sgn} \sim)$. We know from §4.2 that all diagrams of the form stated do carry $\text{sgn} \sim$ -eigenvectors for D_n . To show that they exhaust all representations with $\text{sgn} \sim$ -eigenvectors, it is enough to show that the number of such representations is at least as big as the dimension of $C(S_{2n}/D_n; \text{sgn} \sim)$. As explained in §4.2 the number of these representations is the number of partitions of n into disjoint parts. In part a) of the theorem, we have described $\dim C(S_{2n}/D_n; \text{sgn} \sim)$ as the number of partitions of n into odd parts. It is a classic fact ([And], p.5), in the theory of partitions, that these two numbers are the same. Thus, if we prove the dimension formula, the whole theorem follows. In fact, we only need to show that the dimension is at most the number of partitions of n into odd parts. Consider a D_n double coset defined by two partitions P_1, P_2 of $2n$ into pairs. Let $\Pi(P_1, P_2)$ be the partition of $2n$ into P_1, P_2 -stable sets. Let X be a minimal set in $\Pi(P_1, P_2)$ and suppose that X is divisible by 4, so that the numerical partition of n corresponding to $\Pi(P_1, P_2)$ has an even part. Consider the cycle of pairs, alternately from P_1 and P_2 , as described in (A1.2.1). Let w be the reflection of this cycle through the middle, i.e.,

$$w(a_j) = b_{k-j+1} \quad w(b_j) = a_{k-j+1}.$$

Here $2k$ is the cardinality of X . One can check that w preserves both P_1 and P_2 . If we take k even, then as a permutation of P_1 , the reflection w leaves no pairs fixed, while as a permutation of P_2 , it leaves the pairs $\{a_1, b_k\}$ and $\{a_\ell, b_{\ell+1}\}$, where $2\ell = k$, fixed. Thus, the permutation of P_1 defined by w consists of ℓ transpositions of pairs, and the permutations of the pairs of P_2 consists of $\ell - 1$ transpositions. It follows that the character $\text{sgn} \sim$ takes on different values on w , depending on which partition w is considered to be acting. By a standard result of Mackey [CR], this implies that, if $P_1 = \Pi_0$ is the standard partition, then the D_n double coset corresponding to P_2 cannot support a non-zero function in $C(S_{2n}/D_n; \text{sgn} \sim)$. It follows that only the P_2 such that the partitions $\Pi(P_2(P_1))$ consist of only odd parts can define double cosets which support elements of $C(S_{2n}/D_n; \text{sgn} \sim)$. Thus $\dim C(S_{2n}/D_n; \text{sgn} \sim)$ is not larger than what is stated in part a). Since we have already seen from §4.2 that is at least this large, we have proved Theorem A1.4.

Appendix 2: Weyl Algebra, Clifford Algebra.

A2.1: Polynomial coefficient differential operators.

A2.1.1: Polarized symbol

Let V denote a finite dimensional complex vector space. Let $\mathcal{PD}(V)$ be the algebra of polynomial-coefficient differential operators on V . $\mathcal{PD}(V)$ is of course a non-commutative algebra, but it contains the commutative algebras $\mathcal{P}(V)$ of polynomials and $\mathcal{D}(V)$ of constant-coefficient differential operators. Further, the map

$$(A2.1.1.1) \quad m : \mathcal{P}(V) \otimes \mathcal{D}(V) \rightarrow \mathcal{PD}(V)$$

given by multiplication inside $\mathcal{PD}(V)$:

$$m(p \otimes L) = pL$$

is a linear isomorphism. The algebra $\mathcal{P}(V)$ of multiplication operators is generated by the linear functionals on V , i.e., by V^* . Also, $\mathcal{D}(V)$ is generated by the directional derivatives ∂_v , which are defined by the classical formula

$$(A2.1.1.2) \quad \partial_v(f)(u) = \lim_{t \rightarrow 0} \frac{f(u + tv) - f(u)}{t}, \quad v \in V$$

for any smooth function f on V . The map $v \rightarrow \partial_v$ defines an embedding of V in $\mathcal{D}(V)$, and by virtue of the universal property of the symmetric algebra $\mathcal{S}(V)$, this leads to the algebra homomorphism

$$(A2.1.1.3) \quad \partial : \mathcal{S}(V) \rightarrow \mathcal{D}(V).$$

The map ∂ is in fact an isomorphism, which we use to identify $\mathcal{S}(V)$ and $\mathcal{D}(V)$ when convenient. Combining ∂ with m gives us an isomorphism

$$(A2.1.1.4) \quad \hat{s} : \mathcal{PD}(V) \cong \mathcal{P}(V) \otimes \mathcal{D}(V) \cong \mathcal{P}(V) \otimes \mathcal{S}(V) \cong \mathcal{P}(V) \otimes \mathcal{P}(V^*) \cong \mathcal{P}(V \oplus V^*)$$

between $\mathcal{PD}(V)$ and $\mathcal{P}(V \oplus V^*)$. The isomorphism $\hat{s}$ is called the (total) polarized symbol mapping. The polarized symbol mapping does not preserve the algebra structures, but it almost does. The last isomorphism gives the algebra $\mathcal{P}(V \otimes V^*)$ a bigrading, i.e., a grading by $\mathbb{Z}^{+2}$, by taking $\mathcal{P}^{r,s}(V \oplus V^*) = \mathcal{P}^r(V) \otimes \mathcal{P}^s(V^*)$. The subspaces

$$(A2.1.1.5) \quad \mathcal{PD}^{(r,s)}(V) = \hat{s}^{-1} \left(\sum_{a \leq r, b \leq s} \mathcal{P}^{(a,b)}(V \oplus V^*) \right)$$

of $\mathcal{PD}(V)$ define a bifiltration of $\mathcal{PD}(V)$, that is,

$$(A2.1.1.6) \quad \mathcal{PD}(V)^{(r,s)} \mathcal{PD}(V)^{(t,u)} = \mathcal{PD}(V)^{(r+t, s+u)}.$$

The associated graded algebra $[\text{Sr}]$ is isomorphic to $\mathcal{P}(V \oplus V^*)$, and the graded symbol mapping (known as the polarized symbol mapping) is an isomorphism of algebras.

A2.1.2: Conjugation by $GL(V)$

The group $GL(V)$ acts on $\mathcal{PD}(V)$ by conjugation:

$$(A2.1.2.1) \quad g(L)(f) = g(L(g^{-1}(f))) \quad g \in GL(V), \quad L \in \mathcal{PD}(V), \quad f \in \mathcal{P}(V).$$

An easy calculation shows that the polarized symbol mapping $\hat{s}$ is a $GL(V)$ -module isomorphism. In particular, this action preserves the subalgebras $\mathcal{P}(V)$ and $\mathcal{D}(V) \cong \mathcal{S}(V)$, and the action on each of these is the standard action. Furthermore, this action is "infinitesimally inner" in a sense which we will explain in the next paragraph.

A2.1.3: Unnormalized $\mathfrak{gl}(V)$

Let $\mathfrak{gl}(V)$ denote the Lie algebra of $GL(V)$. We know that

$$(A2.1.3.1) \quad \mathfrak{gl}(V) \cong \text{End}(V) \cong V^* \otimes V.$$

The map

$$(A2.1.3.2) \quad (\lambda, v) \rightarrow \lambda \partial_v \quad \lambda \in V^*, v \in V$$

embeds $V^* \otimes V \cong \mathfrak{gl}(V)$ as a Lie subalgebra of $\mathcal{PD}(V)$. We note that, in fact, the image of the map (A2.1.3.2) is $\hat{s}^{-1}(\mathcal{PD}^{(1,1)}(V))$. We will refer to the map (A2.1.3.2) as the unnormalized embedding of $\mathfrak{gl}(V)$ in $\mathcal{PD}(V)$, and to the Lie subalgebra $\hat{s}^{-1}(\mathcal{PD}^{(1,1)}(V))$ as "unnormalized $\mathfrak{gl}(V)$ ". We also refer to the elements of $\hat{s}^{-1}(\mathcal{PD}^{(1,1)}(V))$ as "linear coefficient vector fields". It is easy to check that the linear coefficient vector fields are the infinitesimal generators of the action (A2.1.2.1) of $GL(V)$ on $\mathcal{P}(V)$ by conjugation. This is what we mean when we call that action infinitesimally inner.

A2.2: Weyl algebra

A2.2.1: Generators and relations for $\mathcal{PD}(V)$

We have an embedding

$$(A2.2.1.1) \quad \begin{aligned} \tilde{j} : V \oplus V^* &\rightarrow \mathcal{PD}(V) \\ \tilde{j}(\lambda, v) &= \lambda + \partial_v, \quad \lambda \in V^*, v \in V. \end{aligned}$$

It is implicit in formula (A2.1.1.1), and easy to check, that $\tilde{j}(V \oplus V^*)$ generates $\mathcal{PD}(V)$ as associative algebra. The non-commutativity of $\mathcal{PD}(V)$ is captured in the commutation relations satisfied by the elements of $\tilde{j}(V \oplus V^*)$:

$$(A2.2.1.2) \quad \begin{aligned} [\tilde{j}(\lambda_1, v_1), \tilde{j}(\lambda_2, v_2)] &= \tilde{j}(\lambda_1, v_1) \tilde{j}(\lambda_2, v_2) - \tilde{j}(\lambda_2, v_2) \tilde{j}(\lambda_1, v_1) \\ &= \lambda_2(v_1) - \lambda_1(v_2). \end{aligned}$$

These are known as the Canonical Commutation Relations (CCR). It is not hard to show that $\mathcal{PD}(V)$ is exactly the algebra generated by $\tilde{j}(V \oplus V^*)$ subject to the CCR.

A2.2.2: Heisenberg Lie subalgebra

The formulas of the previous paragraph may be interpreted as follows. Write

$$(A2.2.2.1) \quad W = \tilde{j}(V \oplus V^*).$$

Then

$$(A2.2.2.2) \quad H(W) = W \oplus \mathbf{C} = \mathcal{PD}(V)^{(1,0)} \oplus \mathcal{PD}(V)^{(0,1)} \oplus \mathcal{PD}(V)^{(0,0)}$$

is a Lie subalgebra of $\mathcal{PD}(V)$. As an abstract Lie algebra, $H(W)$ is two-step nilpotent, with one-dimensional commutator subalgebra and center, which in the decomposition (A2.2.2.2) is equal to $\mathcal{PD}(V)^{(0,0)} = \mathbf{C}$. We call $H(W)$ the Heisenberg Lie algebra attached to W . The commutator operation in $H(W)$ thus defines on W a skew-symmetric, non-degenerate bilinear form, i.e., a symplectic form. We refer to W endowed with this form as a symplectic vector space. Denote the symplectic form by sharp brackets: $\langle \cdot, \cdot \rangle$. With this terminology and notation, we may say that $\mathcal{PD}(V)$ is the algebra generated by a symplectic vector space W , subject to the relations

$$(A2.2.2.3) \quad [w, w'] = \langle w, w' \rangle, \quad w, w' \in W.$$

In this equation, the left hand side indicates the commutator of w and w' , and the right hand side indicates the scalar given by the value of the symplectic form at w and w' .

A2.2.3: Weyl algebra

When we think of $\mathcal{PD}(V)$ in terms of its generators and relations as described in the previous paragraph, we call it the Weyl algebra associated to the symplectic vector space W and denote it $\mathcal{W}(W)$. It is the same algebra as before, but we look at it somewhat differently, and emphasize different aspects of its structure. In particular, we give it a simple $\mathbf{Z}^+$ -filtration instead of the bifiltration of formula (A2.1.1.4). We set $\mathbf{W}(W)^{(1)} = W \oplus \mathbf{C}$, and for $k > 0$, we let $\mathcal{W}(W)^{(k)}$ denote the span of the k -fold products of elements of $\mathbf{W}(W)^{(1)}$. It then follows trivially that

$$(A2.2.3.1) \quad \mathcal{W}(W)^{(k)} \mathcal{W}(W)^{(1)} = \mathcal{W}(W)^{(k+1)}.$$

We can also verify the somewhat more concrete description

$$(A2.2.3.2) \quad \mathcal{W}(W)^{(k)} = \sum_{r+s \leq k} \mathcal{PD}(V)^{(r,s)}.$$

The fact that commutators in $H(W)$ are scalars is expressed by the formula

$$(A2.2.3.3) \quad [\mathcal{W}(W)^{(1)}, \mathcal{W}(W)^{(1)}] \subseteq \mathcal{W}(W)^{(0)}.$$

The extension of this to arbitrary level is

$$(A2.2.3.4) \quad [\mathcal{W}(W)^{(k)}, \mathcal{W}(W)^{(l)}] \subseteq \mathcal{W}(W)^{(k+l-2)}.$$

It follows from this that the associated graded algebra $Gr \mathcal{W}(W)$ is commutative. Knowing this, one sees easily that

$$(A2.2.3.5) \quad Gr \mathcal{W}(W) \cong S(W).$$

A2.2.4: Action of $Sp(W)$; isotropic symbol

Let $Sp(W)$ denote the isometry group of the symplectic form $\langle \dots \rangle$. It consists of all linear transformations g such that

$$(A2.2.4.1) \quad \langle g(w), g(w') \rangle = \langle w, w' \rangle.$$

The discussion of §A2.2.2 above makes clear that the action of $Sp(W)$ on W extends to an action of $Sp(W)$ on $\mathcal{W}(W)$ by filtration-preserving automorphisms. These automorphisms pass to automorphisms of $Gr \mathcal{W}(W)$, and the isomorphism (A2.2.3.5) is then an isomorphism of $Sp(W)$ modules. The homogeneous components $S^k(W)$ are irreducible $Sp(W)$ -modules and mutually non-isomorphic. It follows that there is a unique linear isomorphism

$$(A2.2.4.2) \quad \tilde{s} : \mathcal{W}(W) \cong S(W)$$

of $Sp(W)$ -modules such that the graded version of $\tilde{s}$ is the isomorphism (A2.2.3.5). We call $\tilde{s}$ the total isotropic symbol. We can characterize $\tilde{s}^{-1}(S^k(W))$ as the span of the k -th powers of elements of W . We will denote it by $\mathcal{W}^k(W)$. (Note the lack of parentheses on the superscript k .)

A2.2.5: The Lie subalgebra $\mathfrak{sp}$

The action of $Sp(W)$ on $\mathcal{W}(W)$ described in §A2.2.4 is infinitesimally inner, as we will now explain.

The Lie algebra $\mathfrak{sp}(W)$ of the group $Sp(W)$ is the set of "infinitesimal isometries" of the symplectic form $\langle \dots \rangle$. A linear transformation A of W is in $\mathfrak{sp}(W)$ if it satisfies

$$(A2.2.5.1) \quad \langle Aw, w' \rangle + \langle w, Aw' \rangle = 0 \quad w, w' \in W.$$

Consider the span in $\mathcal{W}(W)$ of the squares of elements in W . In the notation of the end of the previous paragraph, this is $\mathcal{W}^2(W)$. By polarizing, we see it is also the span of the elements $(1/2)(ww' + w'w)$ for $w, w' \in W$. We compute that

$$(A2.2.5.2) \quad [w^2, w'] = 2\langle w, w' \rangle w.$$

Thus, taking commutators with $\mathcal{W}^2(W)$ preserves W . It follows that $[\mathcal{W}^2(W), \mathcal{W}^k(W)] = \mathcal{W}^k(W)$. In particular, $\mathcal{W}^2(W)$ is a Lie algebra. Its action on $\mathcal{W}^k(W)$ by commutators is, of course, a representation by derivations. In particular,

the action on W will consist of infinitesimal isometries of the commutator symplectic form on W . In other words, the action

$$(A2.2.5.3) \quad ad : \mathcal{W}^2(W) \rightarrow End(W)$$

defined by commutators, as in formula (2.5.1), maps $\mathcal{W}^2(W)$ to $\mathfrak{sp}(W)$. It is not hard to check that, in fact, this gives an isomorphism between $\mathcal{W}^2(W)$ and $\mathfrak{sp}(W)$. Therefore, if we exponentiate the operators $ad z, z \in \mathcal{W}^2(W)$, we will get the action of $Sp(W)$ described in §A.2.2.4.

A2.2.6: Grading on $\mathfrak{sp}(W)$

The decomposition $W \cong \tilde{j}(V) \oplus \tilde{j}(V^*)$ of §A2.2.1 gives rise to a three-fold decomposition of $\mathcal{W}^2(W)$. Let $\mathfrak{sp}^{(2,0)}(W)$ be the span of w^2 for w in $\tilde{j}(V)$. Let $\mathfrak{sp}^{(2,0)}(W)$ be the span of w^2 for w in $\tilde{j}(V^*)$, and let $\mathfrak{sp}^{(1,1)}(W)$ be the span of the symmetrized products $ww' + w'w$, for $w \in V$ and $w' \in V^*$. Then we have a decomposition

$$(A2.2.6.1) \quad \mathfrak{sp}(W) \cong \mathcal{W}^2(W) = \mathfrak{sp}^{(2,0)}(W) \oplus \mathfrak{sp}^{(1,1)}(W) \oplus \mathfrak{sp}^{(0,2)}(W).$$

All three of the subspaces $\mathfrak{sp}^{(2,0)}(W)$, $\mathfrak{sp}^{(1,1)}(W)$ and $\mathfrak{sp}^{(0,2)}(W)$ are Lie subalgebras of $\mathfrak{sp}(W)$. Both $\mathfrak{sp}^{(2,0)}$ and $\mathfrak{sp}^{(0,2)}$ are commutative and are normalized by $\mathfrak{sp}^{(1,1)}$. Commutators of the form $[z, w] : z$ in $\mathfrak{sp}^{(2,0)}(W)$ and w in $\mathfrak{sp}^{(0,2)}(W)$ belong to $\mathfrak{sp}^{(1,1)}(W)$, and span it.

A2.2.7: Normalized $g\ell(V)$

The subalgebra $\mathfrak{sp}^{(1,1)}$ is isomorphic to $g\ell(V)$. Indeed, the subgroup $Sp^{(1,1)}(W)$ of $Sp(W)$ which preserves the decomposition $W \cong \tilde{j}(V) \oplus \tilde{j}(V^*)$ is identified to $GL(V)$ by restricting its action to the first summand. The subalgebra $\mathfrak{sp}^{(1,1)}$ is the Lie algebra of $Sp^{(1,1)}(W)$.

Under the action of $Sp(W)$ on $\mathcal{W}(W)$ by automorphisms, the action of $Sp^{(1,1)}(W)$ coincides with the action of $GL(V)$ by conjugation, described above in §A2.1.2. However, the Lie algebra $\mathfrak{sp}^{(1,1)}$ is not exactly the Lie algebra "unnormalized $g\ell(V)$ " of linear coefficient vector fields described in §A2.1.3. Indeed, the linear vector fields are the operators $\lambda\partial_w + \partial_w\lambda$, i.e., they are a symmetrized version of $\mathfrak{sp}^{(1,1)}$ have the form $(1/2)(\lambda\partial_w + \partial_w\lambda)$, i.e., they are a symmetrized version of the linear vector fields. From the commutation relations (A2.2.1.2), we can derive the simple relationship

$$(A2.2.7.1) \quad (1/2)(\lambda\partial_w + \partial_w\lambda) = \lambda\partial_w + (1/2)\lambda(v).$$

Thus, the elements of $\mathfrak{sp}^{(1,1)}$ differ from the elements of unnormalized $g\ell(V)$ by constants. These constants make no difference when taking commutators, so when $\mathfrak{sp}^{(1,1)}$ and unnormalized $g\ell(V)$ act on $\mathcal{W}(W) = \mathcal{PD}(V)$ by the commutator action, they give the same operators. However, the operators on $\mathcal{P}(V)$ obtained by exponentiating the elements of $\mathfrak{sp}^{(1,1)}$ will be slightly different from the operators of $GL(V)$ acting via their standard permutation action (section 2.1.4). The exponentials of elements of $\mathfrak{sp}^{(1,1)}$ will have the form

$$\tilde{g}(f)(v) = |\det g|^{-1/2} f(g^{-1}(v)).$$

Here, g is in $GL(V)$ and f is a function on V . Thus, the operators $\tilde{g}$ are obtained from the standard operators of $GL(V)$ by multiplying by the scalar factor $|\det g|^{-1/2}$. The effect of the scalar factor is to make the operators $\tilde{g}$, thought of as operators on $L^2(V)$ (the Hilbert space of square-integrable functions on V), into unitary operators. Thus, the factor $|\det g|^{-1/2}$ can be thought of as a normalization factor. We will refer to $\mathfrak{sp}^{(1,1)}$ as *normalized* $\mathfrak{gl}(V)$. This modification of the unnormalized $\mathfrak{gl}(V)$ is necessary in order to get it to sit inside the larger Lie algebra $\mathcal{W}^2(W)$.

A2.3: $\text{End}(\Lambda(V))$ and Clifford algebras.

The polynomial coefficient differential operators provide a well-behaved algebra of endomorphisms of the algebra $\mathcal{P}(V)$ of polynomials on V , or alternatively of the symmetric algebra $S(V)$. We can give a description of the algebra $\text{End}(\Lambda(V))$ quite analogous to the description of $\mathcal{PD}(V)$ in terms of the CCR.

A2.3.1: Inner multiplication

Of course, $\Lambda(V)$ acts on itself by multiplication. There is also an action of $\Lambda(V^*)$ on $\Lambda(V)$. This is analogous to the action of constant coefficient differential operators on $\mathcal{P}(V)$, and it is commonly called inner multiplication. It may be defined by the formula

$$(A2.3.1.1) \quad \downarrow_{\lambda}(v_1 \wedge v_2 \wedge v_3 \wedge \dots \wedge v_k) = \sum_{a=1}^k (-1)^a \lambda(v_a) v_1 \wedge v_2 \wedge \dots \wedge v_{a-1} \wedge v_{a+1} \wedge \dots \wedge v_k,$$

for λ in V^* and v_j in V . An alternative way of expressing this is

$$(A2.3.1.2) \quad \downarrow_{\lambda}(v \wedge \omega + \eta) = \lambda(v)\omega$$

for ω in $\Lambda^{k-1}(\ker \lambda)$ and η in $\Lambda^k(\ker \lambda)$.

The parallel between formula (A2.3.1.1) and the product rule for differentiation should be clear. In fact, the operation of $\downarrow_{\lambda}$ is determined by specifying that

$$(A2.3.1.3) \quad \downarrow_{\lambda v} = \lambda(v),$$

and that $\downarrow_{\lambda}$ is a graded derivation, that is,

$$(A2.3.1.4) \quad \downarrow_{\lambda}(\omega \wedge \eta) = \downarrow_{\lambda}(\omega) \wedge \eta + (-1)^r \omega \wedge (\downarrow_{\lambda}(\eta))$$

for ω in $\Lambda^r(V)$ and η in $\Lambda^s(V)$.

Pursuing the analogy between $\downarrow_{\lambda}$ and ∂_v , we see that, while two directional derivatives commute with each other, inner multiplications anticommute:

$$(A2.3.1.5) \quad \downarrow_{\lambda} \downarrow_{\mu} = -\downarrow_{\mu} \downarrow_{\lambda} \quad \lambda, \mu \in V^*.$$

This is easy to check using formulas (A2.3.1.3) and (A2.3.1.4). It follows that the subalgebra of $\text{End}(\Lambda(V))$ generated by the $\downarrow_{\lambda}$ is a homomorphic image of $\Lambda(V^*)$. It is not hard to check that it is actually isomorphic to $\Lambda(V^*)$. Indeed, if $\{b_j : 1 \leq j \leq n\}$ is any basis for V , then $\prod_{j=1}^n b_j$ is a non-zero element of $\Lambda^n(V)$, where

$n = \dim V$, and it is easy to check that inner multiplication of $\prod_{j=1}^n b_j$ by elements of $\Lambda(V^*)$ maps $\Lambda(V^*)$ isomorphically onto $\Lambda(V)$.

A2.3.2: Clifford presentation of $\text{End}(\Lambda(V))$

From §A2.3.1, we know we have subalgebras $\Lambda(V)$, given by wedge multiplication, and $\Lambda(V^*)$, generated by inner multiplications, of $\text{End}(\Lambda(V))$. We can use multiplication inside $\text{End}(\Lambda(V))$ to define a map

$$(A2.3.2.1) \quad \tilde{m} : \Lambda(V) \otimes \Lambda(V^*) \rightarrow \text{End}(\Lambda(V)).$$

This map is an isomorphism. This will be a corollary of the discussion below.

We observe we have an embedding

$$(A2.3.2.2) \quad \begin{aligned} \tilde{j} : V \oplus V^* &\rightarrow \text{End}(\Lambda(V)) \\ \tilde{j}(v + \lambda) &= \Lambda_v + \downarrow_{\lambda}, \end{aligned}$$

where Λ_v indicates the operation of wedge multiplication by $v \in V$. The elements of $\tilde{j}(V \oplus V^*)$ satisfy anticommutation relations, as follows:

$$(A2.3.2.3) \quad \begin{aligned} \{\Lambda_v, \Lambda_{v'}\} &= (\Lambda_v)(\Lambda_{v'}) + (\Lambda_{v'})(\Lambda_v) = 0, & v, v' \in V; \\ \{\downarrow_{\lambda}, \downarrow_{\mu}\} &= 0, \{\downarrow_{\lambda}, \Lambda_v\} = \lambda(v), & \lambda, \mu \in V^*. \end{aligned}$$

Here the curly brackets are used to denote anticommutator:

$$\{A, B\} = AB + BA.$$

In the third equation of formula (A2.3.2.3), the $\lambda(v)$ indicates the scalar operator which multiplies every vector by the number $\lambda(v)$. The first two of these equations reaffirm what we already have observed, viz., that $\tilde{j}(V)$ generates an algebra isomorphic to $\Lambda(V)$ and, similarly, for $\tilde{j}(V^*)$. The third equation, which follows directly from formulas (A2.3.1.3) and (A2.3.1.4), shows us how to change the order of application of Λ_v and $\downarrow_{\lambda}$. It shows that a product of Λ_v 's and $\downarrow_{\lambda}$'s in any order can be expressed as a sum of such products where the $\downarrow_{\lambda}$'s all appear before (i.e., to the right of) the Λ_v 's. This implies that the image of $\tilde{m}$ in formula (3.2.1) is a subalgebra of $\text{End}(\Lambda(V))$. To be sure that it is all of $\text{End}(\Lambda(V))$, it suffices to know that it acts irreducibly on $\Lambda(V)$. This is very easy to see. For example, given ω in $\Lambda(V)$, we can multiply it by an appropriate η to make the product a nonzero element of $\Lambda^n(V)$, where $n = \dim V$. As we have remarked above, inner multiplication applied to $\Lambda^n(V)$ maps $\Lambda(V^*)$ isomorphically onto $\Lambda(V)$. Thus, an appropriate multiplication followed by an appropriate inner multiplication allows us to transform any element of $\Lambda(V)$ into any other; this means $\tilde{m}(\Lambda(V) \otimes \Lambda(V^*))$ acts irreducibly on $\Lambda(V)$.

A2.3.3: $GL(V)$ -module structure; unnormalized $\mathfrak{gl}(V)$

We should take note of the $GL(V)$ action on each of $\Lambda(V)$, $\Lambda(V^*)$ and $End(\Lambda(V))$. These actions are all *via* automorphisms. The action of $GL(V)$ on $\Lambda(V)$ embeds $GL(V)$ in $End(\Lambda(V))$, and the natural action of $GL(V)$ on $End(\Lambda(V))$ is conjugation via this embedding. It is easy to see that when $\Lambda(V)$ and $\Lambda(V^*)$ are regarded as subalgebras of $End(\Lambda(V))$, conjugation by $GL(V)$ preserves them and induces the appropriate action on them. To verify this, it is enough to check that conjugation by $GL(V)$ gives the correct action of $V \subset \Lambda(V)$ and on $V^* \subset \Lambda(V^*)$, since all the actions are by automorphisms. One concludes that the mapping (A2.3.2.1) is an isomorphism of $GL(V)$ -modules.

We can define a bifiltration on $End(\Lambda(V))$ by setting

$$(A2.3.3.1) \quad End(\Lambda(V))^{(j,k)} = \sum_{a \leq j, b \leq k} \tilde{m}(\Lambda^a(V) \otimes \Lambda^b(V^*)).$$

The relations (A2.3.2.3) imply that this is an algebra bifiltration, that is,

$$(A2.3.3.2) \quad End(\Lambda(V))^{(r,s)} End(\Lambda(V))^{(t,u)} \subseteq End(\Lambda(V))^{(r+t, s+u)}.$$

The relations (A2.3.3.2) further imply that taking anticommutators moves one down in the filtration:

$$\{End(\Lambda(V))^{(r,s)}, End(\Lambda(V))^{(t,u)}\} \subseteq End(\Lambda(V))^{(r+t-1, s+u-1)}.$$

It follows that the bigraded algebra associated to this bifiltration is exactly $\Lambda(V) \otimes \Lambda(V^*)$, and that the grading of the map $\tilde{m}$ is the identity map from $\Lambda(V) \otimes \Lambda(V^*)$ to itself. Thus, the inverse of the map $\tilde{m}$ is an analog of the polarized symbol introduced in A2.3.2.

The subspace $\tilde{m}(\Lambda^1(V) \otimes \Lambda^1(V^*)) = \tilde{m}(V \otimes V^*)$, which is spanned by operator products $\wedge_v \wedge_{\lambda}$, is the analog of the linear coefficient differential operators. It is a Lie algebra, isomorphic to $\mathfrak{gl}(V)$; it is the Lie algebra of infinitesimal generators for the action of $GL(V)$ on $\Lambda(V)$. As in the case of symmetric variables, we call it unnormalized $\mathfrak{gl}(V)$.

A2.3.4: Clifford algebras

The above discussion effectively gives a presentation of $End(\Lambda(V))$ as an algebra generated by $\tilde{j}(V \oplus V^*)$ and subject to the anticommutation relations (A2.3.2.3). As in the transition from $\mathcal{PD}(V)$ to the Weyl algebra, one can take a structural viewpoint toward this process. This results in the notion of *Clifford algebra*. Let Y be a vector space equipped with an inner product – a nondegenerate symmetric bilinear form. Denote the inner product with parentheses: $(\cdot, \cdot)$. Let $\mathcal{C}(Y)$ denote the algebra generated by Y subject to the relations

$$(A2.3.4.1) \quad \{y, y'\} = 2(y, y') \quad y, y' \in Y.$$

This is called the Clifford algebra associated to Y and the inner product $(\cdot, \cdot)$.

If V is any vector space, set $Y = V \oplus V^*$. Define an inner product $(\cdot, \cdot)$ on Y by symmetrizing the canonical pairing between V and V^* : (see Proposition A3.8)

$$(A2.3.4.2) \quad ((v, \lambda), (u, \mu)) = \lambda(u) + \mu(v), \quad u \in V; \lambda, \mu \in V^*.$$

Comparison of formulas (A2.3.4.1) and (A2.3.4.2) with formulas (A2.3.2.3) shows that the Clifford algebra associated to $V \oplus V^*$ with this inner product (actually, one should divide this inner product by 2) is isomorphic to $End(\Lambda(V))$. In this, the situation is just like that for the Weyl algebra. However, there is also an important difference. Every symplectic form is isomorphic to the antisymmetric extension to $V \oplus V^*$ of the canonical pairing between V and V^* , for some vector space V . (In fact, this is true over any field, not just $\mathbb{C}$.) However, not every inner product arises as the symmetric extension to $V \oplus V^*$ of the canonical pairing between V and V^* . We can observe that, with respect to the form (A2.3.4.2), V is an isotropic subspace of $V \oplus V^*$. Hence in order for Y to be isomorphic to $V \oplus V^*$, Y must have isotropic subspaces of dimension $\dim Y/2$. In particular, Y must be even dimensional. Over an algebraically closed field such as $\mathbb{C}$, this is sufficient: if $\dim Y = 2n$, then one can find isotropic subspaces V of dimension n , and for any such V , one has $Y \cong V \oplus V^*$, where $V \oplus V^*$ is equipped with the form (A2.3.4.2). (See Proposition A3.8.) However, even over $\mathbb{C}$, one has the possibility that $\dim Y$ might be odd, in which case $\mathcal{C}(Y)$ cannot be of the form $End(\Lambda(V))$. Consider this case. Take Y of odd dimension. Select an orthonormal basis $\{b_j : 1 \leq j \leq \dim Y\}$ for Y . Consider the element

$$(A2.3.4.3) \quad \gamma = \prod_{j=1}^{\dim Y} b_j.$$

Using the relations (3.4.1), we may check that γ commutes with each b_j , i.e., is a central element in $\mathcal{C}(Y)$. Further, we have $\gamma^2 = \pm 1$. Hence the elements

$$(A2.3.4.4) \quad e^{\pm} = (1 \pm \gamma)/1$$

are central idempotents in $\mathcal{C}(Y)$. If Y_1 is a hyperplane in Y ($\dim Y_1 = \dim Y - 1$), then according to our discussion above, $\mathcal{C}(Y_1) \subseteq \mathcal{C}(Y)$ is a matrix algebra. The two-sided ideal $\mathcal{C}(Y)e^{\pm}$ has dimension at most $\dim \mathcal{C}(Y)/2$, and contains the algebra $\mathcal{C}(Y_1)e^{\pm}$ which, if it is non-zero (which it is), will be a matrix algebra of dimension at least $\dim \mathcal{C}(Y)/2$. We may conclude that when $\dim Y$ is odd, the algebra $\mathcal{C}(Y)$ is the direct sum of two matrix algebras $\mathcal{C}(Y)e^{\pm}$.

A2.3.5: Action of $O(Y)$

Let Y be as in §3.4, a vector space equipped with an inner product $(\cdot, \cdot)$. Denote by $O(Y)$ the orthogonal group of Y : the group of linear transformations which preserve the inner product. It is obvious from the definition of the Clifford algebra $\mathcal{C}(Y)$ that the action of $O(Y)$ on Y extends to an action on $\mathcal{C}(Y)$ by algebra automorphisms. Let $\mathcal{C}^{(k)}(Y)$ denote the span of products of k or fewer elements of Y . The $\mathcal{C}^{(k)}(Y)$ clearly form an algebra filtration of $\mathcal{C}(Y)$. The relations (A2.3.4.1) imply that

$$(A2.3.5.1) \quad \{\mathcal{C}^{(r)}(Y), \mathcal{C}^{(s)}(Y)\} \subseteq \mathcal{C}^{(r+s-2)}(Y).$$

This formula says that elements of $\mathcal{C}(Y)$ anticommute modulo lower order terms, just as differential operators commute modulo lower order terms. It follows that the associated graded algebra is isomorphic to $\Lambda(Y)$:

$$(A2.3.5.2) \quad \text{Gr } \mathcal{C}(Y) = \sum_{k=1}^{\dim Y} \mathcal{C}^{(k)}(Y) / \mathcal{C}^{(k-1)}(Y) \cong \sum_{k=1}^{\dim Y} \Lambda^{(k)}(Y) = \Lambda(Y).$$

These isomorphisms are all $O(Y)$ -module isomorphisms.

The action of $SO(Y)$ on $\mathcal{C}(Y)$ is inner. We can see this by general principles. If $\dim Y$ is even, then $\mathcal{C}(Y)$ is a matrix algebra, and all automorphisms of a matrix algebra are inner. (Thus, in this case, all of $O(Y)$ acts by inner automorphisms, not just $SO(Y)$.) Then n is odd, $\mathcal{C}(Y)$ is the sum of two matrix algebras, and an automorphism of $\mathcal{C}(Y)$ will be inner if and only if it preserves each of the simple summands. Since $SO(Y)$ is a connected group and the stabilizer of the two summands must be open, it is all of $SO(Y)$.

We can also describe explicitly some elements of $\mathcal{C}(Y)$, conjugation by which yields automorphisms in $O(Y)$. Let u be a unit vector in Y . Then formula (A2.3.4.1) says that $u^2 = 1$, i.e., u is an involution in $\mathcal{C}(Y)$. We can further compute that

$$(A2.3.5.3) \quad uyu = u(2(y, u) - uy) = -(y - 2(y, u)u) = -R_u(y),$$

where we have used R_u to denote the reflection in the hyperplane perpendicular to u . If $\dim Y$ is even, the elements $-R_u$ have determinant -1 , and they generate all of $O(Y)$ [Art]. If $\dim Y$ is odd, the elements $-R_u$ have determinant $+1$, and they generate $SO(Y)$.

When $\dim Y$ is odd, the automorphism of $\mathcal{C}(Y)$ which equals -1 on Y will transform the element γ of formula (A2.3.4.3) to its negative, and therefore will switch the two simple summands $\mathcal{C}(Y)e^\pm$ of $\mathcal{C}(Y)$. Thus, this automorphism can not be inner.

A2.3.6: The Lie subalgebra $\mathfrak{so}(Y)$

Denote by $\mathcal{C}^2(Y)$ the span of the commutators of elements in $Y \subseteq \mathcal{C}(Y)$. (Note the lack of parentheses in the superscript.) We may verify the identity

$$(A2.3.6.1) \quad [[u, v], y] = u\{y, v\} + \{y, v\}u - v\{y, u\} - \{y, u\}v,$$

valid in any associative algebra. In the context of $\mathcal{C}(Y)$, it says that taking commutators with $\mathcal{C}^2(Y)$ preserves Y inside $\mathcal{C}(Y)$. Since commutators are infinitesimal automorphisms, we know (or we can verify directly) that for u, v in Y , the operator $ad[u, v]$ on Y is an infinitesimal isometry of the inner product $(\cdot, \cdot)$, that is $ad[u, v]$ belongs to $\mathfrak{so}(Y)$, the Lie algebra of $O(Y)$. It is further easy to verify that

$$(A2.3.6.2) \quad ad : \mathcal{C}^2 \rightarrow \mathfrak{so}(Y)$$

is an isomorphism of Lie algebras. Thus, exponentiating the elements of $\mathcal{C}^2(Y)$ will give rise to the subgroup of $\mathcal{C}(Y)$ which acts via conjugation as $SO(Y)$ on $Y = \mathcal{C}^1(Y)$.

A2.3.7: Grading on $\mathcal{C}(V \oplus V^*)$; normalized $\mathfrak{gl}(V)$

If Y is of the form $Y = V \oplus V^*$, then we can decompose $\mathcal{C}^2(Y)$ in three pieces. We set

$$\begin{aligned} \mathcal{C}^{2,0}(Y) &= \mathfrak{so}^{(2,0)}(Y) = \text{span of } \{\wedge_v \wedge_{v'}; v, v' \in V\} \\ \mathcal{C}^{0,2}(Y) &= \mathfrak{so}^{(0,2)}(Y) = \text{span of } \{[\lambda]_\mu; \lambda, \mu \in V^*\} \\ \mathcal{C}^{1,1}(Y) &= \mathfrak{so}^{(1,1)}(Y) = \text{span of } \{[\wedge_v, \lambda]_\lambda; v \in V, \lambda \in V^*\}. \end{aligned}$$

Each of the $\mathcal{C}^{a,b}(Y)$ is a Lie subalgebra of $\mathcal{C}^2(Y)$. The subalgebras $\mathcal{C}^{2,0}(Y)$ and $\mathcal{C}^{0,2}(Y)$ are abelian and are normalized by $\mathcal{C}^{1,1}(Y)$ which is the span of the commutators $[z, w]$, with z in $\mathcal{C}^{2,0}(Y)$ and w in $\mathcal{C}^{0,2}(Y)$.

The subalgebra $\mathcal{C}^{1,1}(Y)$ is isomorphic to $\mathfrak{gl}(V)$. In fact, we can see from the relations (A2.3.2.3) that

$$(A2.3.7.2) \quad [\wedge_v, \lambda]_\lambda = \wedge_v \lambda - \lambda(v).$$

In other words, the elements of $\mathcal{C}^{1,1}(Y)$ differ by constants from the elements of the Lie algebra dubbed unnormalized $\mathfrak{gl}(V)$ in §A2.3.3. The constants make no difference in the commutation relations, so $\mathcal{C}^{1,1}(Y)$ must also be a copy of $\mathfrak{gl}(V)$. We refer to it as normalized $\mathfrak{gl}(V)$. On the other hand, the constants do affect the action of $\mathfrak{gl}(V)$ on $\Lambda(V)$. The effect of replacing $\wedge_v \lambda$ with $[\wedge_v, \lambda]_\lambda$ is to twist the action of $GL(V)$ on $\Lambda(V)$ by $\det^{-1/2}$. Since $(\Lambda^k(V))^* \cong \Lambda^{n-k}(V) \otimes \det^{-1}$, we see that one effect of this change is to make the action of $GL(V)$ (actually, it is now the action of the two-fold cover of $GL(V)$) on $\Lambda(V)$ self-contragredient.

Appendix 3: Varieties of Isotropic Subspaces.

Let V be a vector space equipped with a non-degenerate bilinear form $\langle \cdot, \cdot \rangle$, either symmetric (an inner product) or antisymmetric (a symplectic form). If $U \subseteq V$ is a subspace, we define $U^\perp$, the orthogonal complement of U , by the condition

$$(A3.1) \quad U^\perp = \{v \in V, \langle v, u \rangle = 0, \text{ all } u \in U\}.$$

Orthogonal complementation satisfies the familiar properties of a duality between subspaces:

$$(A3.2) \quad (U^\perp)^\perp = U \quad \dim U + \dim U^\perp = \dim V.$$

(See [Art], [Jac 1], Chapter 6 or [La], Chapter 15 for this and other facts stated below about the basic geometry of spaces with forms. Also, a substantial part of the discussion below can be found in [GH]). If $U^\perp \cap U = \{0\}$, then the restriction of the form $\langle \cdot, \cdot \rangle$ to U is non-degenerate, and conversely. In this case, we have $V = U^\perp \oplus U$. The opposite situation occurs when $U \subseteq U^\perp$. This is equivalent to the condition that the pairing $\langle u, u' \rangle$ between any two elements of U is zero. We call such subspaces isotropic.

We wish to study the collection of all isotropic subspaces of V . Let $G(V, \langle \cdot, \cdot \rangle)$ be the isometry group of the form $\langle \cdot, \cdot \rangle$. We will abbreviate $G(V, \langle \cdot, \cdot \rangle) = G(V)$ or $G(\langle \cdot, \cdot \rangle)$, or even just G , if the omitted data is clear from context. If $U \subseteq V$ is isotropic and $g \in G$, then clearly the transform $g(U)$ will be isotropic, and of the same dimension as U . Thus, the set $J_d(V)$ of isotropic subspaces of V of a given dimension d allows an action of G . We will describe the action of G on $J_d(V)$.

(A3.3) PROPOSITION. G acts transitively on $J_d(V)$.

PROOF. This is an immediate corollary of Witt's Theorem [Art], [Jac 1], [La] which asserts that an isometric linear isomorphism between any two subspaces of V can be extended to an isometry of V to itself, i.e., the isometry between subspaces is implemented by an element of G . But any linear isomorphism between two isotropic subspaces of the same dimension is an isometry.

Thus, $J_d(V)$ is a homogeneous space for G . To get at the finer structure of $J_d(V)$, we need to work with pairs of complementary isotropic subspaces.

Let x in V be an isotropic vector: $\langle x, x \rangle = 0$. Since $\langle \cdot, \cdot \rangle$ is non-degenerate, we can find a vector y such that $\langle y, x \rangle = 1$. We would like to find such a y which is also isotropic. This is automatic if $\langle \cdot, \cdot \rangle$ is symplectic, and can be achieved as follows if $\langle \cdot, \cdot \rangle$ is symmetric. Note that since x is isotropic, we have $\langle y + ax, x \rangle = 1$ for any scalar a . We compute $\langle y + ax, y + ax \rangle = \langle y, y \rangle + 2a\langle y, x \rangle + \langle y, y \rangle + 2a$. Here, we have presumed the symmetry of $\langle \cdot, \cdot \rangle$. We conclude from this that if we replace y by $y - (\langle y, y \rangle/2)x$, we will have the desired isotropic vector. Thus, given an isotropic vector x , we can find a second isotropic vector y such that $\langle y, x \rangle = 1$. The plane H spanned by such a pair of isotropic vectors x, y is called a hyperbolic plane. Note that H is non-degenerate for the form $\langle \cdot, \cdot \rangle$. We call the pair x, y a standard basis for H . We now extend this construction to arbitrary isotropic subspaces.

(A3.4): PROPOSITION. Let $X \subseteq V$ be an isotropic subspace. We can find another isotropic subspace Y complementary to $X^\perp$: $V = X^\perp \oplus Y$. Given such a Y , the subspace $X \oplus Y$ is non-degenerate, and $V_1 = (X \oplus Y)^\perp$ is a non-degenerate complement to X in $X^\perp$. Thus, we have a decomposition

$$(A3.5) \quad V = X \oplus V_1 \oplus Y.$$

PROOF. Choose any non-zero vector x in X , and find a second isotropic vector y such that x and y form a standard basis for a hyperbolic plane H . Then $V = H \oplus H^\perp$ and also $X = (X \cap H) \oplus (X \cap H)^\perp$. The proposition thus follows by an obvious induction.

Given a vector space X , denote by X^* the dual of X , and set $V(X) = X \oplus X^*$. Define on $V(X)$ a symmetric (respectively, skew-symmetric) form $\langle \cdot, \cdot \rangle_X$ by the recipe

$$(A3.6) \quad \langle x + \lambda, x' + \lambda' \rangle_X = \lambda(x') \pm \lambda'(x) \quad x, x' \in X, \lambda, \lambda' \in X^*.$$

Here we use the $+$ sign if we want a symmetric form, and the $-$ sign if we want a symplectic form. It is easy to check that $\langle \cdot, \cdot \rangle_X$ is non-degenerate. We call it the standard form on $V(X)$. We have an obvious action of $GL(V)$ on $V(X)$:

$$(A3.7) \quad g(x + \lambda) = g(x) + g^*{}^{-1}(\lambda), \quad g \in GL(X), x \in X, \lambda \in X^*.$$

It is trivial to check that this action embeds $GL(X)$ in the isometry group $G(V(X), \langle \cdot, \cdot \rangle_X)$. Further, it identifies $GL(X)$ to the subgroup of $G(V(X), \langle \cdot, \cdot \rangle_X)$ which preserves both of the subspaces X and X^* of $V(X)$.

Suppose we can find in V a subspace X such that $X^\perp = X$. Clearly, such an X is isotropic. Further, if $X \subset Z$, then $Z^\perp \subset X^\perp = X$, so that Z cannot be isotropic unless $Z = X$. Hence X is a maximal isotropic subspace. We call an X such that $X^\perp = X$ a polarization of V . Given a polarization $X \subset V$, we can find an isotropic subspace Y giving a decomposition of V as in Proposition A3.4. The assumption that X is a polarization implies that $V_1 = \{0\}$, so that we simply have $V = X \oplus Y$. For this reason, a space V which contains a polarization X is said to be split. The space Y complementary to X is also a polarization, and we call the pair (X, Y) a complete polarization.

(A3.8) PROPOSITION. Let (X, Y) be a complete polarization of V . Then there is a unique isomorphism $\alpha: Y \rightarrow X^*$, such that the map

$$1_X \oplus \alpha: V = X \oplus Y \rightarrow X \oplus X^* = V(X)$$

is an isometry.

PROOF. Indeed, the map $\alpha: Y \rightarrow X^*$ defined by

$$\alpha(y)(x) = \langle y, x \rangle \quad y \in Y, x \in X$$

is easily checked to satisfy the proposition.

(A3.9) COROLLARY. Let $X \subset V$ be an isotropic subspace. Fix a decomposition of $V = X \oplus V_1 \oplus Y$ as in Proposition (A3.4). There is a unique embedding $j_Y = j$ of $GL(X)$ in $G = G(V, \langle, \rangle)$ such that $j(GL(X))$ preserves X, Y and V_1 , acts in the standard way on X , and fixes V_1 pointwise. Let $M_X = M$ be the subgroup of G which preserves each summand in the decomposition $V = X \oplus V_1 \oplus Y$. Then the restriction of elements of M to $X \oplus V_1$ defines an isomorphism

$$(A3.10) \quad r_{X \oplus V_1} : M_X \cong GL(X) \times G(V_1)$$

Similarly, restriction to $V_1 \oplus Y$ defines an isomorphism of M_X with $G(V_1) \times GL(Y)$.

PROOF. This is clear.

Let P_X denote the subgroup of G which preserves X , let U_X be the subgroup of P_X which acts trivially on X and on $X^\perp/X$, and let ZU_X be the subgroup of U_X which acts trivially on $X^\perp$. A simple calculation shows that elements of U_X will act as the identity on $V/X^\perp$, and elements of ZU_X will act as the identity on V/X .

(A3.11) PROPOSITION. a) The group U_X acts simply transitively on the set of all isotropic subspaces Y complementary to $X^\perp$.

b) The group ZU_X is central in U_X ; in particular, ZU_X is abelian. For any decomposition of type (A3.5) of V , there is a natural isomorphism of ZU_X with the space of bilinear forms on $Y \cong V/X$ of parity opposite to the parity of $\langle, \rangle$.

c) The map defined by restriction of U_X to $X^\perp$ defines a surjective map to the subgroup of $GL(X^\perp)$ which acts trivially on X and on $X^\perp/X$. This group is abelian and isomorphic to $\text{Hom}(X^\perp/X, X)$.

d) The group U_X is two-step nilpotent and sits in an exact sequence

$$(A3.12) \quad 1 \rightarrow ZU_X \rightarrow U_X \rightarrow \text{Hom}(X^\perp/X, X) \rightarrow 1.$$

e) P_X is a semidirect product:

$$(A3.13) \quad P_X \cong M_X \ltimes U_X.$$

PROOF. Since the factor V_1 in the decomposition (A3.5) is determined by the factors X and Y , any element of G which preserves X and Y also preserves V_1 , hence belongs to M_X . By the isomorphism (A3.10), any element of M_X which acts as the identity on X and on $X^\perp/X \cong V_1$ must be the identity. This proves that the stabilizer of Y in U_X is trivial. This is the "simple" part of simple transitivity in part a).

We take for granted the fact that if $U_{X^\perp, X}$ is the subgroup of $GL(X^\perp)$ which preserves X , and acts trivially on X and on $X^\perp$, then the map $u \rightarrow u - 1$ (where 1 here indicates the identity map of X) defines an isomorphism of $U_{X^\perp, X}$ with $\text{Hom}(X^\perp/X, X)$. It is trivial to check that $U_{X^\perp, X}$ consists of isometries of the restriction of $\langle, \rangle$ to $X^\perp$. Witt's Theorem guarantees that the elements of $U_{X^\perp, X}$ are the restrictions to $X^\perp$ of isometries of V . By definition of U_X , these extensions will belong to U_X . This establishes part c).

Consider now the transitivity statement of part a). In the decomposition (A3.5), the summand V_1 is equal to $(X \oplus Y)^\perp$, and vice-versa. Consider another such decomposition $V = X \oplus V_1' \oplus Y'$. By part c), we can find an element of U_X which takes V_1' to V_1 . Assuming this done, we take $V_1' = V_1$. Then we have $X \oplus Y' = X \oplus Y$. Suppose we know the transitivity statement for $U_2 = X \oplus Y$. Then we can find an element u of $G(V_2)$ taking Y' to Y and leaving X pointwise fixed. We may extend u to V by letting it fix V_1 pointwise. This extension clearly is in U_X ; in fact, it is in ZU_X . This reasoning reduces the transitivity statement of part a) to the case when $V_1 = \{0\}$, i.e., when V is split. In this case, it is a corollary of Proposition A3.8: one uses Y and Y' to get isometries of V with $X \oplus X^*$, and combines one of these isometries with the inverse of the other. Thus, part a) is proven.

Let $U_{V, X^\perp, X}$ denote the subgroup of $GL(V)$ which preserves X and $X^\perp$, and acts trivially on X , on $X^\perp/X$, and on $V/X^\perp$. This is easily seen to be a two-step nilpotent group whose center consists of elements which act trivially on V/X and on $X^\perp$. These conditions are equivalent for elements of G . It follows that U_X is two-step nilpotent and that ZU_X is central in U_X . To finish proving part b), it is enough to consider the case when $V_1 = \{0\}$. Then $ZU_X = U_X$. Take U in U_X and consider the bilinear form B_u on Y given by

$$(A3.14) \quad B_u(y, y') = \langle u(y), y' \rangle = \langle (u - 1)(y), y' \rangle.$$

The second equality is valid because Y is isotropic. Since u acts as the identity on V/X and on X , we have $\langle u - 1 \rangle^2 = 0$, or $u^2 - 2u + 1 = 0$, or $u(2 - u) = 1$, so that $u^{-1} = 2 - u$. Hence we have

$$\begin{aligned} B_u(y, y') &= \langle u(y), y' \rangle = \langle y, u^{-1}(y') \rangle = \langle y, (2 - u)(y') \rangle \\ &= -\langle y, u(y') \rangle = -\langle B_u(y', y) \rangle. \end{aligned}$$

This shows B_u is a bilinear form on $Y \cong V/X$, of the opposite parity from $\langle, \rangle$. Conversely, given such a form B , we can define a map $z_B : Y \rightarrow X$ by the recipe $\langle z_B(y), y' \rangle = B(y, y')$. It is then straightforward to check that $u = 1 + z_B$ belongs to U_X and that this gives an inverse to the map $u \rightarrow B_u$ just defined. This proves statement b) of the proposition.

It remains only to prove part e). This is essentially a corollary of part a). It is obvious that U_X is a normal subgroup of P_X . Fix the decomposition (A3.5) of V . Given p in P_X , let $Y' = p(Y)$. Part a) tells us we can find u in U_X such that $Y' = u(Y)$, or $u^{-1}p(Y) = Y$, which means that $u^{-1}p$ belongs to M_X . This shows that $P_X = M_X U_X$. The fact that an expression $p = um$ is unique follows from the simple transitivity of U_X on the isotropic complements to X . This finishes the proof of the proposition.

We now pay particular attention to the variety of polarizations. Assume V is split, and let $V = X \oplus Y$ be a complete polarization of V . Let Z be a third polarization. Given z in Z , we can write $z = x + y$ with x in X and y in Y . The set of possible y which arise in this decomposition of elements of Z is the projection of Z into Y along X , which is

$$(A3.15) \quad Y_Z = (X + Z) \cap Y.$$

Given y in Y_Z , the set of x such that $z = x + y$ is in Z is a coset of $Z \cap X$. Since Z and X are their own orthogonal complements, we have $Z \cap X = (X + Z)^\perp$. It follows that for y, y' in Y_Z , the pairing

$$(A3.16) \quad B_Z(y, y') = \langle y, z' \rangle = \langle y, x' \rangle,$$

where $z' = x' + y'$ is any element in Z which projects to y' , is well-defined. Since Z is isotropic, we see that

$$\begin{aligned} 0 &= \langle z, z' \rangle = \langle x + y, x' + y' \rangle = \langle x, x' \rangle + \langle x, y' \rangle + \langle y, x' \rangle + \langle y, y' \rangle \\ &= \pm B_Z(y, y') + B_Z(y, y'). \end{aligned}$$

This says that B_Z is a bilinear form of symmetry type opposite that of $\langle, \rangle$.

(A3.17) PROPOSITION. The map which associates to Z in $J_m(V)$, where $\dim V = 2m$, the pair (Y_Z, B_Z) establishes a bijection between $J_m(V)$ and the set of pairs (U, B) where $U \subseteq Y$ is any subspace of Y and B is a bilinear form on U , of symmetry type opposite that of $\langle, \rangle$.

PROOF. We have just seen that Y_Z and B_Z are objects of the sort stated by the proposition. We need to be sure that the pair (Y_Z, B_Z) determines Z , and that any pair (U, B) of the stated sort arises from some Z . Given such a pair, and a vector u in U , we can find an element $x = x(u, B)$ in X such that $\langle u', x \rangle = B(u', u)$ for all u' in U . This x is defined up to addition of an element of $U^\perp \cap X = (U \oplus X')^\perp$. If one looks at the space $Z = Z(U, B)$ spanned by the vectors $u + x(u, B)$, then reading backwards the calculations after formula (A3.16) shows that Z is isotropic. It has dimension $\dim U + \dim(U \oplus X)^\perp = \dim U + (2m - (m + \dim U)) = m$. It is further easy to check that $(Y_Z, B_Z) = (U, B)$. This finishes the proposition.

According to Proposition A3.11, the stabilizer P_X of the polarization X has the decomposition $P_X \cong GL(Y) \cdot U_X$, where U_X is isomorphic to the vector space of bilinear forms on Y of symmetry type opposite to $\langle, \rangle$. There is an obvious action of $GL(Y)$ on pairs (U, B) where $U \subseteq Y$ is a subspace and B is a form of opposite type to $\langle, \rangle$. One sets

$$(A3.18) \quad g(U, B) = (g(U), B \circ g^{-1}) \quad g \in GL(V).$$

(A3.19) PROPOSITION. a) The action of $M_X \cong GL(Y)$ on $J_m(V)$ transfers via the parametrization map of (A3.17) to the action (A3.18) of $GL(Y)$ on pairs (U, B) .

b) The action of U_X on $J_m(V)$ transfers via parametrization (A3.17) to addition of forms:

$$z(U, B) = (U, B + B_z), \quad z \in U_X,$$

where B_z is the form defined in (A3.14).

PROOF. Straightforward verification.

The apparently uniform statement of Proposition A3.19 hides a significant difference between the case when $\langle, \rangle$ is symmetric and when it is symplectic. When $\langle, \rangle$ is symmetric, the isometry group G , an orthogonal group, is disconnected, and

the variety $J_m(V)$ is also disconnected. Indeed, for G in $GL(X)$, the determinant of g^* acting on X^* satisfies $\det g^* = \det g$. It follows from the formula (A3.7) for the action of g on V that P_X is contained in SG , the special orthogonal group. It is clear from formula (A3.10) that if $d < m$, then M_X intersects both components of G , and when $\langle, \rangle$ is symplectic, G itself is connected, so the only case when $P_d(V)$ can be disconnected is when the form $\langle, \rangle$ is symmetric and $d = m$. Given two polarizations Z, Z' of V , we will say Z and Z' are of the same type if they belong to the same connected component of $P_m(V)$, and that they are of opposite type if they belong to different connected components.

(A3.20) PROPOSITION. When $\langle, \rangle$ is symmetric and $\dim V = 2m$, two members Z, Z' of $P_m(V)$ are of the same type if and only if $\dim Z/(Z \cap Z') = \dim(Z + Z')/Z$ is even. In particular, the members (X, Y) of a complete polarization of V are of the same type if and only if $m = \dim V/2$ is even.

PROOF. Indeed, given a complete polarization (X, Y) of V , we can choose a basis $\{e_j\}$ for X , and a dual basis $\{f_j\}$ for Y , so that $\langle e_j, f_k \rangle = \delta_{jk}$, where δ_{jk} is Kronecker's delta. Then X and Y can be interchanged by sending e_j to f_j , and f_j to e_j . This has determinant $(-1)^m$. In the general case, when X and Z are not transverse, we can apply the same reasoning to $X/(X \cap Z)$ and $Z/(X \cap Z)$, which form a complete polarization in $(X + Z)/(X \cap Z)$.

(A3.21): Inherited pairings.

We observe that if $X \subseteq V$ is a subspace which is isotropic with respect to the form $\langle, \rangle$, then we may define on the quotient $X^\perp/X$ a pairing of the same type as $\langle, \rangle$, by restricting $\langle, \rangle$ to $X^\perp$, and then observing that the pairing of two elements in $X^\perp$ depends only on their images in $X^\perp/X$. It is easy to check that the pairing on $X^\perp/X$ obtained in this way is non-degenerate. Usually we will denote this "subquotient" pairing with the same symbol as for the original pairing. We will call it the pairing inherited from $\langle, \rangle$.

We now consider some relationships between varieties of maximal isotropic spaces in the symmetric case. Let $(V, \langle, \rangle)$ continue as in Proposition A3.20: V is even dimensional and $\langle, \rangle$ is symmetric. Let $\tilde{V} \subseteq V$ be a non-degenerate subspace of codimension 1. The orthogonal complement $\tilde{V}^\perp$ is a non-isotropic line, and we have $V = \tilde{V} \oplus \tilde{V}^\perp$. Let $X \subseteq V$ be a maximal isotropic subspace. If $\dim V = 2m$, then $\dim X = m$. Consider the intersection $X \cap \tilde{V}$. It will have dimension at least $\dim X - \text{codim } \tilde{V} = m - 1$. On the other hand, it will certainly be isotropic, and isotropic subspaces of $\tilde{V}$ can have dimension at most $m - 1$. Hence $\dim X \cap \tilde{V} = m - 1$, and $X \cap \tilde{V}$ is a maximal isotropic subspace of $\tilde{V}$.

Let $\tilde{p}: V \rightarrow \tilde{V}$ be orthogonal projection. Since X is isotropic, it cannot contain the line $\tilde{V}^\perp$, so that $\tilde{p}$ is an isomorphism when restricted to X . Thus, $\dim \tilde{p}(X) = m$. We may also describe $\tilde{p}(X)$ as $(X \oplus \tilde{V}^\perp) \cap \tilde{V}$. From this it is clear that $\tilde{p}(X) \subseteq (X \cap \tilde{V})^\perp$, whence $\tilde{p}(X) = (X \cap \tilde{V})^{\perp\sim}$, where $\perp\sim$ indicates orthogonal complementation inside $\tilde{V}$.

In the statement of the next proposition, the isometry group $G(\tilde{V}, \langle, \rangle)$ is understood to be embedded in $G(V, \langle, \rangle)$ as the subgroup which leaves $\tilde{V}^\perp$ pointwise fixed.

$$\tilde{p}: X \rightarrow X \cap \tilde{V}$$

defines a $G(\tilde{V}, \langle, \rangle)$ -equivariant bijection from each of the connected components of $P_m(V)$ onto $P_{m-1}(\tilde{V})$.

PROOF. Given a maximal isotropic subspace $\tilde{X}$ of $\tilde{V}$, consider the possible maximal isotropic X such that $\tilde{X} = X \cap \tilde{V}$. We must have $X \subseteq (\tilde{X})^\perp$. The quotient $(\tilde{X})^\perp / X$ has dimension 2, and hence when equipped with the inner product inherited from $\langle, \rangle$, is a hyperbolic plane. A maximal isotropic subspace of V containing $\tilde{X}$ will project to an isotropic line in $(\tilde{X})^\perp / \tilde{X}$. But it is well known that a hyperbolic plane contains exactly two isotropic lines. Further, the criterion of Proposition A3.20 tells us that the two elements of $J_m(V)$ defined by these lines belong to different connected components. This proves the proposition.

(A.23) COROLLARY. The action of $G(\tilde{V}, \langle, \rangle)$ on each connected component of $P_m(V)$ is transitive.

(A3.24) Remark: It is quite rare for a proper subgroup H of a group G to act transitively on a flag manifold of G . The only other cases are Sp_{2n} acting on projective $2n-1$ space, and G_2 acting on the Grassmannian of isotropic vectors in $\mathbb{C}^7$ [Dem], [On].

We can refine this corollary to describe the action of certain subgroups of $G(\tilde{V}, \langle, \rangle)$. Let $V = X \oplus Y$ be a complete polarization of V . We claim that

$$(A3.25) \quad \tilde{V} = (X \cap \tilde{V}) \oplus (\tilde{p}(X) \cap \tilde{p}(Y)) \oplus (Y \cap \tilde{V})$$

is a decomposition of $\tilde{V}$ of the type of formula (A3.5), with $\tilde{V}_1 = \tilde{p}(X) \cap \tilde{p}(Y)$ of dimension 1 and $X \cap \tilde{V}$ and $Y \cap \tilde{V}$ maximal isotropic. Indeed, we already know that $X \cap \tilde{V}$ and $X \cap \tilde{V}$ are maximal isotropic subspaces of $\tilde{V}$, and it is obvious that they have trivial intersection. Thus, $(X \cap \tilde{V}) \oplus (Y \cap \tilde{V})$ has codimension 1 in $\tilde{V}$. Further,

$$\tilde{p}(X) \cap \tilde{p}(Y) = ((X \cap \tilde{V})^\perp \cap ((Y \cap \tilde{V})^\perp)^\perp) = ((X \cap \tilde{V}) \oplus (Y \cap \tilde{V}))^\perp$$

is the one-dimensional orthogonal complement to $(X \cap \tilde{V}) \oplus (Y \cap \tilde{V})$ in $\tilde{V}$. This establishes the claim.

(A3.26) COROLLARY. Given a complete polarization (X, Y) of V , the subgroup $U_{X \cap \tilde{V}}$ of $G(\tilde{V}, \langle, \rangle)$ acts simply transitively on the set of polarizations of V transverse to X .

PROOF. The set of polarizations of V transverse to a fixed polarization all belong to the same component of $P_m(V)$, by the criterion of Proposition A3.20. The corollary follows from this, from the discussion of formula (A3.25), from Proposition A3.22, and from Proposition A3.11 a).

Now, we reverse the situation and consider an odd dimensional V , say $\dim V = 2m+1$, containing an even dimensional non-degenerate hyperplane $\tilde{V}$. The inner

product on V is still denoted $\langle, \rangle$. Consider a maximal isotropic subspace X of V . We know that $\dim X = m$. It is possible that $X \subseteq \tilde{V}$, but if not, the intersection $X \cap \tilde{V}$ will have dimension $m-1$. Then $\dim(X \cap \tilde{V})^\perp = m+2$. Since $\tilde{V}^\perp \subseteq (X \cap \tilde{V})^\perp$, if we set $(X \cap \tilde{V})^\perp \cap \tilde{V} = (X \cap \tilde{V})^\perp \cap \tilde{V}$, we have $\dim((X \cap \tilde{V})^\perp \cap \tilde{V}) = m+1$. Thus, the space $(X \cap \tilde{V})^\perp \cap \tilde{V}$ is three dimensional, and is equipped with the inner product inherited from $\langle, \rangle$. We have the orthogonal decomposition

$$(X \cap \tilde{V})^\perp \cap \tilde{V} = \tilde{V}^\perp \oplus (X \cap \tilde{V})^\perp \cap (X \cap \tilde{V}).$$

If X' is a maximal isotropic subspace of V such that $X' \cap \tilde{V} = X \cap \tilde{V}$, then the quotient $X'/(X \cap \tilde{V})$ is an isotropic line in $(X \cap \tilde{V})^\perp \cap (X \cap \tilde{V})$. This gives us a bijection between maximal isotropic subspaces of V with fixed intersection $X \cap \tilde{V}$ with $\tilde{V}$, and isotropic lines in the three-dimensional space $(X \cap \tilde{V})^\perp \cap (X \cap \tilde{V})$.

We digress briefly to discuss the variety of isotropic lines in 3-space. On $\mathbb{C}^3$ consider the inner product:

$$(A3.27) \quad ((x_1, x_2, x_3), (y_1, y_2, y_3)) = x_1 y_3 - 2x_2 y_2 + x_3 y_1$$

Consider the (Severi) map from $\mathbb{C}^2$ to $\mathbb{C}^3$, given by the formula

$$(A3.28) \quad S(x_1, x_2) = (x_1^2, x_1 x_2, x_2^2).$$

It is trivial to check that $S(x_1, x_2)$ is isotropic for the form (A3.27), and easy to verify that any isotropic vector in $\mathbb{C}^3$ must be of the form $S(x_1, x_2)$, with the pair (x_1, x_2) determined up to ± 1 . Thus, the Severi map factored modulo scalars defines a bijection between the projective line $\mathbb{P}^1(\mathbb{C})$ and the variety $J_1(\mathbb{C}^3)$ of isotropic lines in $\mathbb{C}^3$.

Consider the actions of $\mathbb{C}^\times$ on $\mathbb{C}^2$ and on $\mathbb{C}^3$ defined by the matrix groups

$$(A3.29) \quad \alpha_2(a) = \begin{bmatrix} a & 0 \\ 0 & a^{-1} \end{bmatrix} \quad \alpha_3(a) = \begin{bmatrix} a^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & a^{-2} \end{bmatrix} \quad a \in \mathbb{C}^\times.$$

It is trivial to check that $S(\alpha_2(a)(x_1, x_2)) = \alpha_3(a)(S(x_1, x_2))$ and that $\alpha_3(a)$ is an isometry of the form (A3.27). It is easy to see that the lines through $(1, 0)$ and $(0, 1)$ are invariant under α_2 , and that $\mathbb{C}^\times$ acts simply transitively via α_2 on the rest of the lines in $\mathbb{C}^2$. Because of the connection between α_2, α_3 and S , similar statements hold for the action of $\mathbb{C}^\times$ on $J_1(\mathbb{C}^3)$. Thus, α_3 preserves the (isotropic) lines through $(1, 0, 0)$ and $(0, 0, 1)$, and acts simply transitively on the remaining points in $J_1(\mathbb{C}^3)$.

Now, return to consideration of maximal isotropic subspaces of V . Let $V = X \oplus V_1 \oplus Y$ be a decomposition of type (A3.5), with X and Y maximal isotropic, so that V_1 is just a line. Suppose that neither X nor Y is contained in the hyperplane $\tilde{V}$. Then we have decompositions

$$(A3.30) \quad \tilde{V} = (X \cap \tilde{V}) \oplus V_2 \oplus (Y \cap \tilde{V}) \quad \text{and} \quad V = (X \cap \tilde{V}) \oplus V_3 \oplus (Y \cap \tilde{V})$$

also of the type of (A3.5). In these decompositions, the space V_2 is 2-dimensional, and the space V_3 is 3-dimensional. Also we have

$$(A3.31) \quad V_3 = V_1 \oplus V_2 \quad \text{and} \quad V = V_1 \oplus \bar{V}.$$

Furthermore, the space V_3 is a complement to $X \cap \bar{V}$ in $(X \cap \bar{V})^\perp$, and likewise is a complement to $Y \cap \bar{V}$ in $(Y \cap \bar{V})^\perp$.

In the context of the space V_3 , our discussion just above of 3-dimensional geometry tells us that the special (= determinant one) isometry group $SG(V_2)$ of the plane V_2 acts transitively on the set of isotropic lines in $V_3 - V_2$. We can, of course, regard $G(V_2)$ as the subgroup of $G(V)$ which leaves $V_2^\perp$ pointwise invariant. Doing so, and recalling the relation between isotropic lines in V_3 and maximal isotropic subspaces of V containing $X \cap \bar{V}$, we see that $SG(V_2)$ acts simply transitively on the maximal isotropic subspaces X' of V such that $X \cap \bar{V} = X \cap \bar{V}$. A similar statement holds with Y in place of X . Combining this fact with Proposition A3.11 a), we can make the following conclusion, analogous to Corollary A3.26 in the case of even dimensional V .

(A3.32) PROPOSITION. *Use the notation of equations (A3.30) and (A3.31). The group $SG(V_2) \cdot U_{X \cap \bar{V}}$ acts simply transitively on the set of maximal isotropic spaces transverse to X and not contained in $\bar{V}$.*

Appendix 4: Examples of Skew Duality.

In this appendix, we describe the cases of skew duality when one of the groups is small, meaning it is a variant of GL_1 or GL_2 , whose representations are very well understood. These cases include some of the classical lore about fundamental representations, the Hodge decomposition, etc.

A4.1: (GL_n, GL_1) .

The pair (GL_n, GL_1) is realized by the standard action of GL_n on $\Lambda(C^n)$. The action of GL_1 coincides with the center of GL_n . The eigenspaces of GL_1 are the homogeneous components $\Lambda^k(C^n)$, for $0 \leq k \leq n$. They define irreducible representations for GL_n . Their highest weight vectors are $\prod_{j=1}^k \wedge e_j$, (where $\{e_j\}$ is the standard basis of C^n), and their highest weights are 1_k (see §1.2.7 for notation). For $1 \leq k \leq n-1$, the restriction of these representations to SL_n define the fundamental representations of SL_n . This representation is almost self-dual. We have

$$(A4.1.1) \quad \Lambda^{n-k}(C^n) \cong \det \otimes (\Lambda^k(C^n))^*$$

as GL_n -module. Thus, if we tensor by $\det^{-1/2}$ (this involves replacing GL_n by a two-fold cover, but we will ignore this in our notation), the action of GL_n on $\Lambda(C^n)$ becomes self-dual. This is the normalized GL_n (cf. §A2.3.7). Alternatively, we can note that $\Lambda^{n-k}(C^n)$ and $\Lambda^k(C^n)$ are mutually contragredient as SL_n -modules.

A4.2: (GL_n, GL_2) .

We can realize the pair (GL_n, GL_2) on $\Lambda(C^n \otimes C^2)$. As a basis for $C^n \otimes C^2$ we take the elements $e_{jk} = e_j \otimes \bar{e}_k$, where $\{e_j : 1 \leq j \leq n\}$ is the standard basis for C^n , and $\{\bar{e}_k : k = 1, 2\}$ is the standard basis for C^2 . Then the Lie algebra of GL_2 has a basis made of the four operators

$$(A4.2.1) \quad d_1 = \sum_{j=1}^n \wedge_{e_{j1}} \rfloor e_{j1}, \quad d_2 = \sum_{j=1}^n \wedge_{e_{j2}} \rfloor e_{j2},$$

$$e^+ = \sum_{j=1}^n \wedge_{e_{j1}} \rfloor e_{j2}, \quad \text{and} \quad e^- = \sum_{j=1}^n \wedge_{e_{j2}} \rfloor e_{j1}$$

(These are the unnormalized operators.) The operator d_1 measures total degree in the e_{j1} 's, and likewise d_2 measures total degree in the e_{j2} 's. The operator e^+ replaces e_{j2} with e_{j1} , and the operator e^- replaces e_{j1} with e_{j2} .

Let us denote by $\Lambda^{a,b}$ the representation of GL_n corresponding to the diagram with two columns of length a and b , with $a \geq b$. Then in the decomposition of $\Lambda(C^n \otimes C^2)$ as $GL_n \times GL_2$ -module, the representation $\Lambda^{a,b}$ of GL_n is paired with the representation $\rho_2^{(a,b)}$ of GL_2 . The restriction of $\rho_2^{(a,b)}$ to the diagonal torus contains weights (c, d) such that $c + d = a + b$ and $b \leq c, d \leq a$. On the other hand, the joint eigenspaces for the operators d_k , which span the Lie algebra of the

diagonal torus of gl_2 , are the tensor products $\Lambda^c(C^n) \otimes \Lambda^d(C^n)$. As an example of the reciprocity law of §5.7.1, we have the decomposition

$$(A4.2.2) \quad \Lambda^c(C^n) \otimes \Lambda^d(C^n) \cong \sum_{\substack{a+b=c+d \\ b \leq c, d \leq a}} \Lambda^{a,b} \cong \sum_{r=0}^{\min(d, n-c)} \Lambda^{c+r, d-r}.$$

The last expression assumes that $d \leq c$.

Alternatively, we may realize the action of (GL_n, GL_2) on $\Lambda(C^n \oplus C^{n*})$. This involves tensoring the previous action of GL_n by $\det^{-1}$, or no change at all if we were using the normalized action. Let $\{e_j : 1 \leq j \leq n\}$ be the standard basis for C^n , and let $\{e_j^* : 1 \leq j \leq n\}$ be the dual basis for C^{n*} . Now, the Lie algebra of GL_2 is spanned by the operators

$$d = \sum_{j=1}^n \Lambda_{e_j} \rfloor e_j - n/2, \quad d^* = \sum_{j=1}^n \Lambda_{e_j^*} \rfloor e_j^* - n/2,$$

$$e^+ = \sum_{j=1}^n \Lambda_{e_j} \wedge e_j^*, \quad \text{and} \quad e^- = \sum_{j=1}^n \rfloor e_j \rfloor e_j^*.$$

The operator d records the degree in the elements e_j , and d^* records degree in the e_j^* . The operator e^- reduces each degree (i.e., degree in the e_j and degree in the e_j^*) by one, and e^+ raises each degree by one. The kernel of e^- is the space of lowest weight vectors for GL_2 . Its intersection with $\Lambda^a(C^n) \otimes \Lambda^b(C^{n*})$ is called the space of primitive vectors in $\Lambda^a(C^n) \otimes \Lambda^b(C^{n*})$, and is denoted $(\Lambda^a(C^n) \otimes \Lambda^b(C^{n*}))_{prim}$. By skew (GL_n, GL_2) -duality, $(\Lambda^a(C^n) \otimes \Lambda^b(C^{n*}))_{prim}$ is an irreducible module for GL_n . Its highest weight vector is $e_1 \wedge e_2 \wedge \dots \wedge e_a \wedge e_c^* \wedge e_{c+1}^* \wedge \dots \wedge e_n^*$, where $a < c = n + 1 - b$. In particular, $(\Lambda^a(C^n) \otimes \Lambda^b(C^{n*}))_{prim} = \{0\}$ if $a + b > n$. If we take into account the structure of GL_2 -modules, we can conclude that

$$(A4.2.4) \quad \Lambda^a(C^n) \otimes \Lambda^b(C^{n*}) \cong \sum_{k=0}^{\min(a,b)} e^{+k} ((\Lambda^{a-k}(C^n) \otimes \Lambda^{b-k}(C^{n*}))_{prim})$$

is a decomposition into irreducible GL_n -modules. This is essentially the same formula as (A4.2.2). In this form it is called the Hodge decomposition. This example is also discussed in [Hw 1]. It also provides an example of a reducible symplectic geometrically decomposable action (cf. Remark 4.7.4c). In that connection, note that $C \oplus C^{n*}$ does occur in a Heisenberg nilradical (in GL_{n+2} ; see List 5.5.2).

A4.3: (Sp_{2n}, Sp_2) .

If we put on $C^n \oplus C^{n*}$ the symplectic extension of the natural pairing between C^n and C^{n*} (see formulas (A3.6) and (A3.7)), then GL_n is seen as part of the symplectic group $Sp(C \oplus C^{n*}) = Sp_{2n}$. Since Sp_{2n} mixes C^n and C^{n*} together, the degree operators d and d^* of formulas (A4.2.3) do not commute with Sp_{2n} . However, the operators e^+ and e^- , since they are built out of the symplectic form, do commute with Sp_{2n} , and so does their commutator $d + d^* - n$, the (normalized)

total degree operator. These span a copy of the Lie algebra of $SL_2 = Sp_2$, which is the group dual to Sp_{2n} in this situation. Again, we consider the kernel of e^- , but now we cannot separate the various $(\Lambda^a(C^n) \otimes \Lambda^b(C^{n*}))_{prim}$. Instead the space $\Lambda^r(C^n \oplus C^{n*})_{prim}$, for $r \leq n$, is an irreducible Sp_{2n} -module, and the Hodge decomposition says

$$(A4.3.1) \quad \Lambda^r(C^n \oplus C^{n*})_{prim} \cong \sum_{0 \leq 2k \leq r} e^{+k} (\Lambda^{r-2k}(C^n \oplus C^{n*}))_{prim}.$$

Comparison of this formula with formula (A4.2.4) describes the decomposition of the $\Lambda^r(C^n \oplus C^{n*})_{prim}$ when restricted to GL_n .

We also know from the representation theory of the Lie algebra sl_2 that $e^{+\ell} : \Lambda^r(C^n \oplus C^{n*})_{prim} \rightarrow \Lambda^{r+2\ell}(C^n \oplus C^{n*})$ is injective for $\ell \leq n - r$, and is zero for $\ell > n - r$. The analogy with the theory of spherical harmonics for O_n in the symmetric case is clear. Something more interesting about the skew-symmetric situation for Sp_{2n} is that the spaces $\Lambda^r(C^n \oplus C^{n*})_{prim}$, for $1 \leq k \leq n$, realize the fundamental representations of Sp_{2n} .

The Hodge decomposition (A4.3.1) allows us to describe the Lie algebra cohomology of the Heisenberg Lie algebra. We recall from [Jac 3] or [Kna] the basics of Lie algebra cohomology. If $\mathfrak{g}$ is a Lie algebra, then the cohomology of $\mathfrak{g}$ (with trivial coefficients) is the cohomology of a complex based on $\Lambda(\mathfrak{g}^*)$. The differential of this complex is

$$(A4.3.2) \quad \delta = (1/2) \sum_j \Lambda_{b_j}^* ad_{b_j}^*,$$

where $\{b_j\}$ is a basis for $\mathfrak{g}$ and $\{b_j^*\}$ is the dual basis for $\mathfrak{g}^*$. Thus, if $Z^k = \ker \delta \cap \Lambda^k(\mathfrak{g}^*)$, and $B^k = \text{im } \delta \cap \Lambda^k(\mathfrak{g}^*)$, then $B^k \subset Z^k$ since $\delta^2 = 0$, and the cohomology $H^k(\mathfrak{g})$ is the quotient

$$(A4.3.3) \quad H^k(\mathfrak{g}) = Z^k / B^k.$$

Let $\mathfrak{h}$ be the Heisenberg Lie algebra. This is the vector space

$$(A4.3.4) \quad \mathfrak{h} = V \oplus Z,$$

where V is a vector space equipped with a symplectic form $\langle \cdot, \cdot \rangle$, and Z is a line with a given basis element z . The Lie bracket on $\mathfrak{h}$ is given by

$$(A4.3.5) \quad [v + tz, v' + t'z] = \langle v, v' \rangle z, \quad v, v' \in V.$$

Let $\{e_j, f_j\}$ be a standard symplectic basis for V (see Appendix 3, from below (A3.3) through (A3.8)). Let $\{e_j^*, f_j^*\}$ be the dual basis for V^* , and let z^* be the basis element of Z^* dual to z . Plugging in to formula (A4.3.2), we find the formula

$$(A4.3.6) \quad \delta = e^+ \rfloor_{z^*},$$

where e^+ is the operator in the $\mathfrak{sl}_2$ dual to $Sp(V)$ acting on V^* (see formula (A4.2.3)). We have

$$\Lambda^k(\eta^*) = \Lambda^k(V^*) \oplus \mathbf{z}^* \wedge (\Lambda^{k-1}(V^*)).$$

Applying d to this decomposition, we find

$$Z^k = \Lambda^k(V^*) \oplus \mathbf{z}^* \wedge (\ker e^+ \cap (\Lambda^{k-1}(V^*))),$$

and

$$B^k = e^+(\Lambda^{k-2}(V^*)).$$

From formula (A4.3.1) and the accompanying remark, we know that $\ker e^+ \cap (\Lambda^{k-1}(V^*)) = \{0\}$ for $k < n$, while $B^k = \Lambda^k(V^*)$ for $k > n$. From these facts we arrive at the following description of the cohomology of $\mathfrak{h}$.

$$(A4.3.7) \quad H^k(\mathfrak{h}) \cong \Lambda^k(V)_{prim}$$

for $k \leq n$, and

$$H^k(\mathfrak{h}) \cong \mathbf{z}^* \wedge (\ker e^+ \cap (\Lambda^{k-1}(V^*))) = \mathbf{z}^* \wedge (e^{+\tau}(\Lambda^s(V)_{prim})) \text{ for } k > n,$$

where $\tau = k - n - 1$ and $s = 2n + 1 - k$.

These formulas are special cases of the result of Kostant [Kos 1].

(A4.4): (O_n, SO_1) ; the spin representations.

We have seen in §A2.3.6 that the orthogonal Lie algebra $\mathfrak{so}(V \oplus V^*)$ embeds in $End(\Lambda(V))$. Thus, $\Lambda(V)$ is a module for $\mathfrak{so}(V \oplus V^*)$, hence for the universal cover $Spin(V \oplus V^*)$ of $SO(V \oplus V^*)$. Likewise, by using the operators coming from elements of norm 1 in $V \oplus V^*$, as in formula (A2.3.5.3), we can get an action of $Pin(V \oplus V^*)$, a covering of $O(V \oplus V^*)$ whose identity component is $Spin(V \oplus V^*)$. According to the Clifford algebra commutant version of the FFT for O_n (Theorem 3.3.4.2), the commutant of $O(V \oplus V^*)$ in $End(\Lambda(V))$ is reduced to the scalars. Hence $\Lambda(V)$ is an irreducible module for $Pin(V \oplus V^*)$. It is called the spin module.

On the other hand, under $Spin(V \oplus V^*)$, equivalently under $\mathfrak{so}(V \oplus V^*)$, $\Lambda(V)$ is clearly not irreducible, since the subspaces $\Lambda^{even}(V)$ and $\Lambda^{odd}(V)$, consisting of the $\Lambda^k(V)$ for k even (respectively, k odd), are invariant under $\mathfrak{so}(V \oplus V^*)$, since this is spanned by the spaces $\mathfrak{so}(C \oplus C^{m*})^{(2,0)}$, $\mathfrak{so}(C^m \oplus C^{m*})^{(0,2)}$ and $\mathfrak{so}(C^n \oplus C^{n*})^{(1,1)}$, which change degree by $+2, -2$, and 0 respectively. On the other hand, since $Spin(V \oplus V^*)$ has index two in $Pin(V \oplus V^*)$, the $Pin(V \oplus V^*)$ -module $\Lambda(V)$ can break up into at most two components for $Spin(V \oplus V^*)$, so $\Lambda^{even}(V)$ and $\Lambda^{odd}(V)$ must be irreducible for $Spin(V \oplus V^*)$. They are called the half-spin modules $spin^+(V \oplus V^*)$ and $spin^-(V \oplus V^*)$. We adopt the convention that $spin^+(V \oplus V^*)$ is the subspace of containing $\Lambda^n(V)$, and $spin^-(V \oplus V^*)$ is the other one. If for a Borel subalgebra of $\mathfrak{so}(V \oplus V^*)$ we take the abelian subspace $\mathfrak{so}(V \oplus V^*)^{(2,0)}$ together with the standard Borel of $\mathfrak{so}(V \oplus V^*)^{(1,1)} \cong \mathfrak{gl}(V \oplus V^*)$, then the highest weight vector for $spin^+(V \oplus V^*)$ is $\prod_{j=1}^n \Lambda_{e_j}$, where $\{e_j\}$ is the standard basis for

C^n . It has weight $(1/2)\mathbf{1}_m$ (see §1.2.7 for notation). The highest weight vector of $spin^-(V \oplus V^*)$ is $\prod_{j=1}^{m-1} \Lambda_{e_j}$, and its weight is $(1/2)(\mathbf{1}_m - 2(\mathbf{1}_m - \mathbf{1}_{m-1}))$.

We now turn to spin representations for odd orthogonal groups. Let $W = (V \oplus V^*) \oplus \mathbb{C}$. Endow W with the direct sum of the standard inner product on $V \oplus V^*$ and z^2 on $\mathbb{C}$. We can embed the Lie algebra $\mathfrak{so}(W)$ in $End(\Lambda(V))$. Precisely, it is the span of the Lie algebra $\mathfrak{so}(V \oplus V^*)$ of degree two elements, plus $\bar{j}(V \oplus V^*)$. (See formula (A2.3.2.2) for notation.) This clearly acts irreducibly on $\Lambda(V)$, so it defines an irreducible representation of $\mathfrak{so}(W)$. It is called the spin representation of $\mathfrak{so}(W)$, denoted $spin(W)$. The Cartan subalgebra of $\mathfrak{so}(C^n \oplus C^{n*})$ remains a Cartan subgroup in $\mathfrak{so}(W)$, and we can arrange that $\Lambda^{top}(V)$ is again the highest weight vector for $\mathfrak{so}(W)$.

A4.5: (O_n, SO_2) .

Consider the action of O_n on $\Lambda(C^n)$. According to Theorem 3.3.4.1, the commutant of O_n in $End(\Lambda(C^n))$ is generated by the centralizer of O_n in the copy of $\mathfrak{so}_{2n}$ described in §A2.3.6 and §A2.3.7. This coincides with the center of (normalized) $\mathfrak{gl}_n$ and is spanned by the normalized degree operator $d - n/2$, where d acts on $\Lambda^k(C^n)$ by multiplication by k . Thus, we conclude that the homogeneous components $\Lambda^k(C^n)$ define distinct irreducible modules for O_n . On the other hand, we know that as modules for GL_n , $\Lambda^k(C^n)$ and $\Lambda^{n-k}(C^n)$ are related by formula (A4.1.1). Since the inner product on C^n gives rise to an O_n -invariant inner product on all of $\Lambda(C^n)$, each $\Lambda^k(C^n)$ is self-dual as an O_n -module. Hence we have

$$(A4.5.1) \quad \Lambda^{n-k}(C^n) \cong \Lambda^k(C^n) \otimes \det$$

as O_n -modules. However, it is easy to compute that for $k \leq n/2$, they define mutually distinct modules, even for SO_n . Indeed, if we coordinatize C^n as in §2.6.6.1, then the highest weight vector for O_n is the same as the highest weight vector for GL_n , and when $k \leq n/2$, its weight is $\mathbf{1}_k$. When $n = 2m + 1$ is odd, the $\Lambda^k(C^n)$ are fundamental representations for SO_n when $k < m$, and $\Lambda^m(C^n)$ is the Cartan square of the spin representation. When $n = 2m$ is even, the $\Lambda^k(C^n)$ are fundamental representations for $k < m - 1$. The space $\Lambda^m(C^n)$ decomposes into two SO_n -submodules, denoted $\Lambda^{m+}(C^n)$ and $\Lambda^{m-}(C^n)$. These are the Cartan squares of the half-spin modules for $Spin_n$. The space $\Lambda^{m-1}(C^n)$ is irreducible, and defines the Cartan product of the two half-spin modules. Then $\Lambda^k(C^n)$ which are fundamental, together with the spin representation(s), constitute the fundamental representations of $Spin_n$.

The Hodge star operator [GH] interchanges Λ^k and Λ^{n-k} , and commutes with the action of SO_n . Together with the scalar matrices, it generates a copy of Pin_2 which spans the full commutant of SO_n in $End(\Lambda(C^n))$.

A4.6: (O_n, SO_4) .

Next, consider the action of O_n on $\Lambda((C^n)^2) \cong \Lambda(C^n \otimes C^2)$. By Theorem 3.3.4.1, the commutant of O_n is generated by a Lie algebra isomorphic to $\mathfrak{so}_4$, which is itself isomorphic to $\mathfrak{sl}_2 \times \mathfrak{sl}_2$. This copy of $\mathfrak{so}_4$ is spanned by the operators of (normalized) $\mathfrak{gl}_2$ acting on the factor C^2 in the tensor product $C^n \otimes C^2$, together with two other operators. After we identify C^{n*} with C^n by means of the inner product, these

are essentially the operators e^+ and e^- of formula (A4.2.3). Also, the Cartan subalgebra in $\mathfrak{so}_4$ is spanned by the operators d and d^* of formula (A4.2.3). The elements belonging to the simple factors $\mathfrak{sl}_2$ of $\mathfrak{so}_4$ are $d - d^*$ and $d + d^*$.

Let us parametrize a representation of $\mathfrak{so}_4$ by the highest weights for $d - d^*$ and the highest weights for $d + d^*$. However, we will look for the corresponding representation of $O(V)$ among the highest weight vectors for $d - d^*$, but the lowest weight vectors for $d + d^*$. The eigenspaces of d and d^* are the tensor products $\Lambda^c(\mathbb{C}^n) \otimes \Lambda^d(\mathbb{C}^n)$, with corresponding $(d - d^*, d + d^*)$ -eigenvalue $(c - d, c + d - n)$. In order for this to be a highest weight for $d - d^*$, we must have $c - d \geq 0$, or $c \geq d$. Similarly, to have a lowest weight for $d + d^*$, we need $c + d - n \leq 0$, or $c + d \leq n$. The subspace of $\Lambda^c(\mathbb{C}^n) \otimes \Lambda^d(\mathbb{C}^n)$ consisting of lowest weight vectors for $d + d^*$ is the kernel in $\Lambda^c(\mathbb{C}^n) \otimes \Lambda^d(\mathbb{C}^n)$ of e^- . We denote this by $(\Lambda^c(\mathbb{C}^n) \otimes \Lambda^d(\mathbb{C}^n))_{prim}$ in analogy with §A4.2. If c and d satisfy the inequalities of the previous paragraph, we can find a joint $O_n \times GL_2$ highest weight vector in $\Lambda^c(\mathbb{C}^n) \otimes \Lambda^d(\mathbb{C}^n)$, by the method of §3.3.5. In the coordinate system of Remark 3.3.4.1.1, it is

$$(A4.6.1) \quad (e'_{11} \wedge e'_{21} \wedge \dots \wedge e'_{c_1}) \wedge (e'_{12} \wedge e'_{22} \wedge \dots \wedge e'_{d_2}).$$

We will denote the representation of $O(V)$ with this highest weight vector by $\tilde{\Lambda}^{c,d}(O(V))$. We will denote the corresponding representation of $\mathfrak{so}_4$ by $\sigma_4^{c,f}$, where $e = c - d$ and $f = n - c - d$. Thus (e, f) is the $(d - d^*, d + d^*)$ highest weight of $\sigma_4^{c,f}$. Neither of these notations is our usual notation for these representations.

We observe that the Cartan subalgebra of $\mathfrak{so}_4$ has multiplicity-free spectrum in the irreducible representations of $\mathfrak{so}_4$. In order for a (d, d^*) weight $(a - n/2, b - n/2)$ to appear in the representation $\sigma_4^{c,f}$, we must have

$$|a - b| \leq e, |a + b - n| \leq f,$$

and

$$a - b \equiv e \pmod{2}, a + b - n \equiv f \pmod{2}.$$

Let us write $e = c - d$ and $f = n - c - d$. Also, let us assume that $a \geq b$ and that $a + b \leq n$. Then the conditions above become

$$a - b \leq c - d \quad a + b \geq c + d,$$

and

$$a + b \equiv c + d \pmod{2}.$$

If we require that $a + b = c + d$, then the congruence condition is automatic, and we just need the additional requirement that $a - b \leq c - d$. Applying the reciprocity laws of §5.7, we may conclude that

$$(A4.6.2) \quad \Lambda^a(\mathbb{C}^n) \otimes \Lambda^b(\mathbb{C}^n)_{prim} \cong \sum_{k=0}^{\min(b, n-a)} \tilde{\Lambda}^{a+k, b-k}(O(V)),$$

for $a \geq b$, and $a + b \leq n$ (which is necessary in order that $(\Lambda^a(\mathbb{C}^n) \otimes \Lambda^b(\mathbb{C}^n))_{prim}$ be non-zero). Using equation (A4.2.2), we get a description of the tensor product

$\Lambda^a(\mathbb{C}^n) \otimes \Lambda^b(\mathbb{C}^n)$ as an O_n -module. Note that it is multiplicity-free. Also note that, up to permutation of factors and replacing (a, b) by $(n - a, n - b)$ (which does not change the tensor product, as it twists each factor by $\det$ (see formula (A4.5.1))), our restriction on (a, b) is no loss of generality. On the other hand, summing over (a, b) such that $a + b = k$, we get a decomposition of $\Lambda^k(\mathbb{C}^n \oplus \mathbb{C}^n)_{prim} \cong \Lambda^k(\mathbb{C}^n \oplus \mathbb{C}^{n*})_{prim}$ as O_n -module. It is not multiplicity-free, but it is multiplicity-free as $O_n \times SL_2$ -module. Note that SL_2 is the centralizer of O_n inside Sp_{2n} . Another consequence of our analysis is that $\Lambda(\mathbb{C}^n \otimes \mathbb{C}^2)$ is symplectic geometrically decomposable as an $O_n \times GL_2$ -module. This follows from the form (A4.6.1) of the primitive joint $O_n \times GL_2$ highest weight vectors. This observation is consistent with Remark 4.7.4 c) concerning the connection between symplectic geometrically decomposable modules and Heisenberg nilradicals. Note that the action of $O_n \times A'$, where A' is the torus in $GL_2 \subset SO_4$, is also multiplicity-free, since the A' spectrum of irreducible representations of SO_4 is multiplicity-free, but the action of $O_n \times A'$ is not symplectic geometrically decomposable.

A4.7: (SO_n, SO_3) .

As a final example, we discuss the pair (O_n, SO_3) . This was treated in a geometric context by Slupinsky [Slu 1]. Suppose for simplicity that n is even. Then the sum of the half spin representations of $Spin(\mathbb{C}^n \otimes \mathbb{C}^3)$ can be realized as a tensor product: $spin_{3n} \cong \Lambda(\mathbb{C}^n) \otimes spin_n$. This isomorphism is an isomorphism of O_n modules. As operators commuting with O_n , we can produce the following. Let $\{e_j : 1 \leq j \leq n\}$ be the standard orthonormal basis of $\mathbb{C}^n$. We have the (normalized) degree operator on $\Lambda(\mathbb{C}^n)$:

$$(A4.7.1) \quad d = \sum_{j=1}^n \wedge_{e_j} |e_j - n/2|,$$

as already described in equation (A4.2.3). We also have operators

$$(A4.7.2) \quad \varepsilon^+ = \sum_{j=1}^n \wedge_{e_j} \tilde{j}(e_j), \text{ and } \varepsilon^- = \sum_{j=1}^n |e_j| \tilde{j}(e_j).$$

Here $\tilde{j}(e_j)$ is the standard image of e_j in $C(\mathbb{C}^n)$ (see §A2.3.2 and §A2.3.4). It is easy to see that these three operators span a copy of $\mathfrak{sl}_2 \cong \mathfrak{so}_3$. It follows from Remark 4.3.4.5 b) that this copy of $\mathfrak{sl}_2$ generates the commutant of O_n in $End(spin_{3n})$. It follows that there is a Hodge-type decomposition of $\Lambda(\mathbb{C}^n) \otimes spin_n$. Denote by $(\Lambda^k(\mathbb{C}^n) \otimes spin_n)_{prim}$ the kernel of ε^- in $\Lambda^k(\mathbb{C}^n) \otimes spin_n$. Then the spaces $(\Lambda^k(\mathbb{C}^n) \otimes spin_n)_{prim}$ define distinct irreducible representations for O_n , and

$$(A4.7.3) \quad \Lambda^k(\mathbb{C}^n) \otimes spin_n = \sum_{\ell=0}^k (\varepsilon^+)^{\ell} ((\Lambda^{k-\ell}(\mathbb{C}^n) \otimes spin_n)_{prim}).$$

We refer to [Slu 1] for further details.

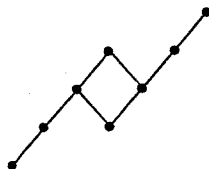
Appendix 5: Tables of Skew Multiplicity-Free Actions.

A5.1: Table of the weight posets for geometrically decomposable modules.

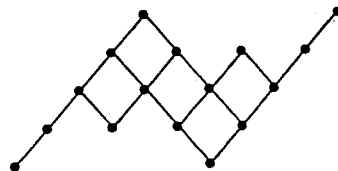
i) $\Lambda_{\text{prim}}^2(C^4)$ for Sp_4 : (This is equivalent to the standard representation of SO_5 .)



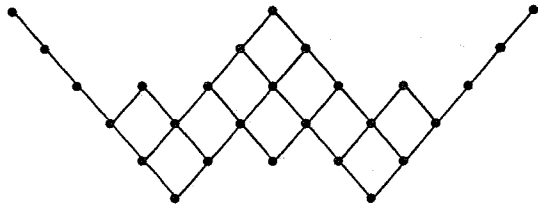
ii) $\text{spin}_8^\pm$: (This is equivalent to the standard representation of SO_8 .)



iii) $\text{Spin}_{10}^\pm$:



iv) V_{27} for E_6 :

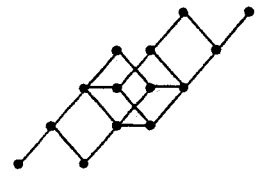


A5.2: Table of the weight posets of symplectic geometrically decomposable modules.

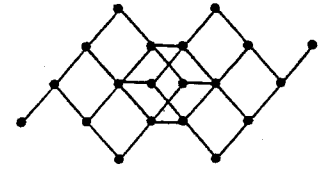
i) $S^3(\mathbb{C}^2)$:



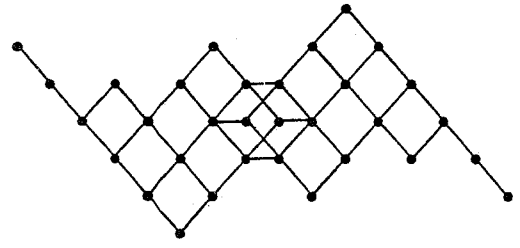
ii) $\Lambda^3(\mathbb{C}^6)_{\text{prim}}$:



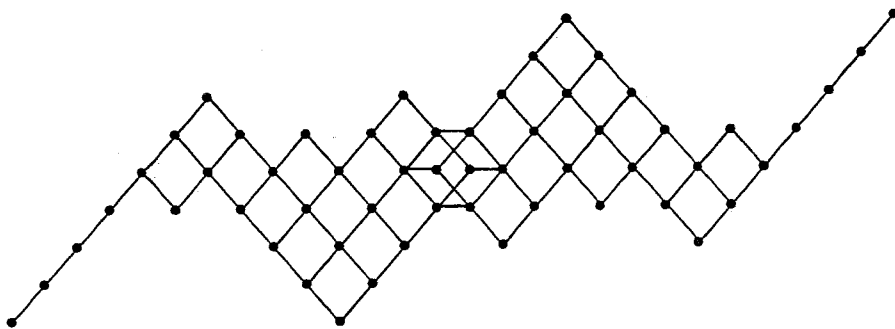
iii) $\Lambda^3(\mathbb{C}^6)$:



iv) spin_{12}^+ :



v) V_{56} for E_7 .

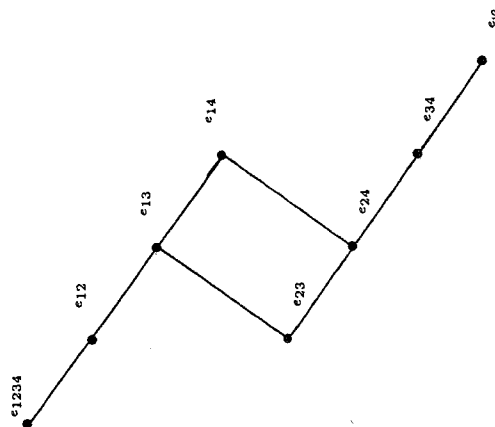


A5.3: Tables of $\Lambda(\text{spin}_{2n}^\pm)$ for $n = 4, 5, 6$, and their restrictions to Spin_{2n-1} .

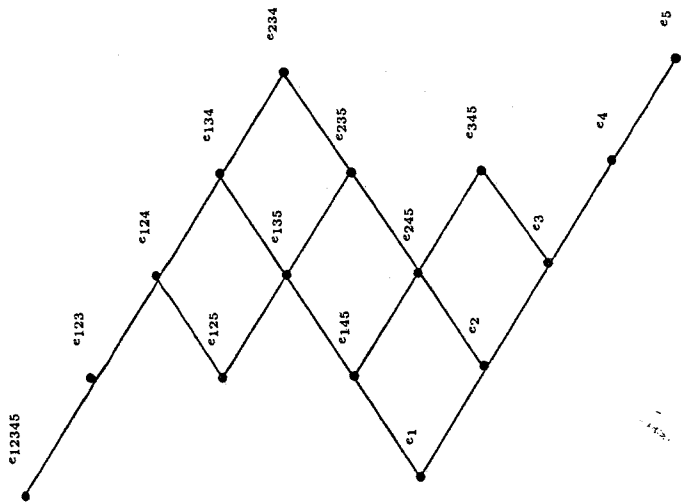
We realize spin_{2n}^+ on $\Lambda^{\text{even}}(\mathbb{C}^n)$ for $n = 4, 6$, and on $\Lambda^{\text{odd}}(\mathbb{C}^5)$ for $n = 5$. We take as Cartan subgroup of Spin_{2n} the standard diagonal torus in GL_n , acting on $\Lambda(\mathbb{C}^n)$ by its normalized action (see §A2.3.7). The weight vectors are the wedge products of sets of the standard basis vectors e_j of $\mathbb{C}^n$. We denote these products by subscripts indicating which vectors enter into the product. Thus, $e_{124} = e_1 \wedge e_2 \wedge e_4$, and so forth. The weights of these vectors have the form $\frac{1}{2}(\pm 1, \pm 1, \dots, \pm 1)$ where a given entry is ± 1 if the corresponding e_j is a factor in the weight vector, and is -1 if not. In this scheme, the highest weight vector is always $e_{12\dots n}$, with weight $\frac{1}{2}(1, 1, \dots, 1)$.

A5.3.1: Our first table exhibits the weight posets of the modules spin_{2n}^+ .

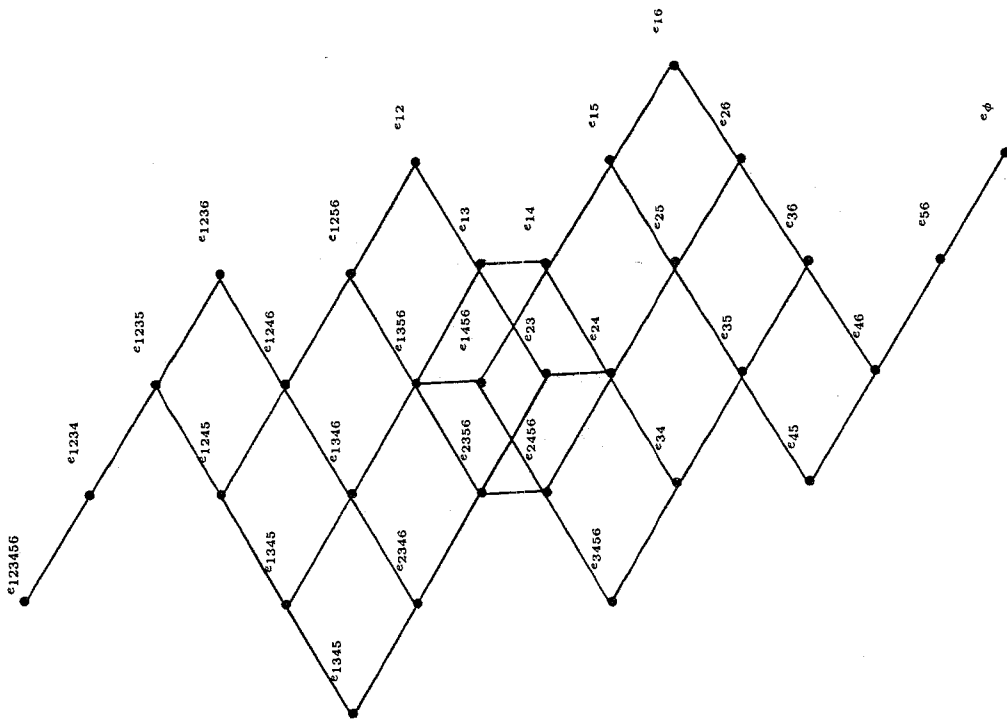
i) spin_8^+ :



ii) spin_{10}^+



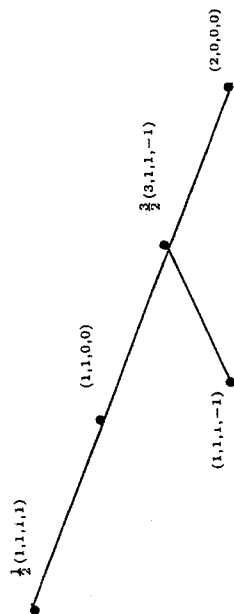
iii) $\text{Spin}_{12}^\pm$



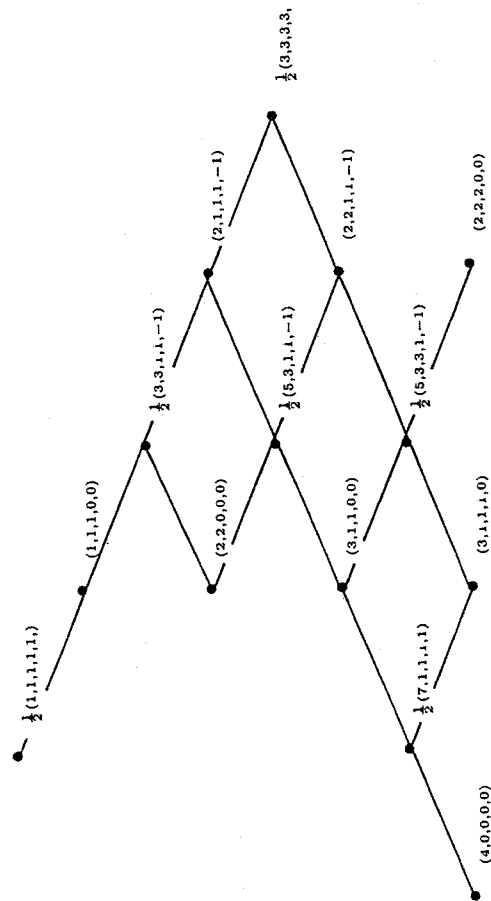
A5.3.2: Next we exhibit the lattices of geometric highest weight vectors, which are the same as the lattices of increasing subsets of the weight posets. For $n = 4, 5$, we write the weights of the geometric highest weight vectors at the points corresponding to them. For $n = 6$, this would create too messy a diagram, so we list the weights separately, preserving the order of the diagram in each row. For $n = 4, 5$, the geometric highest weight vectors are all the highest weight vectors. For $n = 6$, we give all the primitive highest weight vectors.

We exhibit the lattices of increasing sets up to size $2^{n-2} = \frac{1}{2} \dim \text{spin}_{2n}^\pm$, since past there everything is determined by duality.

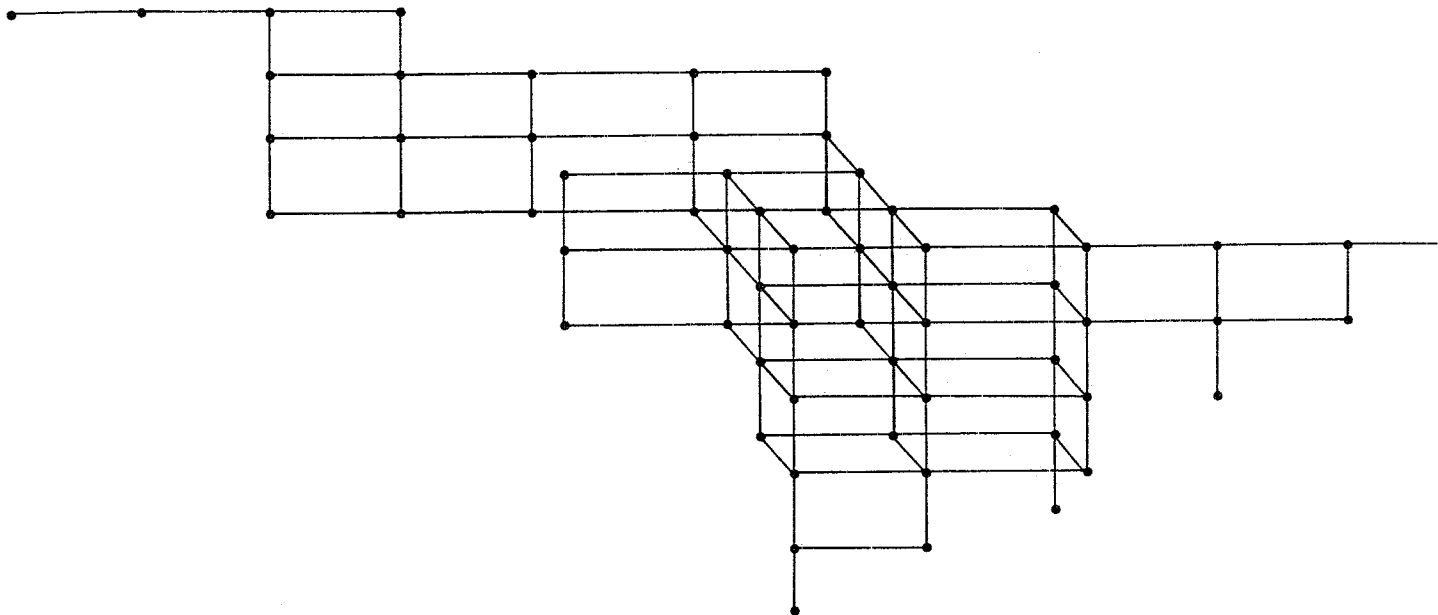
i) spin_8^+ :



ii) spin_{10}^+



iii) $\text{spin}_{12}^\pm$:



Geometric Highest Weights for spin_{12}^+

i) $\Lambda(\text{spin}_8^+) \mid \text{Spin}_7$:

$$\begin{aligned}\Lambda_1 &: \frac{1}{2}(1, 1, 1) \\ \Lambda_2 &: (1, 1, 0)(1, 0, 0) \\ \Lambda_3 &: \frac{1}{2}(3, 1, 1)\frac{1}{2}(1, 1, 1) \\ \Lambda_4 &: (1, 1, 1)(2, 0, 0)(1, 0, 0)(0, 0, 0)\end{aligned}$$

ii) $\Lambda(\text{spin}_{10}^+) \mid \text{Spin}_9$:

$$\begin{aligned}\Lambda_1 &: \frac{1}{2}(1, 1, 1, 1) \\ \Lambda_2 &: (1, 1, 1, 0)(1, 1, 0, 0) \\ \Lambda_3 &: \frac{1}{2}(3, 3, 1, 1)\frac{1}{2}(3, 1, 1, 1) \\ \Lambda_4 &: (2, 2, 0, 0)(2, 1, 0, 0)(2, 0, 0, 0)(2, 1, 1, 1)(1, 1, 1, 1) \\ \Lambda_5 &: \frac{1}{2}(5, 3, 1, 1)\frac{1}{2}(5, 1, 1, 1)\frac{1}{2}(3, 3, 1, 1)\frac{1}{2}(3, 1, 1, 1)\frac{1}{2}(3, 3, 3, 3) \\ \Lambda_6 &: (3, 1, 1, 0)(3, 1, 0, 0)(2, 1, 1, 0)(2, 1, 0, 0)(1, 1, 1, 0)(1, 1, 0, 0)(2, 2, 1, 1) \\ &\quad (2, 1, 1, 1) \\ \Lambda_7 &: \frac{1}{2}(7, 1, 1, 1)\frac{1}{2}(5, 1, 1, 1)\frac{1}{2}(3, 1, 1, 1)\frac{1}{2}(1, 1, 1, 1)\frac{1}{2}(5, 3, 3, 1)\frac{1}{2}(5, 3, 1, 1) \\ &\quad \frac{1}{2}(3, 3, 3, 1)\frac{1}{2}(3, 3, 1, 1) \\ \Lambda_8 &: (4, 0, 0, 0)(3, 0, 0, 0)(2, 0, 0, 0)(1, 0, 0, 0)(0, 0, 0, 0)(3, 1, 1, 1)(3, 1, 1, 0) \\ &\quad (2, 1, 1, 1)(2, 1, 1, 0)(1, 1, 1, 1)(1, 1, 1, 0)(2, 2, 2, 0)(2, 2, 1, 0)(2, 2, 0, 0)\end{aligned}$$

$$\begin{aligned}\Lambda^1 &: \frac{1}{2}(1, 1, 1, 1, 1, 1) \\ \Lambda^2 &: (1, 1, 1, 1, 0, 0) \\ \Lambda^3 &: \frac{1}{2}(3, 3, 3, 1, 1, -1) \\ \Lambda^4 &: (2, 2, 1, 1, -1)(2, 2, 2, 0, 0, 0) \\ \Lambda^5 &: \frac{1}{2}(5, 3, 3, 3, -3)\frac{1}{2}(5, 5, 3, 1, 1, +1) \\ \Lambda^6 &: (2, 2, 2, 2, -2)(3, 2, 2, 1, 1, -1)(3, 3, 1, 1, 0, 0) \\ \Lambda^7 &: \frac{1}{2}(5, 5, 5, 3, 3, -3)\frac{1}{2}(7, 5, 3, 3, 1, -1)\frac{1}{2}(7, 7, 1, 1, 1, 1) \\ \Lambda^8 &: (3, 3, 2, 2, 1, -1)(4, 2, 2, 2, 0, 0)(4, 3, 1, 1, 1, 0)(4, 4, 0, 0, 0, 0) \\ \Lambda^9 &: \frac{1}{2}(7, 5, 5, 5, 1, -1)\frac{1}{2}(7, 7, 3, 3, 3, -1)\frac{1}{2}(9, 5, 3, 3, 1, 1)\frac{1}{2}(9, 7, 1, 1, 1, -1) \\ \Lambda^{10} &: (3, 3, 3, 3, 0, 0)(4, 3, 2, 2, 1, 0)(5, 2, 2, 1, 1, 1)(4, 4, 1, 1, 1, -1)(5, 3, 1, 1, 0, 0) \\ \Lambda^{11} &: \frac{1}{2}(7, 7, 5, 5, 1, 1)\frac{1}{2}(9, 5, 5, 3, 3, 1)\frac{1}{2}(11, 3, 3, 3, 3, 3)\frac{1}{2}(9, 7, 3, 3, 1, -1) \\ &\quad \frac{1}{2}(11, 5, 3, 1, 1, 1) \\ \Lambda^{12} &: (4, 3, 3, 2, 1, 1)(5, 2, 2, 2, 2, 1)(4, 9, 2, 2, 0, 0)(5, 3, 2, 1, 1, 0)(6, 2, 1, 1, 1, 1) \\ &\quad (6, 2, 2, 0, 0, 0) \\ \Lambda^{13} &: \frac{1}{2}(7, 7, 7, 3, 3, 3)\frac{1}{2}(9, 5, 5, 5, 3, 3)\frac{1}{2}(9, 7, 5, 3, 1, 1)\frac{1}{2}(11, 5, 3, 3, 3, 1) \\ &\quad \frac{1}{2}(11, 5, 5, 1, 1, -1)\frac{1}{2}(13, 3, 3, 1, 1, 1) \\ \Lambda^{14} &: (4, 3, 3, 2, 2, 2)(4, 4, 3, 1, 1, 1)(5, 3, 2, 2, 1, 1)(5, 3, 3, 1, 0, 0)(6, 2, 2, 1, 1, 0) \\ &\quad (7, 1, 1, 1, 0, 0) \\ \Lambda^{15} &: \frac{1}{2}(7, 7, 5, 5, 5, 5)\frac{1}{2}(9, 7, 5, 3, 3, 3)\frac{1}{2}(9, 7, 7, 1, 1, 1)\frac{1}{2}(11, 5, 5, 3, 1, 1) \\ &\quad \frac{1}{2}(13, 3, 3, 3, 1, -1)\frac{1}{2}(15, 1, 1, 1, 1, -1) \\ \Lambda^{16} &: (3, 3, 3, 3, 3, 3)(4, 4, 2, 2, 2, 2)(4, 4, 4, 0, 0, 0)(5, 3, 3, 1, 1, 1)(6, 2, 2, 2, 0, 0) \\ &\quad (7, 1, 1, 1, -1)(8, 0, 0, 0, 0, 0)\end{aligned}$$

A5.3.3: Finally, we describe the restriction of $\Lambda^k(\text{spin}_{2n}^+)$ to Spin_{2n-1} , for $k \leq 8$, using the branching law of §5.4. For $n = 4, 5$, this gives the whole story, by duality. For $n = 6$, it is already sufficient to show that $\Lambda^*(\text{spin}_{11})$ is not multiplicity-free.

iii) $\Lambda_{\text{prim}}^+(\text{spin}_{12}^+) | \text{Spin}_{11}$:

$$\Lambda_{\text{prim}}^1 : \frac{1}{2}(1, 1, 1, 1, 1)$$

$$\Lambda_{\text{prim}}^2 : (1, 1, 1, 1, 0)(1, 1, 1, 0, 0)$$

$$\Lambda_{\text{prim}}^3 : \frac{1}{2}(3, 3, 3, 1, 1)\frac{1}{2}(3, 3, 1, 1, 1)$$

$$\Lambda_{\text{prim}}^4 : (2, 2, 1, 1, 1)(2, 1, 1, 1, 1)(2, 2, 2, 0, 0)(2, 2, 1, 0, 0)(2, 2, 0, 0, 0)$$

$$\Lambda_{\text{prim}}^5 : \frac{1}{2}(5, 3, 3, 3, 3)\frac{1}{2}(3, 3, 3, 3, 3)\frac{1}{2}(5, 5, 3, 1, 1)\frac{1}{2}(5, 5, 1, 1, 1)$$

$$\frac{1}{2}(5, 3, 3, 1, 1)\frac{1}{2}(5, 3, 1, 1, 1)$$

$$\Lambda_{\text{prim}}^6 : (2, 2, 2, 2, 2)(3, 2, 2, 1, 1)(3, 2, 1, 1, 1)(2, 2, 2, 1, 1)(2, 2, 1, 1, 1)(3, 3, 1, 1, 0)(3, 3, 1, 0, 0)(3, 2, 1, 1, 0)(3, 1, 1, 1, 0)(3, 1, 1, 0, 0)$$

$$\Lambda_{\text{prim}}^7 : \frac{1}{2}(5, 5, 5, 3, 3)\frac{1}{2}(5, 5, 3, 3, 3)\frac{1}{2}(7, 5, 3, 3, 1)\frac{1}{2}(7, 5, 3, 1, 1)$$

$$\frac{1}{2}(7, 3, 3, 3, 1)\frac{1}{2}(7, 3, 3, 1, 1)\frac{1}{2}(5, 5, 3, 3, 1)\frac{1}{2}(5, 5, 3, 1, 1)$$

$$\frac{1}{2}(5, 3, 3, 3, 1)\frac{1}{2}(5, 3, 3, 1, 1)\frac{1}{2}(7, 7, 1, 1, 1)\frac{1}{2}(7, 5, 1, 1, 1)\frac{1}{2}(7, 3, 1, 1, 1)$$

$$\frac{1}{2}(7, 1, 1, 1, 1)$$

$$\Lambda_{\text{prim}}^8 : (3, 3, 2, 2, 1)(3, 3, 2, 1, 1)(3, 2, 2, 2, 1)(3, 2, 2, 1, 1)$$

$$(4, 2, 2, 2, 0)(4, 2, 2, 1, 0)(4, 2, 2, 0, 0)(3, 2, 2, 2, 0)(3, 2, 2, 1, 0)(3, 2, 2, 0, 0)$$

$$(2, 2, 2, 2, 0)(2, 2, 2, 1, 0)(2, 2, 2, 0, 0)$$

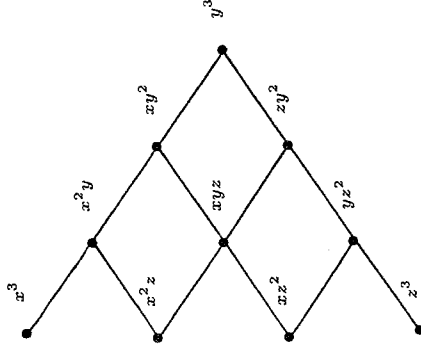
$$(4, 3, 1, 1, 1)(4, 3, 1, 1, 0)(4, 2, 1, 1, 1)(4, 2, 1, 1, 0)(4, 1, 1, 1, 1)(4, 1, 1, 1, 0)$$

$$(3, 3, 1, 1, 1)(3, 3, 1, 1, 0)(3, 2, 1, 1, 1)(3, 2, 1, 1, 0)(3, 1, 1, 1, 1)(3, 1, 1, 1, 0)$$

$$(4, 4, 0, 0, 0)(4, 3, 0, 0, 0)(4, 2, 0, 0, 0)(4, 1, 0, 0, 0)(4, 0, 0, 0, 0)$$

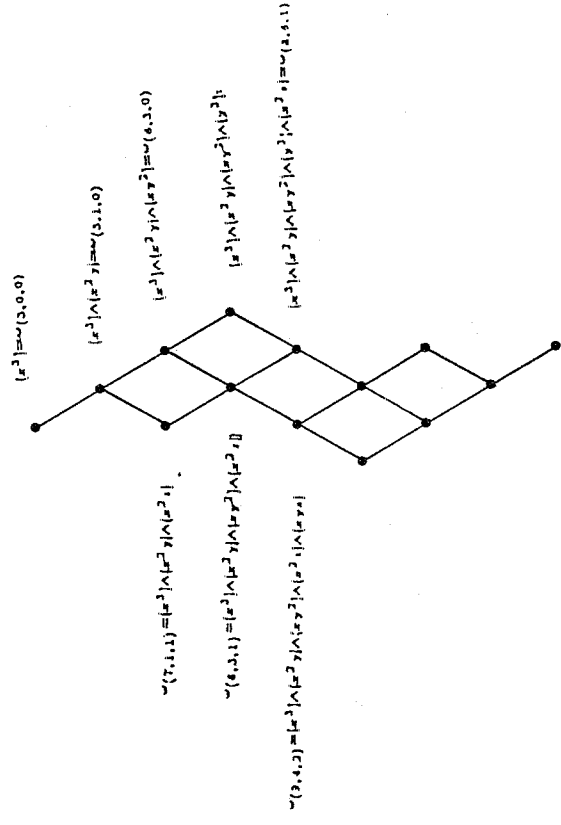
A5.4: Table of $\Lambda(S^3(\mathbb{C}^3))$.

A: The weight poset.



B: The geometric highest weight vectors (through degree 5).

They are denoted ω where $D = (a_1, a_2, a_3)$ is the highest weight.



C: Other highest weight vectors.

$$\begin{aligned}
\Lambda^2 : \omega_{(3,3,0)} &= [x^3] \wedge [y^3] - 3[x^2y] \wedge [xy^2] \\
\Lambda^3 : \omega_{(5,3,1)} &= [x^3] \wedge [x^2y] \wedge [y^2z] \\
&\quad + [x^3] \wedge [y^3] \wedge [x^2z] - [x^3] \wedge [xy^2] \wedge [xyz] \\
&\quad - 2[x^2y] \wedge [xy^2] \wedge [x^2z] \\
\omega_{(3,3,3)} &= [x^3] \wedge [y^3] \wedge [z^3] - 3([x^3] \wedge [y^2z] \wedge [yz^2] \\
&\quad + [x^2z] \wedge [y^3] \wedge [xz^2] + [x^2y] \wedge [xy^2] \wedge [z^3]) \\
&\quad + 3([x^2y] \wedge [y^2z] \wedge [xz^2] + [x^2z] \wedge [xy^2] \wedge [yz^2]) \\
&\quad + 6[xyz] \wedge [x^2z] \wedge [y^2z] - [x^2y][yz^2] + [xy^2] \wedge [xz^2]) \\
\Lambda_4 : (\omega_{7,4,1} &= [x^3] \wedge [x^2y] \wedge [x^2z] - 3[x^3] \wedge [x^2y] \wedge [xyz] \wedge [xy^2] \\
&\quad \omega_{(6,4,2)} = (\omega_{(5,3,1)} \wedge \omega_{(3,3,3)})^* \\
&\quad \omega_{(6,3,3)} = [x^3] \wedge \omega_{(3,3,3)} \\
\Lambda_5 : \omega_{(8,4,3)} &= \omega_{(5,1,0)} \wedge \omega_{(3,3,3)} \quad \omega_{(7,6,2)} = \omega_{(8,4,3)}^* \\
&\quad \omega_{(6,6,3)} = \omega_{(3,3,0)} \wedge \omega_{(3,3,3)} \quad \omega_{(7,4,4)} = \omega_{(6,6,3)}^*
\end{aligned}$$

If ω_D is a highest weight vector of ρ_3^D in $\Lambda^k(S^3(\mathbb{C}^3))$, then ω_D^* is the highest weight vector $\Lambda^{10-k}(S^3(\mathbb{C}^3))$ of ρ_3^D where $\bar{D} = (10 - a_3, 10 - a_2, 10 - a_1)$ if $D = (a_1, a_2, a_3)$. Equation (A.4.1.1) tells us that ω_D^* must exist. Dimension counting tells us that the representations generated by the above highest weight vectors exhaust all of $\Lambda^k(S^3(\mathbb{C}^3))$ for $k \leq 5$. For $k > 5$, the structure of $\Lambda^k(S^3(\mathbb{C}^3))$ follows by duality.

D: Summary of $\Lambda^k(S^3(\mathbb{C}^3))$.

$$\begin{aligned}
\Lambda^1 &= \rho_3^{(3,0,0)} \\
\Lambda^2 &= \rho_3^{(5,1,0)} + \rho_3^{(3,3,0)} \\
\Lambda^3 &= \rho_3^{(6,3,0)} + \rho_3^{(7,1,1)} + \rho_3^{(5,3,1)} + \rho_3^{(3,3,3)} \\
\Lambda^4 &= \rho_3^{(6,6,0)} + \rho_3^{(8,3,1)} + \rho_3^{(7,4,1)} + \rho_3^{(6,4,2)} + \rho_3^{(6,3,3)} \\
\Lambda^5 &= \rho_3^{(8,6,1)} + \rho_3^{(9,4,2)} + \rho_3^{(8,4,3)} + \rho_3^{(6,6,3)} + \rho_3^{(7,6,2)} + \rho_3^{(7,4,4)}
\end{aligned}$$

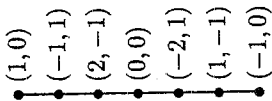
A5.5: Table of $\Lambda(V_7)$ for G_2 .

The dimension of $\Lambda^k(\mathbb{C}^7)$ is $\binom{7}{k}$. The largest is $\binom{7}{3} = 35 = \binom{7}{4}$. Using the dimension formula ([Hmp], page 140) we see that the non-trivial representations of G_2 of dimension 35 or less are:

representation	$\gamma_{(1,0)}$	$\gamma_{(0,1)}$	$\gamma_{(2,0)}$
dimension 7	14	27	

Here, the subscripts indicate the coefficients of the highest weight as a sum of the two fundamental weights. Thus, $\gamma_{(1,0)}$ is the first fundamental representation, $\gamma_{(0,1)}$ is the other fundamental representation (which happens to be the adjoint representation), and $\gamma_{(2,0)}$ is the Cartan square of $\gamma_{(1,0)}$.

The weight poset is linear:



We have written next to each element of the weight poset the weight of the corresponding vector, expressed in terms of the fundamental weights. It follows from this that there is one geometric highest weight vector in each degree, and in degrees 1, 2 and 3, the representations they generate are $\gamma_{(1,0)}$, $\gamma_{(0,1)}$, and $\gamma_{(2,0)}$, respectively. Knowing this, we can determine the rest of the components by dimension counting and symmetry. The result is

$$\begin{array}{ccccccc}
\Lambda^1 & \Lambda^2 & \Lambda^3 & \Lambda^4 & \Lambda^5 & \Lambda^6 & \\
\gamma_{(1,0)} & \gamma_{(0,1)} \oplus \gamma_{(1,0)} & \gamma_{(2,0)} \oplus \gamma_{(1,0)} \oplus \gamma_0 & \gamma_{(2,0)} \oplus \gamma_{(1,0)} \oplus \gamma_0 & \gamma_{(0,1)} \oplus \gamma_{(1,0)} & \gamma_{(1,0)} & \gamma_{(1,0)}
\end{array}$$

The invariant in degree 3 is part of the lore about G_2 : under the action of GL_7 on $\Lambda^3(\mathbb{C}^7)$, there is one open orbit, and the stabilizer of a point in that orbit is a copy of G_2 .

A5.6: Table of the exterior algebras $\Lambda(S^m(\mathbb{C}^2))$ for $k = 3, 4, 5, 6, 7$.

To compute these tables, we use a yoga parallel to the classical method for computing multiplicities of covariants of given degree for binary n -ics [Spr],[Hw2]. Let $b_j = x^{m-j}y^j$ be the familiar monomial basis for the homogeneous polynomials of degree m in two variables x and y . A basis for $\Lambda^r(S^m(\mathbb{C}^2))$ consists of wedge products $B_J = b_{j_1} \wedge b_{j_2} \wedge \dots \wedge b_{j_r}$ with $0 \leq j_1 < j_2 < \dots < j_r \leq m$. The weight of b_j for the standard diagonal element of $\mathfrak{sl}_2$ is $m - 2j$. Thus, the weight of B_J is $rm - 2 \sum_{a=1}^r j_a = r(m - \frac{r-1}{2}) - 2 \sum_{a=1}^r (j_a - a + 1)$. The numbers $j_a - a + 1$ are (weakly) increasing and not larger than $m - r + 1$. Thus, the number of B_J of weight $r(m - \frac{r-1}{2}) - 2\ell$ is the number $\pi(m - r + 1, r, \ell)$ of partitions of ℓ into at most r parts of size at most $m - r + 1$. This observation gives us a nice addendum to Hermite Reciprocity [Spr],[Hw2].

(T.1). There is an isomorphism

$$\Lambda^r(S^m(\mathbb{C}^2)) \simeq S^r(S^{m-r+1}(\mathbb{C}^2))$$

of SL_2 -modules. This fact has been observed in the literature, but does not seem to be widely known.

The multiplicity formula for weights immediately translates into a multiplicity formula for irreducible representations. See for example [Spr],[Hw2]. Let σ_ℓ denote the representation of SL_2 on the homogeneous polynomials of degree ℓ on $\mathbb{C}^2$. Then the multiplicity of σ_ℓ in $\Lambda^r(S^m(\mathbb{C}^2))$ is

$$\pi(m - r + 1, r, d) - \pi(m - r + 1, r, d - 1)$$

where $\ell = r(m - \frac{r-1}{2}) - 2d$. Using this formula, one easily constructs the tables below. In these tables, the module whose exterior algebra is being described is in the first column, and the constituents of each exterior power of this module are listed in the row to the left of the given module. Note that $\Lambda^{m+1-r}(S^m(\mathbb{C}^2)) \simeq \Lambda^r(S^m(\mathbb{C}^2))$, so the tables are symmetric around the middle dimension.

i) $S^3(\mathbb{C}^2)$:

$$\begin{array}{c} \Lambda^1 \\ \sigma_3 \end{array} \quad \begin{array}{c} \Lambda^2 \\ \sigma_4 \oplus \sigma_0 \end{array} \quad \begin{array}{c} \Lambda^3 \\ \sigma_3 \end{array}$$

ii) $S^4(\mathbb{C}^2)$:

$$\begin{array}{c} \Lambda^1 \\ \sigma_4 \end{array} \quad \begin{array}{c} \Lambda^2 \\ \sigma_6 \oplus \sigma_2 \end{array} \quad \begin{array}{c} \Lambda^3 \\ \sigma_6 \oplus \sigma_2 \end{array} \quad \begin{array}{c} \Lambda^4 \\ \sigma_4 \end{array}$$

iii) $S^5(\mathbb{C}^2)$:

$$\begin{array}{c} \Lambda^1 \\ \sigma_5 \end{array} \quad \begin{array}{c} \Lambda^2 \\ \sigma_8 \oplus \sigma_4 \oplus \sigma_0 \end{array} \quad \begin{array}{c} \Lambda^3 \\ \sigma_9 \oplus \sigma_5 \oplus \sigma_3 \end{array} \quad \begin{array}{c} \Lambda^4 \\ \sigma_8 \oplus \sigma_4 \oplus \sigma_0 \end{array} \quad \begin{array}{c} \Lambda^5 \\ \sigma_5 \end{array}$$

iv) $S^6(\mathbb{C}^2)$:

$$\begin{array}{c} \Lambda^1 \\ \sigma_6 \end{array} \quad \begin{array}{c} \Lambda^1 \\ \sigma_{10} \oplus \sigma_6 \\ \oplus \sigma_2 \end{array} \quad \begin{array}{c} \Lambda^3 \\ \sigma_{12} \oplus \sigma_8 \\ \oplus \sigma_6 \oplus \sigma_4 \\ \oplus \sigma_0 \end{array} \quad \begin{array}{c} \Lambda^4 \\ \sigma_{12} \oplus \sigma_8 \\ \oplus \sigma_6 \oplus \sigma_4 \\ \oplus \sigma_0 \end{array} \quad \begin{array}{c} \Lambda^5 \\ \sigma_{10} \oplus \sigma_6 \\ \oplus \sigma_2 \end{array} \quad \begin{array}{c} \Lambda^6 \\ \sigma_6 \end{array}$$

v) $S^7(\mathbb{C}^2)$:

$$\begin{array}{c} \Lambda^1 \\ \sigma_7 \end{array} \quad \begin{array}{c} \Lambda^1 \\ \sigma_{12} \oplus \sigma_8 \\ \oplus \sigma_4 \oplus \sigma_0 \end{array} \quad \begin{array}{c} \Lambda^3 \\ \sigma_{15} \oplus \sigma_{11} \\ \oplus \sigma_9 \oplus \sigma_7 \\ \oplus \sigma_5 \oplus \sigma_3 \end{array} \quad \begin{array}{c} \Lambda^4 \\ \sigma_{16} \oplus \sigma_{12} \oplus \sigma_{10} \\ \oplus 2\sigma_8 \oplus 2\sigma_4 \\ \oplus \sigma_0 \end{array} \quad \begin{array}{c} \Lambda^5 \\ \sigma_{15} \oplus \sigma_{11} \\ \oplus \sigma_9 \oplus \sigma_7 \\ \oplus \sigma_5 \oplus \sigma_3 \end{array} \quad \begin{array}{c} \Lambda^6 \\ \sigma_{12} \oplus \sigma_8 \\ \oplus \sigma_4 \oplus \sigma_0 \end{array} \quad \begin{array}{c} \Lambda^7 \\ \sigma_8 \end{array}$$

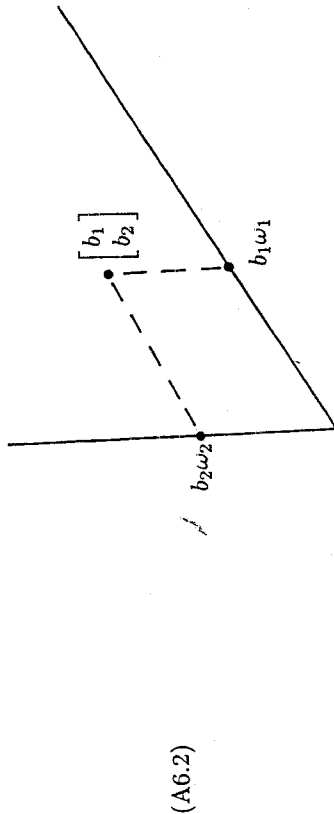
Appendix 6: Geometry of Tensor Products for SL_3 .

Irreducible representations of SL_3 may be parametrized by diagrams with two rows:

$$(A6.1) \quad \begin{array}{|c|c|} \hline a_1 & \\ \hline a_2 & \\ \hline \end{array}$$

We will label the representation corresponding to this vector by the pair $\begin{bmatrix} a_1 - a_2 \\ a_2 \end{bmatrix}$ of non-negative integers. These are the coefficients of the highest weight with respect to the two fundamental weights ω_1, ω_2 . We will write $\rho \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ for the representation with highest weight $b_1\omega_1 + b_2\omega_2$.

We will follow common practice and write the weight lattice of SL_3 as the hexagonal lattice in the plane with ω_1 and ω_2 generating a cone C^+ of opening angle 60° . This is the cone of dominant weights



We want to describe the constituents $\rho \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ of a tensor product $\rho \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \otimes \rho \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ in terms of the geometry of the triple $\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \right) \simeq \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix}$ of highest weights. We use the description of $\mathcal{R}((G/U) \times (G/U))^U$ given in §5.6.5. This algebra has 8 generators listed in formula (5.6.5.6), and there is one relation, stated in formula (5.6.5.7). We use that relation to rewrite any monomial involving both generators and with positive exponent in terms of monomials involving only one or the other of $\Delta(0,2)$ and $\delta_{(1,1)}$. This way to writing an element of $\mathcal{R}((G/U) \times (G/U))^U$ is unique. We will call a monomial in the generators (5.6.5.6) reduced if it involves only one of $\Delta(0,2)$ or $\delta_{(1,1)}$.

Each of the generators (5.6.5.6) is homogeneous with respect to the canonical $(A^+)^3$ -grading on $\mathcal{R}((G/U) \times (G/U))^U$. We record the weights of the generators as 2×3 matrices whose columns describe the coordinate of the 3 relevant highest weights with respect to the fundamental weight vectors ω_1 and ω_2 , as described

above.

$$(A6.3) \quad u_1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, u_2 = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}, v_1 = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix},$$

$$w_{11} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, w_{21} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, w_{12} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, w_{22} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

A monomial in the generators (5.6.5.6) has weight equal to the sum of weights (A6.3), with the exponents as coefficients. The collection of monomials in the generators (5.6.5.6) which yield highest weight vectors in a given tensor product $\rho \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \times \rho \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ are those corresponding to sums $\begin{bmatrix} a_1 & b_1 & x \\ a_2 & b_2 & y \end{bmatrix}$. For given $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, we want to describe the possible $\begin{bmatrix} x \\ y \end{bmatrix}$ and to see which and how many reduced monomials occur for a given $\begin{bmatrix} x \\ y \end{bmatrix}$.

First, consider which sums of the matrices (A6.3) yield a matrix with given two first columns. We will write $\approx$ to indicate that two 2×3 matrices agree in their first two columns. We see that the first four matrices in (A6.3) are independent with respect to $\approx$, and that each of the last four can be expressed in terms of the first four:

$$(A6.4) \quad w_{11} \approx u_1 + v_1 \quad w_{21} \approx u_2 + v_1 \quad w_{12} \approx u_1 + v_2 \quad w_{22} \approx u_2 + v_2.$$

Suppose that

$$(A6.5) \quad \begin{bmatrix} a_1 & b_1 & x \\ a_2 & b_2 & y \end{bmatrix} = \alpha_1 u_1 + \alpha_2 u_2 + \beta_1 v_1 + \beta_2 v_2 + \gamma_{11} w_{11} + \gamma_{21} w_{21} + \gamma_{12} w_{12} + \gamma_{22} w_{22}$$

$$\approx (\alpha_1 + \gamma_{11} + \gamma_{12})u_1 + (\alpha_2 + \gamma_{21} + \gamma_{22})u_2 + (\beta_1 + \gamma_{11} + \gamma_{21})$$

$$+ (\beta_2 + \gamma_{12} + \gamma_{22})v_2$$

$$= a_1 u_1 + a_2 u_2 + b_1 v_1 + b_2 v_2$$

We may think of this sum as arising by starting with the last sum, which involves only u_j and v_k , and gradually using w s to replace the appropriate combinations of u s and v s. We can continue with this replacement process as long as we don't force the α_j and β_k to become negative. Thus, the coefficients γ_{ij} are arbitrary subject to the conditions

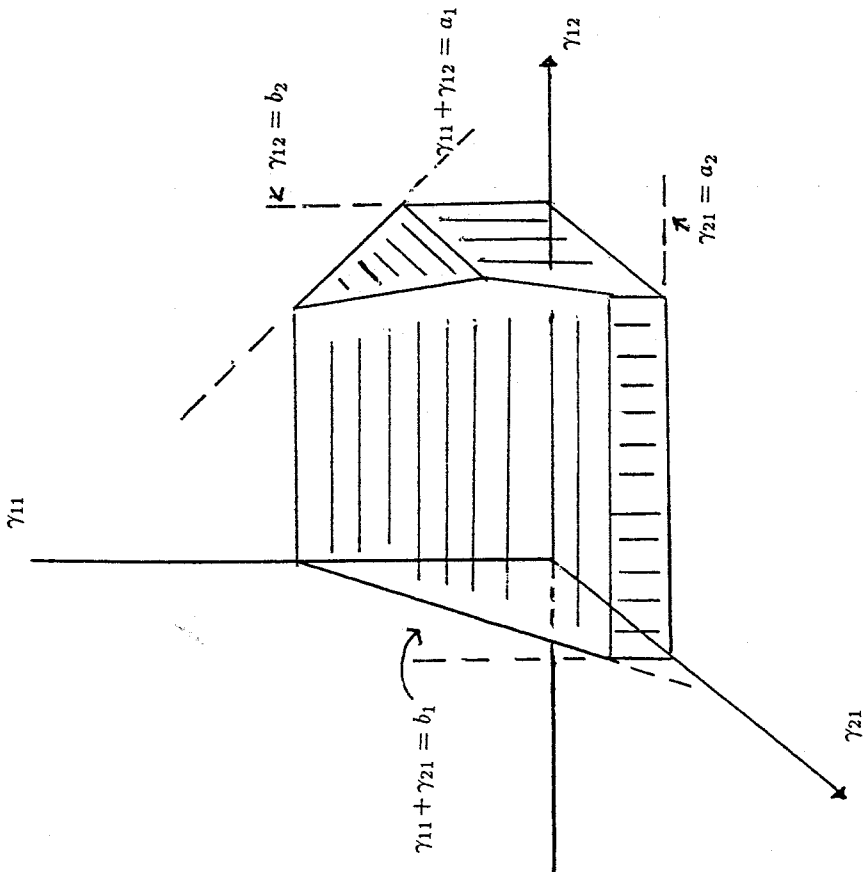
$$(A6.6) \quad \gamma_{11} + \gamma_{12} \leq a_1 \quad \gamma_{21} + \gamma_{22} \leq a_2$$

$$\gamma_{11} + \gamma_{21} \leq b_1 \quad \gamma_{12} + \gamma_{22} \leq b_2$$

We should only consider sums corresponding to reduced monomials. This means we should take either $\gamma_{11} = 0$ or $\gamma_{22} = 0$. First, take $\gamma_{22} = 0$. Then the system (A6.6) reduces to

$$(A6.7) \quad \gamma_{11} + \gamma_{12} \leq a_1 \quad \gamma_{11} + \gamma_{21} \leq b_1 \quad \gamma_{21} \leq a_2 \quad \gamma_{12} \leq b_2$$

The solid in $(\gamma_{11}, \gamma_{12}, \gamma_{21})$ -space defined by these inequations (together with $\gamma_{jk} \geq 0$), is sketched in Figure (A6.8):



Given a triple $(\gamma_{11}, \gamma_{12}, \gamma_{21})$ satisfying (A6.7) the corresponding matrix is

$$\begin{bmatrix} a_1 & b_1 & x \\ a_2 & b_2 & y \end{bmatrix} \text{ where}$$

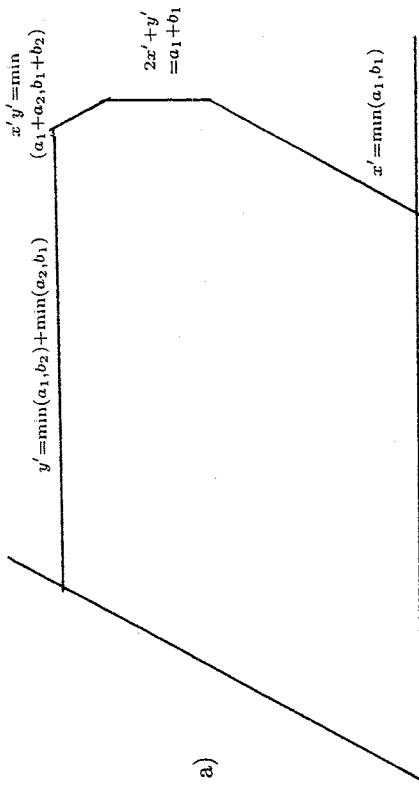
$$(A6.9) \quad \begin{aligned} x &= a_1 - \gamma_{11} - \gamma_{21} + b_1 - \gamma_{11} - \gamma_{21} = a_1 + b_1 - 2\gamma_{11} - \gamma_{12} - \gamma_{21} \\ y &= a_2 - \gamma_{21} + b_2 - \gamma_{12} + \gamma_{11} = a_2 + b_2 - (\gamma_{21} + \gamma_{12}) + \gamma_{11}. \end{aligned}$$

Thus, we see that the "defect"

$$(A6.10) \quad \begin{bmatrix} a_1 + b_1 - x \\ a_2 + b_2 - y \end{bmatrix} = \begin{bmatrix} 2\gamma_{11} + (\gamma_{12} + \gamma_{21}) \\ -\gamma_{11} + (\gamma_{12} + \gamma_{21}) \end{bmatrix} = \gamma_{11} \begin{bmatrix} 2 \\ -1 \end{bmatrix} + (\gamma_{12} + \gamma_{21}) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

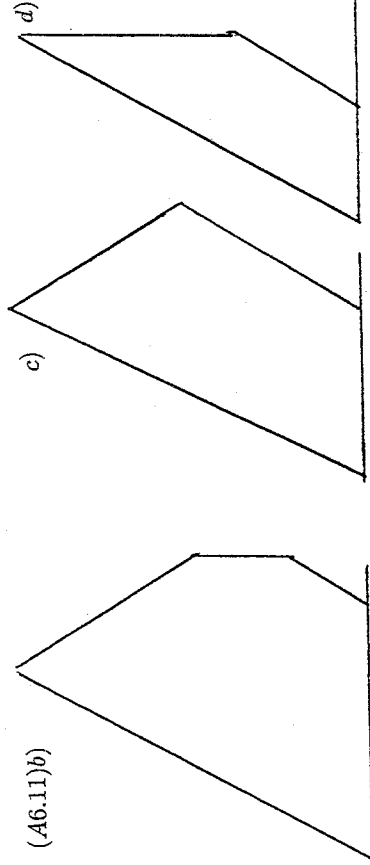
only depends on $\gamma_{11} = x'$ and on the sum $\gamma_{12} + \gamma_{21} = y'$. The projection of the region (A6.8) to the weight plane (A6.2) via the "defect map" (A6.10) looks like this:

(A6.11) a)



Of course, depending on the relative sizes of a_1, a_2, b_1 , and b_2 , one or more of the boundary lines pictured in Figure (A6.11)a) may become irrelevant, or "slack" in the language of linear programming. In particular, if $(a_1 - b_2)(a_2 - b_1) \geq 0$ the segment $y' = \min(a_1, b_2) + \min(a_2, b_1)$ intersects the defect region only one point on the y' -axis, and the defect region has one of these shapes.

(A6.11) b)



We can perform a similar analysis when $\gamma_{11} = 0$. Then the system (A6.6) of inequalities reduces to

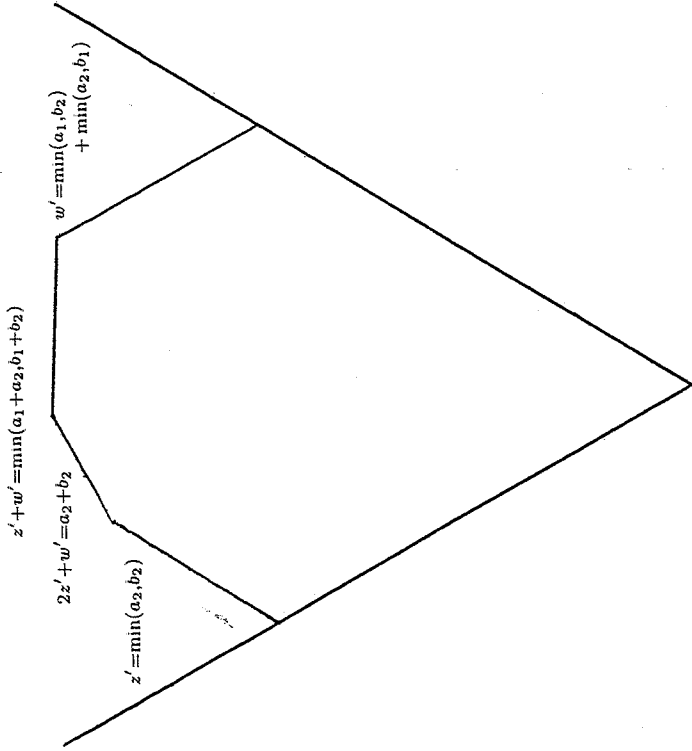
$$(A6.12) \quad \gamma_{12} \leq a_1 \quad \gamma_{21} \leq b_1 \quad \gamma_{21} + \gamma_{22} \leq a_2 \quad \gamma_{12} + \gamma_{22} \leq b_2$$

This is essentially the same system as (A6.7), but with γ_{11} replaced by γ_{22} , and a_1 switched with b_2 , a_2 switched with b_1 . The defect for this case is

$$(A6.13) \quad \begin{bmatrix} \gamma_{12} + \gamma_{21} - \gamma_{22} \\ \gamma_{12} + \gamma_{21} + 2\gamma_{22} \end{bmatrix} = \gamma_{22} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + (\gamma_{12} + \gamma_{21}) \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

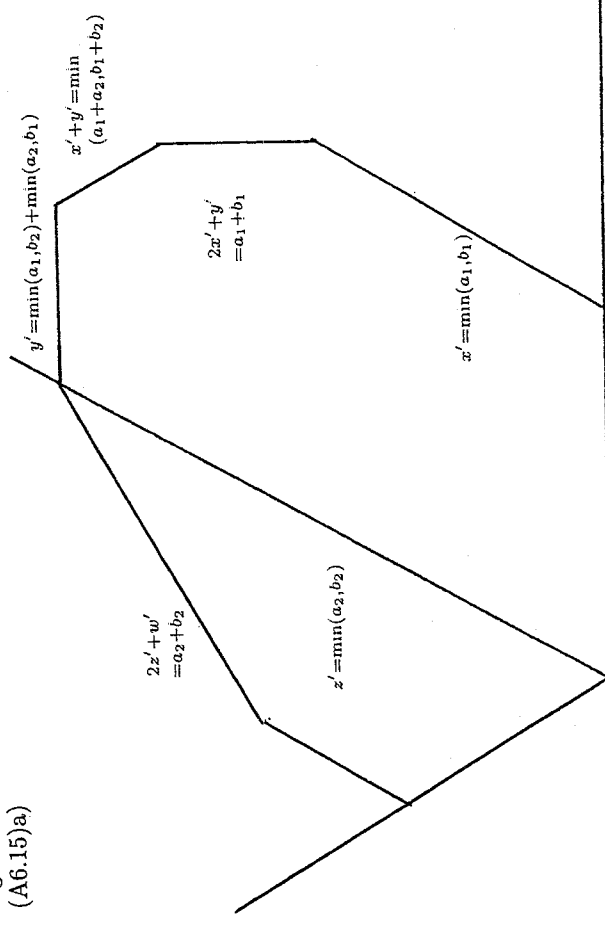
The defect region, i.e., the image of the region (A6.12) under the defect map

(A6.13), is illustrated in Figure (A6.14). Here, $z' = \gamma_{22}$ and $w' = \gamma_{12} + \gamma_{21}$. (A6.14)



Again, the region (A6.14) may degenerate into regions analogous to (A6.11)b), c), d).

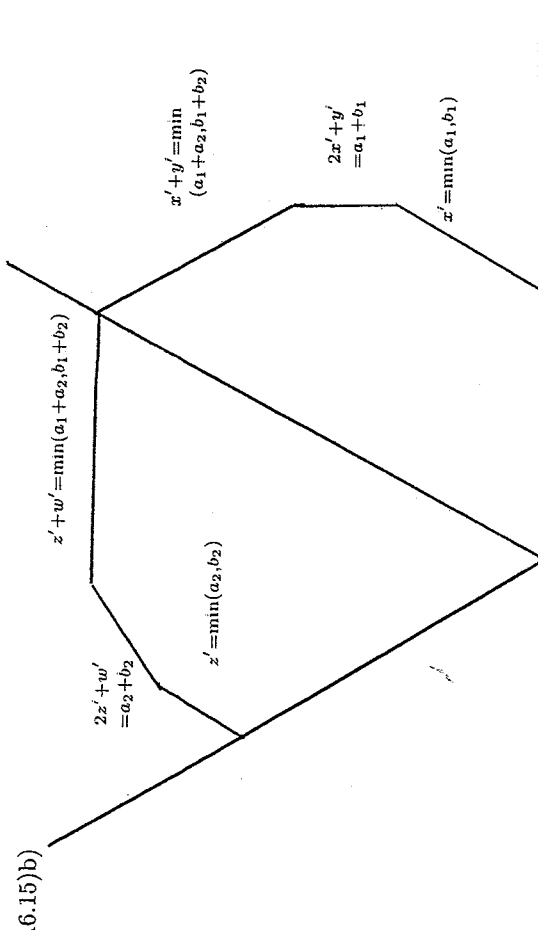
The total defect region is the union of the regions (A6.11) and (A6.14). We note that along their common boundary line, e.g., the line $x' = 0$ in Figure (A6.11)a), and the line $z' = 0$ in Figure (A6.14), the two regions match up: both include exactly the segment $\{t \begin{bmatrix} 1 \\ 1 \end{bmatrix} : 0 \leq t \leq \min(a_1, b_2) + \min(a_2, b_1)\}$. We can say more: the union of the two regions is convex. For example, either by inspection of Figure (A6.8) and its analog for the system (A6.13), or by analysis of the inequalities defining the regions (A6.11)a) and (A6.14), we can see that if the line $y' = \min(a_1, b_2) + \min(a_2, b_1)$ forms a non-trivial part of the boundary of region (A6.11)a), then region (A6.14) must degenerate to the analog of (A6.11)d). Thus, in this case we get a defect region which looks like this: (A6.15)a).



If the line $w' = \min(a_1, b_2) + \min(a_2, b_1)$ forms a non-trivial part of the boundary of the region (A6.14), then the appropriate form of the other region is (A6.11)d). The union of the two regions looks like the reflection of (A6.15)a) through the common boundary. (This is the line $x' = 0$ or $z' = 0$; we will call it the central axis.)

The Figure (A6.15)a) and its reflection through the central axis correspond to the situation $(a_1 - b_2)(a_2 - b_2) < 0$. If $(a_1 - b_2)(a_2 - b_1) \geq 0$, then the defect regions take the shape (A6.11)b) (or possibly (A6.11)c)), and the total defect region looks like this.

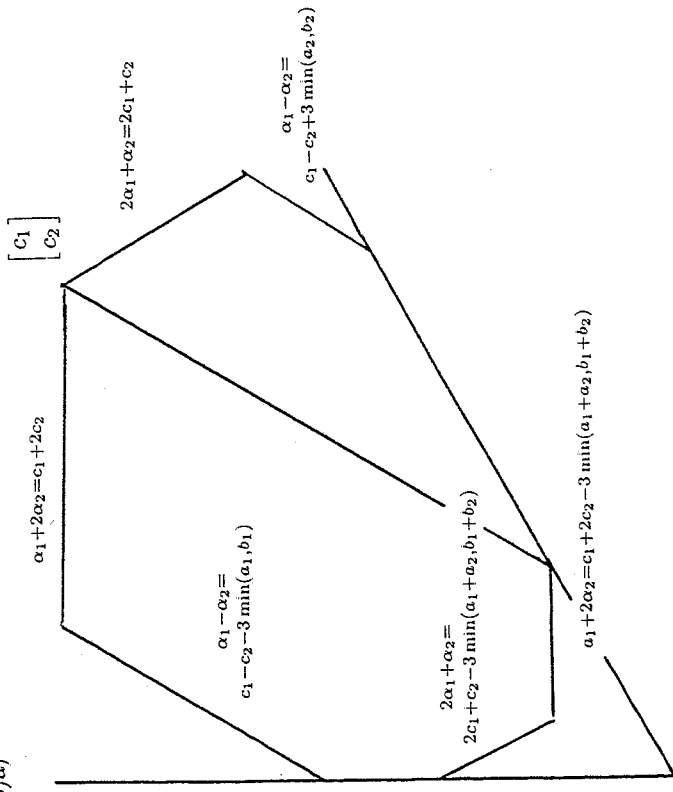
(A6.15)b)



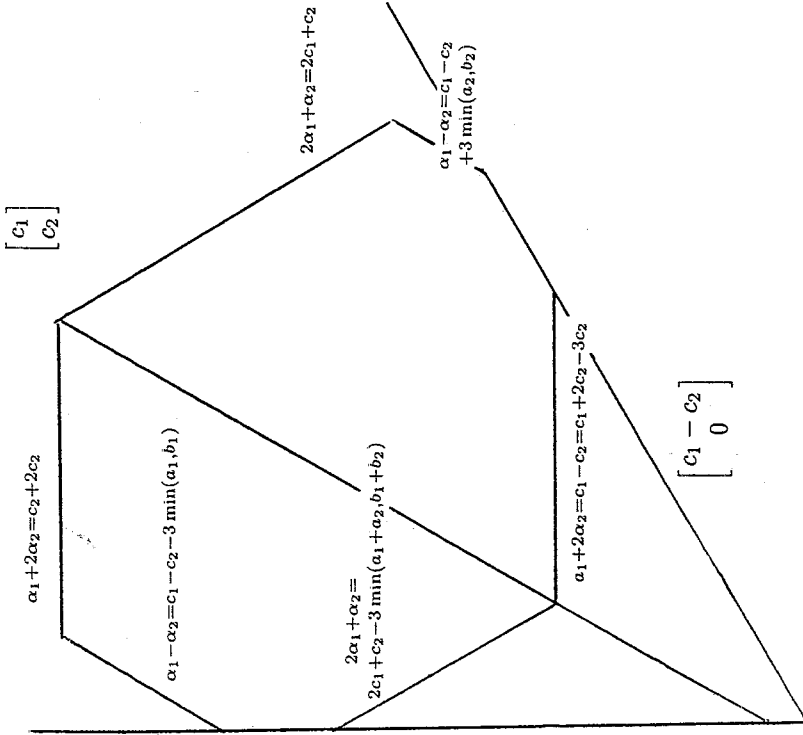
To find the set of highest weights in $\rho \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \otimes \rho \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, we subtract the appropriate one of the defect regions (A6.15) from the highest weight. To do this explicitly, we must convert the coordinate systems used in constructing Figures (A6.15)a), b) to the coordinate system with respect to the fundamental weights ω_1 and ω_2 , illustrated in Figure (A6.2). This may be done by means of formulas (A6.10) and (A6.13). The results are given in Figure (A6.16)a), b), which arise from Figures (A6.15)a), b), respectively. We use α_1, α_2 as coordinates with respect to ω_1, ω_2 .

We write $\begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$.

(A6.16)a)



(b)



Remarks: A standard result about tensor products [HMP] says that a highest weight of $\rho \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \otimes \rho \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$ has the form $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$ where $\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$ is a weight of $\rho \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$. Thus, the region of highest weights must be contained in $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} +$

$wr \left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right)$, where $wr \left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right)$ is the convex hull of the set of weights of $\rho \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$. Similarly, the region of highest weights is contained in $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + wr \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \right)$. A standard calculation gives the following description of $wr \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \right)$.

(A6.17)

$wr \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \right)$ is the set of $\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ satisfying the inequalities

- $-(2a_1 + a_2) \leq \alpha_1 + 2\alpha_2 \leq a_1 + 2a_2$
- $-(a_1 + 2a_2) \leq 2\alpha_1 + \alpha_2 \leq 2a_1 + a_2$
- $-(2a_1 + a_2) \leq \alpha_1 - \alpha_2 \leq a_1 + 2a_2$

It follows that the region $\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + wr \left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right) \right) \cap \left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + wr \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \right) \right)$ is described by the inequalities

$$(A6.18) \quad \begin{aligned} c_1 + 2c_2 - 3\gamma &\leq \alpha_1 + 2\alpha_2 \leq c_1 + 2c_2 \\ 2c_1 + c_2 - 3\gamma &\leq 2\alpha_1 + \alpha_2 \leq 2c_1 + c_2 \\ c_1 - c_2 - 3 \min(a_1, b_1) &\leq \alpha_1 - \alpha_2 \leq c_1 - c_2 + 3 \min(a_2, b_2) \end{aligned}$$

where $\gamma = \min(a_1 + a_2, b_1 + b_2)$.

Of course, highest weights must also belong to the cone C^+ of dominant weights.

Thus, set

$$(A6.19) \quad R \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right) = \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + wr \left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right) \right) \cap \left(\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + wr \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \right) \right) \cap C^+.$$

We may regard $R \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right)$ as the "trivial estimate" for the region of highest

weight vectors of $\rho \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \otimes \rho \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$. Comparison with Figures (A6.16) shows that

when $(a_1 - b_2)(a_2 - b_1) \geq 0$, illustrated in (A6.16)b), the highest weight region

coincides with $R \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right)$. However, when $(a_1 - b_2)(a_2 - b_1) < 0$, illustrated

in (A6.16)a), the highest weight region is strictly smaller than $R \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right)$. It

coincides in seven of eight sides with $R \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right)$, but the constraint

$\alpha_1 + 2\alpha_2 \geq c_1 - c_2 = c_1 + 2c_2 - 3c_2$, defining the bottom horizontal face of the highest weight region, is strictly stronger than the corresponding constraint

$$\alpha_1 + 2\alpha_2 \geq c_1 + 2c_2 - 3 \min(a_1 + a_2, b_1 + b_2) \text{ of } R \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \right).$$

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ARITHMETIC QUANTUM CHAOS*

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Abstract. Quantum Chaos is concerned with the study of the quantization of a classically chaotic Hamiltonian. In these notes we investigate this problem for the Hamiltonian which is the geodesic flow on an arithmetic hyperbolic manifold. We report on recent numerical computations of the eigenvalues and eigenfunctions of the Laplacian for such manifolds. We then study their fine structure by relating these problems to ones about L -functions, exponential sums and especially the theory of correspondences (which such manifolds carry).

Section 1. Background

These lectures are aimed at a general audience. A detailed and up to date account of Quantum Chaos (Q.C.) can be found in the proceedings of the recent conference "Chaos and Quantum Physics" [G-V-Z] and in Gutzwiller's article [GU]. The mathematical problems associated with Q.C. are very difficult and one must be satisfied with proving partial results and gaining insight into the truth. Our main focus will be on problems of Q.C. associated with arithmetic hyperbolic manifolds. For these we can bring to bear techniques from analytic number theory and arithmetic. This allows us to resolve some of the problems and also gain understanding of the key issues. It turns out that some of these key issues are intimately connected with well known problems in the analytic theory of L -functions. In much of theoretical physics, the theory revolves around certain well understood examples (harmonic oscillator, hydrogen atom...). With time, as more is established about the spectra of arithmetic hyperbolic manifolds, we hope that they become central models for Q.C. The new results (see Section 3) described in these lectures have been obtained in separate collaborations with H. Iwaniec, W. Luo and Z. Rudnick.

The basic problem of Q.C. is to understand the quantization of a classical Hamiltonian system whose dynamics are "chaotic." By this we mean that the flow when

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